



EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB, 639-94441, JP, ULTRA)	EEEE_J08R	CRITICAL	EEEE_ULTIMATE
825-7691	1	EEEE FOR (MLB, 639-93377, JP, EXTREME)	EEEE_HNV8	CRITICAL	EEEE_EXTREME
825-7691	1	EEEE FOR (MLB, 639-94442, ROW, ULTRA)	EEEE_J08X	CRITICAL	EEEE_ULTIMATE_ROW
825-7691	1	EEEE FOR (MLB, 639-93389, ROW, EXTREME)	EEEE_HNVF	CRITICAL	EEEE_EXTREME_ROW

D21x Specific

D21x Specific

D21x Specific

D21x Specific

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00416	1	SoC, H, 3GB 23nm DDR, B1	U1000	CRITICAL	COMMON

D21x Specific

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00415	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00417	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00418	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 1xnm DDR, B1
339S00429	339S00416	BOM_TABLE_ALTS	U1000	SoC, H, 3GB 20nm DDR, B1
339S00430	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00431	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00432	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 18nm DDR, B1

SIPs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339M00011	339M00010	BOM_TABLE_ALTS	M5500	NIKI, STATS SIP
339M00013	339M00012	BOM_TABLE_ALTS	M4100	HUNT, STATS SIP

D21x Specific

SOFT-TERM CAP SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00152	685-00151	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM_MLB, CAP, SOFT, D21
138S00049	138S0031	TYPICAL_CAP	ALL	CAP, 0201, 2.2UF, 6.3V, KYOCERA

To enable alt in SUBBOM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00151	1	SUBBOM_MLB, SOFT-TERM, D21	SUBBOM_CAP		
138S00159	9	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2000, C2001, C2003, C2005, C2007, C2008, C2013		SOFT_CAP
138S0031	9	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2000, C2001, C2003, C2005, C2007, C2008, C2013		TYPICAL_CAP
138S00159	15	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2051, C2052, C2054, C2055, C2057, C2060, C2068		SOFT_CAP
138S0031	15	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2051, C2052, C2054, C2060, C2062, C2066, C2068		TYPICAL_CAP
138S00159	10	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C379, C375, C374, C376, C378, C371, C360, C361		SOFT_CAP
138S0031	10	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C379, C375, C374, C376, C378, C371, C360, C361		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3351, C3352		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3351, C3352		TYPICAL_CAP
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3602, C3611, C3622		SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3602, C3611, C3622		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3791, C3795, C3798		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3791, C3795, C3798		TYPICAL_CAP
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3909, C3925, C3926		SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3909, C3925, C3926		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4250, C4261		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4250, C4261		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4303		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4303		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4885, C4889, C4914, C4926, C4929		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4885, C4889, C4914, C4926, C4929		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5010, C5018, C5134, C5029, C5136		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5010, C5018, C5134, C5029, C5136		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5611		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5611		TYPICAL_CAP
138S00159	4	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5700, C5705, C5716, C5721		SOFT_CAP
138S0031	4	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5700, C5705, C5716, C5721		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5802, C5804, C5811, C5888		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5802, C5804, C5811, C5888		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5890		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5890		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C6320, C6321, C6322, C6323, C6324		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C6320, C6321, C6322, C6323, C6324		TYPICAL_CAP
138S00160	4	CAP, SOFT-TERM, 18UF, 10V, 0402, MURATA	C4907, C4903, C4904, C4931		SOFT_CAP
138S00979	4	CAP, TYPICAL, 18UF, 10V, 0402, MUR/KYO	C4907, C4903, C4904, C4931		TYPICAL_CAP
138S00160	5	CAP, SOFT-TERM, 18UF, 10V, 0402, MURATA	C5680, C5682, C5693, C5702, C5703		SOFT_CAP
138S00979	5	CAP, TYPICAL, 18UF, 10V, 0402, MUR/KYO	C5680, C5682, C5693, C5702, C5703		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4825, C4826		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4825, C4826		TYPICAL_CAP

Default is typical

D21x Specific

D21x Specific

IKTARA DIODES SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00172	685-00171	BOM_TABLE_ALTS	SUBBOM_DS	SUBBOM_MLB, SCHOTTKY DIODES, D21

#31526972: Update Iktara Diodes APNs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00171	1	SUBBOM_MLB, SCHOTTKY DIODES, D100ES, D21	SUBBOM_DS		
371S00133	4	DIODES, SHOTTKY DIODE, 30V, 2A, 9683	D3400, D3401, D3402, D3403		DIODES_DS
371S00132	4	ONSEMI, SHOTTKY DIODE, 30V, 2A, 9683	D3400, D3401, D3402, D3403		ONSEMI_DS

Footprint is ONSMT

Primary is D100ES



NAND BOM Options

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00287	1	NAND, H, ULTRA, TLC	U2600	CRITICAL	ULTIMATE
335S00240	1	NAND, H, EXTREME, TLC	U2600	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00284	335S00287	BOM_TABLE_ALTS	U2600	T, 1z, ULTIMATE
335S00285	335S00287	BOM_TABLE_ALTS	U2600	T, B1CS3, ULTIMATE
335S00288	335S00287	BOM_TABLE_ALTS	U2600	S, B1CS3, ULTIMATE
335S00286	335S00287	BOM_TABLE_ALTS	U2600	SS, 30v4, ULTIMATE
335S00228	335S00240	BOM_TABLE_ALTS	U2600	T, B1CS3, EXTREME
335S00247	335S00240	BOM_TABLE_ALTS	U2600	S, B1CS3, EXTREME
335S00276	335S00240	BOM_TABLE_ALTS	U2600	SS, 30v4, EXTREME

Isolator Switch

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S00126	311S00114	BOM_TABLE_ALTS	ALL	SPDT, TI

Mamba LDO

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00932	353S00956	BOM_TABLE_ALTS	U5890	STMICRO LDO

Load Switch

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00999	353S01039	BOM_TABLE_ALTS	U6100	TI, LOAD SW, TPS22955C
353S01007	353S01039	BOM_TABLE_ALTS	U6100	ONSEMI, LOAD SW, MCP333

Crystal Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0012	197S00118	BOM_TABLE_ALTS	Y1000	NDK, XTAL, 24MHz, 1.6x1.2
197S00120	197S00118	BOM_TABLE_ALTS	Y1000	EPSON, XTAL, 24MHz, 1.6x1.2

CRITICAL PART#	COMMENT
197S00118	XTAL, 24M, 1612, TDK
197S0012	XTAL, 24M, 1612, NDK
197S00120	XTAL, 24M, 1612, EPSON

PMID1 Zener Diode

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S00121	371S00030	BOM_TABLE_ALTS	DZ3300	DIODE, ZENER, 20V

ZRB Caps

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00683	1	1uF, 25V, 0402, XSR, 10%	C3301	CRITICAL	COMMON

CRITICAL PART#	COMMENT
138S00185	MUR, CAP, ZRB, 2.2UF, 25V, 0402
138S00186	MUR, CAP, ZRB, 4.7UF, 25V, 0603

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S00764	118S0717	BOM_TABLE_ALTS	ALL	RES, 3.92K, 0.1%, 0201
138S00648	138S0652	BOM_TABLE_ALTS	ALL	CAP, XSR, 4.7UF, 6.3V, 0.60MM, 0402, TAIYO
138S00024	138S0086	BOM_TABLE_ALTS	ALL	CAP, CER, 3.3UF, 1.5V, 20%, 0402, 1612/10704
138S00706	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 10P, 20V, 20%, 10V, 0301, NIDECITA
138S00945	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 10P, 20V, 20%, 10V, 0301, KYOCERA
138S00739	138S0706	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP, XSR, XSR, 0.22UF, 20%, 6.3V, 0305
138S00133	138S00128	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 0.47UF, 20%, 6.3V, 0305
138S00148	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 15UF, 4V, KYOCERA
138S00149	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 14UF, 4V, MURATA
138S00151	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 15UF, 4V, TAIYO
138S00144	138S00143	BOM_TABLE_ALTS	ALL	CAP, 0402, 28UF, 4V, MURATA
138S00143	138S00144	BOM_TABLE_ALTS	ALL	CAP, 0402, 22UF, 4V, KYOCERA
138S00139	138S00138	BOM_TABLE_ALTS	ALL	CAP, 0201, 1UF, 4V, MURATA
138S00138	138S00139	BOM_TABLE_ALTS	ALL	CAP, 0201, 4.7UF, 4V, KYOCERA
138S00145	138S00146	BOM_TABLE_ALTS	ALL	CAP, 0402, 28UF, 6.3V, KYOCERA
138S00141	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, MURATA
138S00142	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, SAMSUNG
138S00142	138S00141	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, SAMSUNG
138S00049	138S0031	BOM_TABLE_ALTS	ALL	CAP, 0201, 2.2UF, 6.3V, KYOCERA
138S00163	138S00143	BOM_TABLE_ALTS	ALL	CAP, 0402, 22UF, 4V, TY
138S00164	138S00138	BOM_TABLE_ALTS	ALL	CAP, 0201, 1UF, 4V, TY
138S00165	138S00146	BOM_TABLE_ALTS	ALL	CAP, 0402, 28UF, 6.3V, TY
138S00166	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 6.3V, TY
138S00140	138S00141	BOM_TABLE_ALTS	ALL	CAP, 0201, 1UF, 6.3V, KYO
138S00048	138S00003	BOM_TABLE_ALTS	ALL	CAP, XSR, 150V, 0.33UF, 0.30MM, 0402, KYO
132S0639	132S00088	BOM_TABLE_ALTS	ALL	CAP, XSR, 0.47UF, 25V, 0.20MM, 0201, MUR

CRITICAL PART#	COMMENT
118S0717	RES, 3.92K, 0.1%, 0201
138S00652	CAP, XSR, 4.7UF, 6.3V, 0.65MM, 0402
138S0706	CAP, CER, XSR, 1UF, 20%, 6.3V, 0201
132S0400	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 01005
138S0031	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201
138S00128	CAP, CER, XSR, 0.47UF, 20%, 6.3V, 01005

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
155S00610	FERR BD, 1500HM, 01005

CRITICAL PART#	COMMENT
152S00617	IND, MLD, 0.1UH, 20%, 6.1A, 29MOHM, H= .65, 1608
152S00620	IND, MLD, 0.1UH, 20%, 7.2A, 17MOHM, H=0.8, 2012
152S00621	IND, MLD, 0.47UH, 20%, 3.5A, 53MO, H= .65, 2012
152S00622	IND, MLD, 1.0UH, 20%, 2.1A, 100MO, H= .65, 2012
152S00626	IND, MLD, 1.5UH, 20%, 1.1A, 160MO, H= .65, 2012
152S00631	IND, MLD, 1.0UH, 20%, 2.5A, 78MO, H=0.8, 2012
152S00632	IND, MLD, 1.0UH, 20%, 3.2A, 60MO, H=0.8, 2016
152S00640	IND, MLD, 0.47UH, 3.8A, 55MO, H=0.65MM, 2012
152S00641	IND, MLD, 0.47UH, 4A, 48MO, H=0.8MM, 2012
152S00623	IND, MLD, 1UH, 3.6A, 60MO, H=0.8MM, 2016
152S00651	IND, 1.2UH, 3A, 2016, 0.65Z
152S00650	IND, 0.47UH, 6.6A, 3225, 0.8Z

CRITICAL PART#	COMMENT
152S00617	IND, MLD, 0.1UH, 20%, 6.1A, 29MOHM, H= .65, 1608
152S00620	IND, MLD, 0.1UH, 20%, 7.2A, 17MOHM, H=0.8, 2012
152S00621	IND, MLD, 0.47UH, 20%, 3.5A, 53MO, H= .65, 2012
152S00622	IND, MLD, 1.0UH, 20%, 2.1A, 100MO, H= .65, 2012
152S00626	IND, MLD, 1.5UH, 20%, 1.1A, 160MO, H= .65, 2012
152S00631	IND, MLD, 1.0UH, 20%, 2.5A, 78MO, H=0.8, 2012
152S00632	IND, MLD, 1.0UH, 20%, 3.2A, 60MO, H=0.8, 2016
152S00640	IND, MLD, 0.47UH, 3.8A, 55MO, H=0.65MM, 2012
152S00641	IND, MLD, 0.47UH, 4A, 48MO, H=0.8MM, 2012
152S00623	IND, MLD, 1UH, 3.6A, 60MO, H=0.8MM, 2016
152S00651	IND, 1.2UH, 3A, 2016, 0.65Z
152S00650	IND, 0.47UH, 6.6A, 3225, 0.8Z

Multi-Vendor Criticals

CRITICAL PART#	COMMENT	CRITICAL PART#	COMMENT
377S0106	SUPPR, TRANS, VARISTOR, 12V, 33PF, 01005	132S0288	CAP, CER, XSR, 0.1UF, 10%, 16V, 0201
197S0446	XTAL, 24MHZ, 30PPM, 9.5PF, 60 OHM MAX, 1612	132S0275	CAP, CER, XSR, 470PF, 10%, 10V, 01005
155S00576	FERR BD, 10 OHM, 50%, 750MA, 0.07 DCR, 01005	132S0249	CAP, CER, XTR, 220PF, 10%, 10V, 01005
155S00168	FLTR, NOISE, 65 OHMZ, 3.40HM, 0.7-26HZ, 0605	132S0245	CAP, CER, XSR, 0.01UF, 10%, 6.3V, 01005
130S00979	CAP, CER, XSR, 10UF, 20%, 10V, 0402, H=0.65MM	132S00093	CAP, XSR, 0.042UF, 20%, 6.3V, 01005
138S00692	CAP, CER, XSR, 1UF, 20%, 6.3V, 0201	132S00025	CAP, CER, XSR, 0.047UF, 20%, 6.3V, 01005
138S0683	CAP, CER, XSR, 1UF, 10%, 25V, 0402	132S00008	CAP, CER, 0.1UF, 10%, 50V, XTR, 0402
138S0652	CAP, CER, XSR, 4.7UF, 20%, 6.3V, H=0.65MM, 0402	131S00883	CAP, CER, NP0/C0G, 220PF, 2%, 50V, 0201
138S00070	CAP, XSR, 4.7UF, 20%, 25V, 0402	131S00004	CAP, CER, 27PF, 5%, C0G, 25V, 0201
138S00014	CAP, CER, 1UF, 20%, 16V, XSR, 0201, H=0.39MM	131S00307	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005
132S00664	CAP, CER, 0.047UF, 10%, 25V, XSR, 0201	131S00225	CAP, CER, NP0/C0G, 15PF, 5%, 16V, 01005
132S00663	CAP, CER, XSR, 1UF, 10%, 25V, 0402	131S00223	CAP, CER, NP0/C0G, 12PF, 5%, 16V, 01005
132S00534	CAP, CER, XSR, 0.1UF, 10%, 25V, 0201	131S00220	CAP, CER, NP0/C0G, 12PF, 5%, 16V, 01005
132S0436	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 01005	131S00216	CAP, CER, NP0/C0G, 47PF, 5%, 16V, 01005
132S00396	CAP, CER, XSR, 1000PF, 10%, 10V, 01005	131S00053	CAP, CER, C0G, 220PF, 5%, 10V, 01005
132S0316	CAP, CER, XSR, 0.1UF, 20%, 6.3V, 01005	118S00068	RES, MF, 1.3 MOHM, 1%, 200PPM, 1/20W, 0201
132S0304	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 0201	117S00055	RES, MF, 1/20W, 2M OHM, 5, 0201, SMD
132S0296	CAP, CER, XSR, 1000PF, 10%, 6.3V, 01005	107S0257	THERMISTOR, NTC, 10K OHM, 1%, B=3435, 01005

D

D

C

C

B

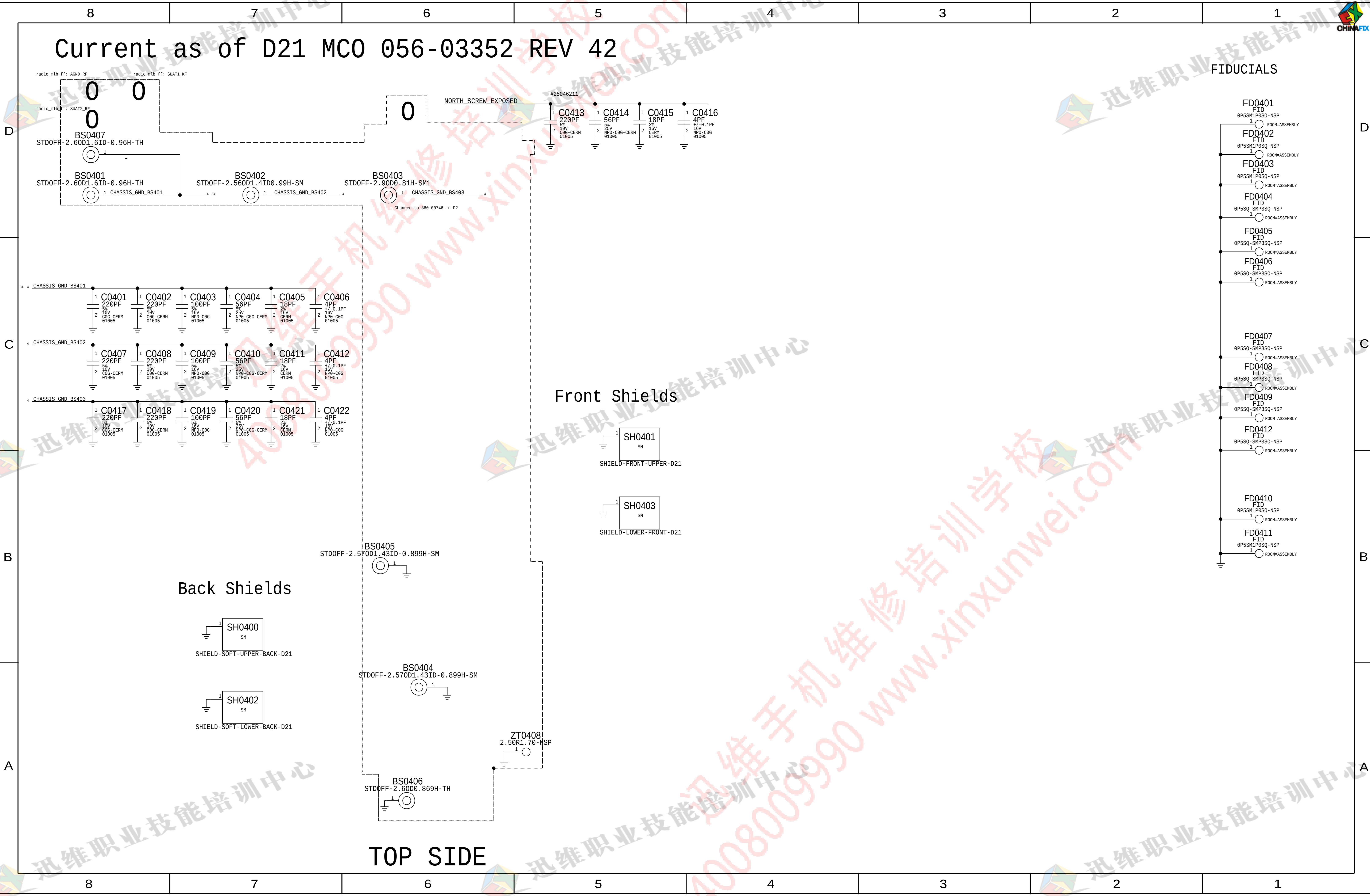
B

A

A



Current as of D21 MC0 056-03352 REV 42

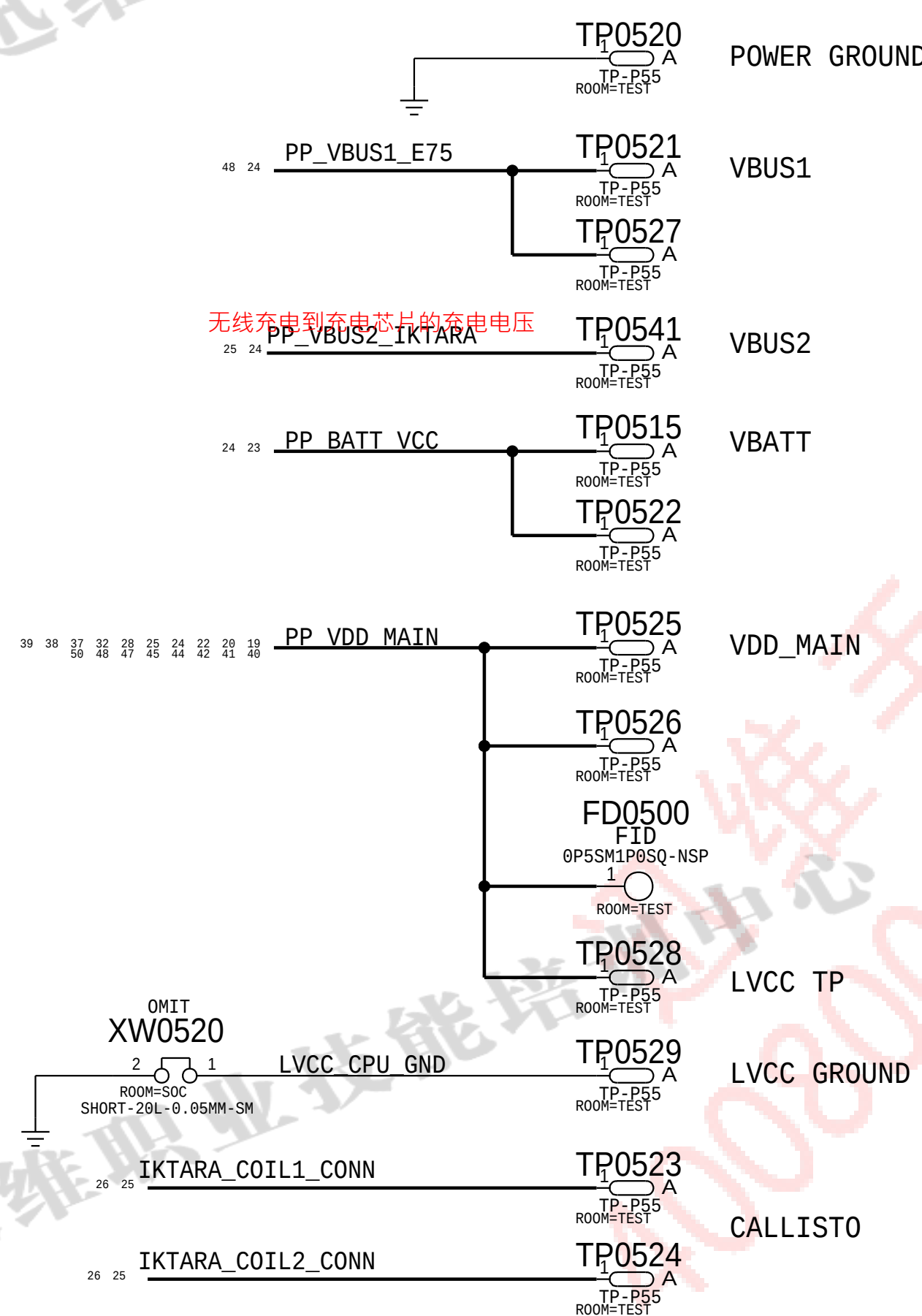




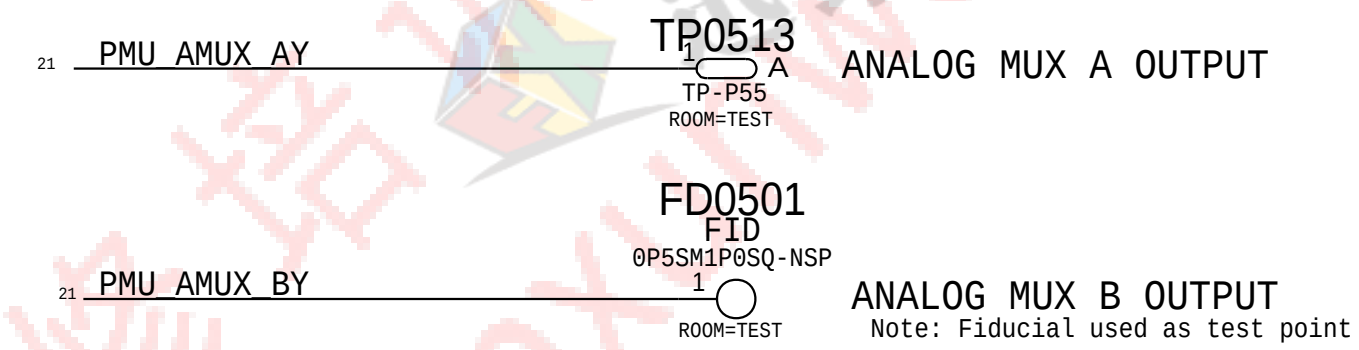
Test Points

Probe Points

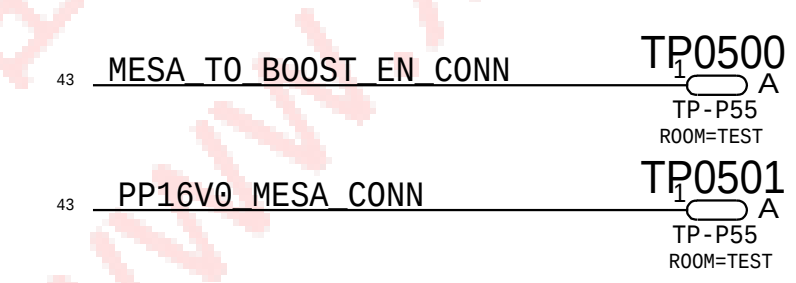
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AMUX



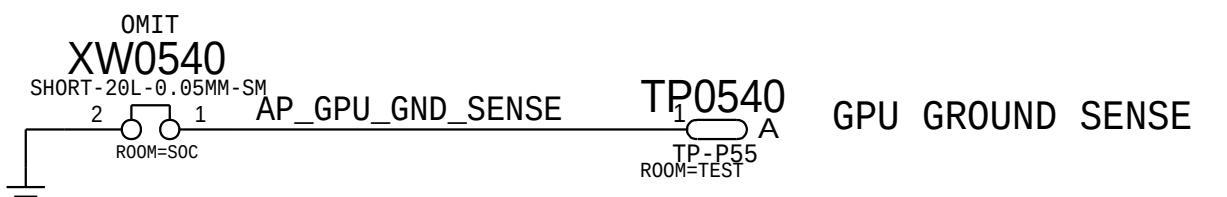
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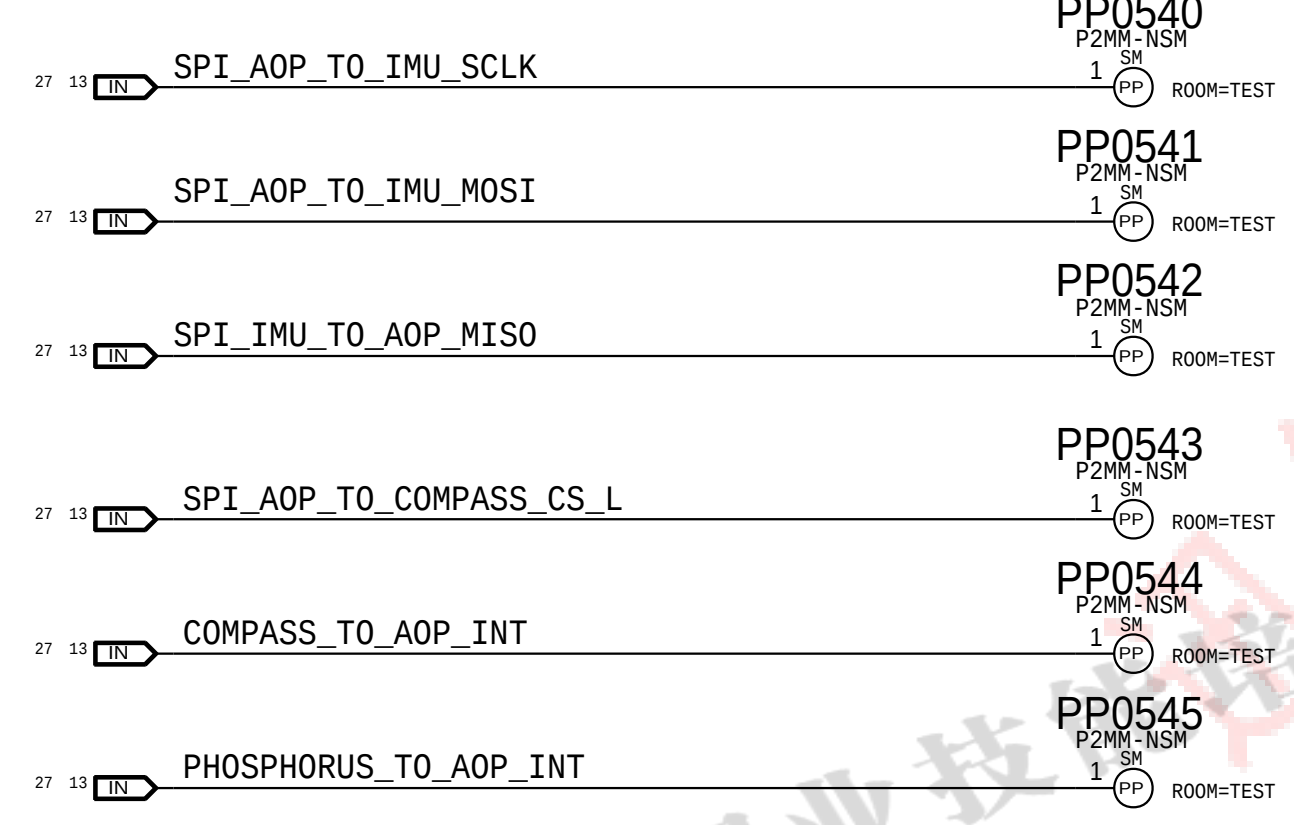
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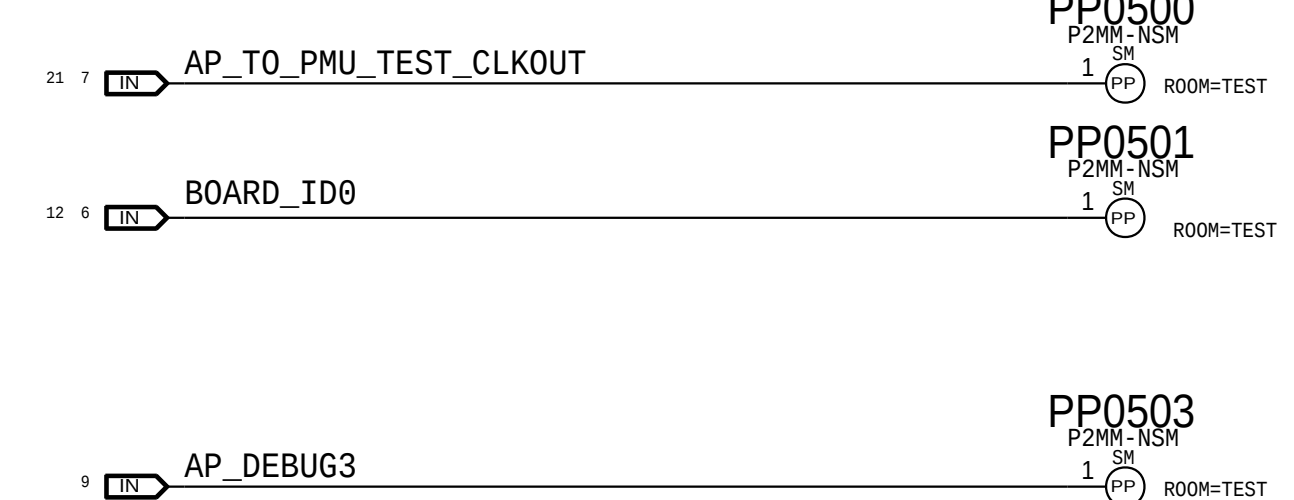
SENSE TP



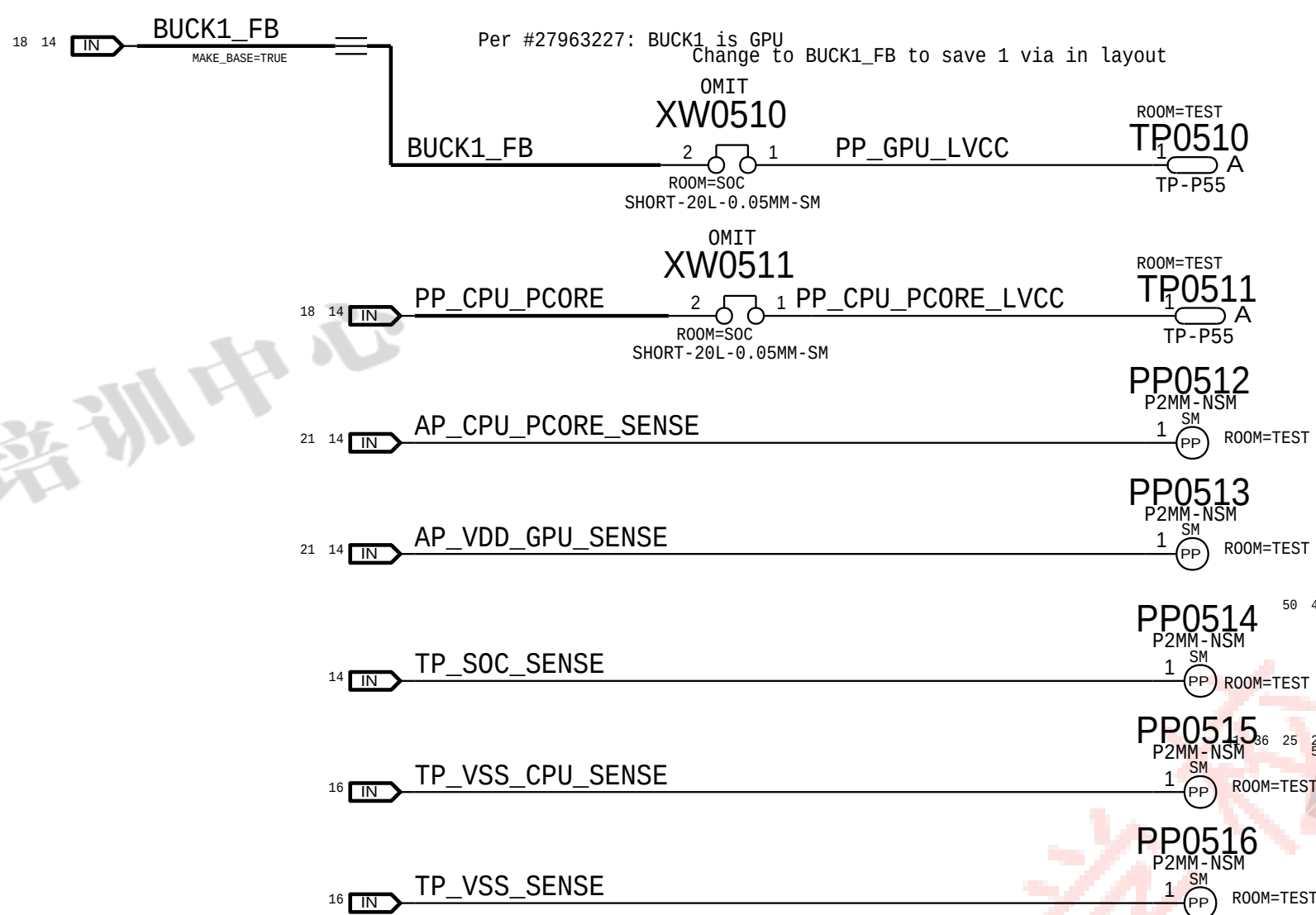
Sensor SPI



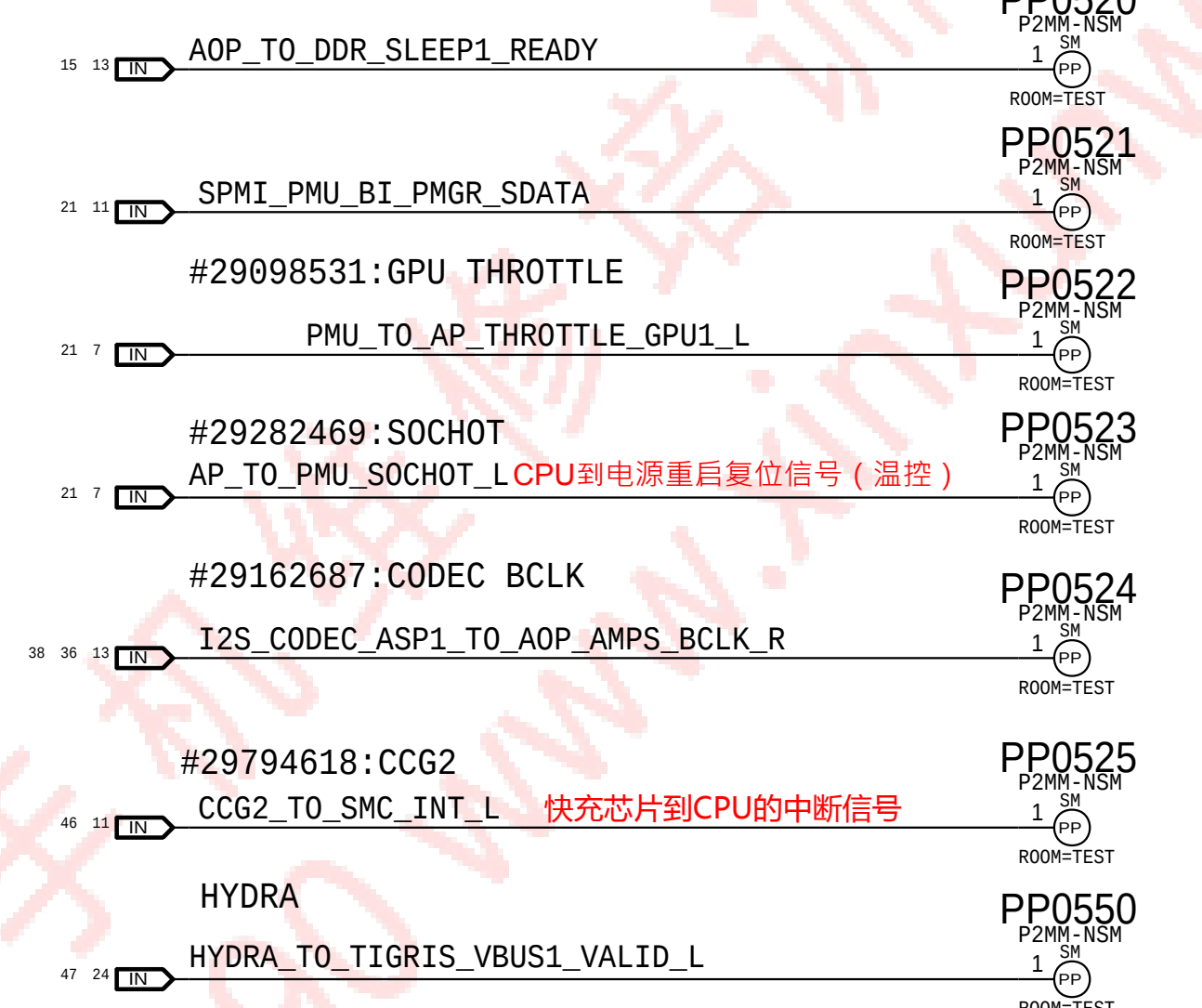
SOC Debug



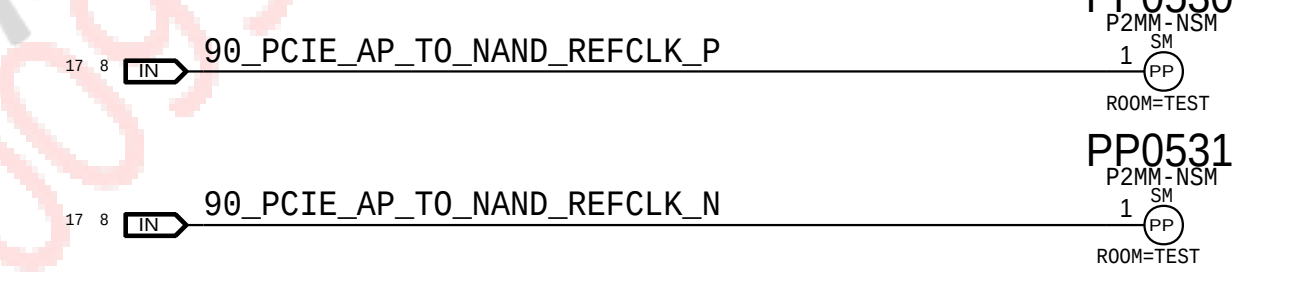
SOC CPU/GPU



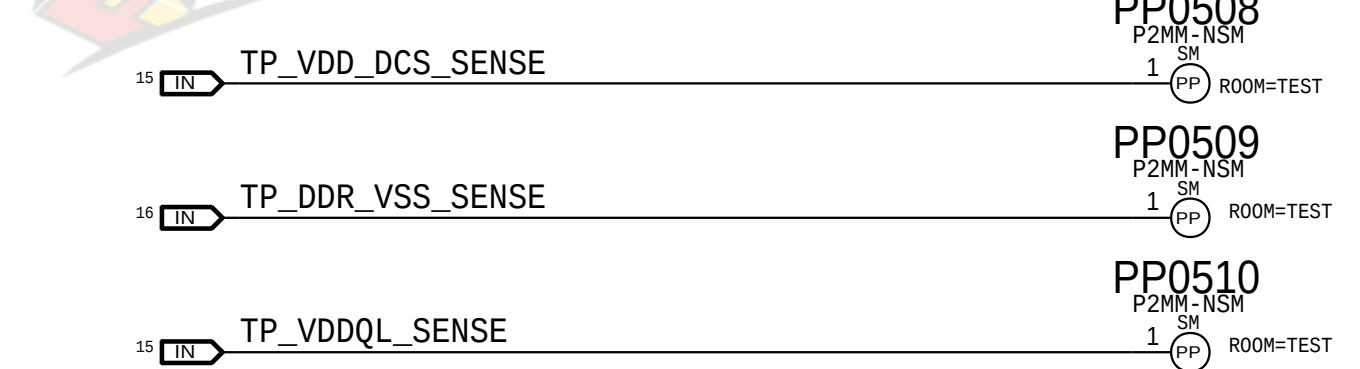
PMU Debug



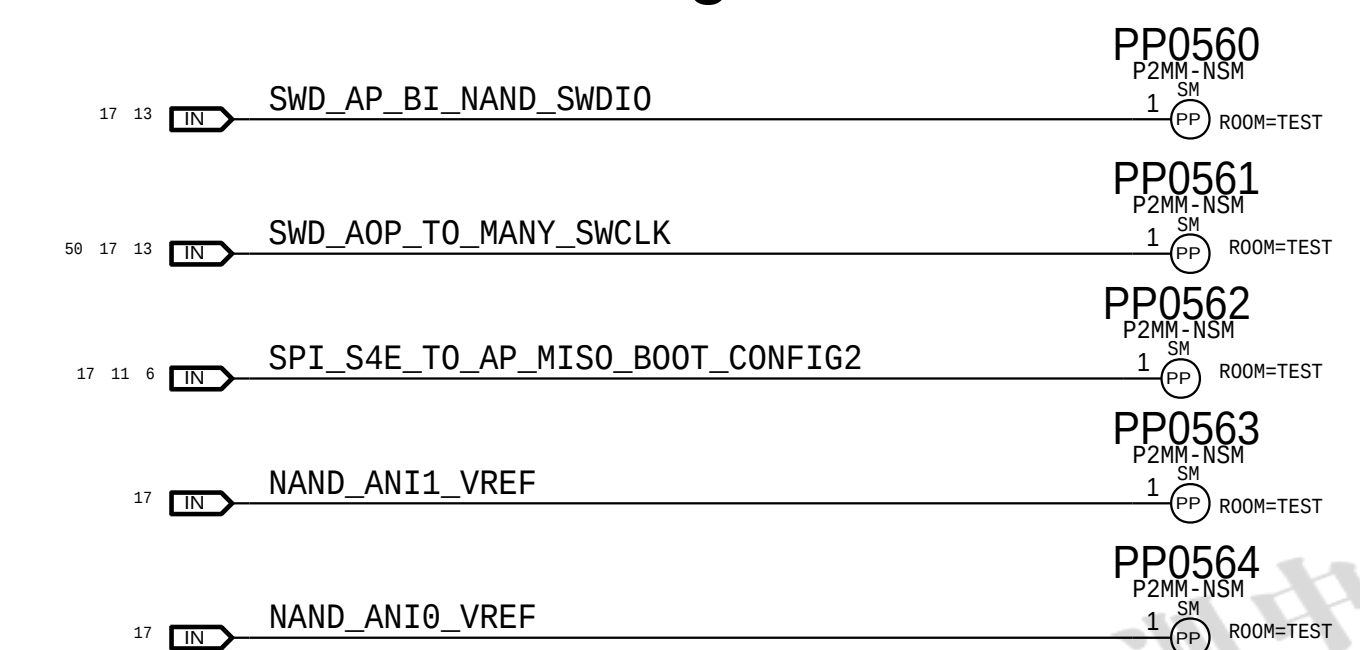
PCIE Refclk



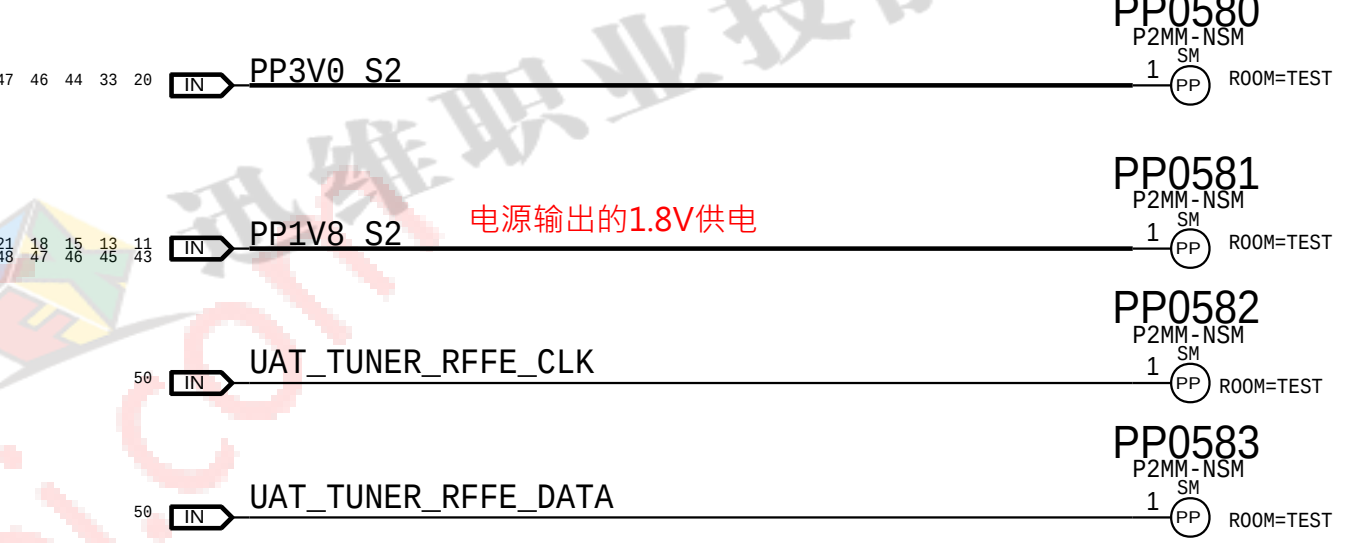
SOC DDR



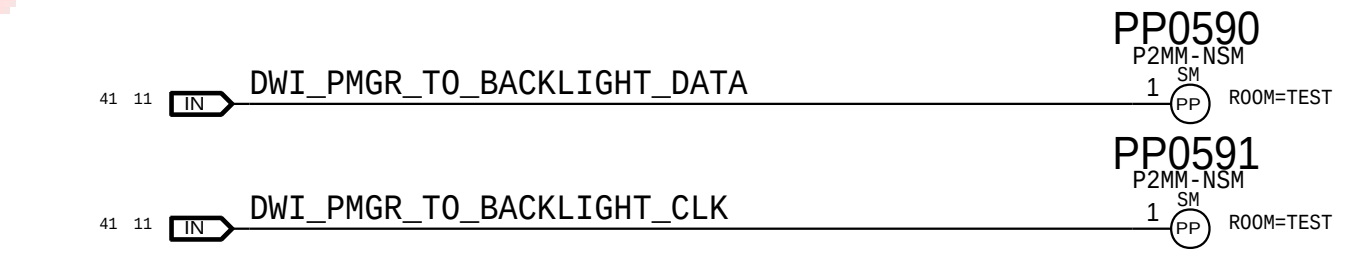
NAND Debug



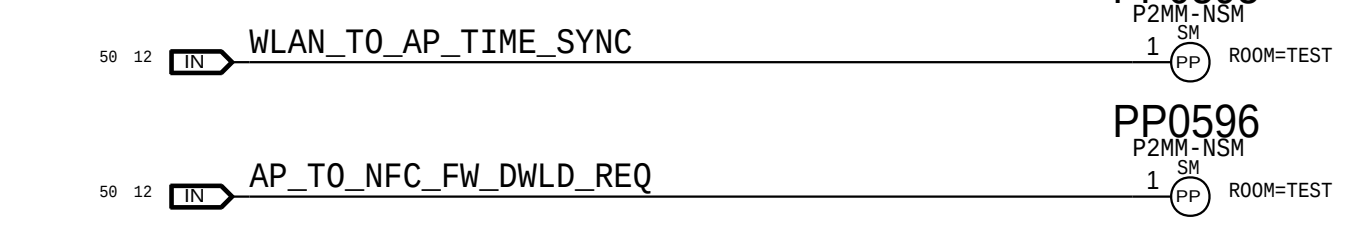
BB UAT Debug



BACKLIGHT Debug



RF Debug



D

C

B

A

D

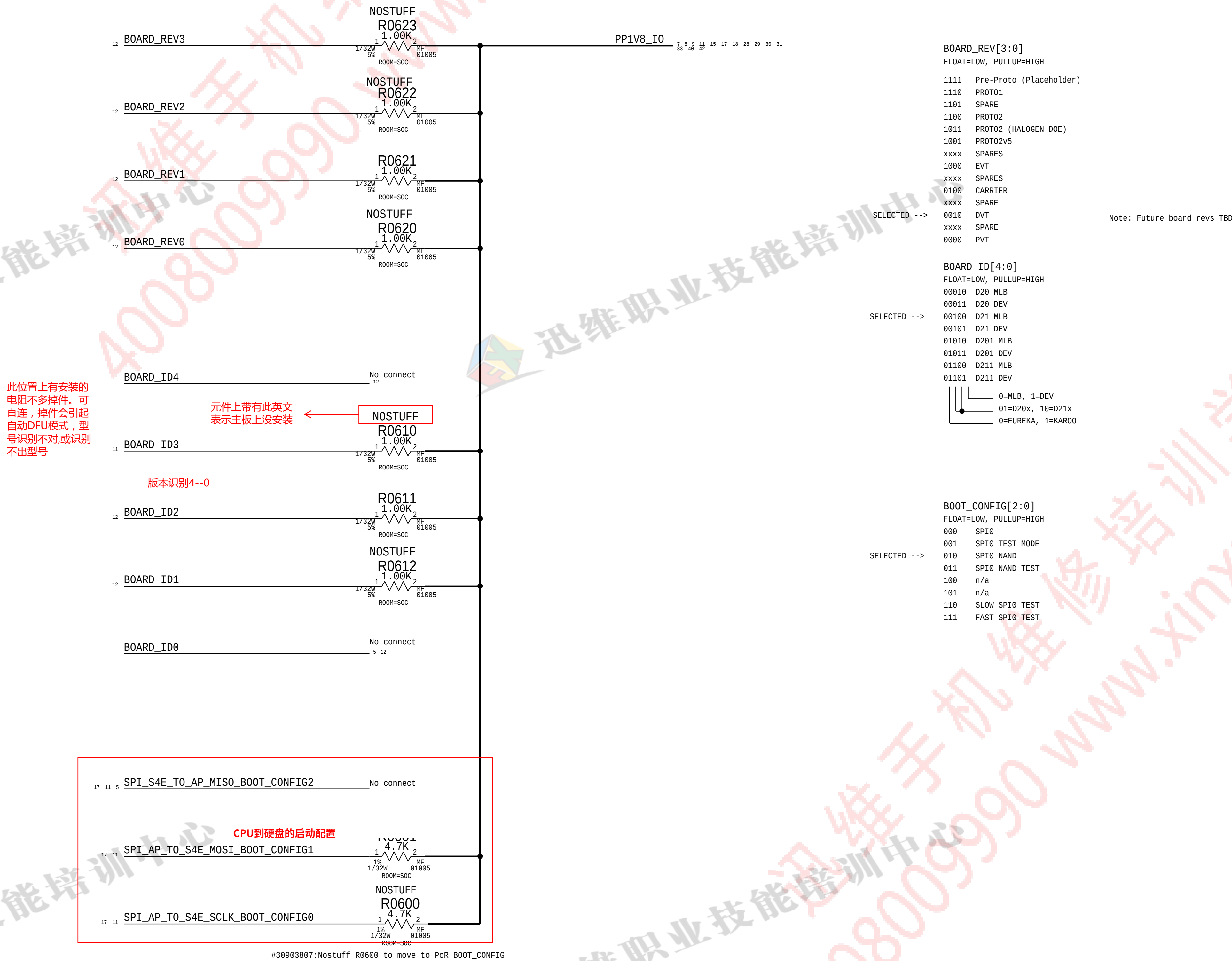
C

B

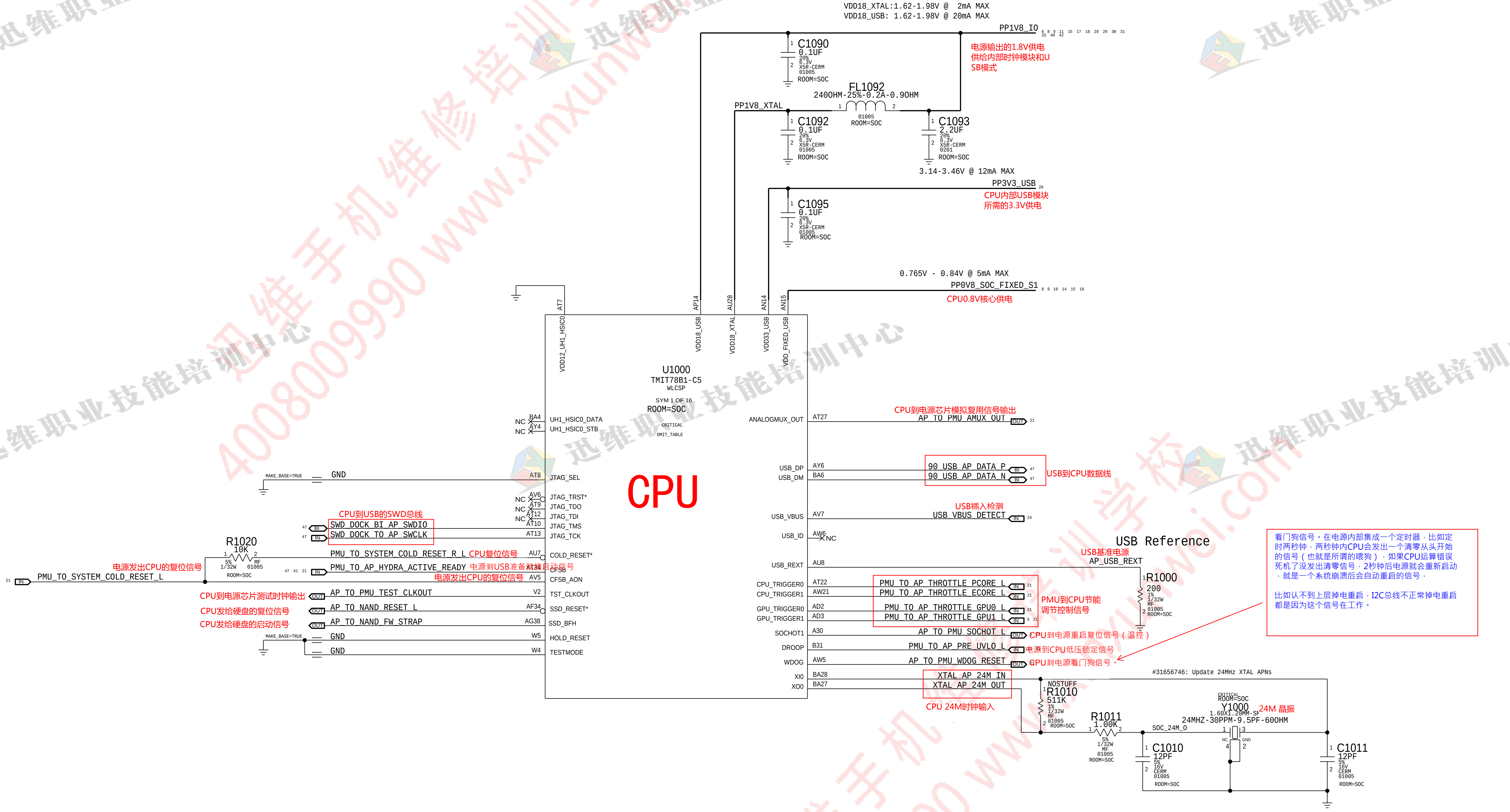
A



BOOTSTRAPPING:BOARD REV BOARD ID BOOT CONFIG



SOC - USB, JTAG, XTAL



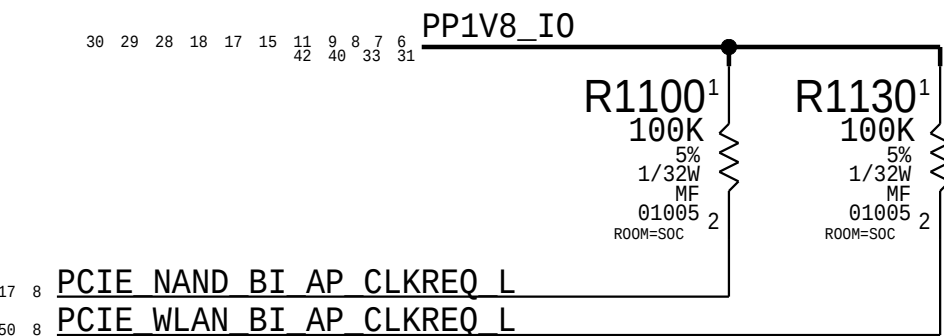
CPU

看门狗信号，在电源内部集成一个定时器，比如定时两秒钟，两秒钟内CPU会发出一个清零从头开始的信号（也就是所谓的喂狗），如果CPU运算错误死机了没发出清零信号，2秒钟后电源就会重新启动，就是一个系统崩溃后会自动重启的信号。

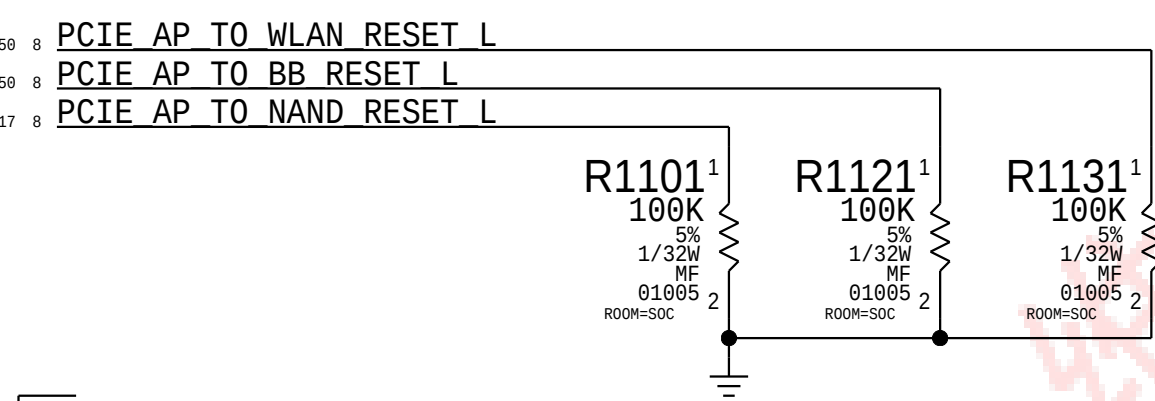
比如认不到上层掉电重启，I2C总线不正常掉电重启都是因为在这个信号在工作。

SOC - PCIE INTERFACES

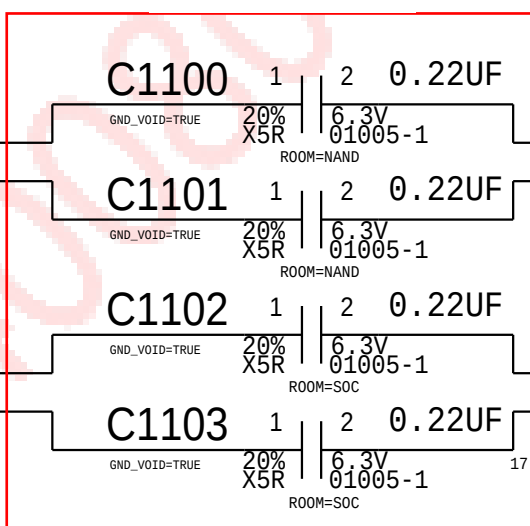
PCie Clock Request Pull-Ups



PCie Reset Pull-Downs



耦合电容



耦合电容：在电路中是电容的两端都是接信号数。作用是隔离滤除信号中的直流信号，避免前级后级之间互相干扰

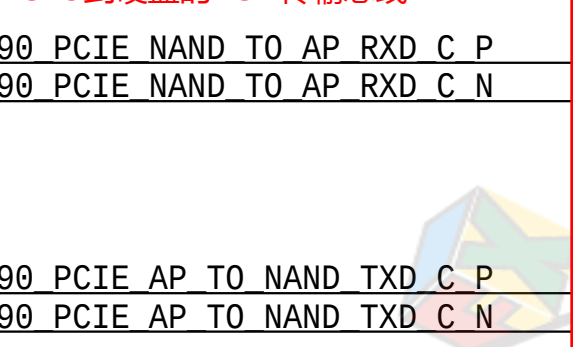
如果耦合电容掉了，尽量使用更换的方式，要直连的话也最好正负都直连，不要一对信号中一个直连一个耦合，

更换必须在相同位置拆的电容才能代换

PCIE总线

一种高速串行传输总线。在6S以后的手机中，CPU和基带、硬盘、WIFI之间的数据传输都是使用PCIE总线
REFCLK：参考时钟，CPU送出的基准时钟频底。为了让设备与CPU之间在相同的频率下运行
CLKREQ：请求运行时钟，也就是请求输出参考时钟
RXD：数据接收，PCIE总线数据传输是成对的，一条后面加P是正的，后面带N是负的
TXD：数据发射，也是一正一负
RESET：复位信号，由CPU发出。

CPU到硬盘的PCIE传输总线

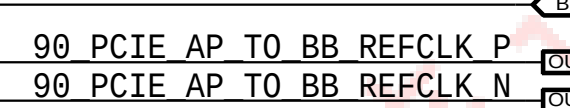


CPU到硬盘的复位信号

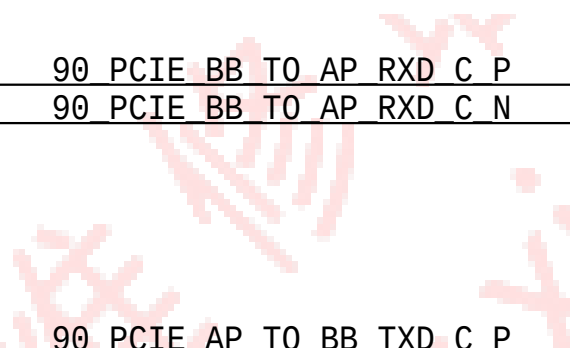


CPU

PCIE_BB_BI_AP_CLKREQ_L



CPU到基带的PCIE通讯总线



PCIE AP TO BB RESET_L

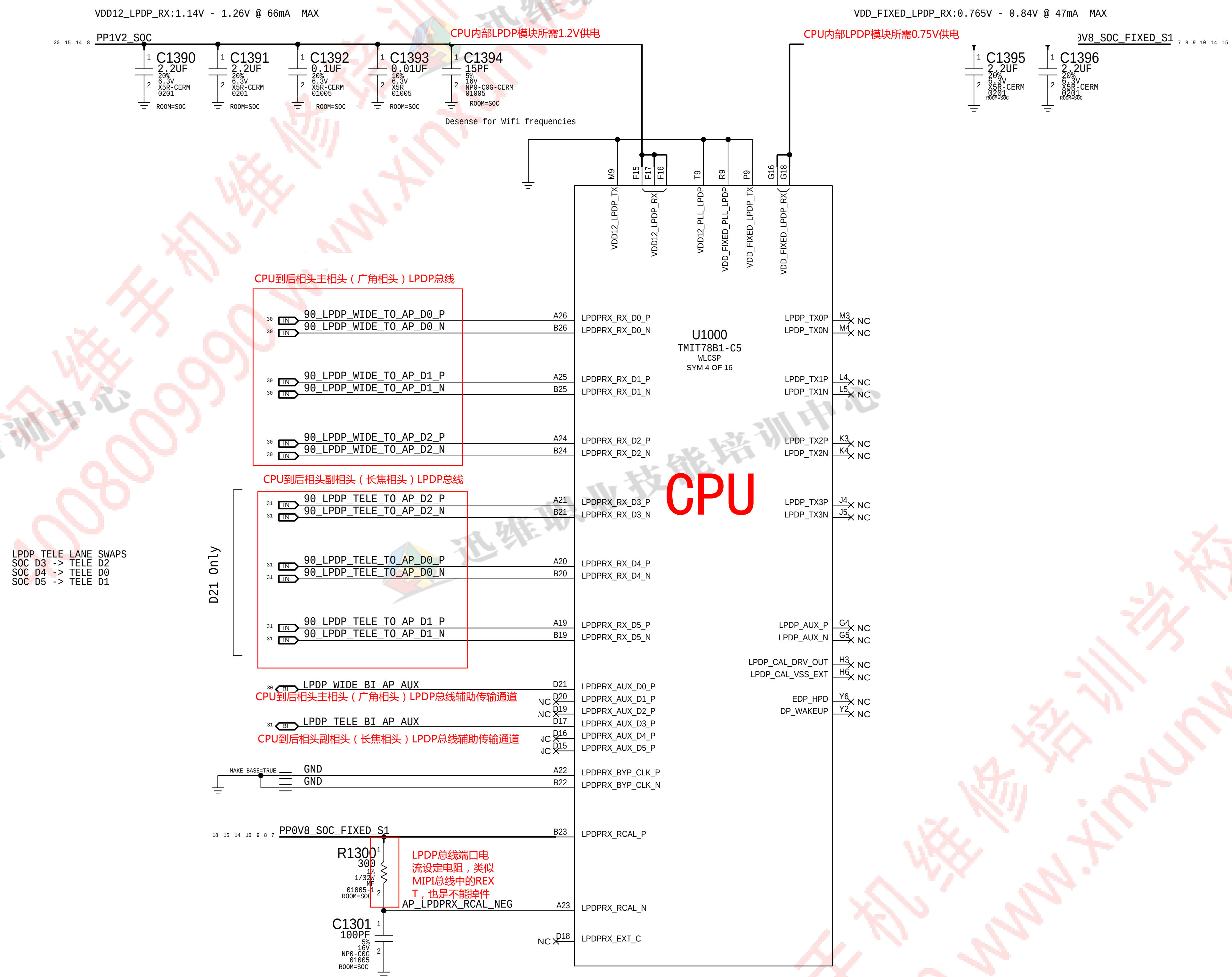


PCIE总线外部电阻
AP_PCIE_RCAL

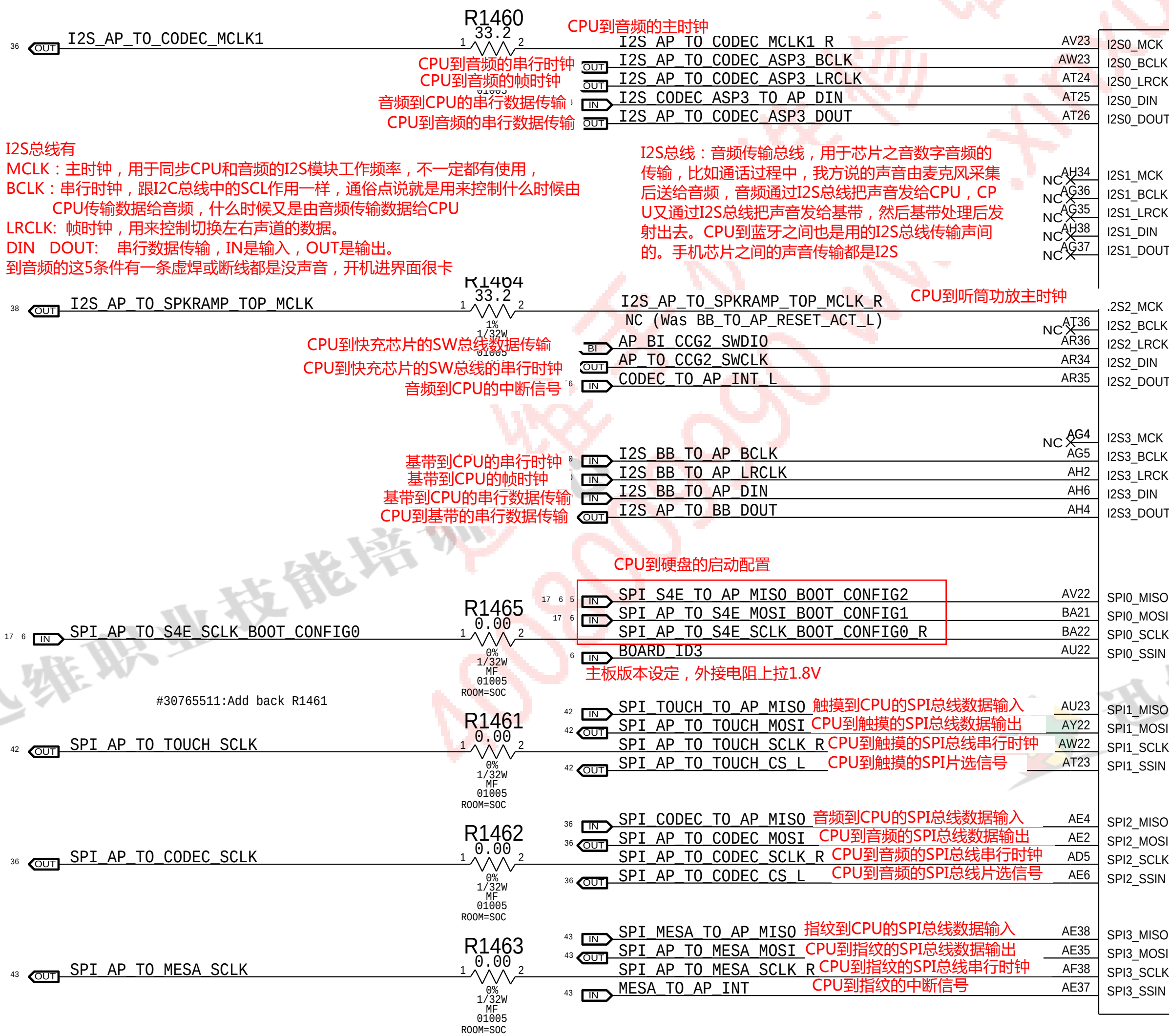


REXT在MIPI总线和PCIE总线都丰富这样的电阻，作用是来设定总线端口电流，如果此电阻掉件会引起该总线不工作。如认不到硬盘，MIPI总线的则引起不照相不显示等

SOC - LPDP INTERFACES



SOC - SERIAL INTERFACES

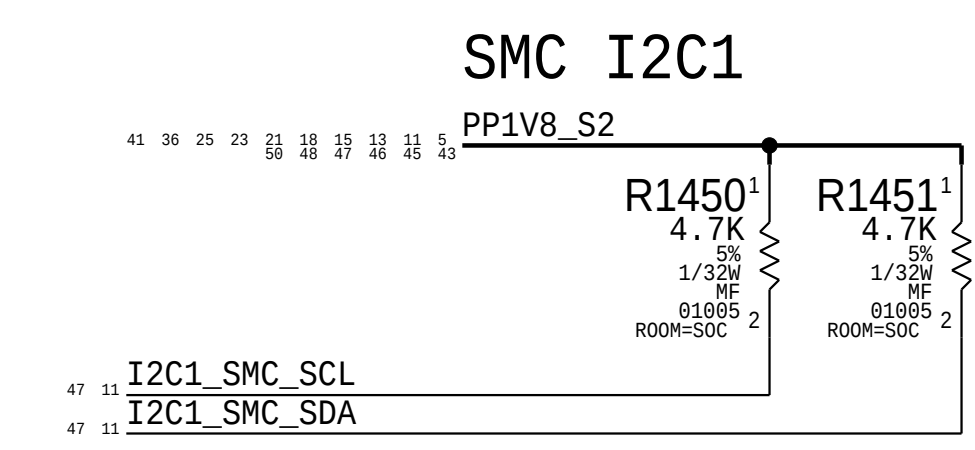
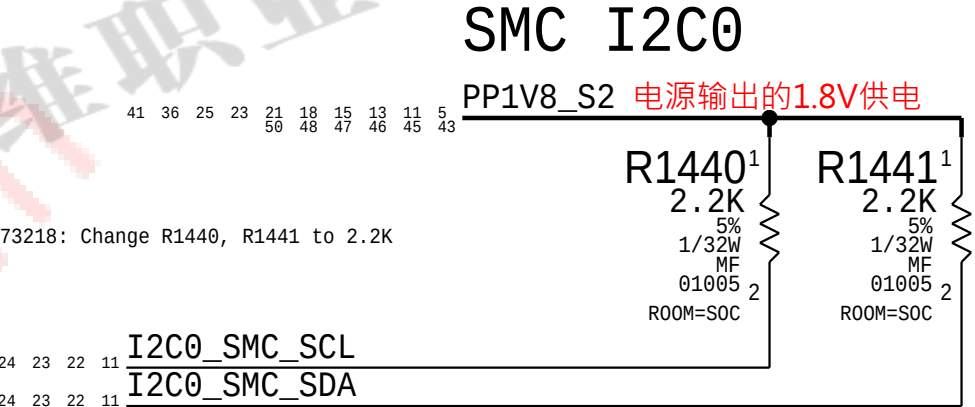
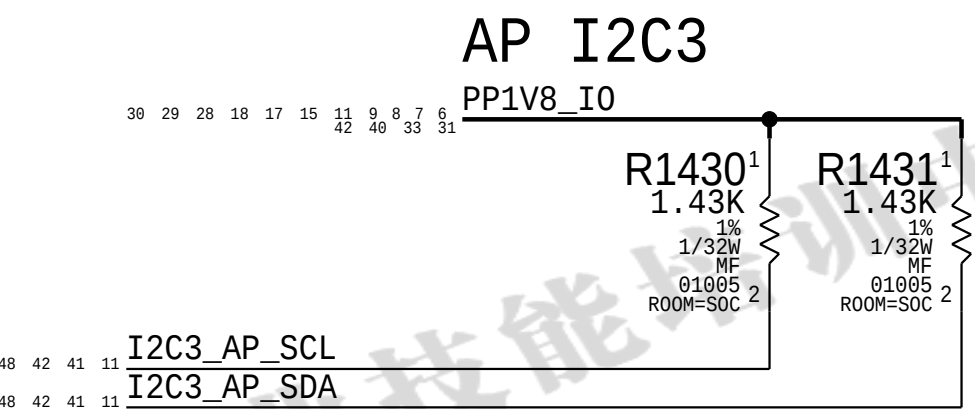
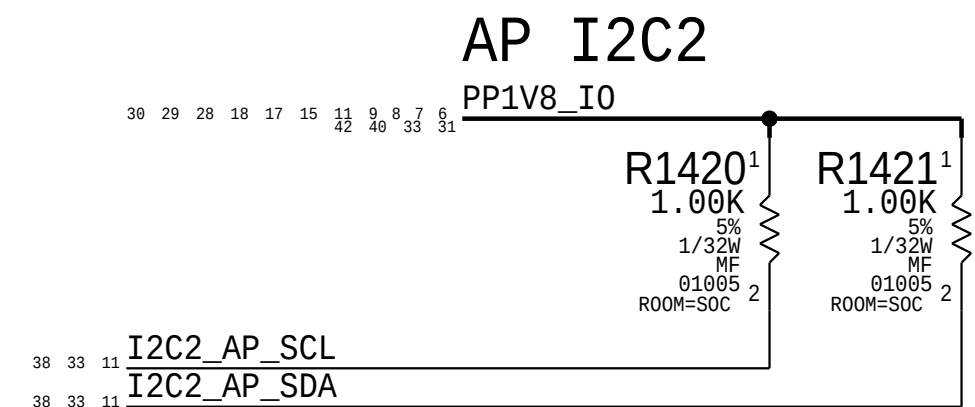
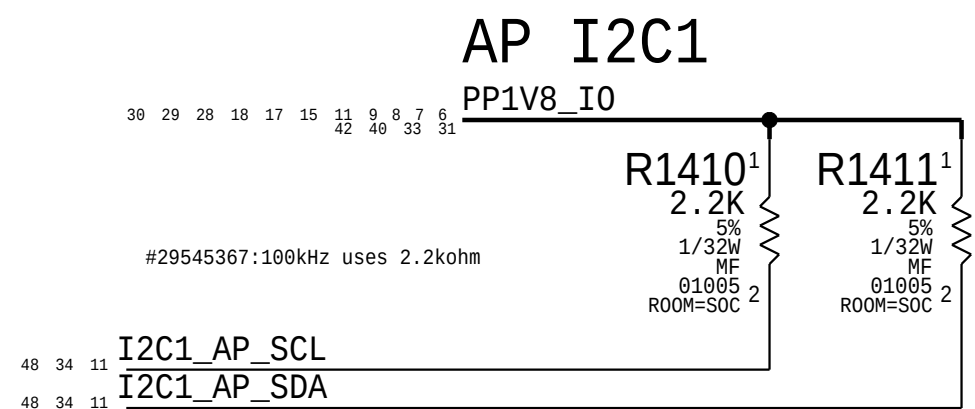
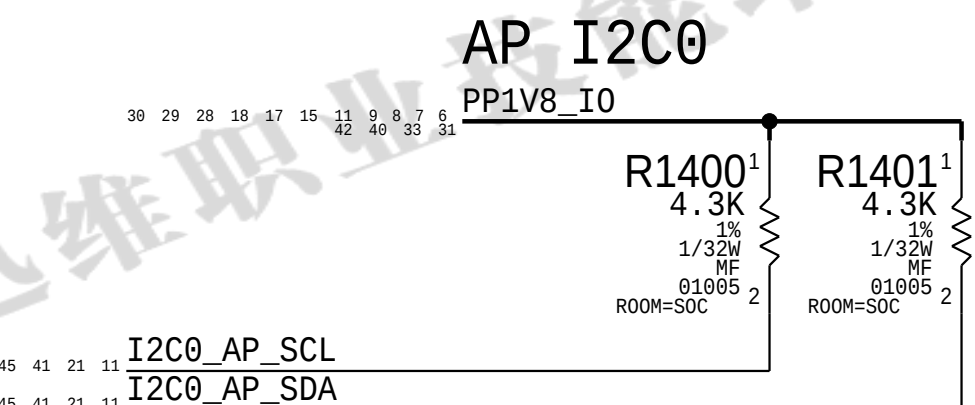
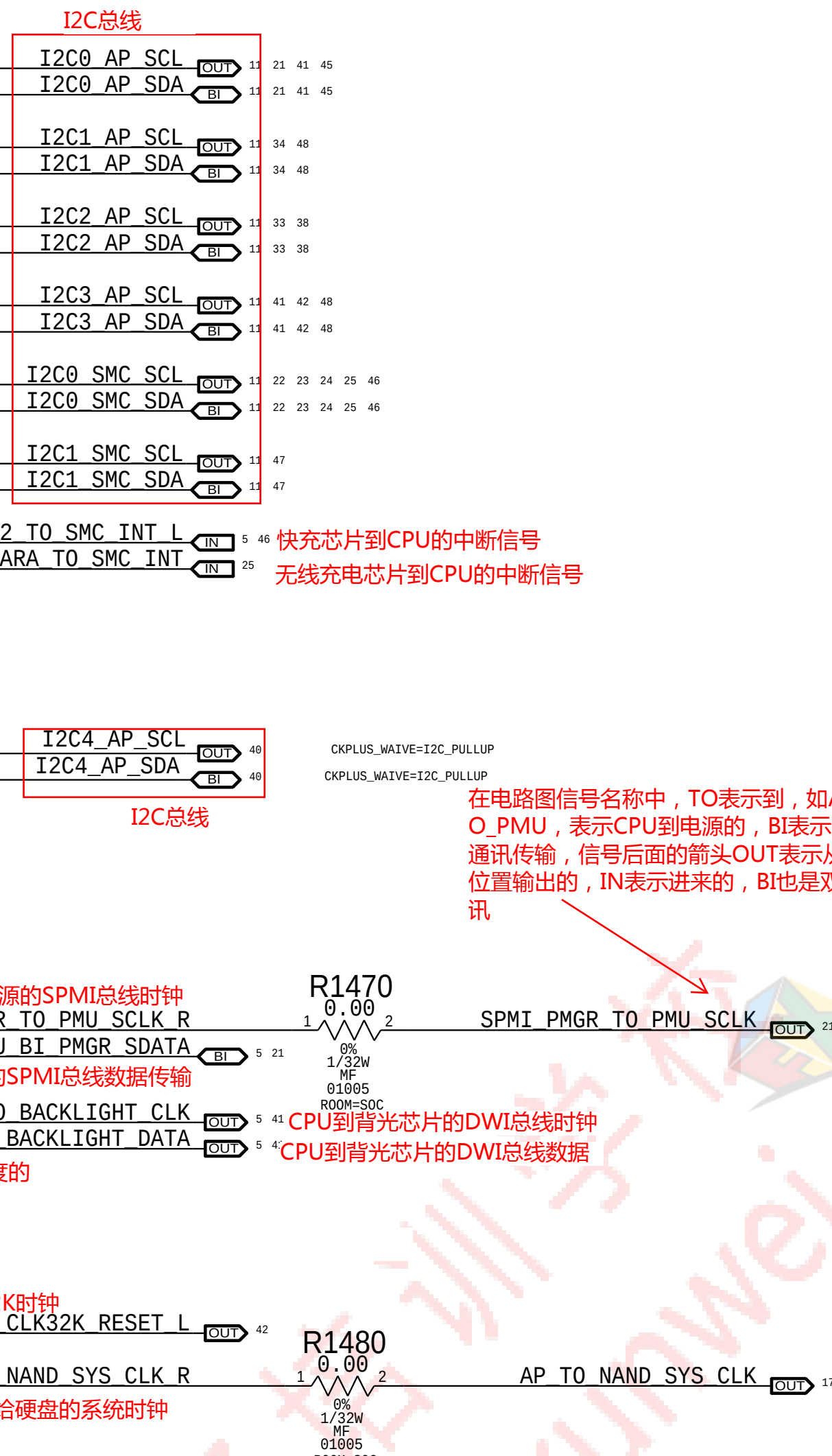


SPI: Route as Daisy-Chain (No T's Allowed)

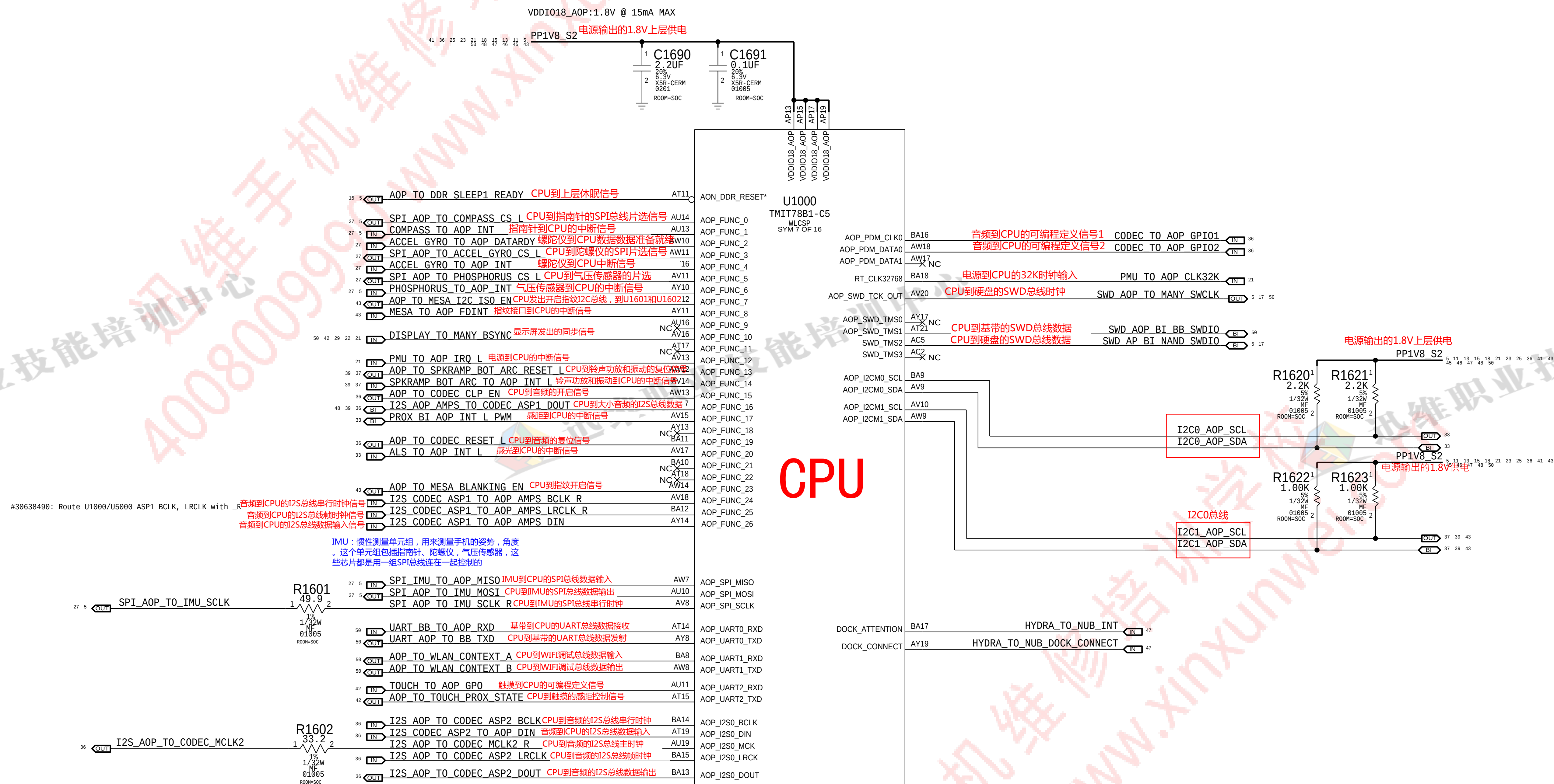
SPI总线：串行高速通讯总线，在手机中一般有4根信号线，
MISO：数据传输，M是主设备，I是IN输入，S是从设备，O是OUT输出，CPU是主设备，相连的芯片是从设备，就是从设备输出数据到CPU
MOSI：CPU输出数据到从设备。
SCLK：串行时钟，用来控制什么时候发送数据，什么时候接收数据，比如高电平时接收，低电平发送
CS或SS：片选信号，SPI总线上可以同时接很多从设备，只有CPU选中的设备才可以接收发送数据，手机中一般都是一组SPI总线只接一个从设备，

I2C总线：串行通讯总线，一般手机维修人员所说的直总线就是I2C总线，基本上I2C总线中间带AP的不管那一组都会引起触发开机后小电流50-80定住或掉电归零。
I2C总线有条信号线：
SCL: 串行时钟
SDA: 串行数据
所有的I2C总线都是1.8V电压上拉的，触发后都能测到1.8V电压。

CPU



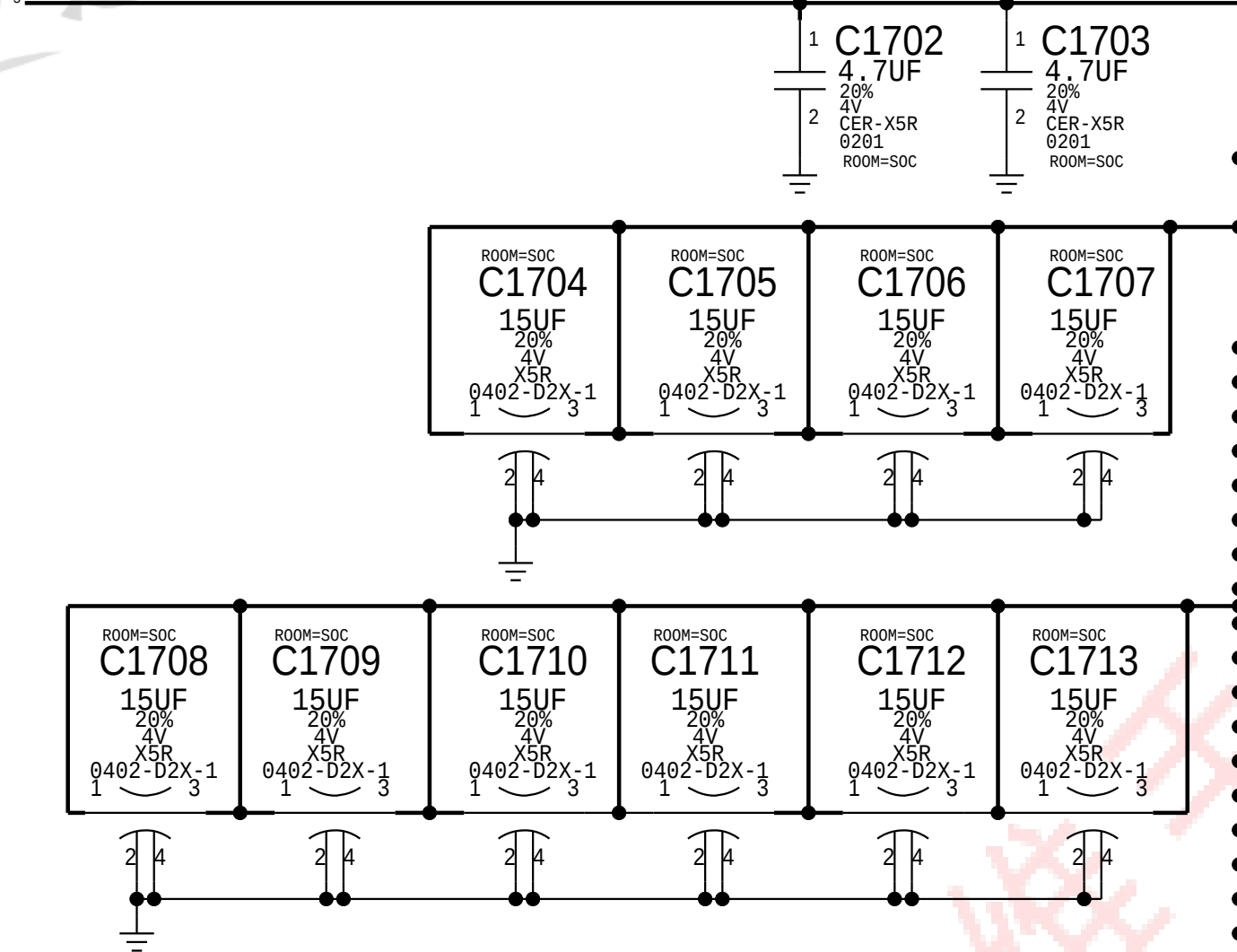
SOC - AOP



SOC - CPU, GPU & SOC RAILS

1.06V @ 11.0A MAX
0.8V @ 6A MAX
0.575V @ 2.7A MAX

PP_CPU_CORE CPU核心供电

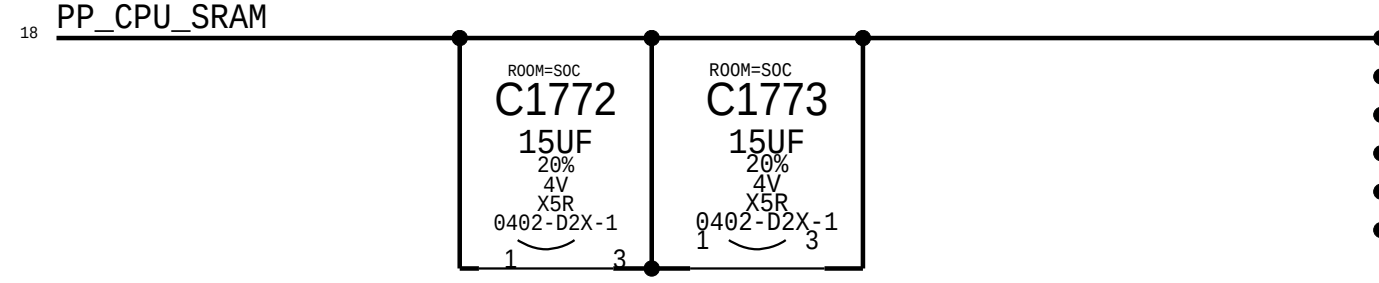


CPU核心供电

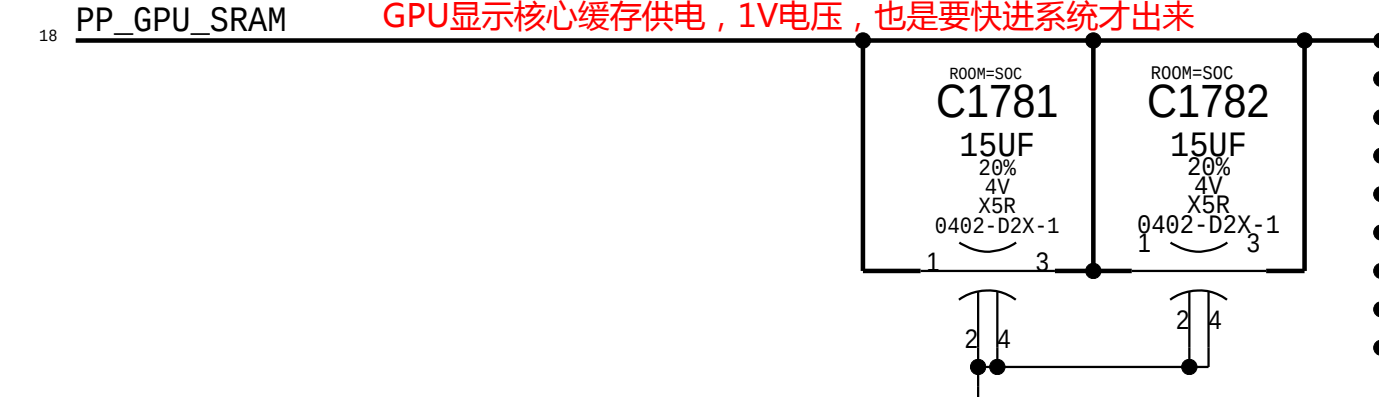
0.7V @ 75mA MAX
PP0V7_VDD_LOW_S2 CPU的0.7v供电



1.01V @ 2.1A MAX
0.735V @ 0.6A MAX
PP_CPU_SRAM



1.06V @ 1.1A MAX
0.80V @ 0.63A MAX
0.675V @ 0.19A MAX
PP_GPU_SRAM



21 5 OUT AP_VDD_GPU_SENSE
GPU电压检测信号，到电源芯片

在苹果电路图中，名称前面P表示供电，PP是正压，PN是负压
PP后面跟的数字表示电压值。
如PP0V7：表示正的0.7V。
PN5V7：表示负的5.7V。

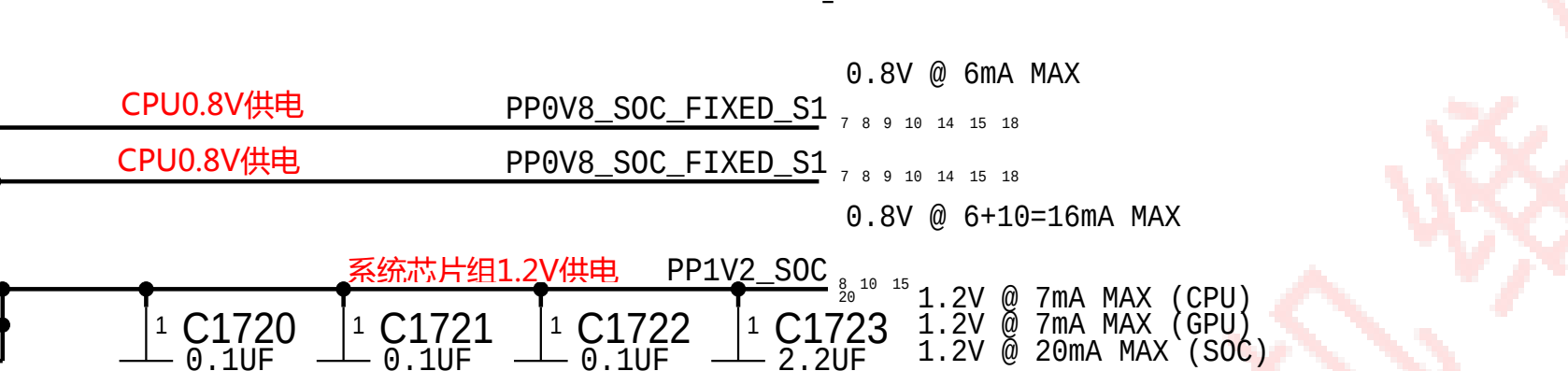
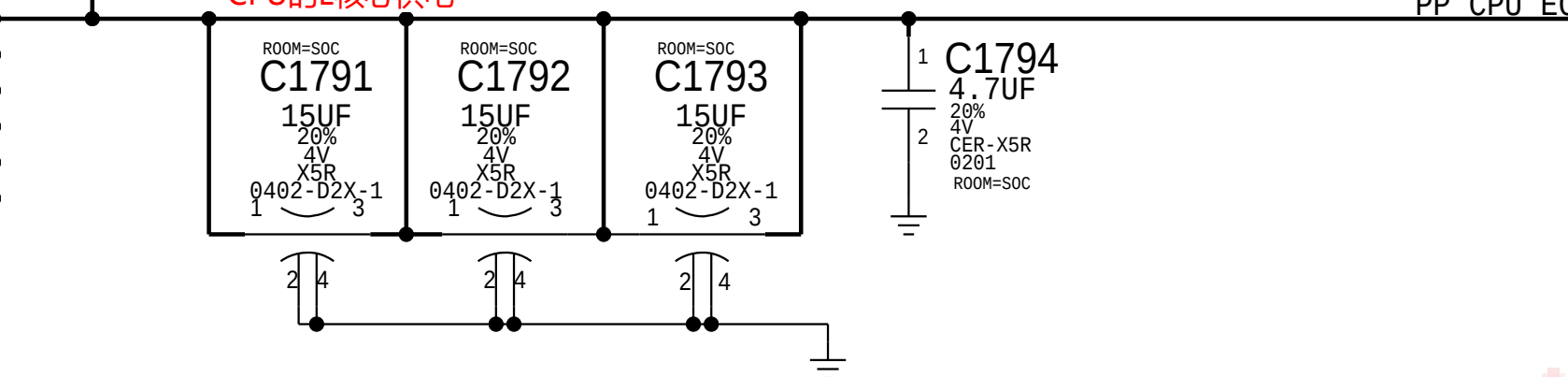
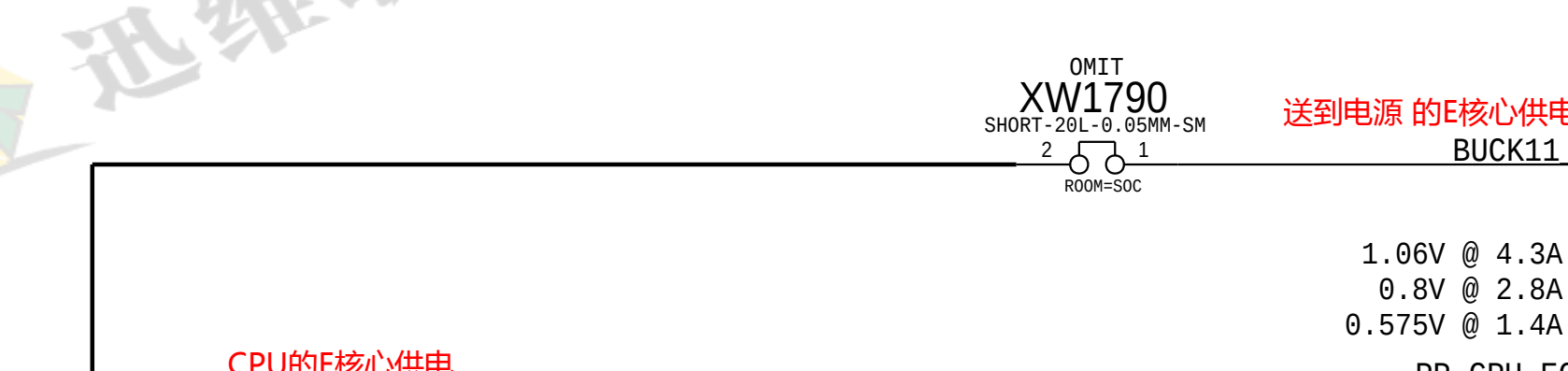
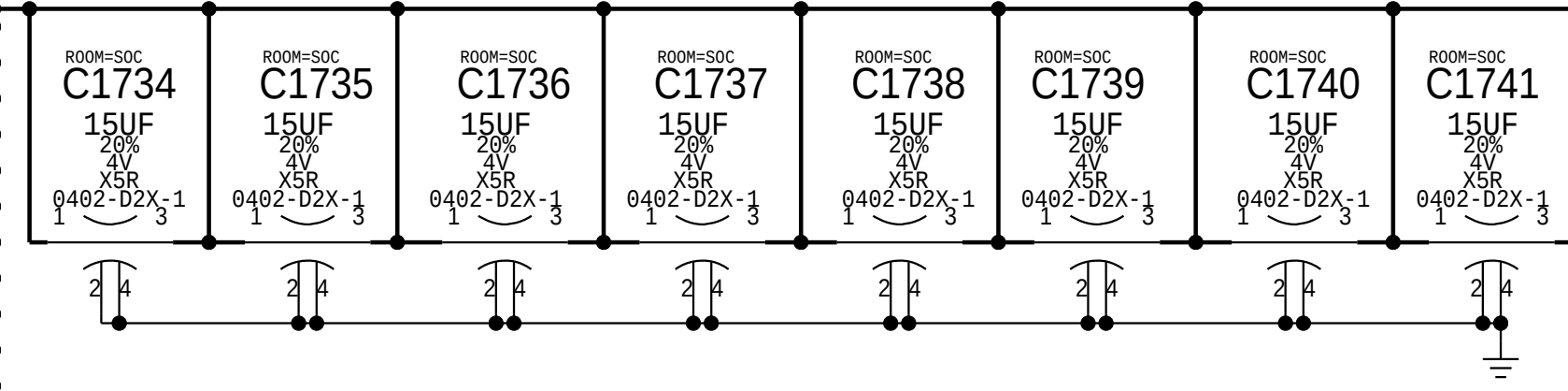
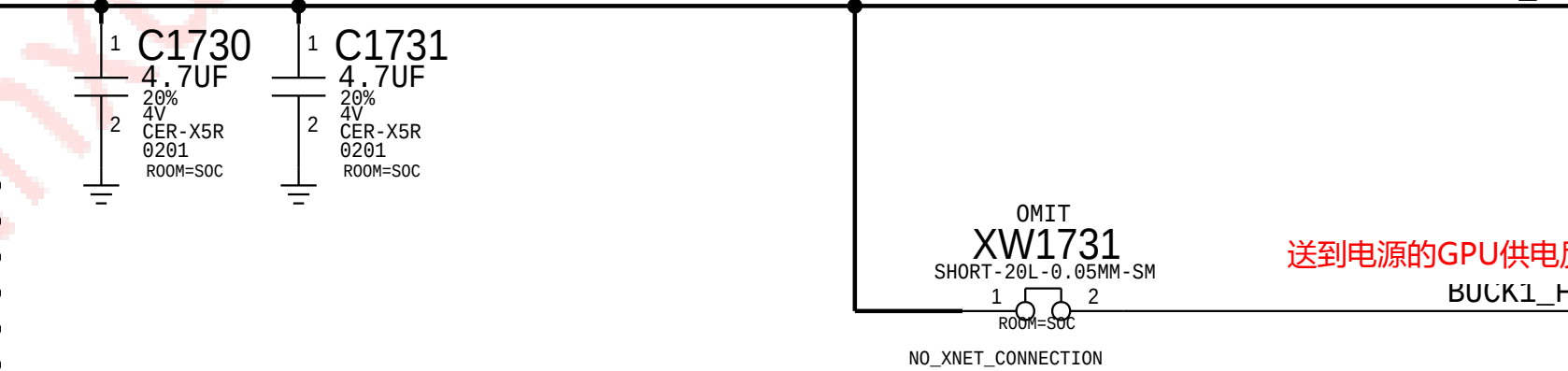
OMIT XW1701
SHORT-20L-0.05MM-SM
ROOM=SOC
到电源的核心供电反馈信号，用来调整核心供电电压

U1000
TMIT78B1-C5
WLCSP
SYM 8 OF 16
ROOM=SOC

VDD_CPU
VDD_GPU
VDD_LOW
VDD_CPU_SRAM
VDD_CPU_SENSE
VDD_GPU_SRAM
VDD_GPU_SENSE

CPU

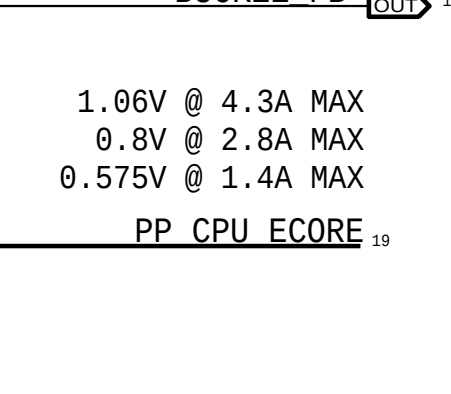
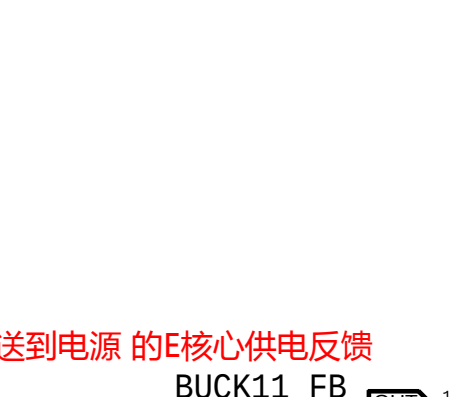
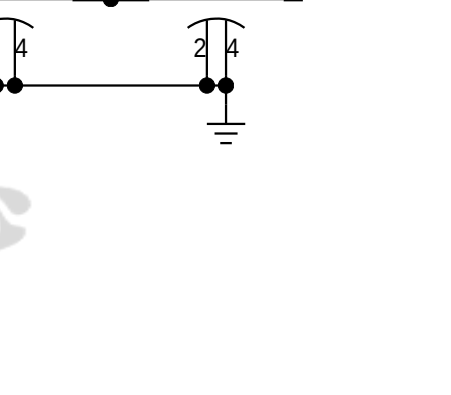
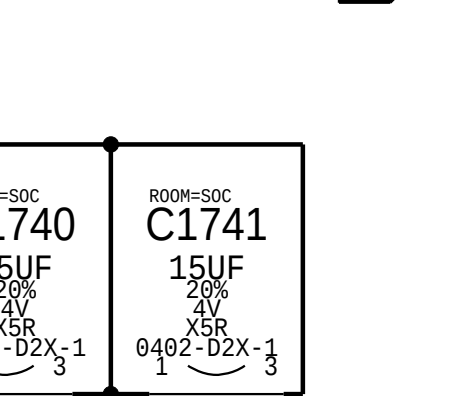
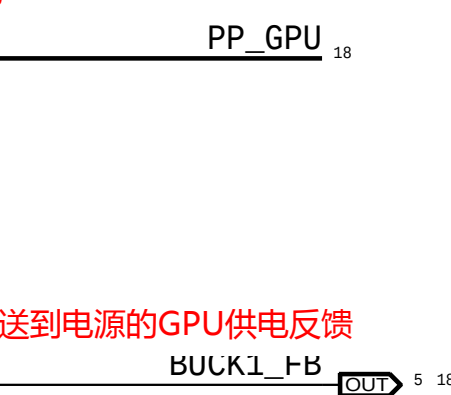
CPU内部显示核心供电，电压0.8V左右此供电
亮白苹果后进系统之前才会出来



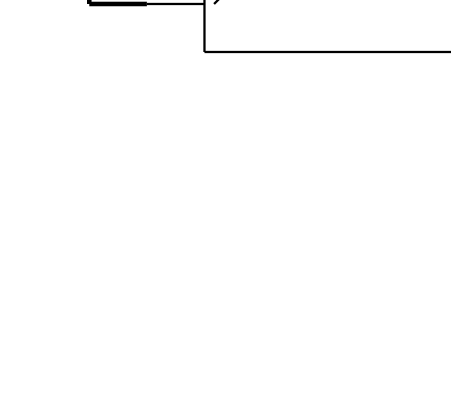
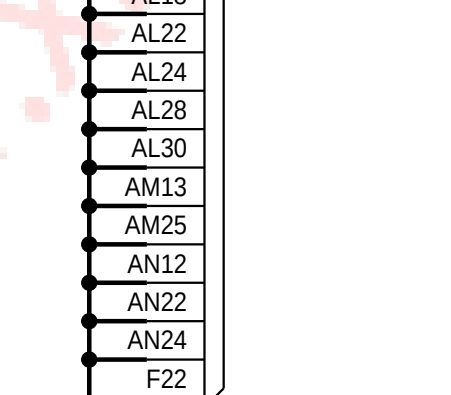
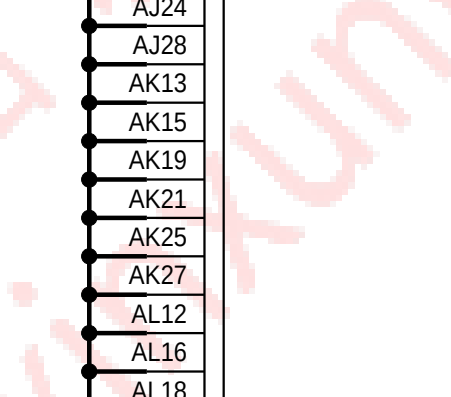
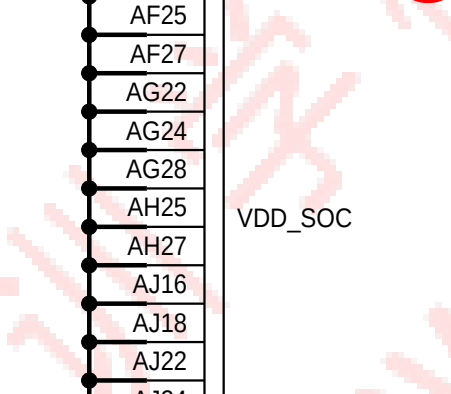
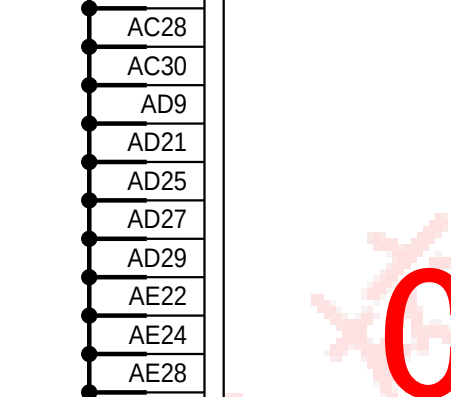
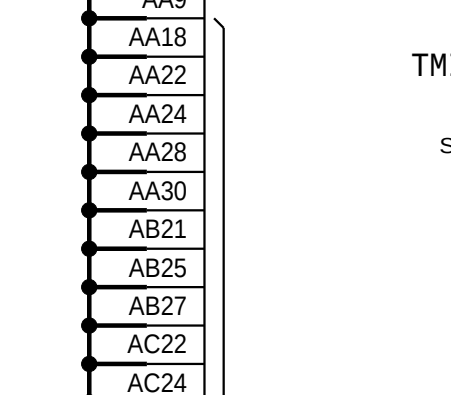
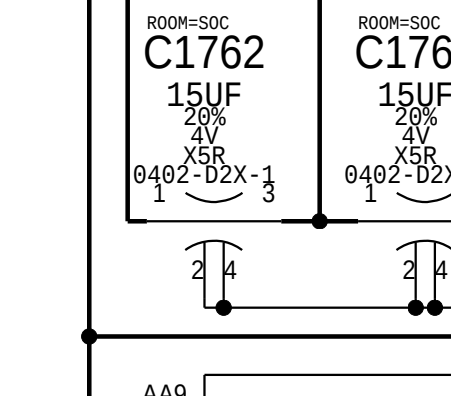
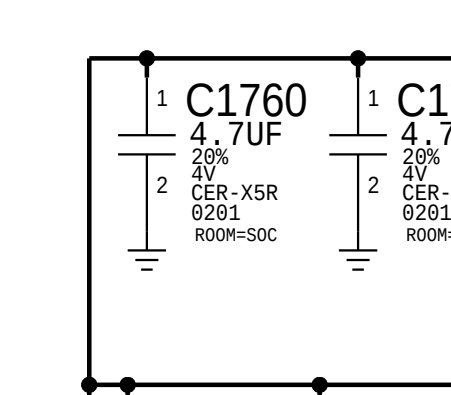
0.8V @ 6mA MAX
PP0V8_SOC_FIXED_S1
0.8V @ 6mA MAX
PP0V8_SOC_FIXED_S1
0.8V @ 6+10=16mA MAX
PP1V2_SOC
1.2V @ 7mA MAX (CPU)
1.2V @ 7mA MAX (GPU)
1.2V @ 20mA MAX (SOC)



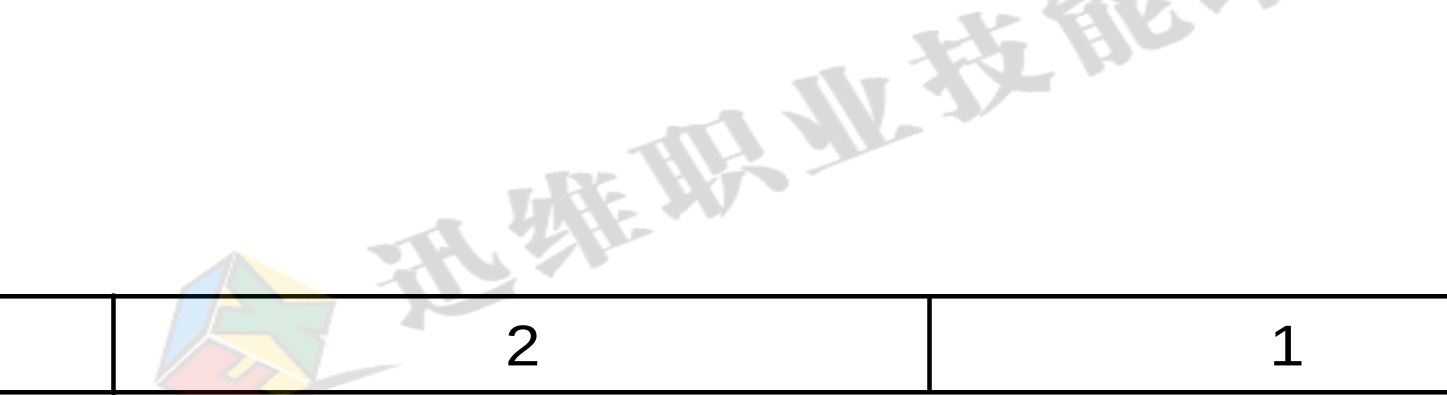
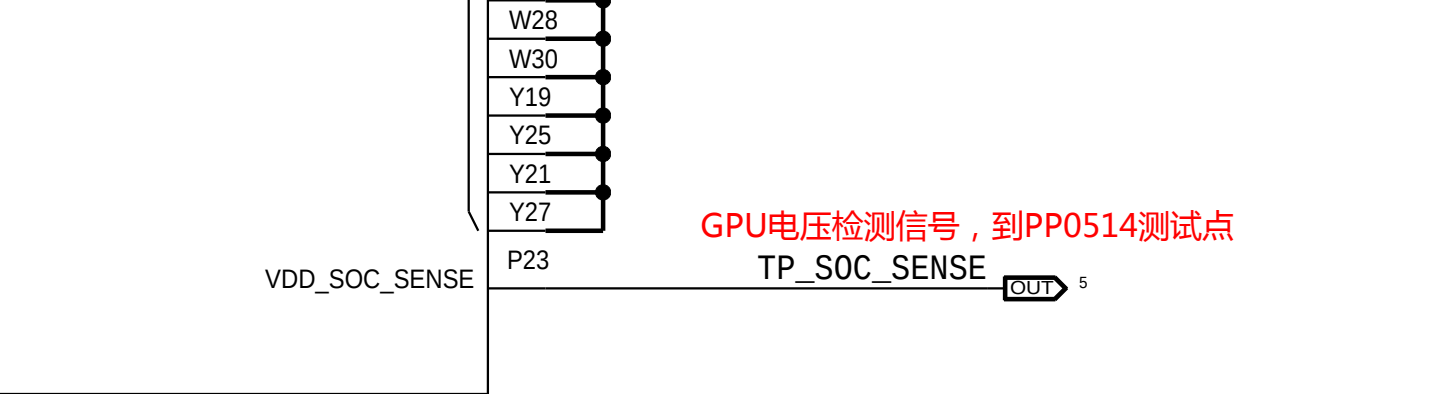
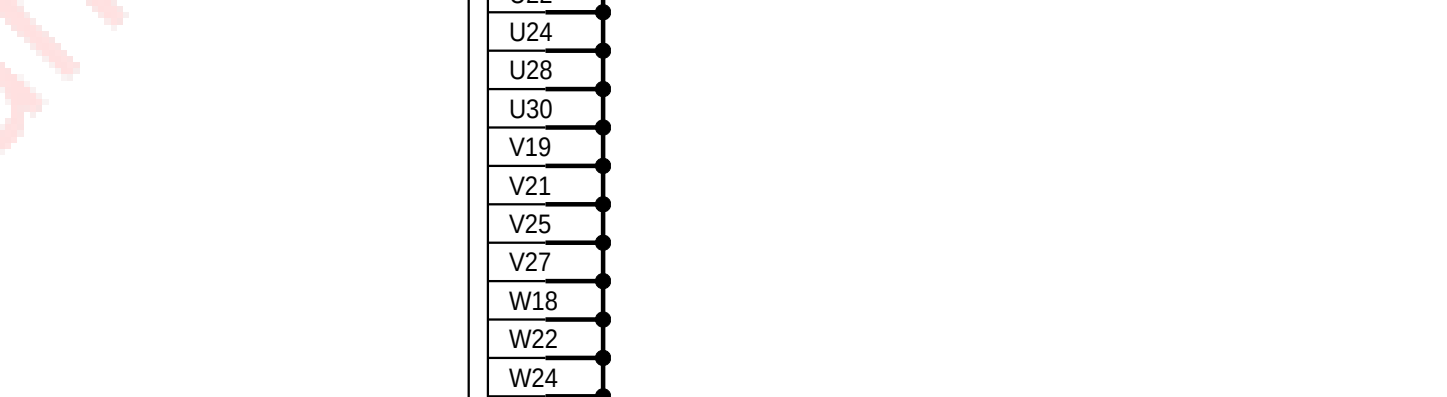
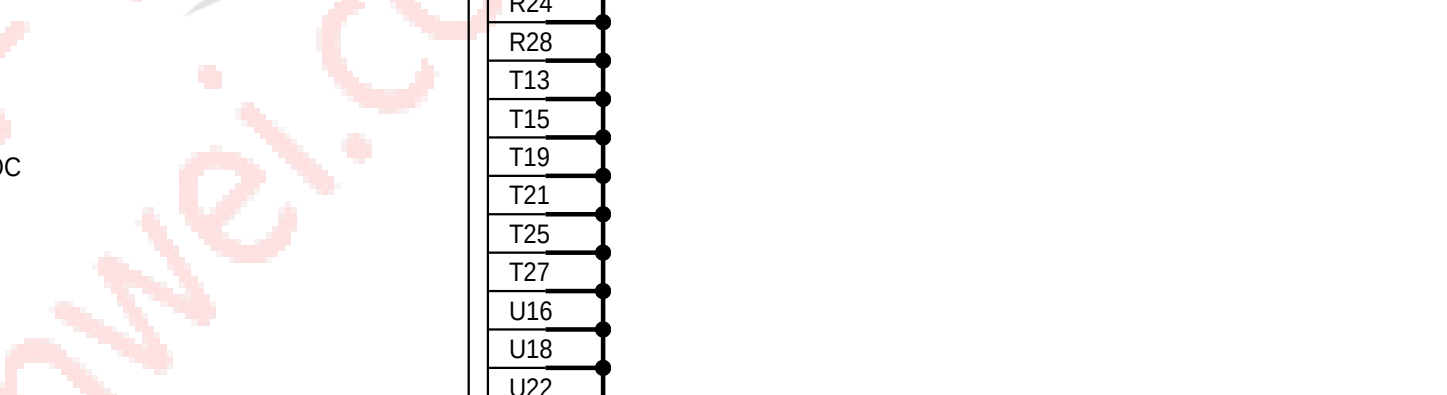
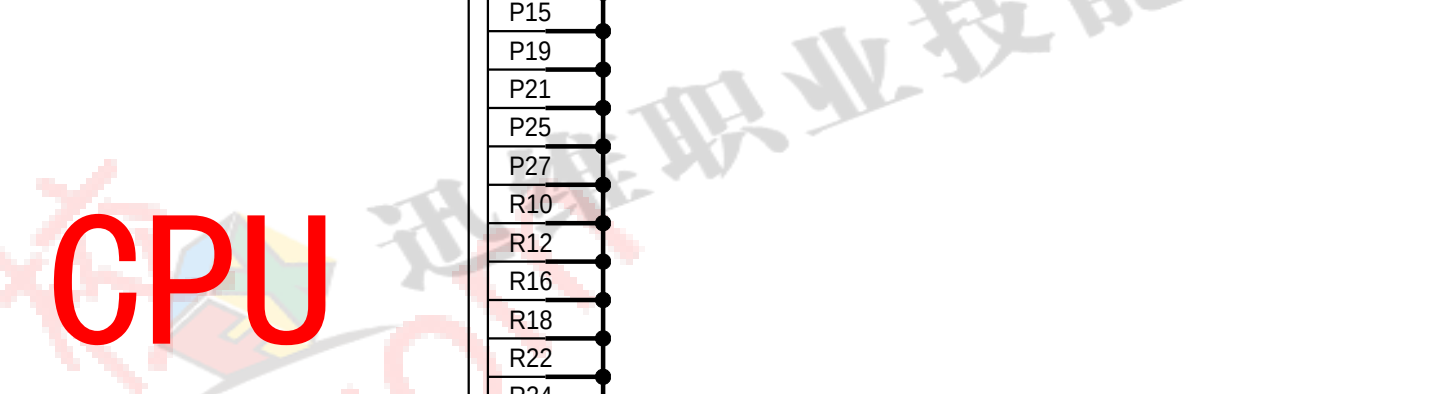
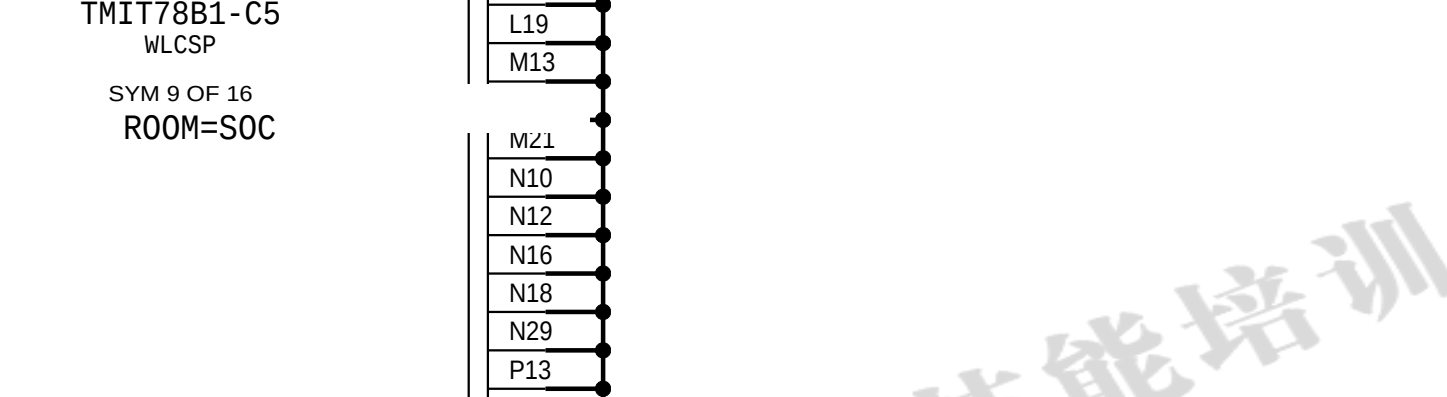
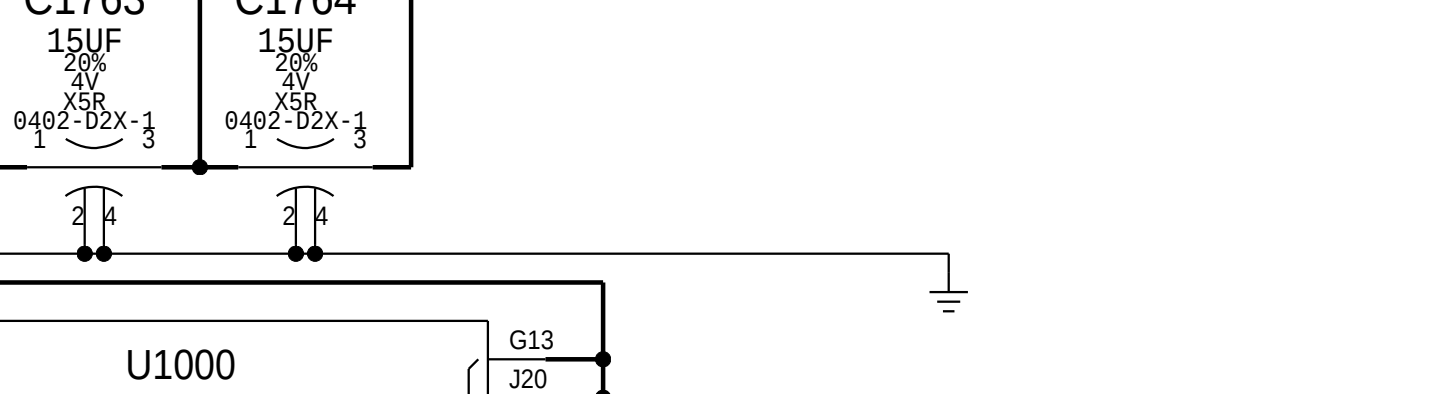
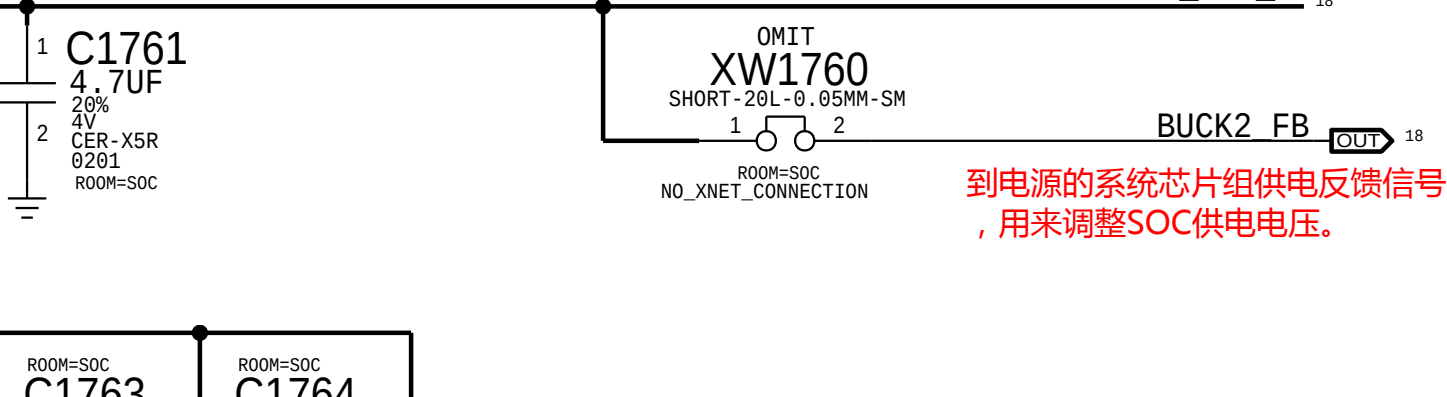
1.06V @ 18.3A MAX
0.8V @ 10.6A MAX
0.575V @ 3.4A MAX
PP_GPU_18



1.06V @ 18.3A MAX
0.8V @ 10.6A MAX
0.575V @ 3.4A MAX
PP_GPU_18



系统芯片组供电，0.8V左右
PP_SOC_S1_18



[illegible]

限流、保险电阻，



SOC - POWER SUPPLIES

D

C

B

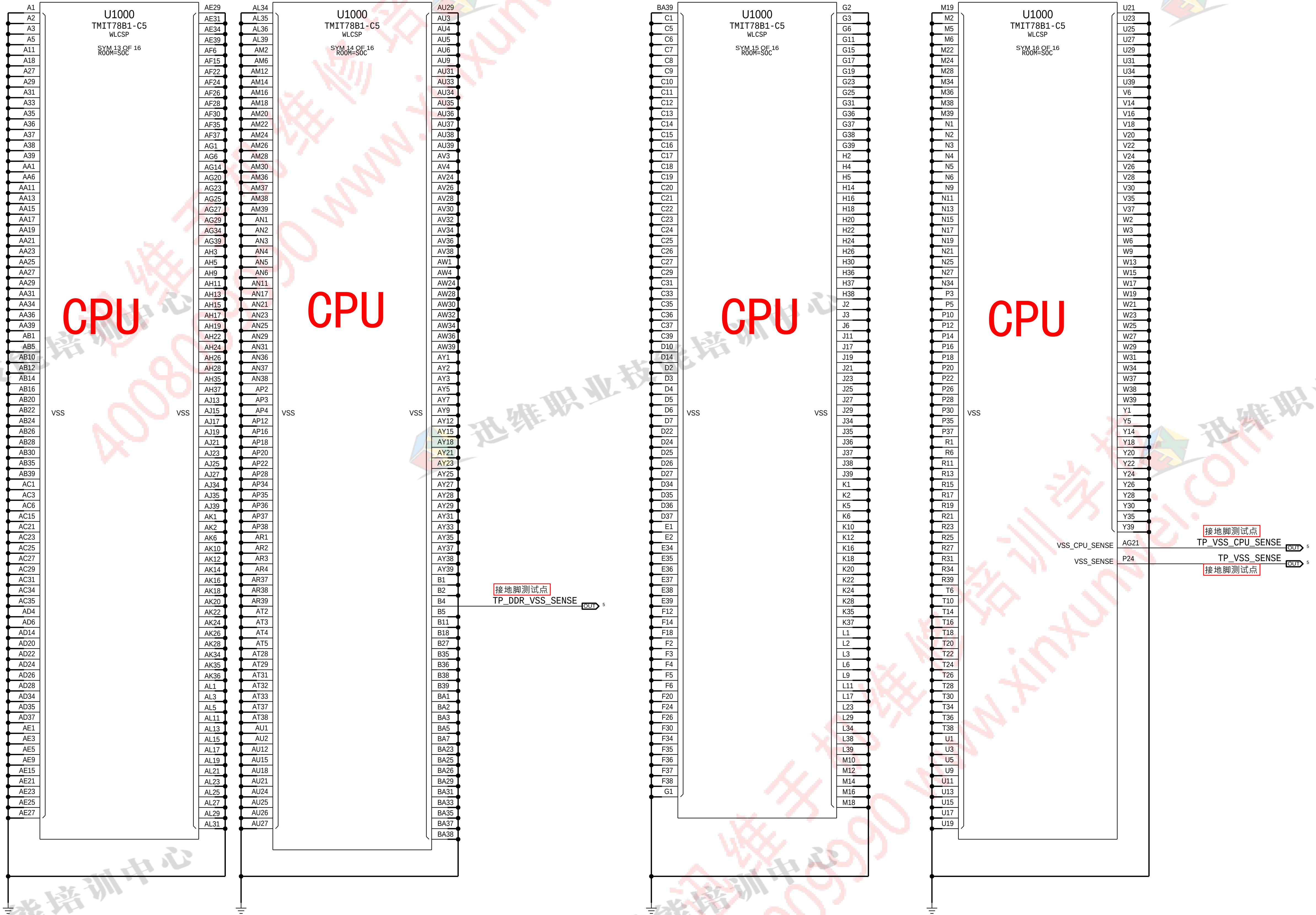
A

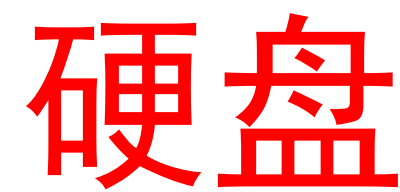
D

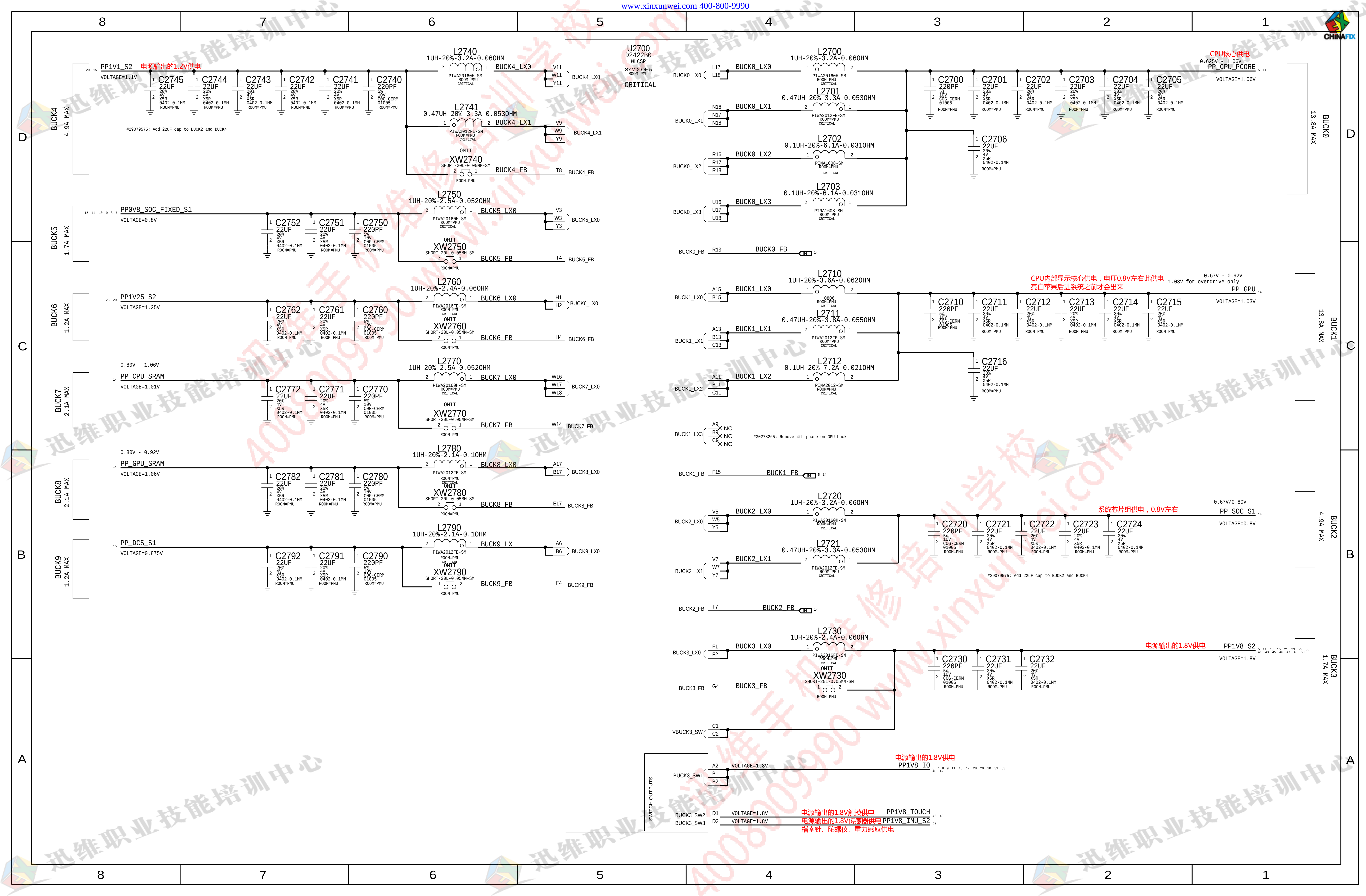
C

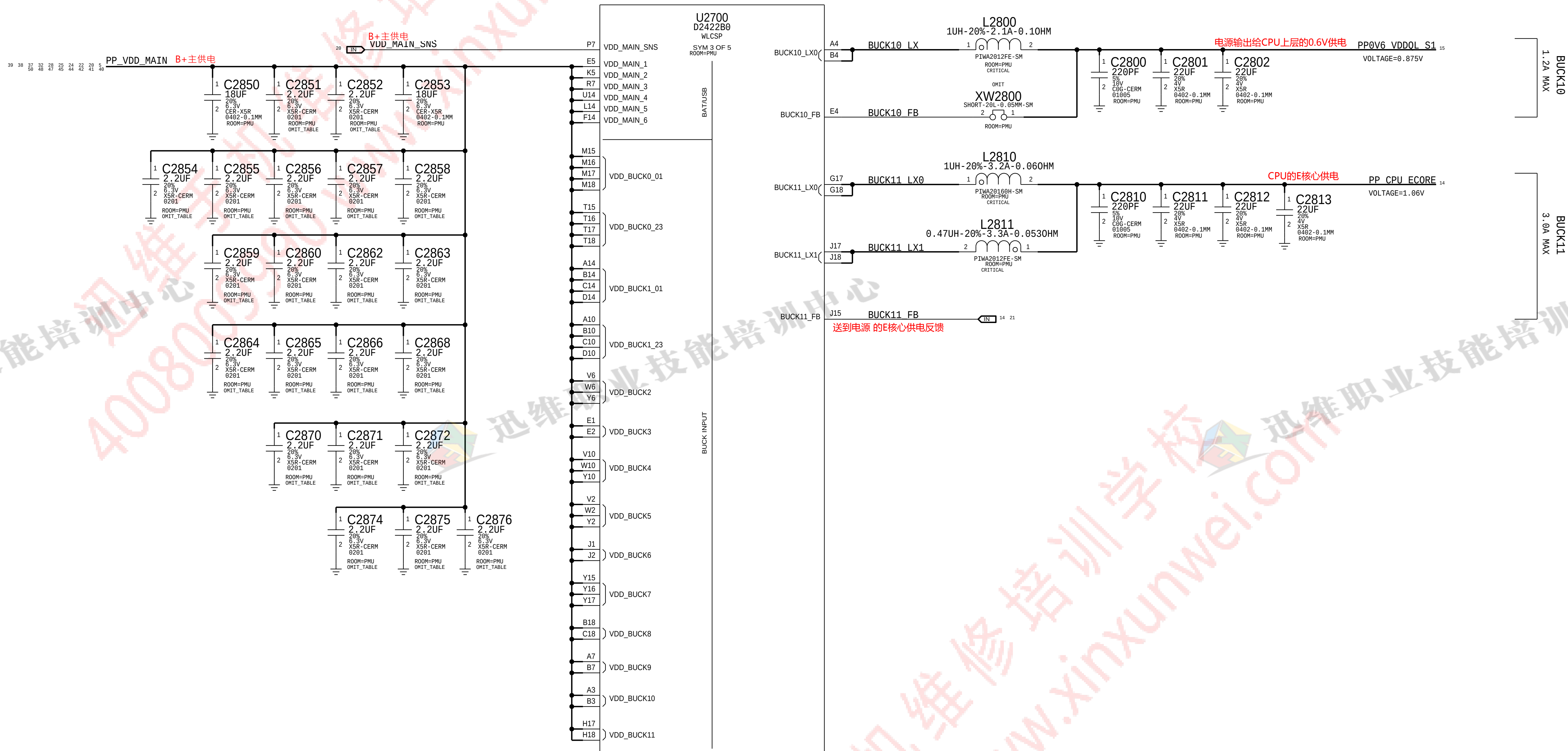
B

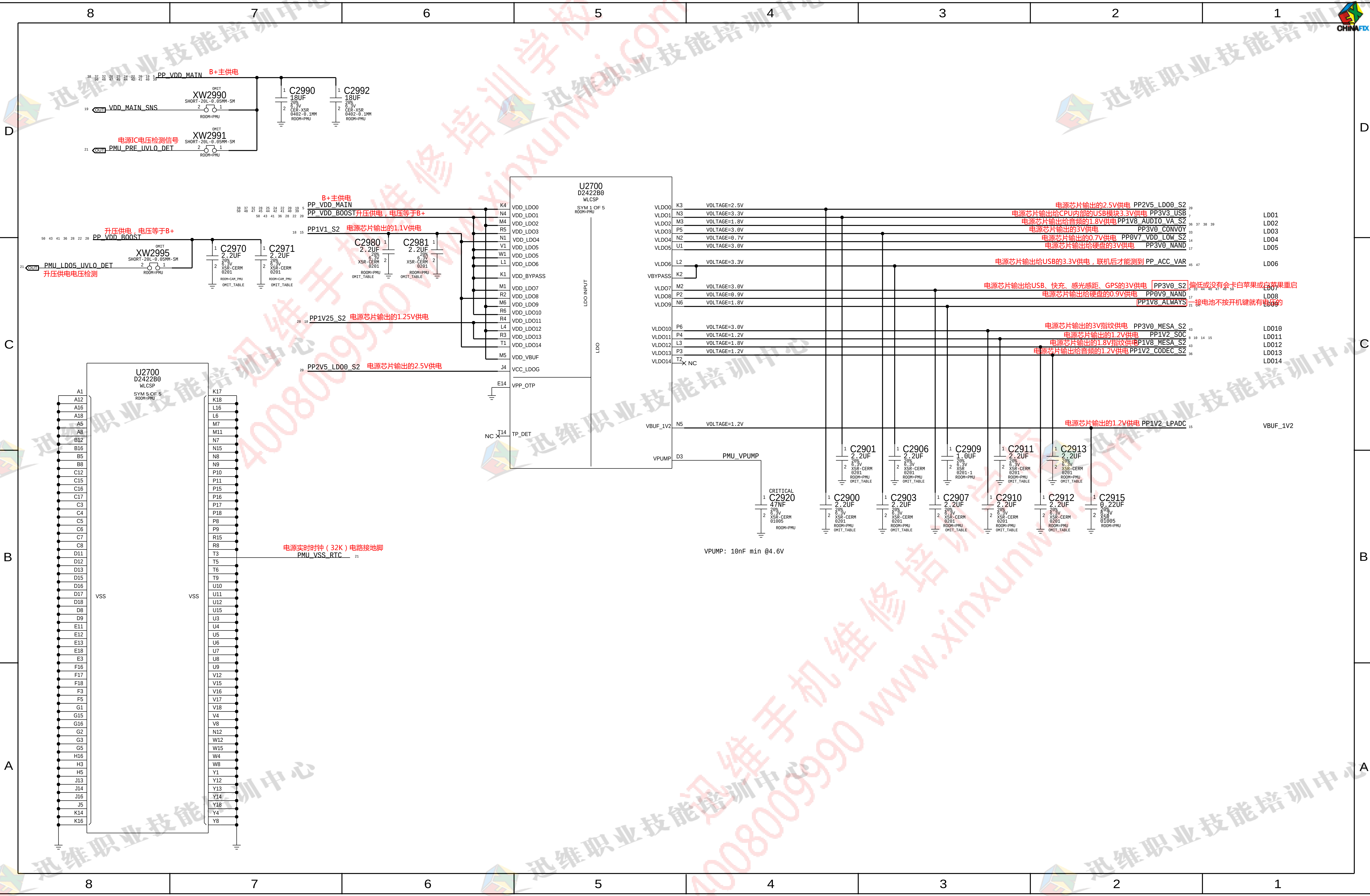
A



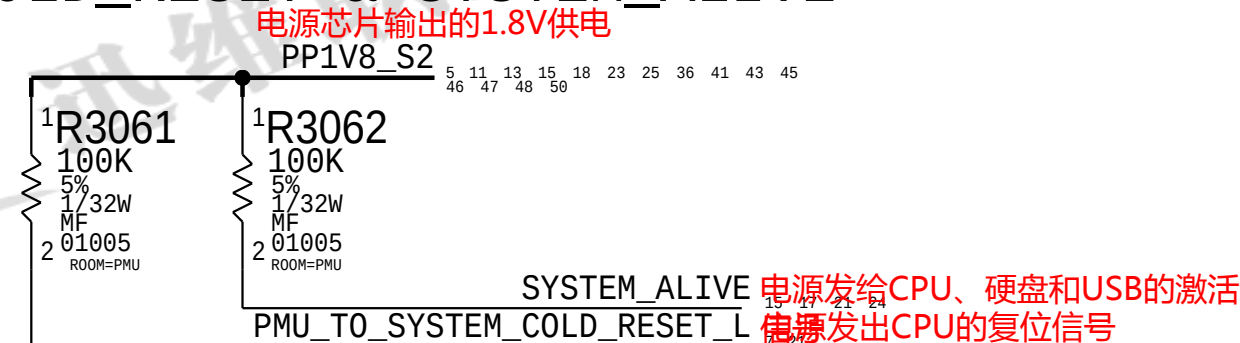




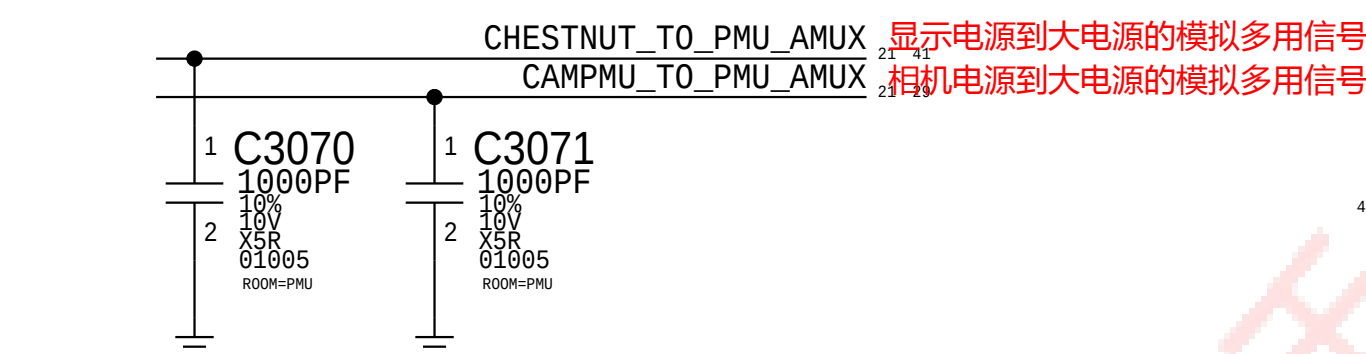




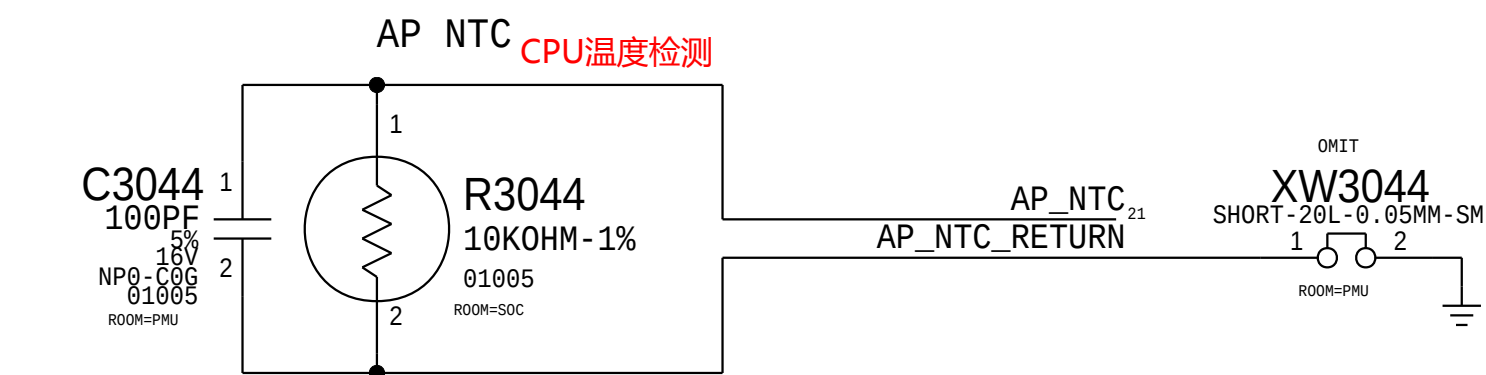
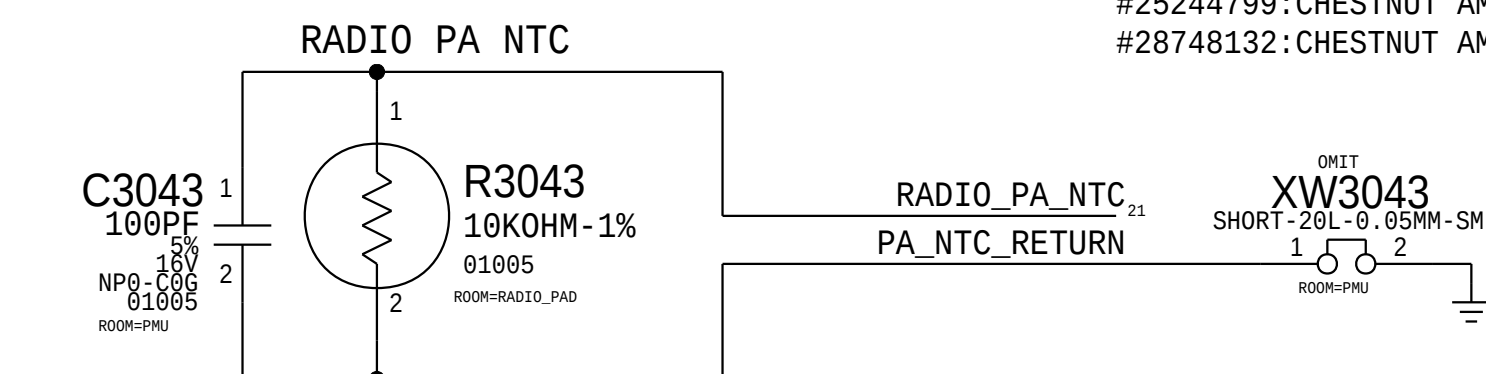
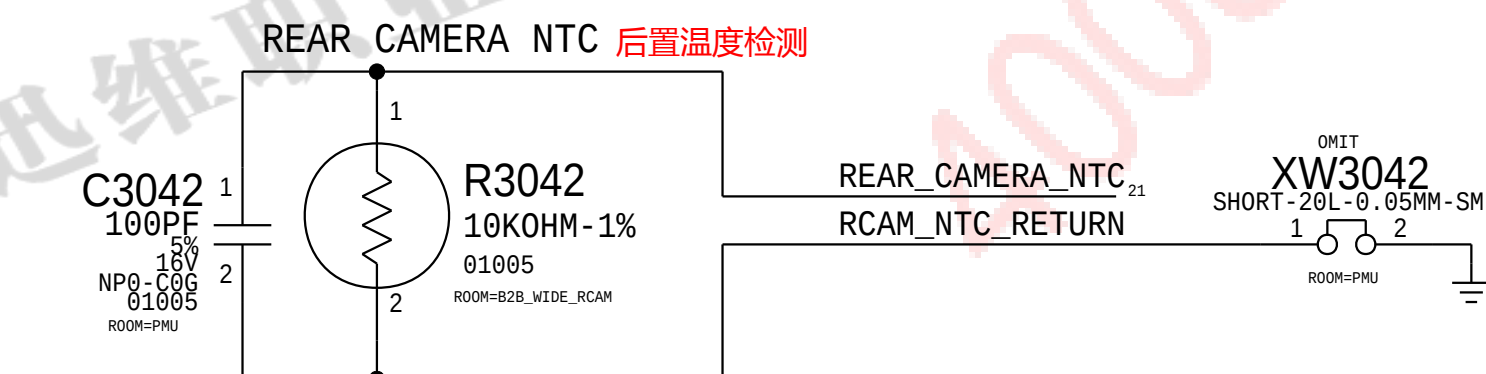
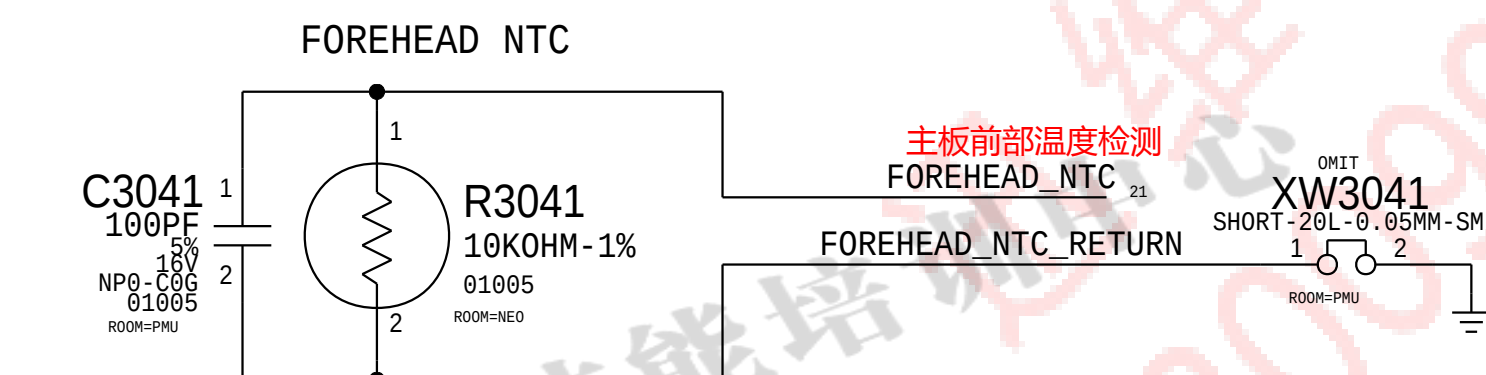
COLD_RESET & SYSTEM_ALIVE



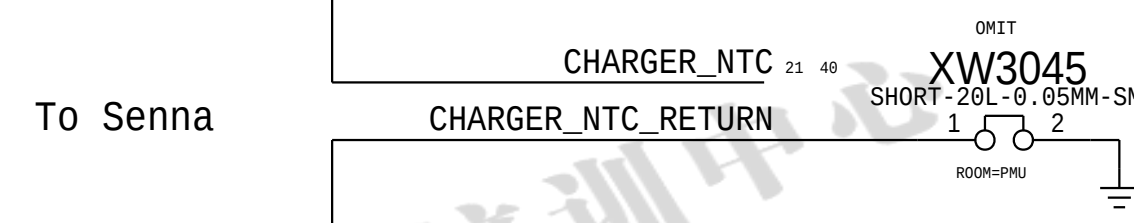
Chestnut / Ansel AMUX



温度检测 NTCs



CHARGER NTC (IN SIP)



NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC

#29404057:Damping R for ACTIVE_RDY

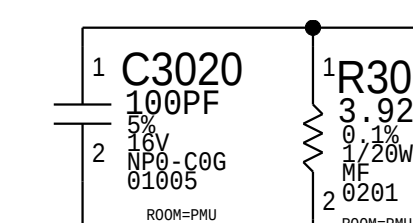
电源到USB准备就绪启动信号
PMU_TO_AP_HYDRA_ACTIVE_READY

#29300440:Connect PMU crash to SoC

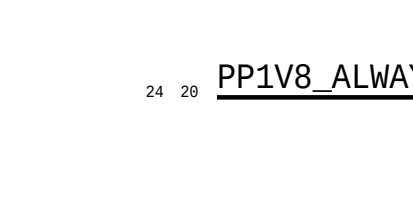
#27852679:No SDA R, PoR is SPMI

#25244799:200K pull-down

#25244799:CHESTNUT AMUX AND PWR_EN ON SAME CHANNEL
#28748132:CHESTNUT AMUX AND PWR_EN SWAP BETWEEN A6/B6



PP1V8_ALWAYS



CHARGER_NTC

CHARGER_NTC_RETURN

To Senna

R3009 100

1/32W

01005

ROOM=PMU

5%

1/32W

01005

ROOM=PMU

5%

1/32W

01005

ROOM=PMU

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1/32W

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1/32W

01005

ROOM=PMU

5%

1/32W

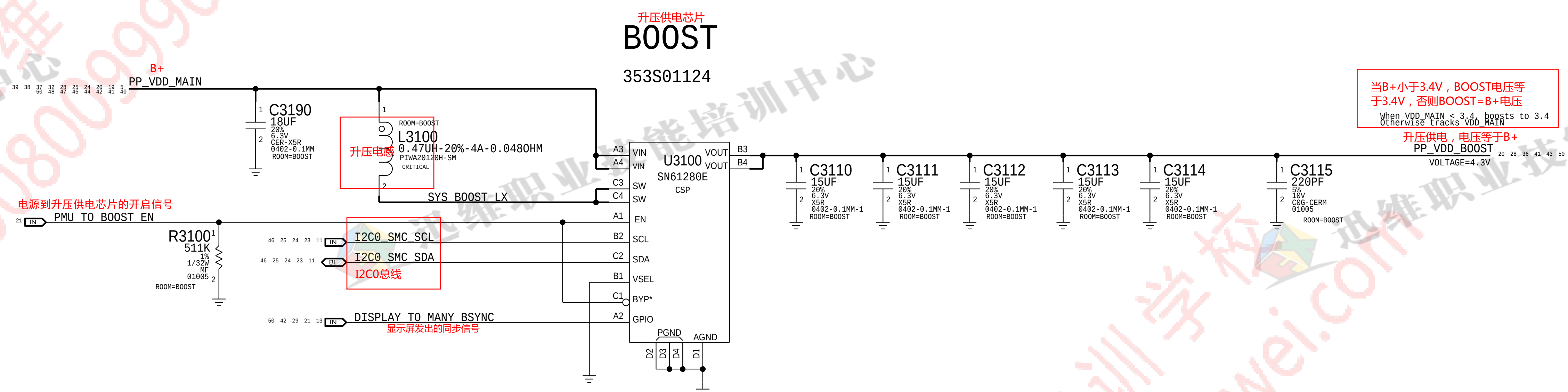
01005

ROOM=PMU

5%

1/32W

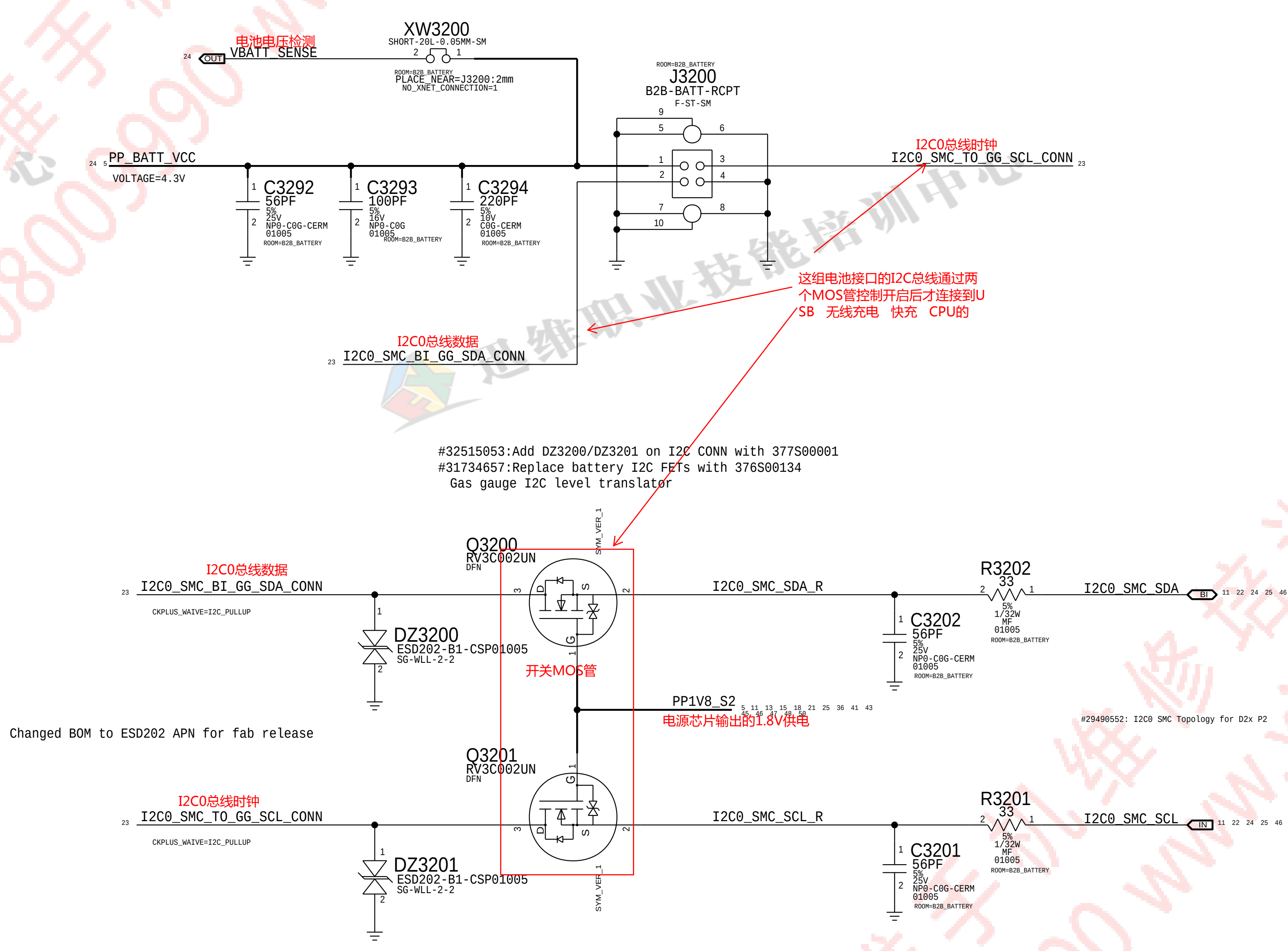
01005





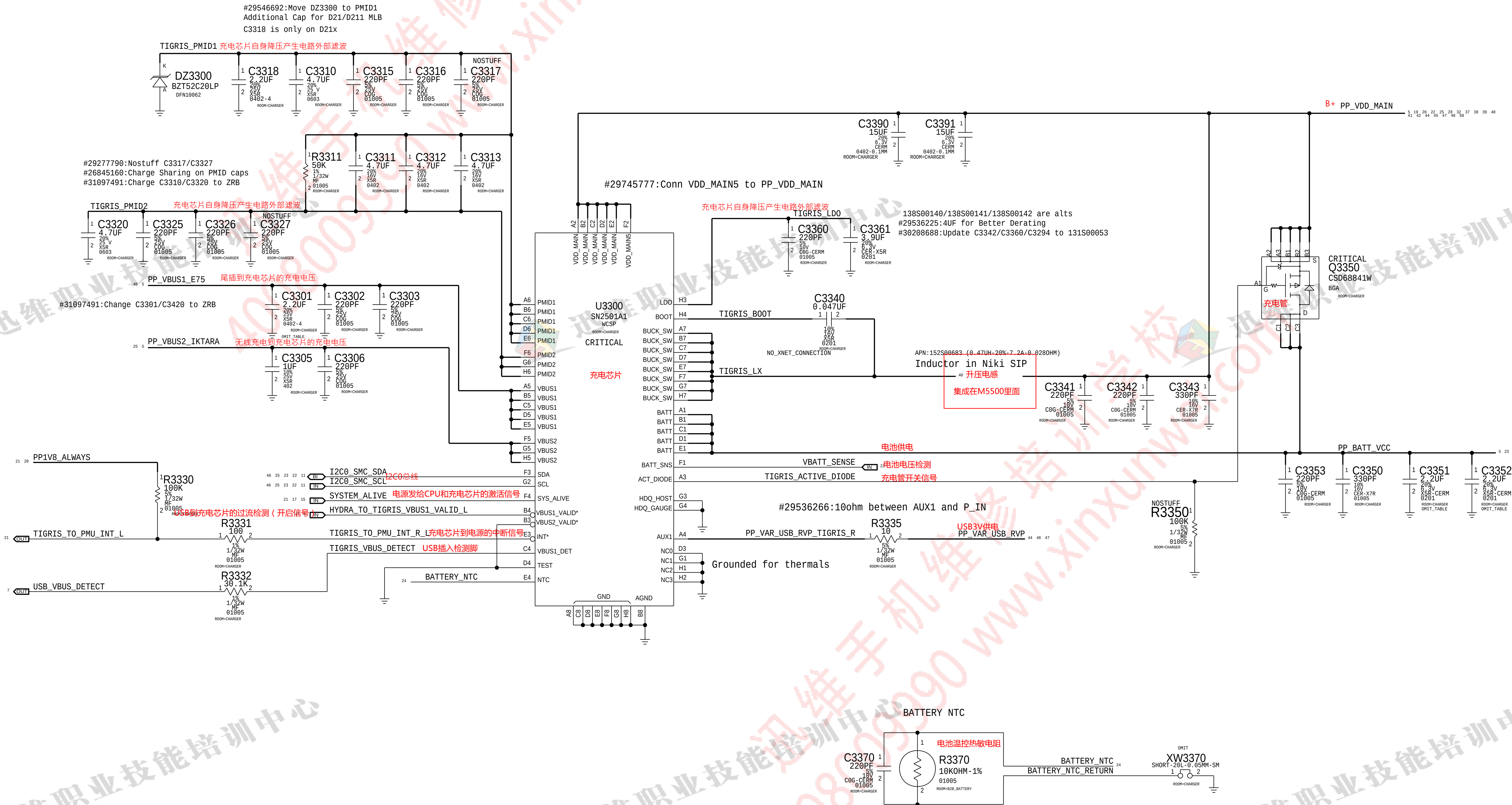
Battery Connector

THIS ONE ON MLB ---> 516S00232



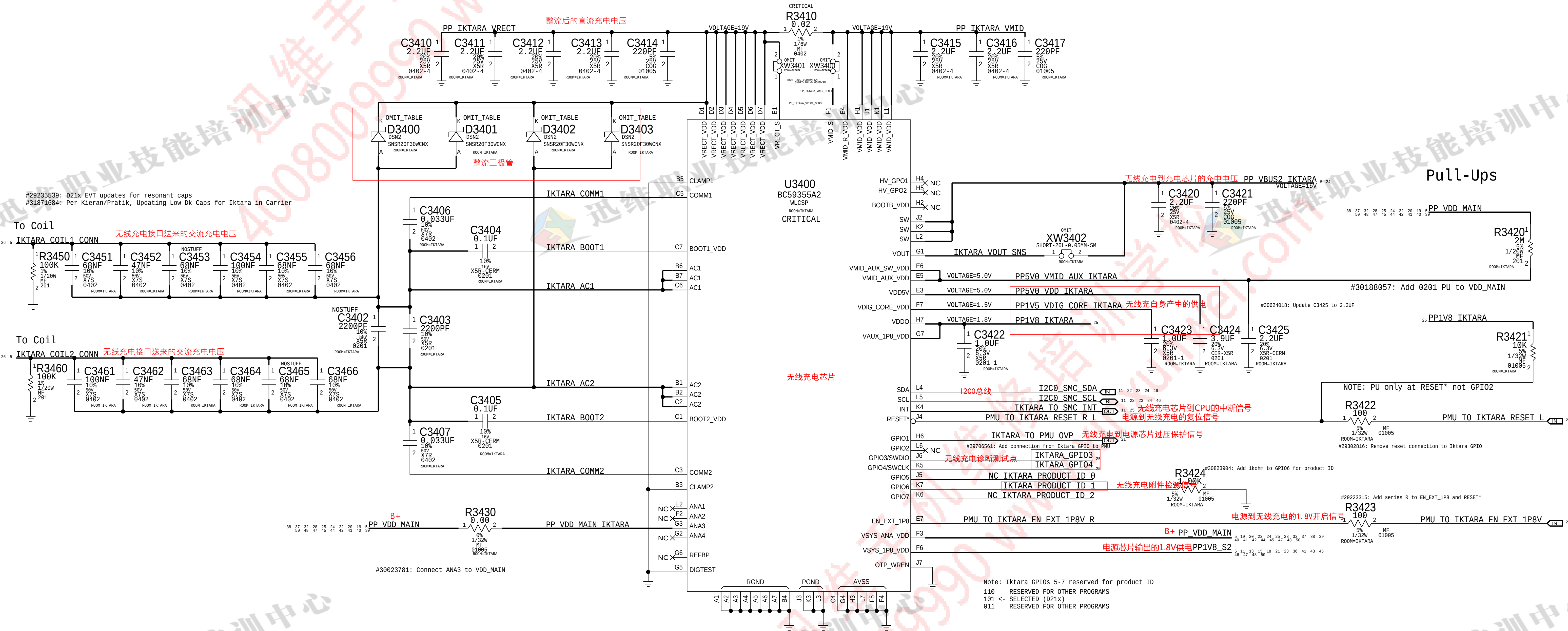
Tigris 2

APN:343S00170



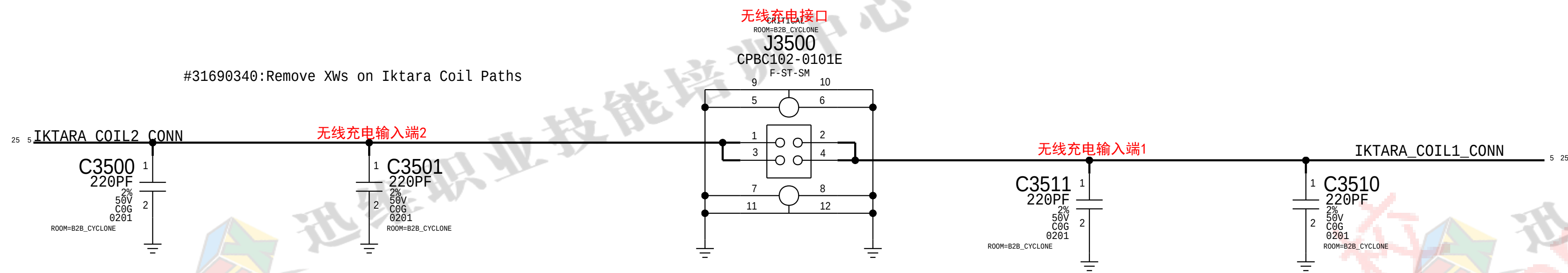
Iktara

Iktara Debug



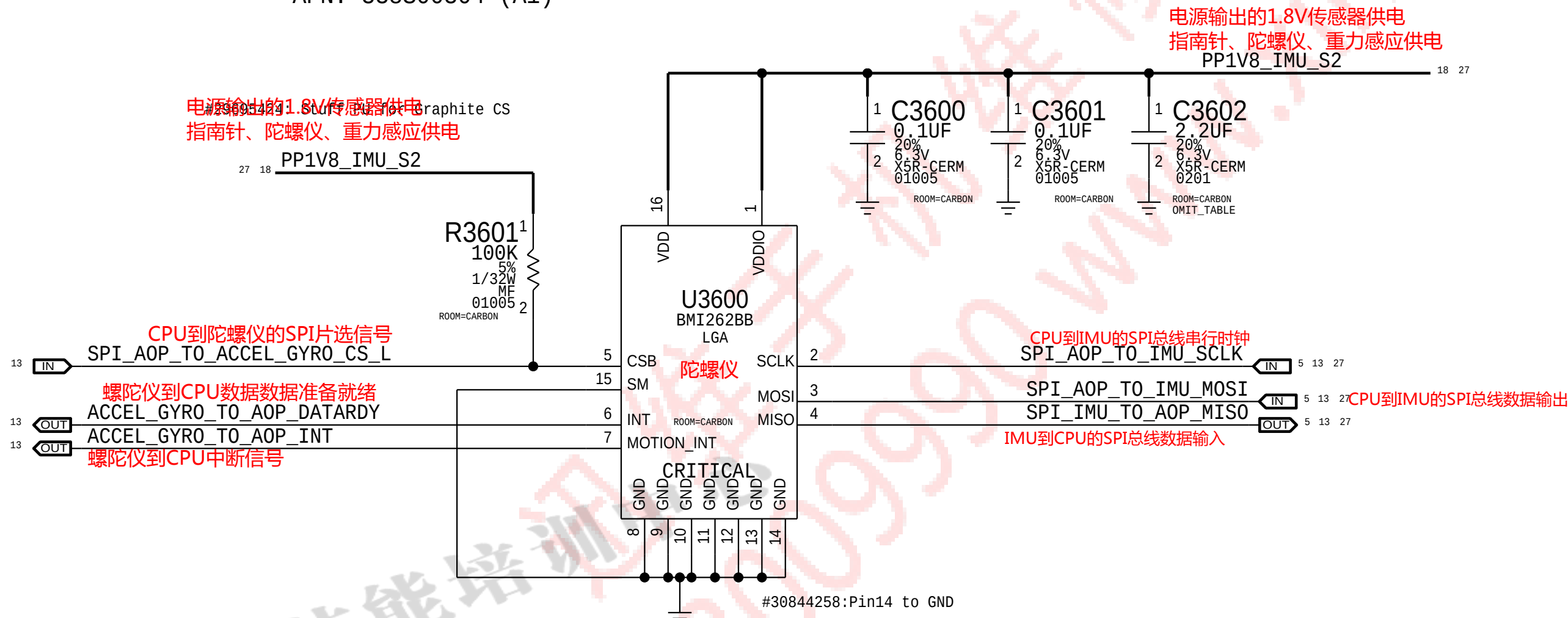
Cyclone connector

THIS ONE ON MLB ----> 516S00311



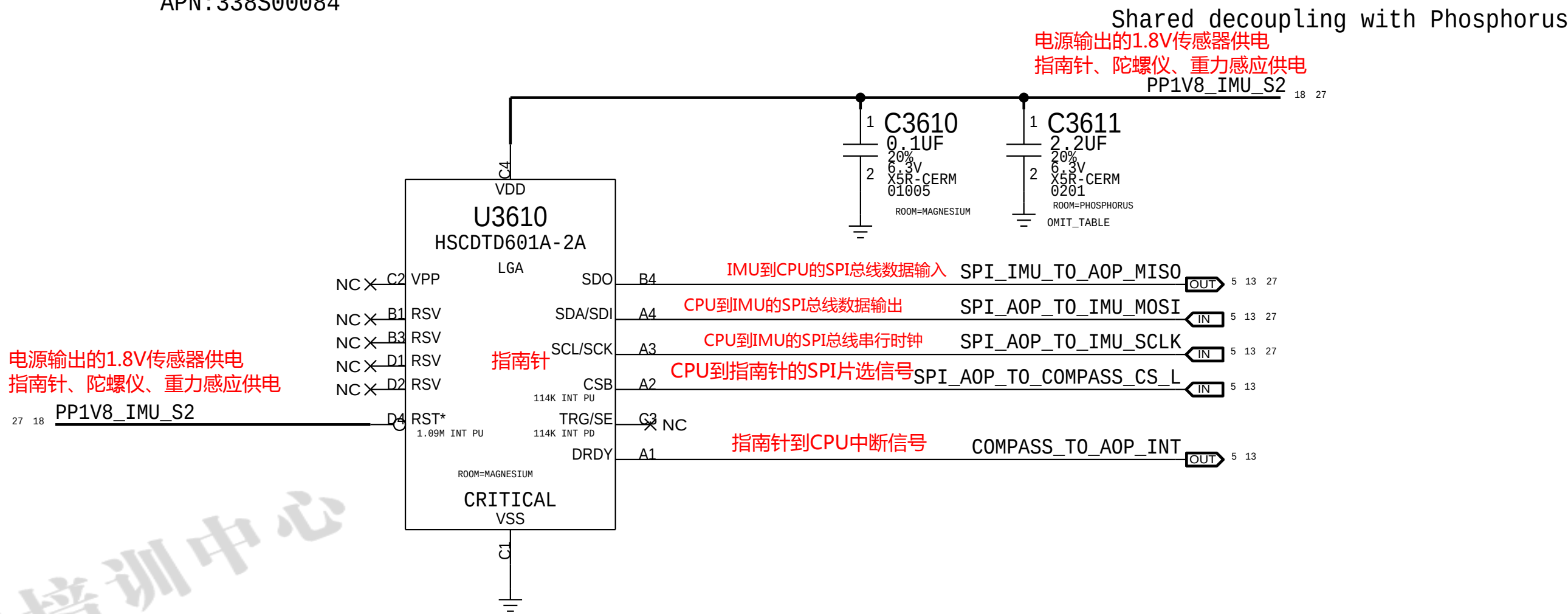
Graphite - Accel & Gyro

APN: 338S00304 (A1)

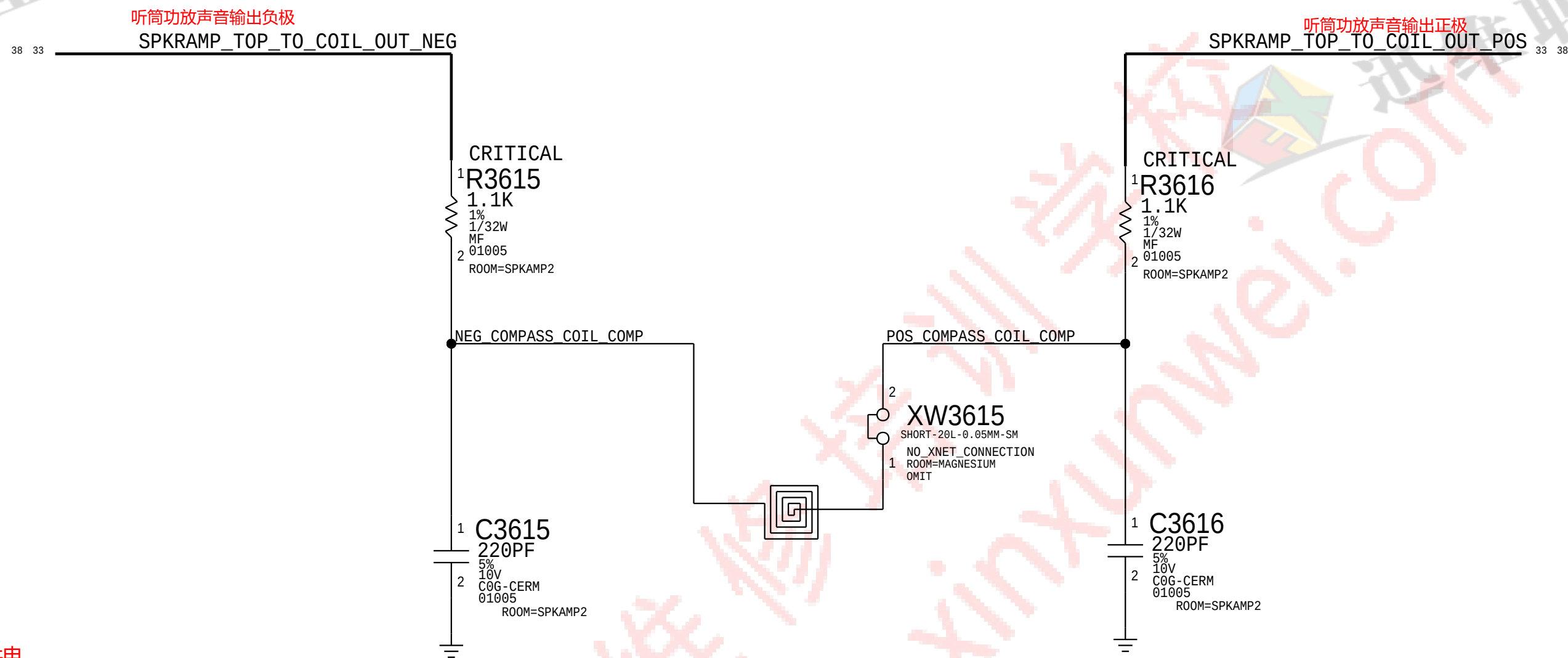


Magnesium - Compass

APN:338S00084

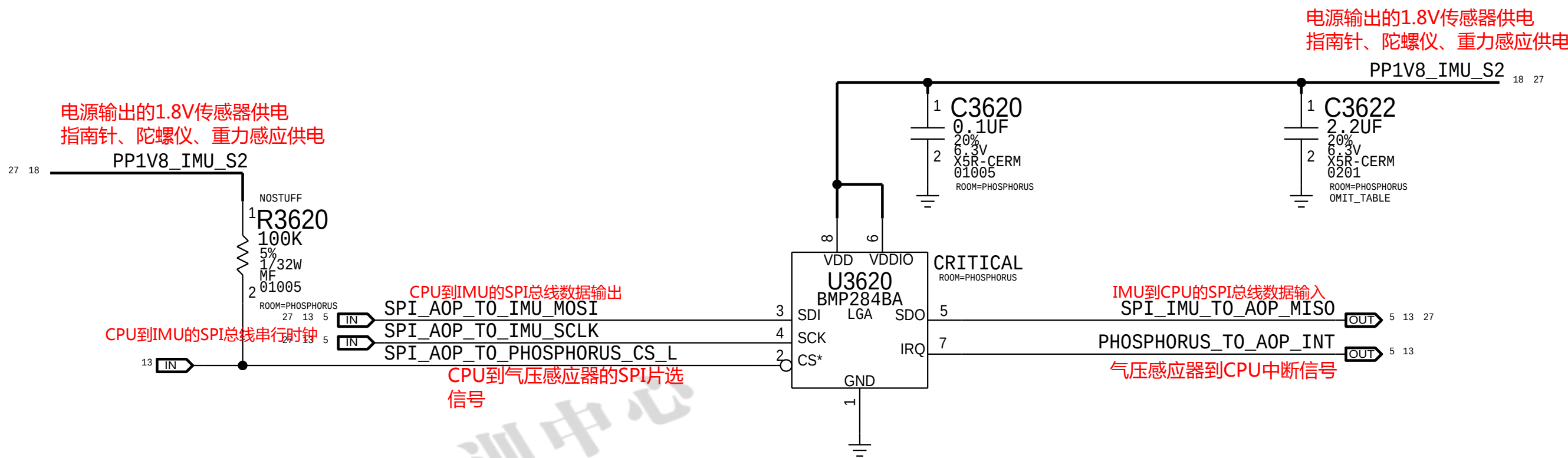


North Speaker Compass Compensation Coil

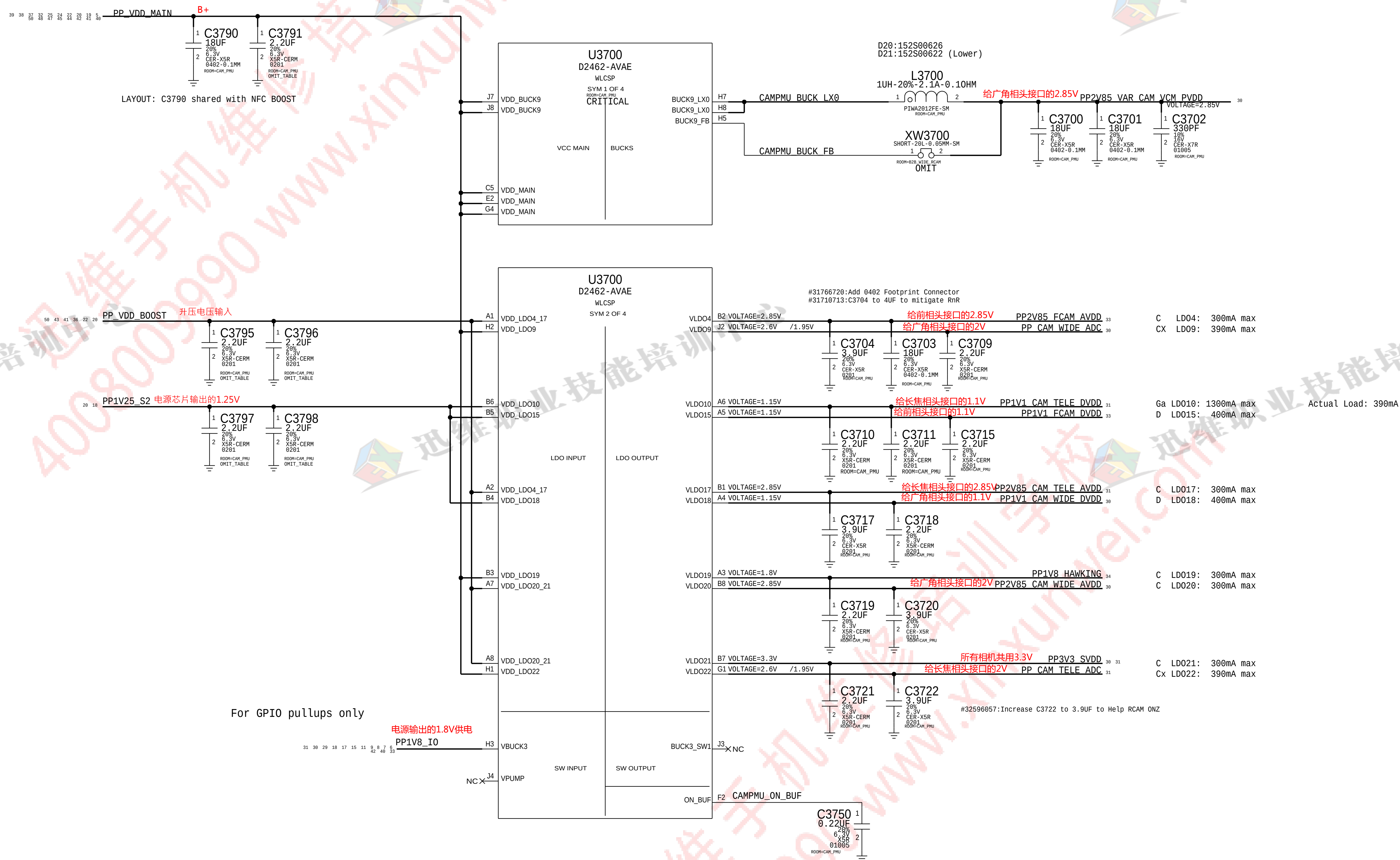


Phosphorus

APN: 338S00334



相机电源



D

D

C

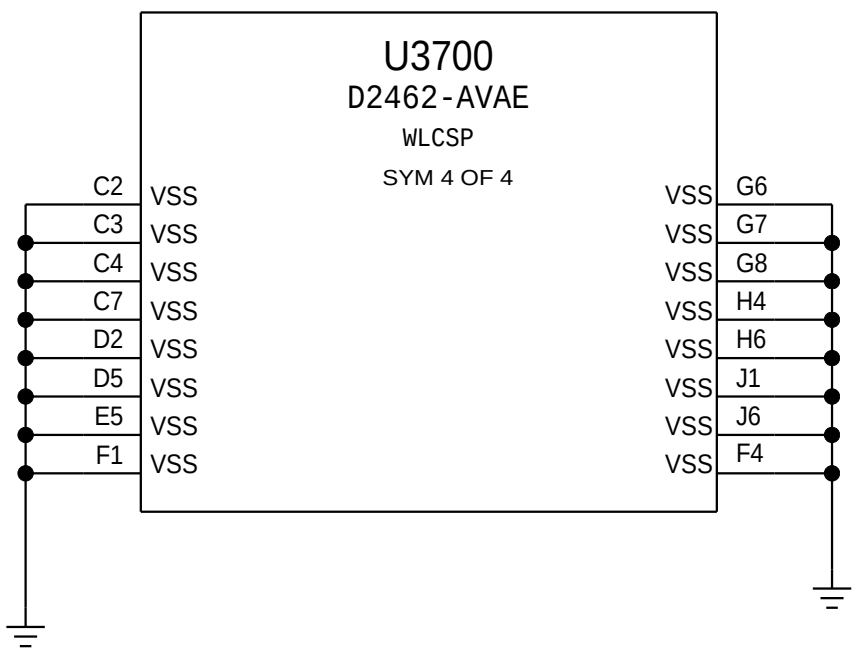
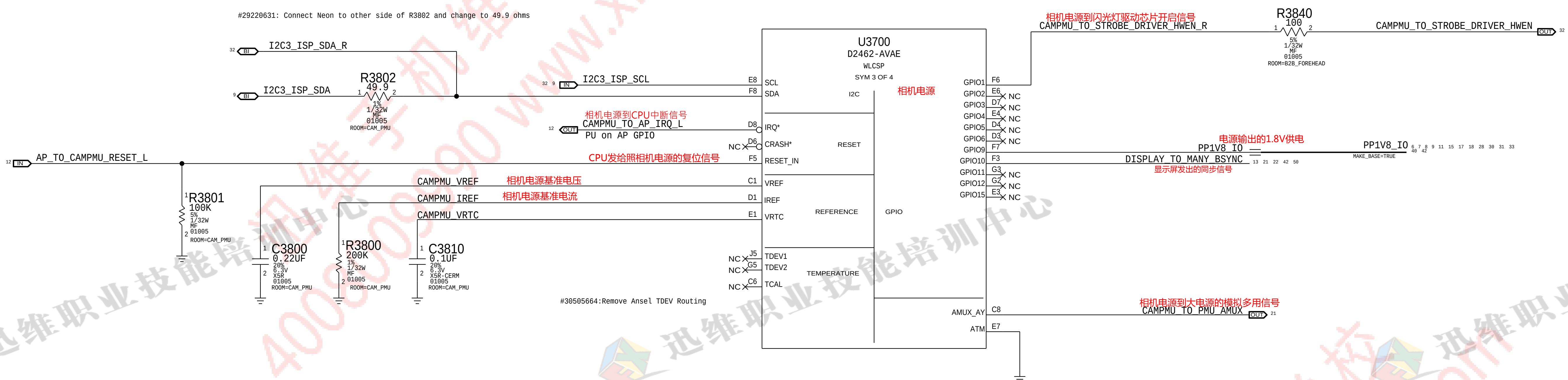
C

B

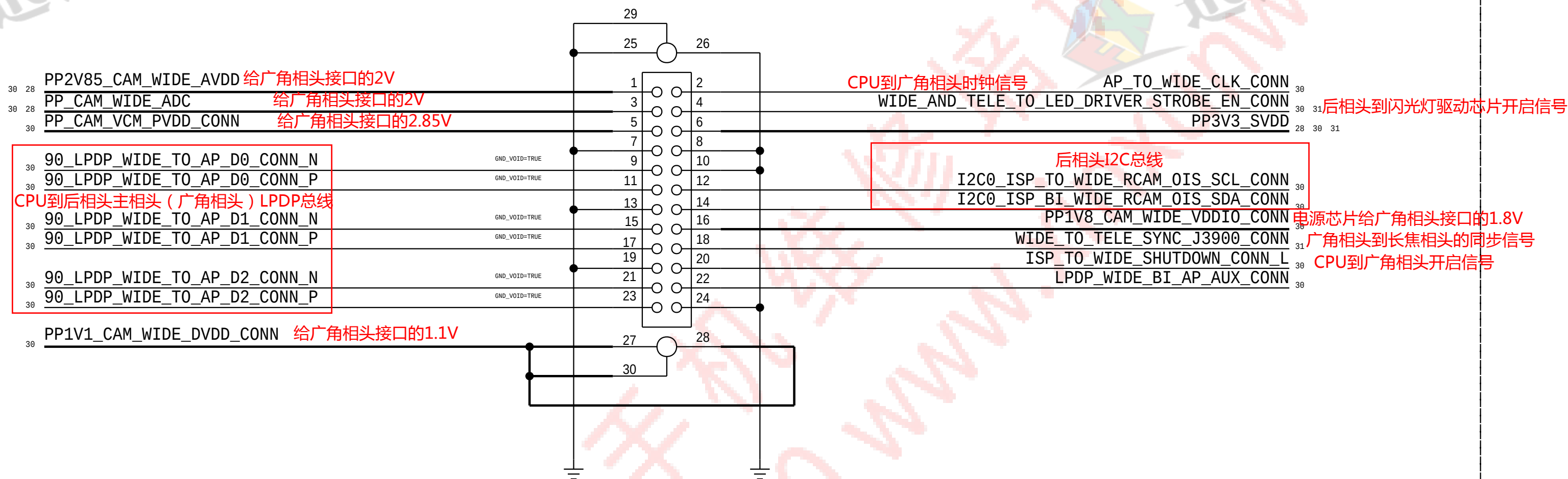
B

A

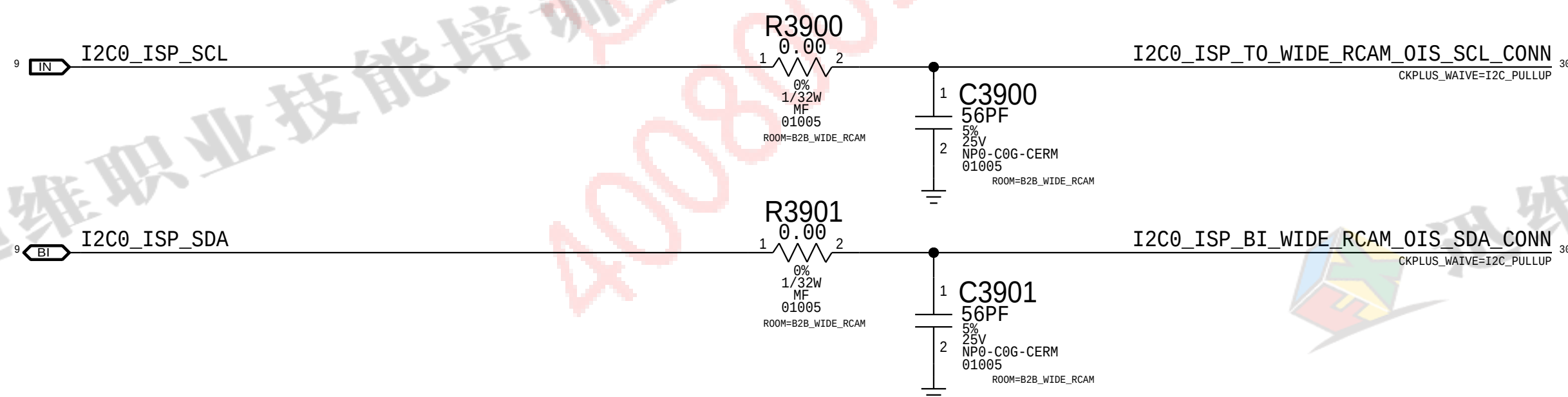
A



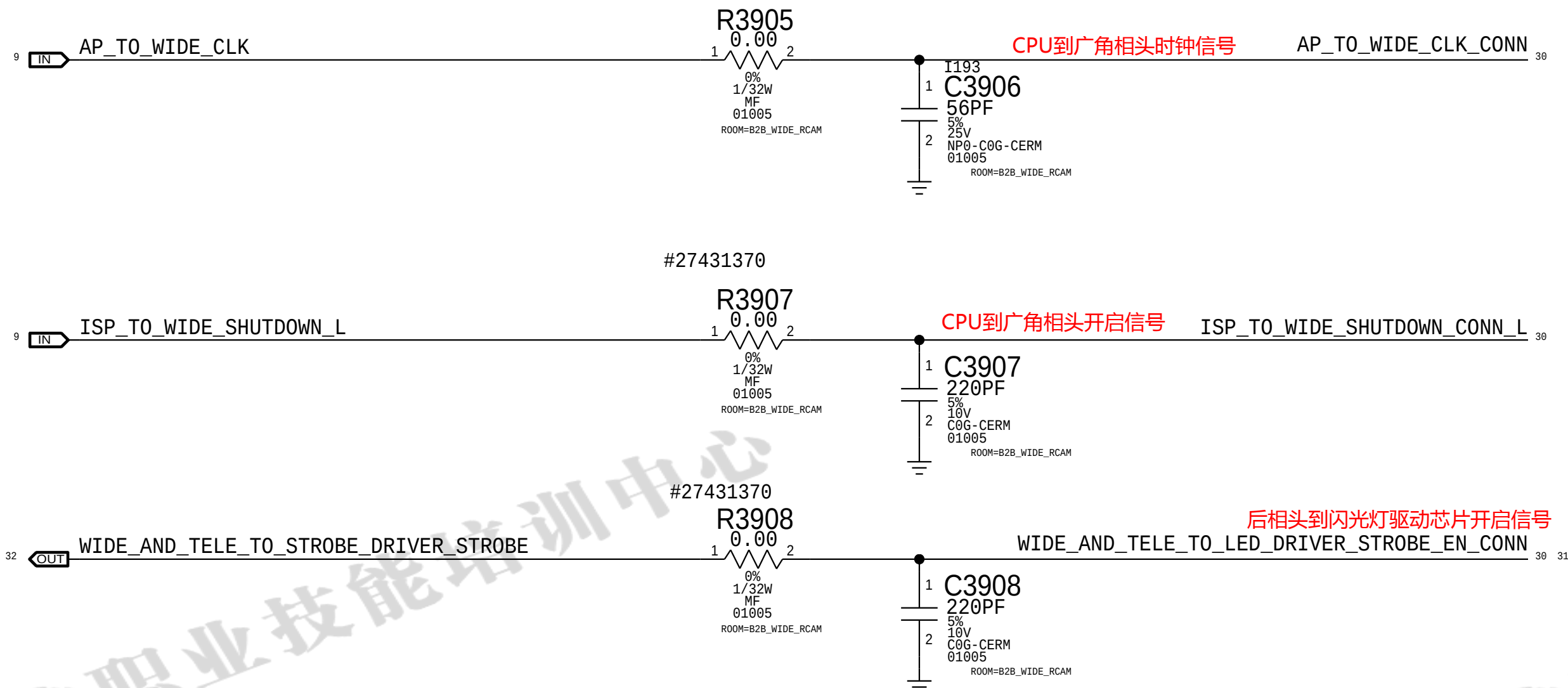
Wide Camera Connector

THIS ONE ----> 516S00310/P2 RCPT (USED ON MLB)
516S00309/P2 PLUG (USED ON FLEX)J3900
AA27DK-S024VA1
F-ST-SM

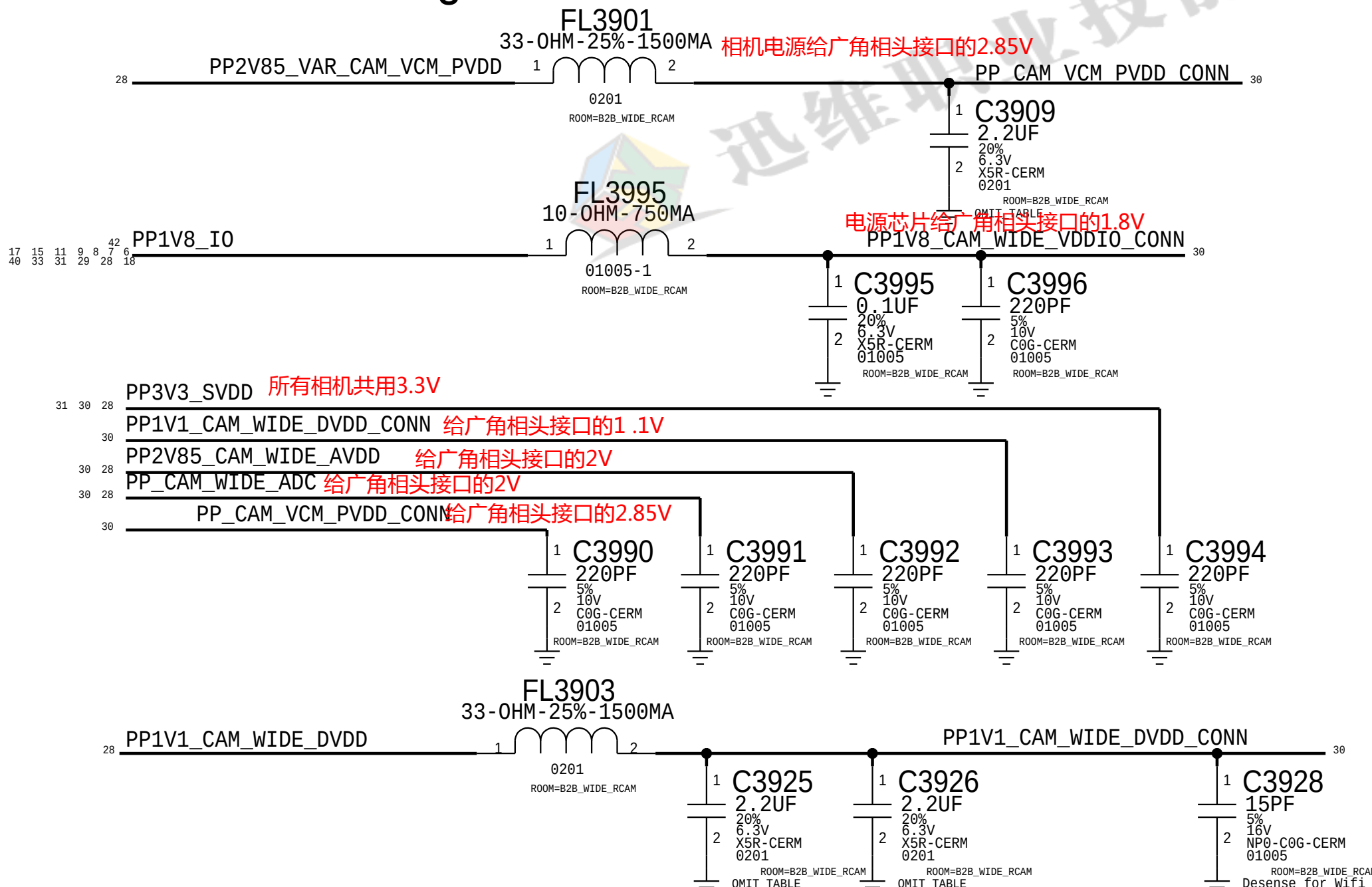
ISP I2C



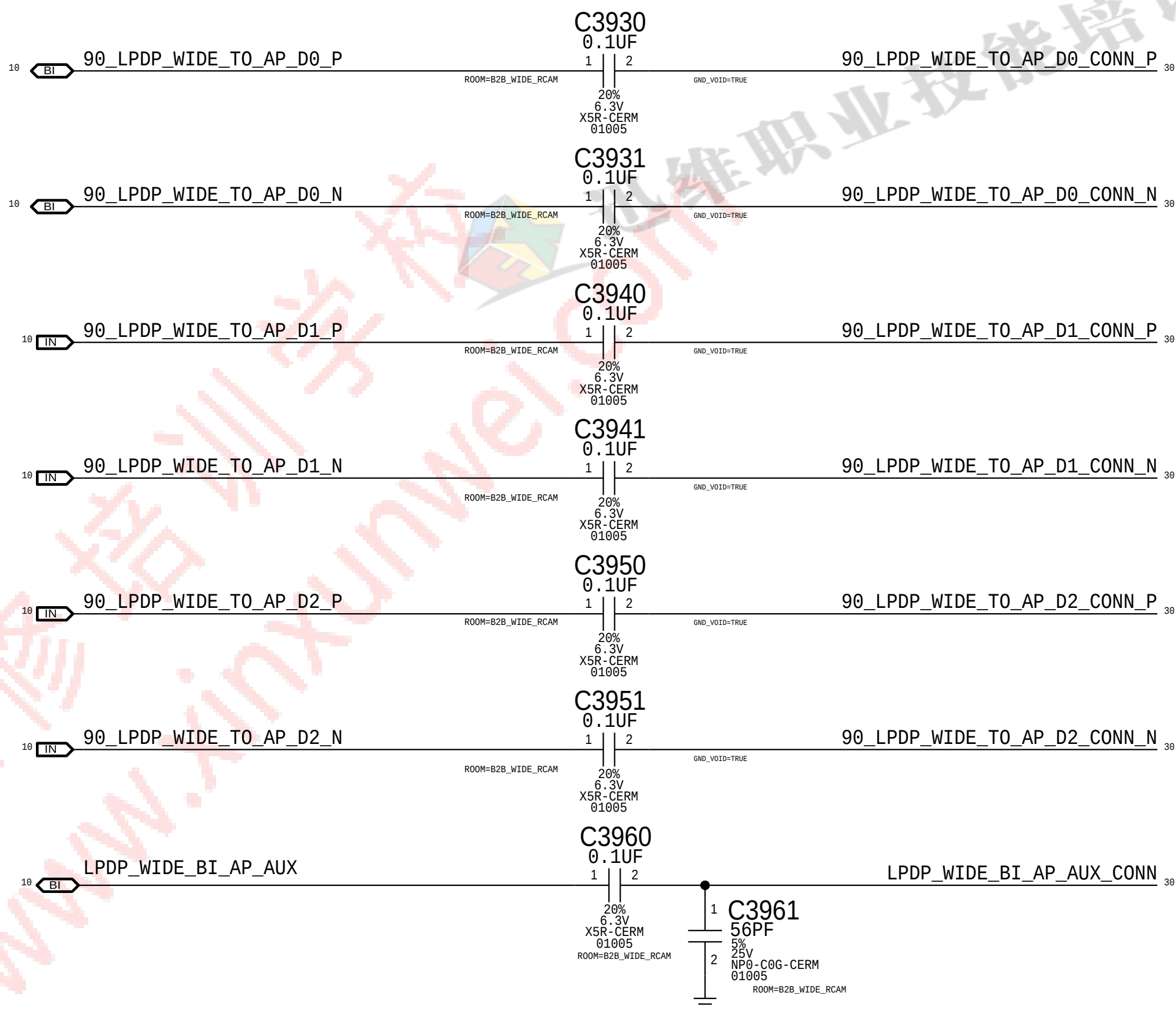
IO Filters



Power Filtering



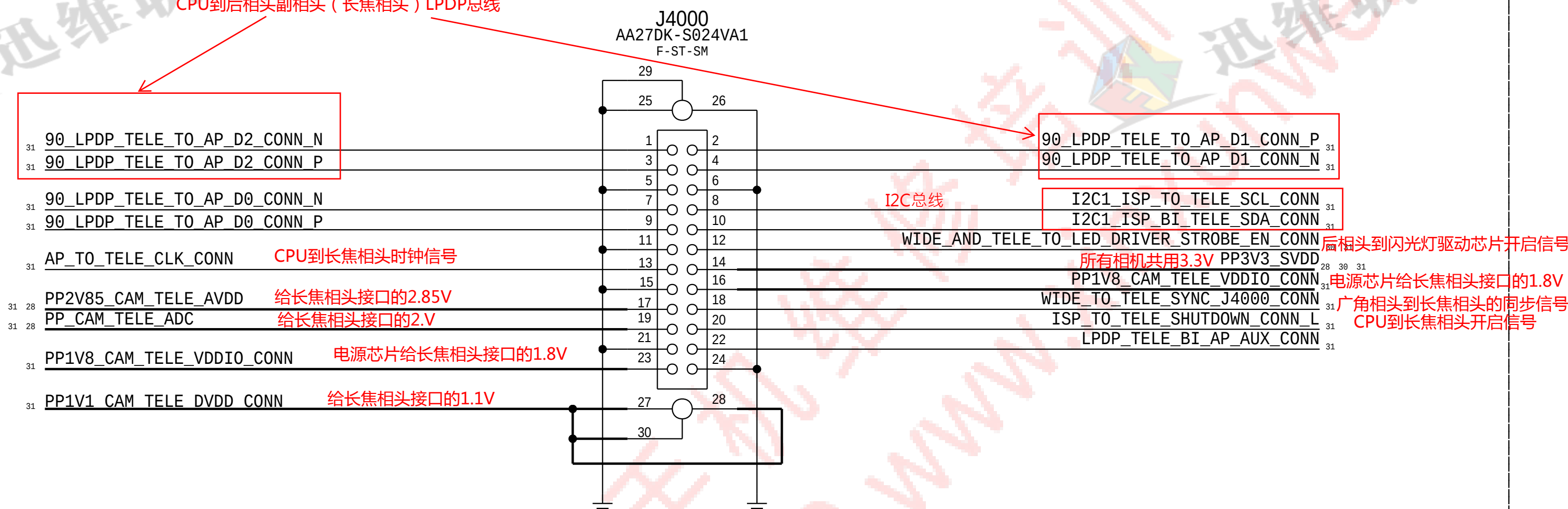
LPDP Filters



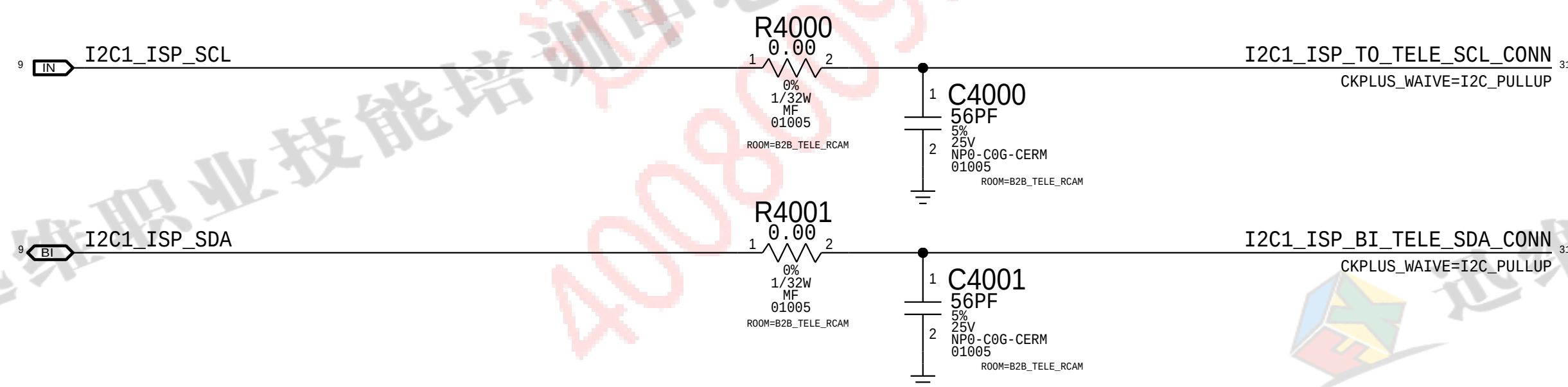
Tele Camera Connector 长焦相机

THIS ONE ---> 516S00310/P2 RCPT (USED ON MLB)
516S00309/P2 PLUG (USED ON FLEX)

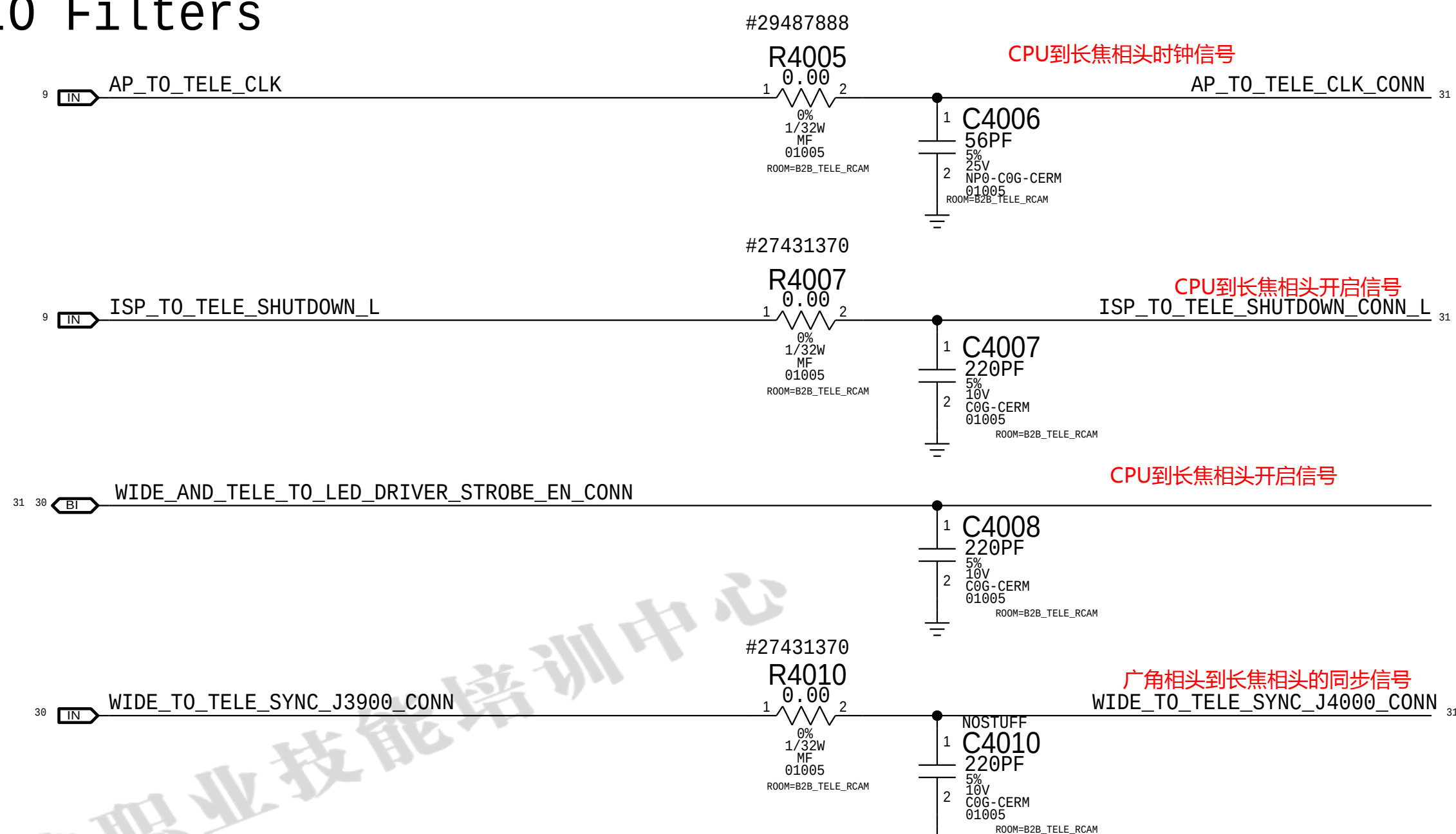
CPU到后相机副相机(长焦相机) LPDP总线



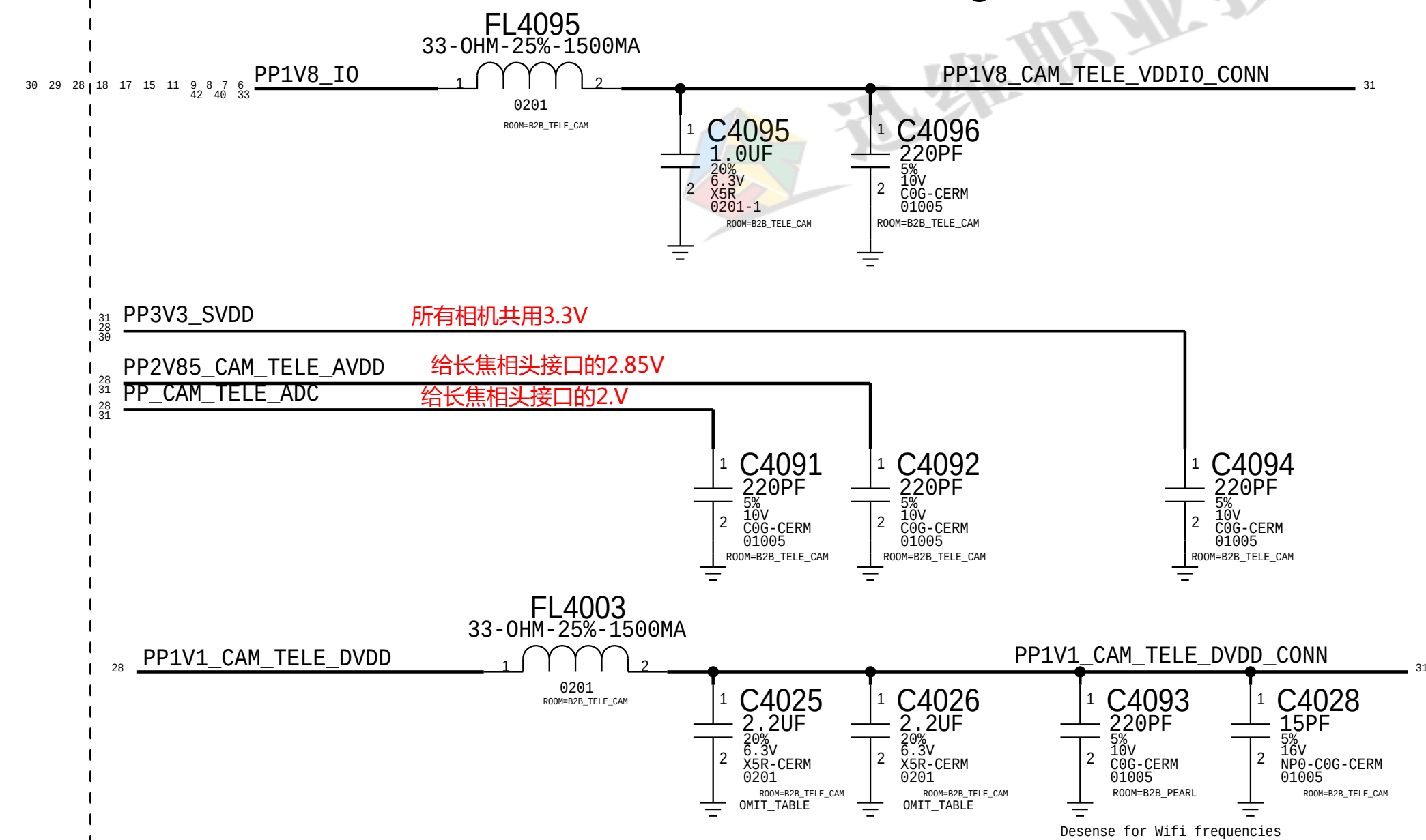
ISP I2C



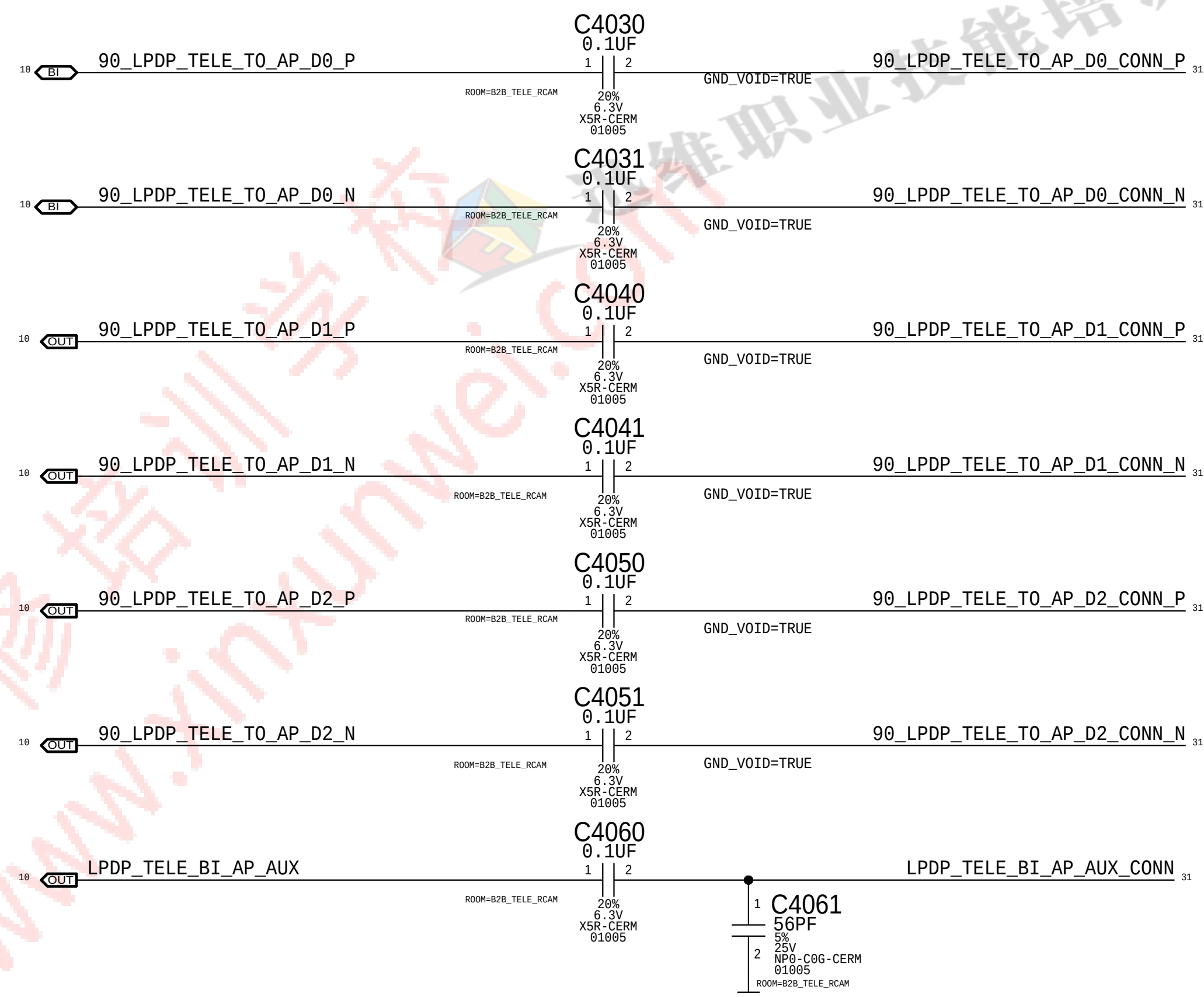
IO Filters



Power Filtering

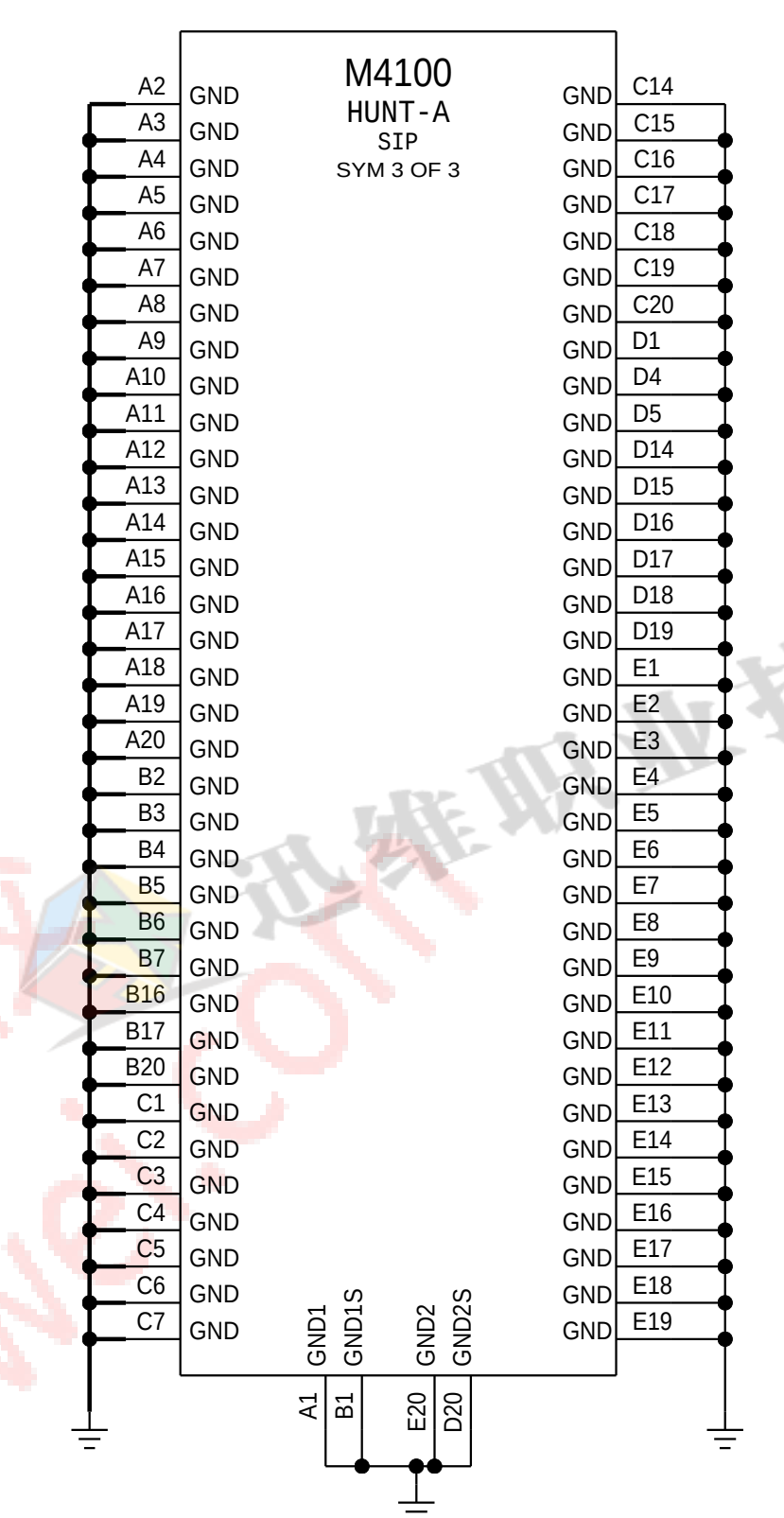
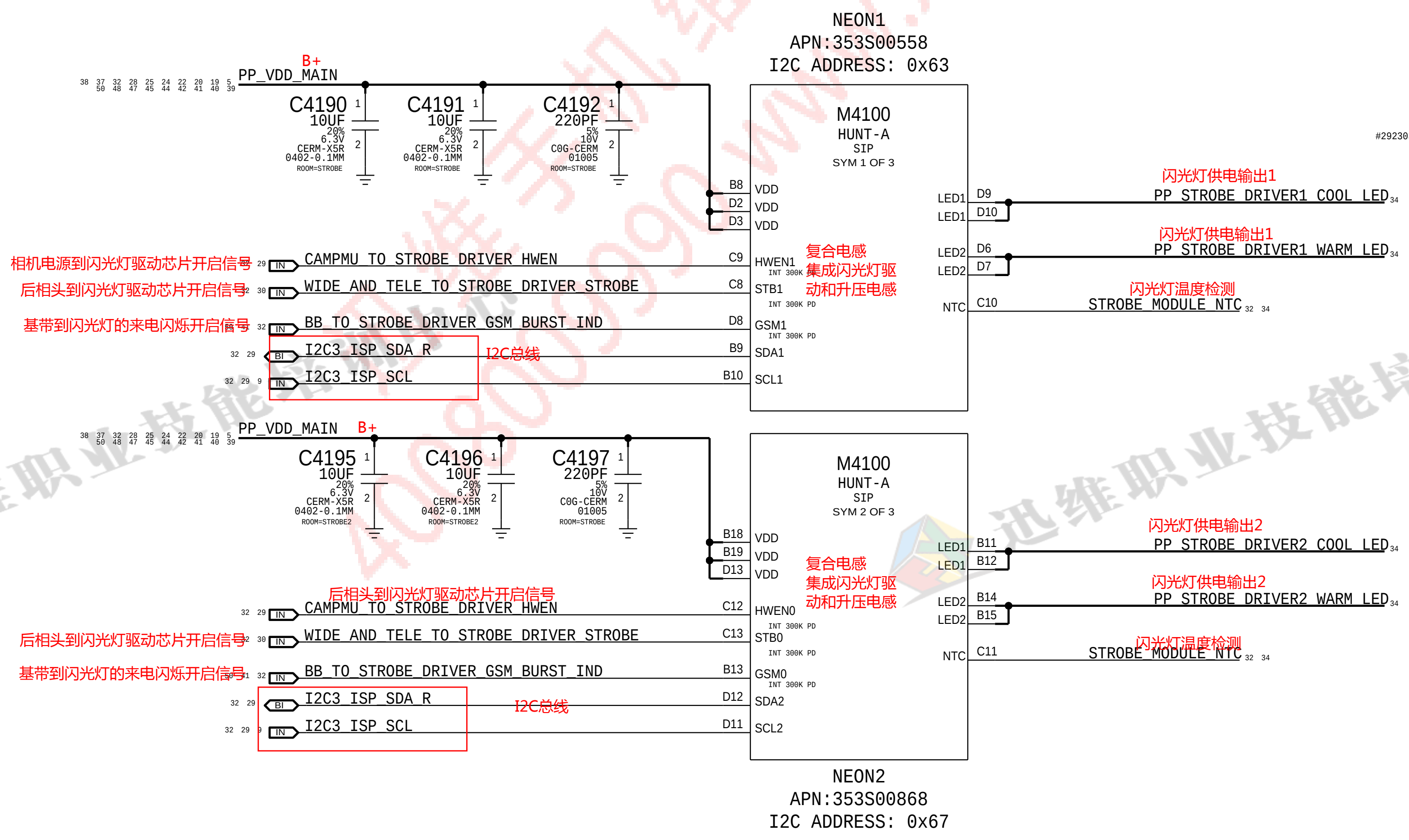


LPDP

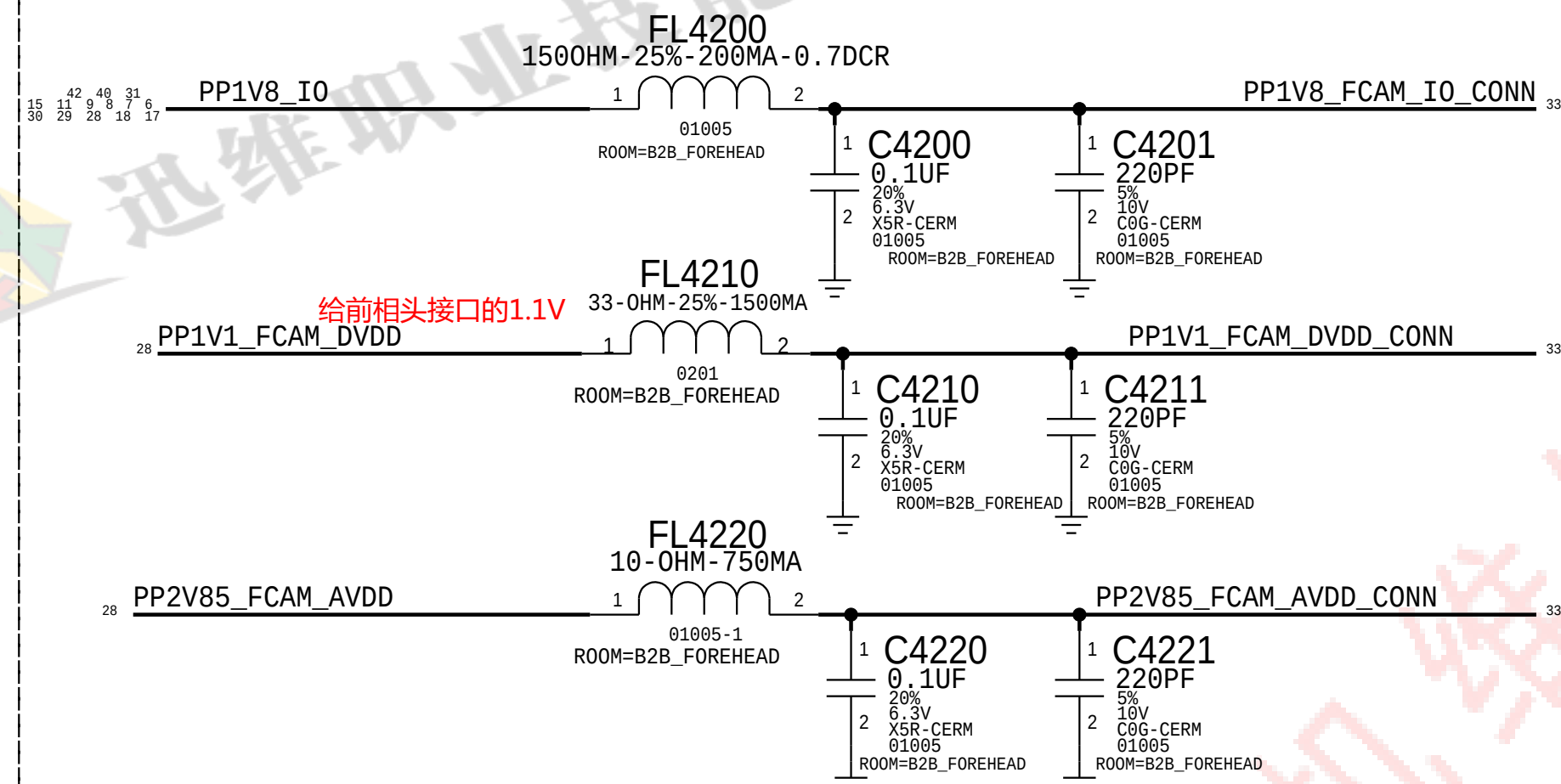


LED STROBE DRIVERS (NEON)

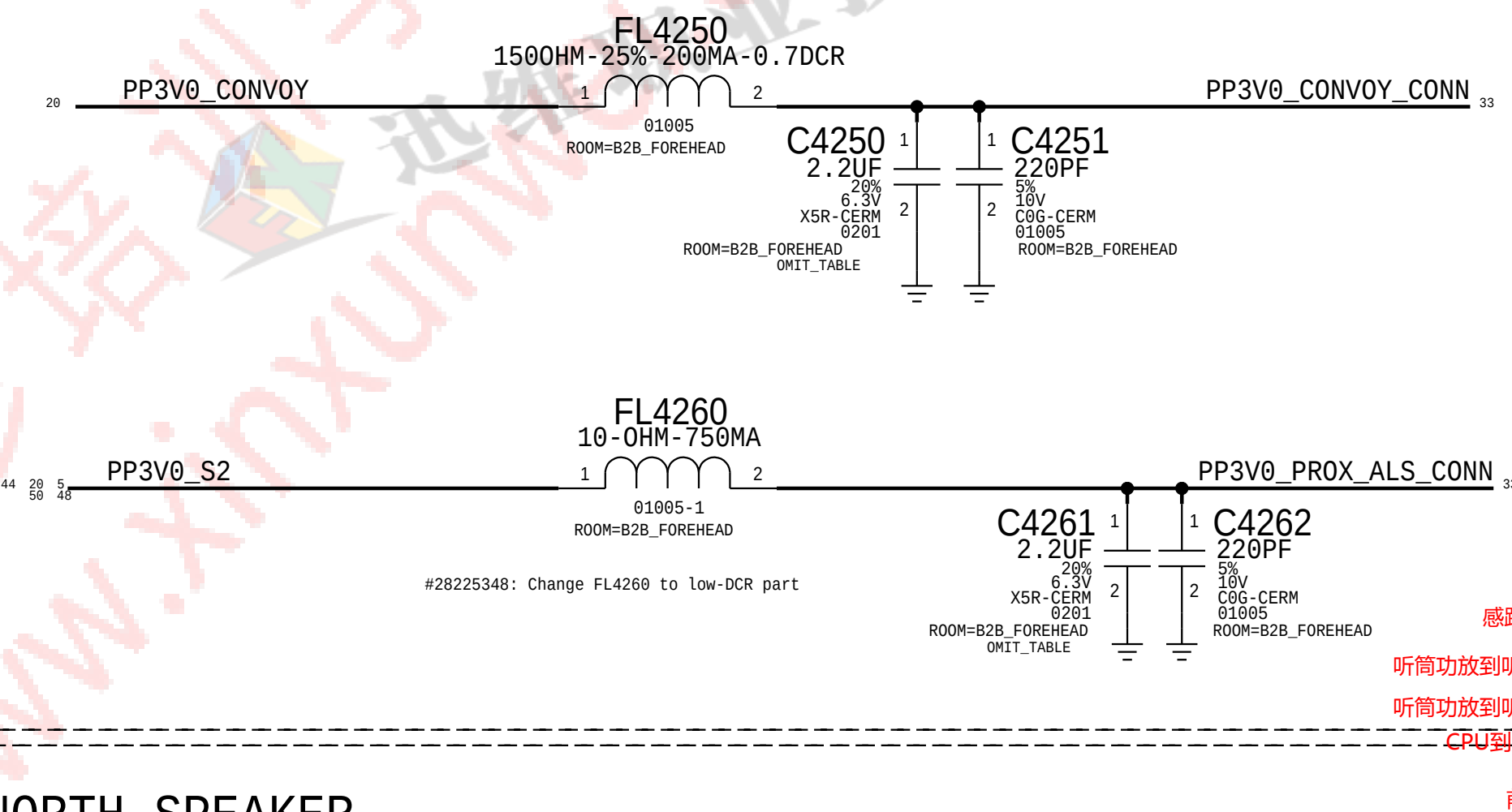
闪光灯驱动芯片



FCAM POWER

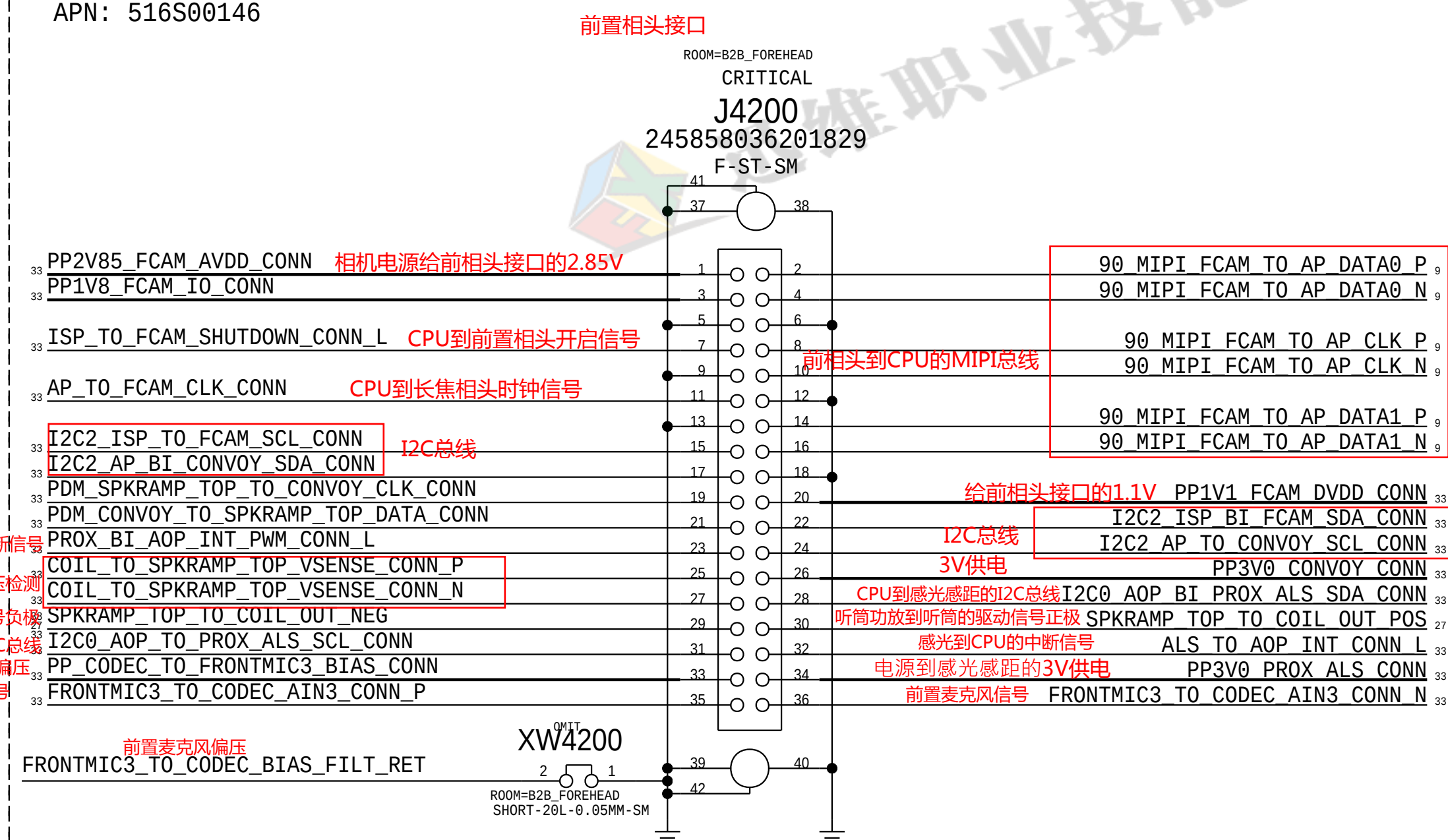


PROX + ALS + CONVOY POWER

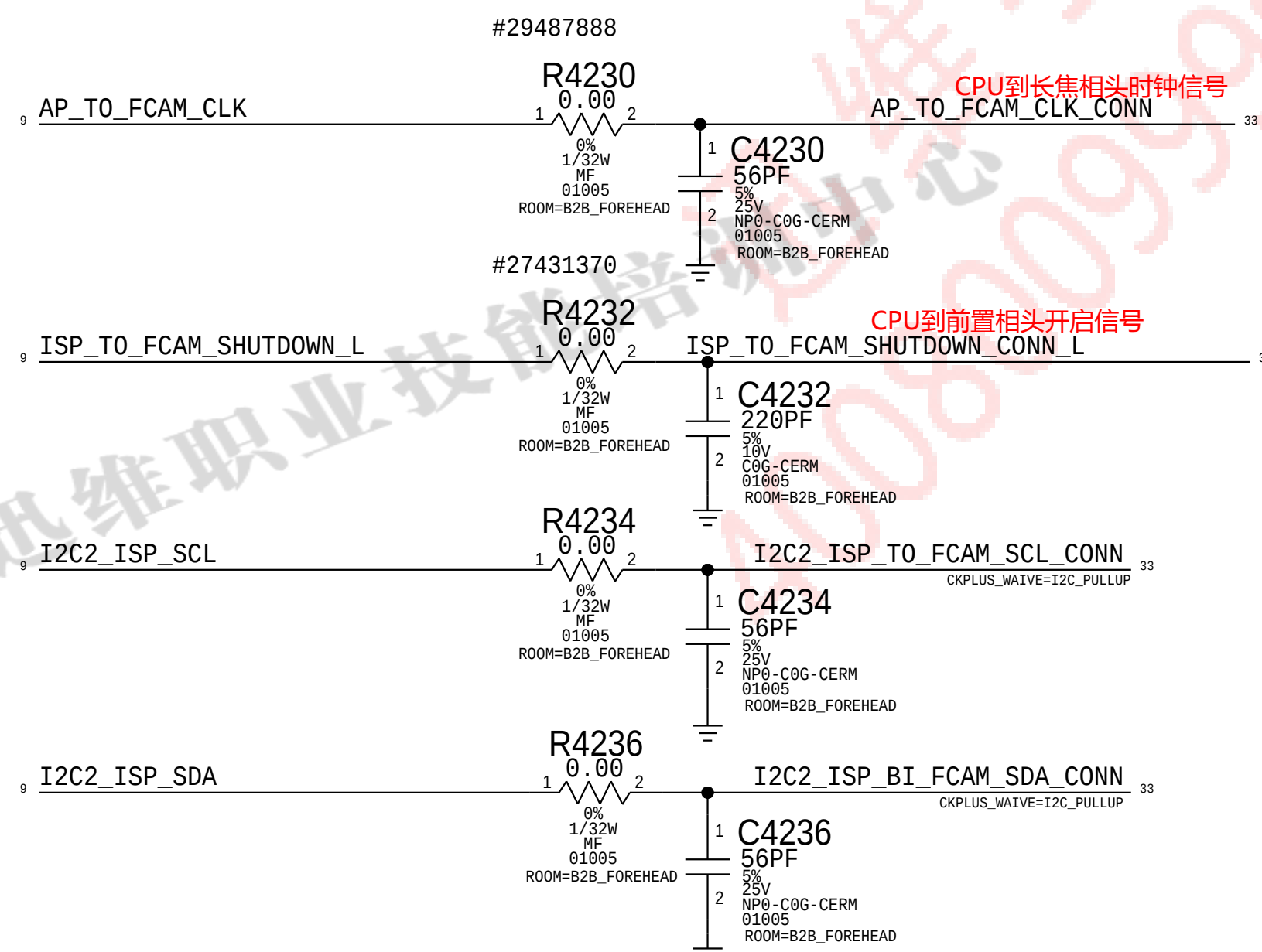


FOREHEAD CONNECTOR

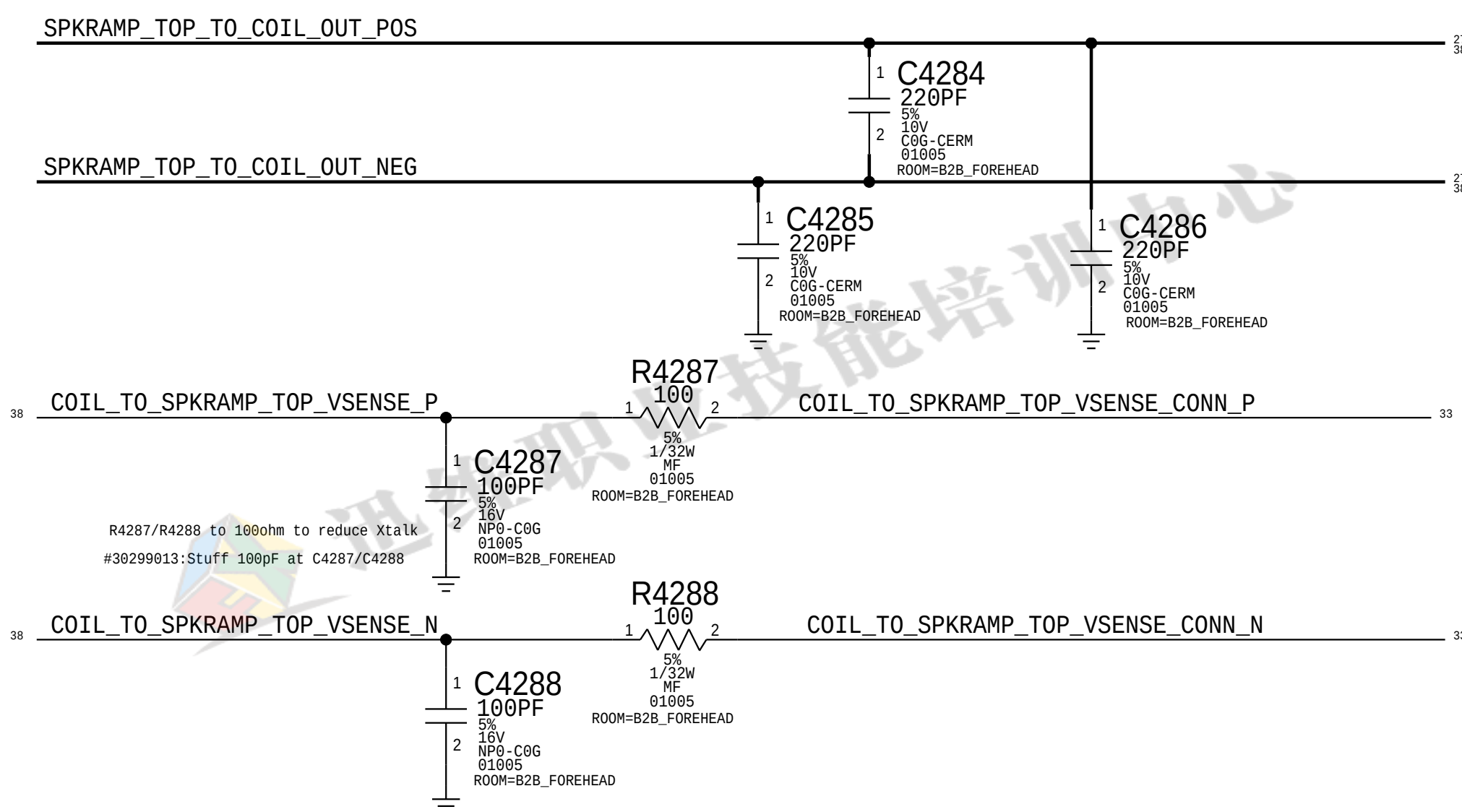
APN: 516S00146



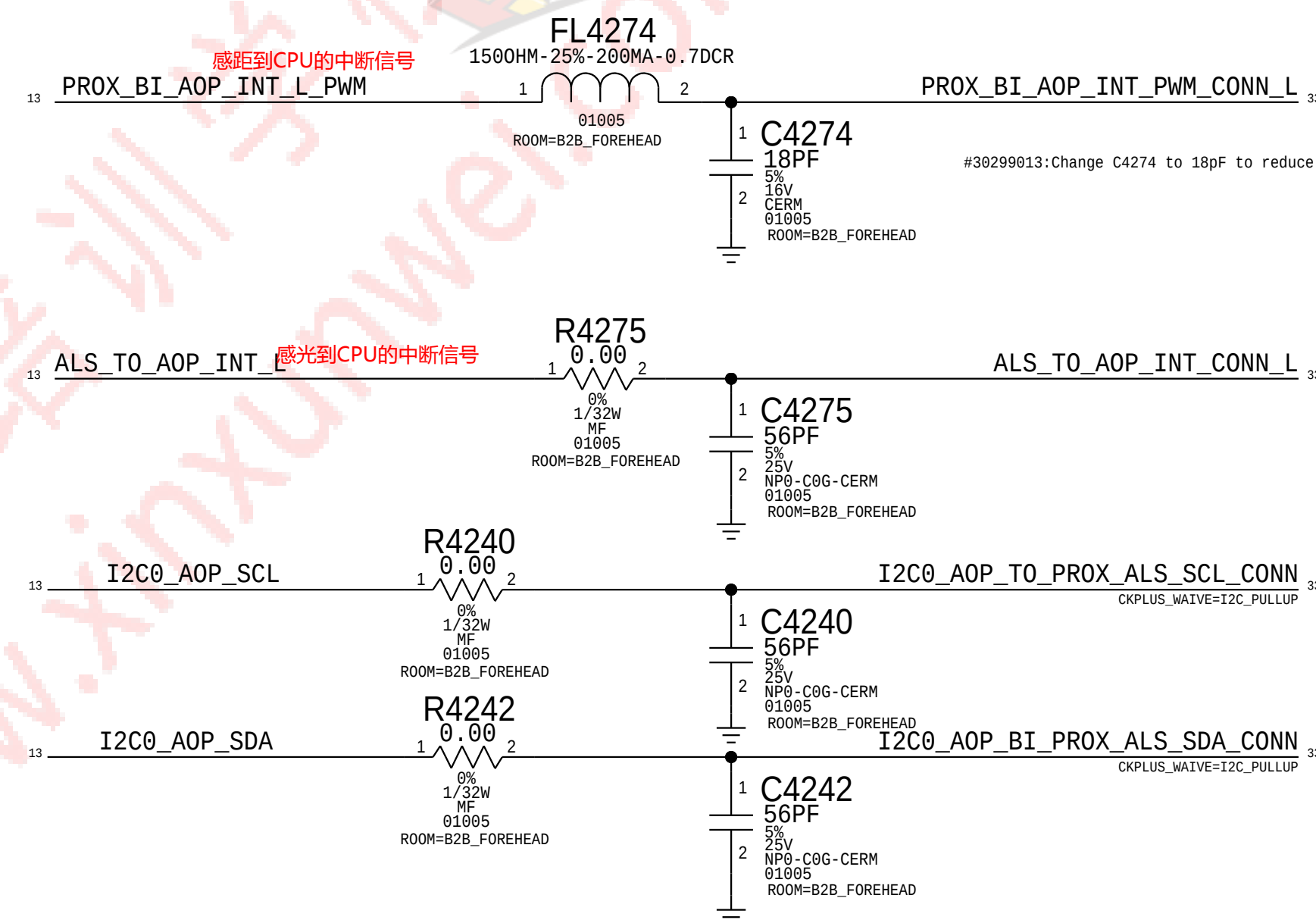
FCAM I/O



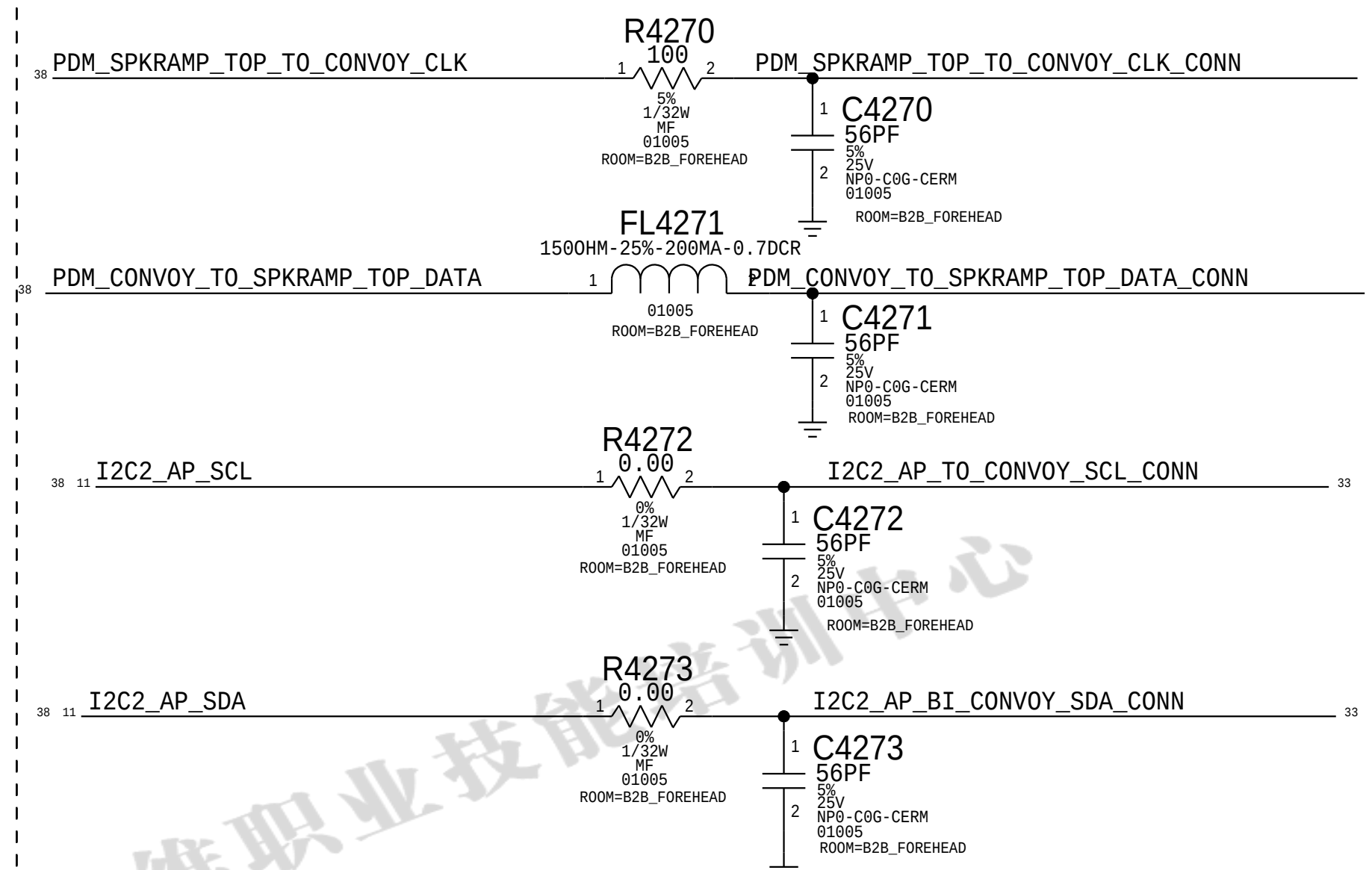
NORTH SPEAKER



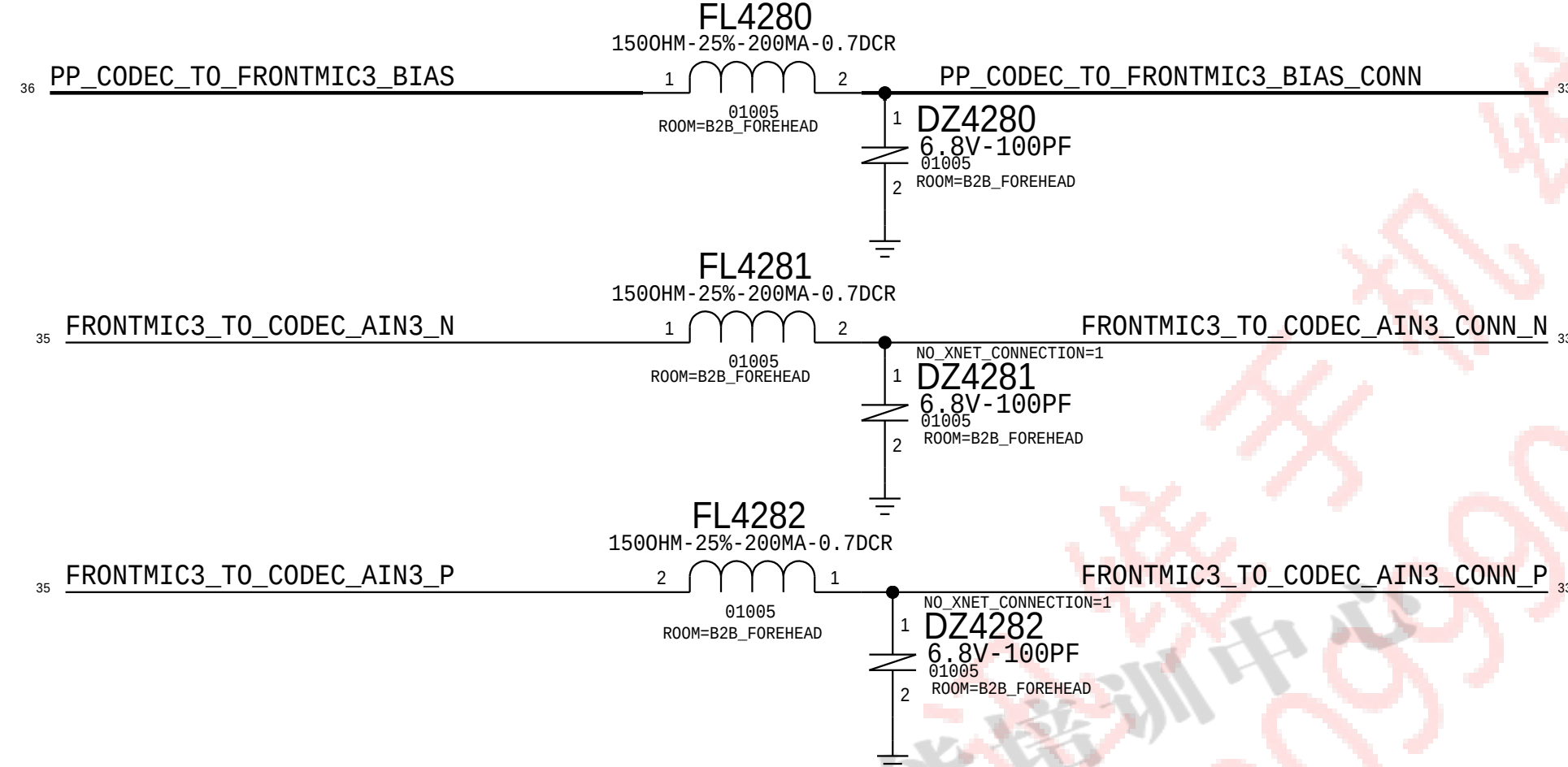
PROX + ALS I/O



CONVOY I/O

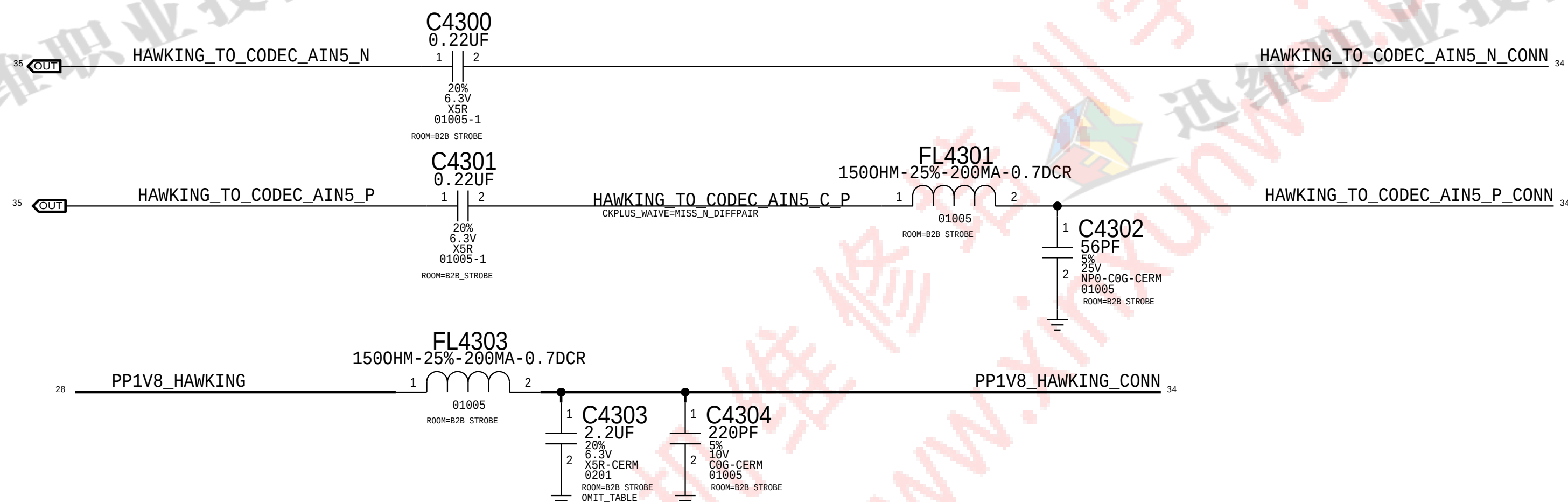


MIC3

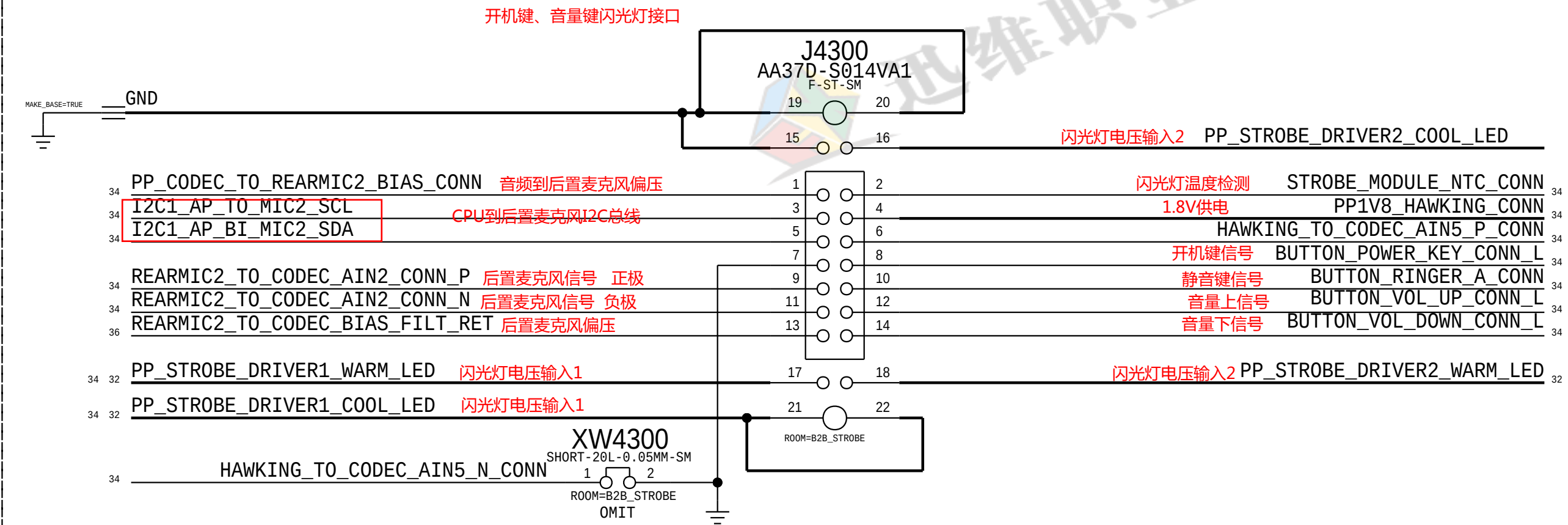




HAWKING



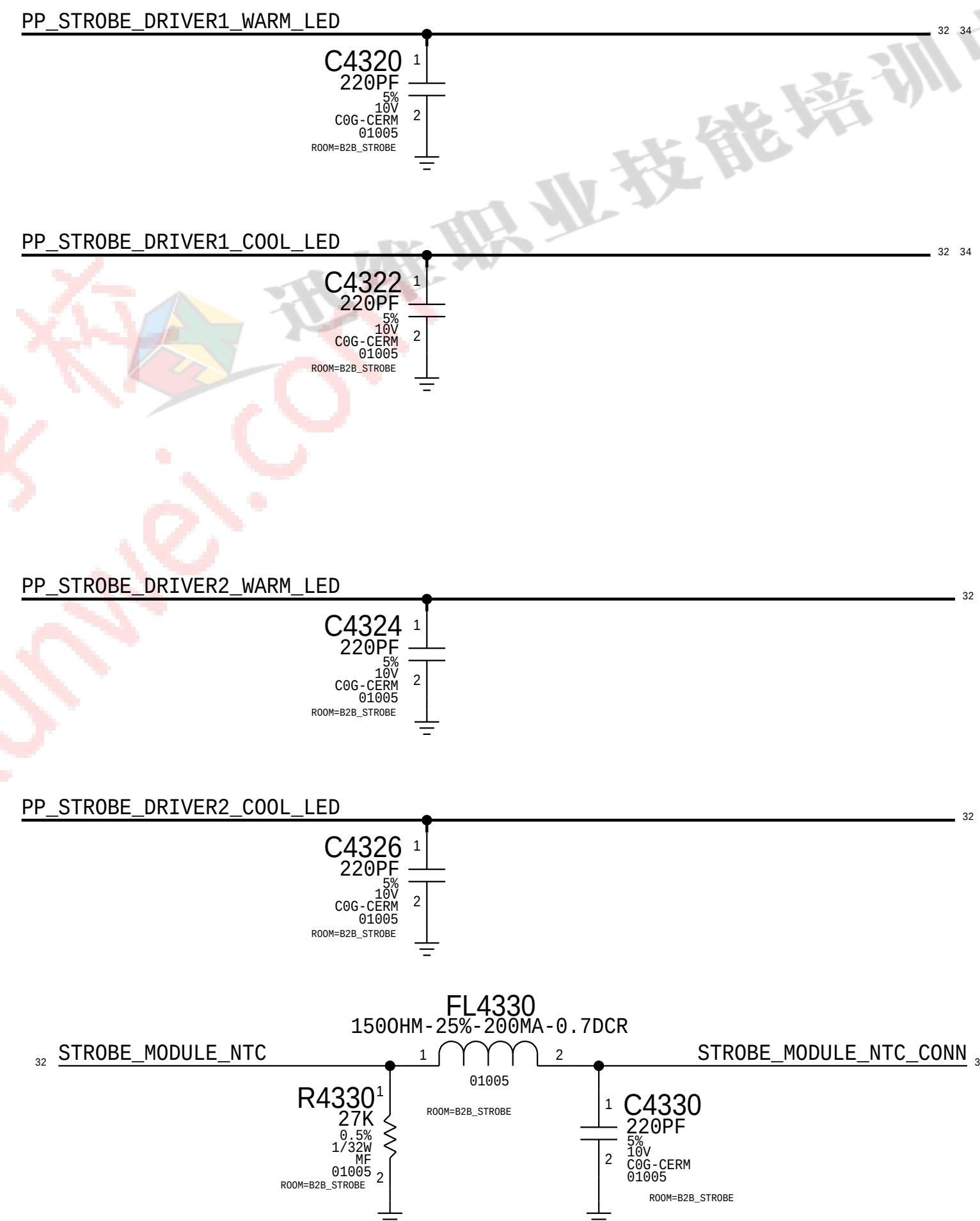
Strobe Connector



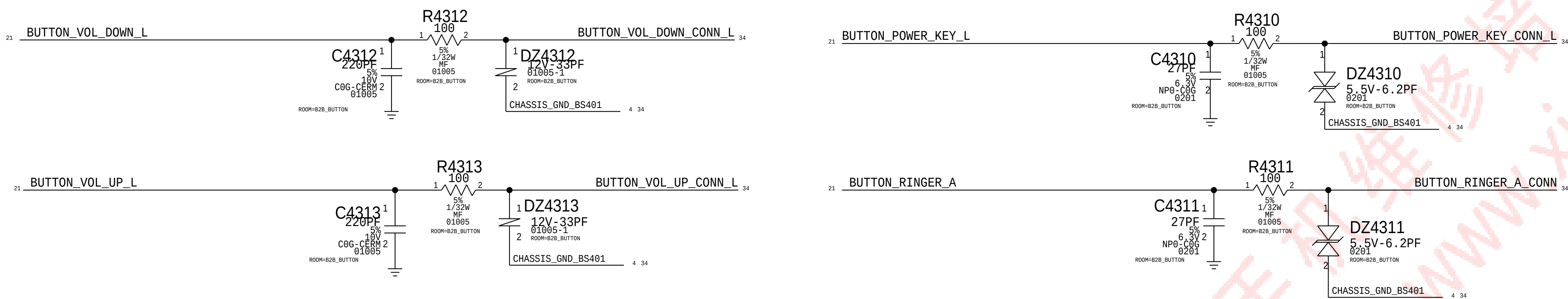
MIC2 (ANC REF)



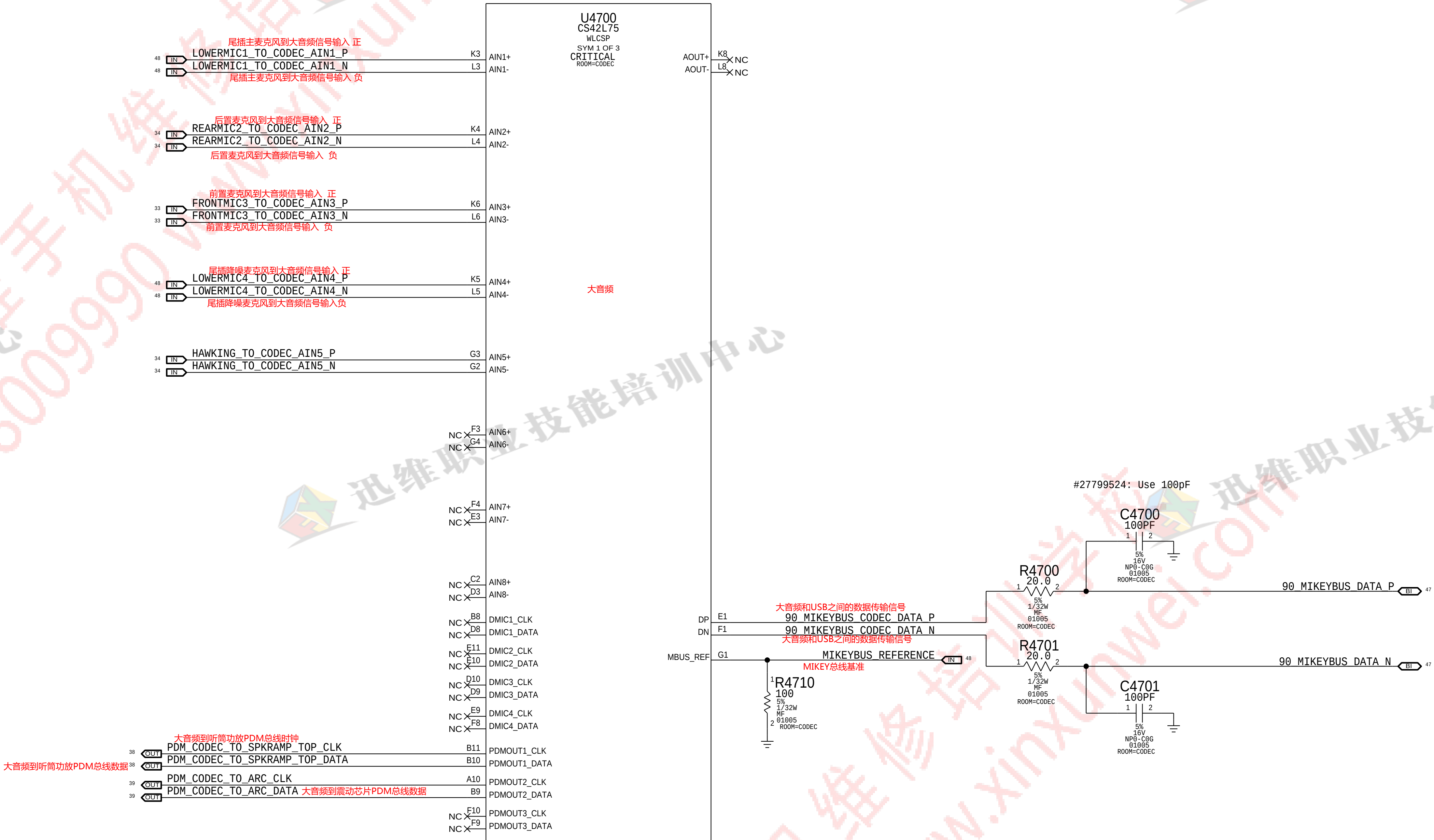
Strobe Filtering



BUTTON



CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)





CALLAN AUDIO CODEC (POWER & I/O)

D

C

B

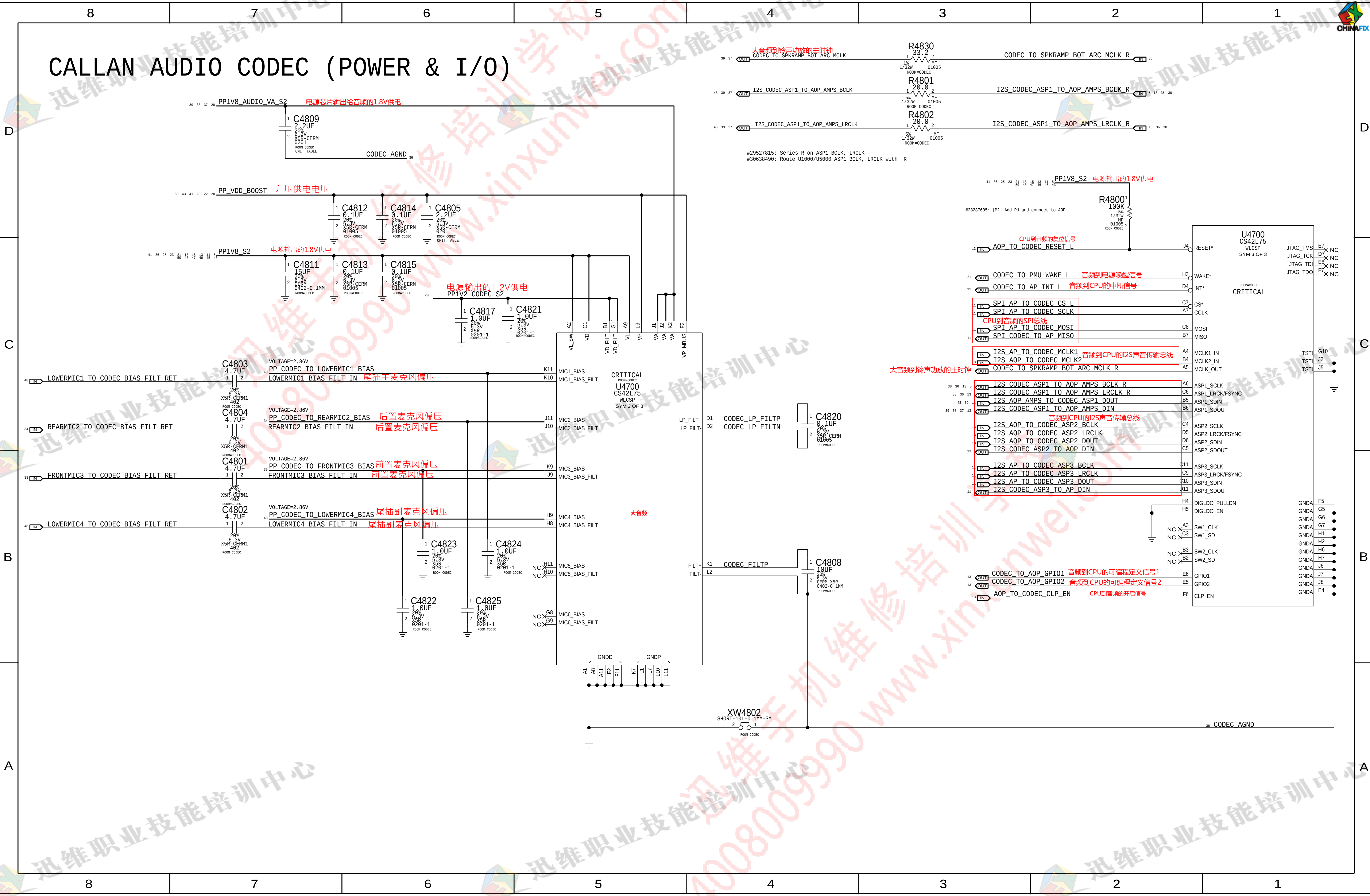
A

D

C

B

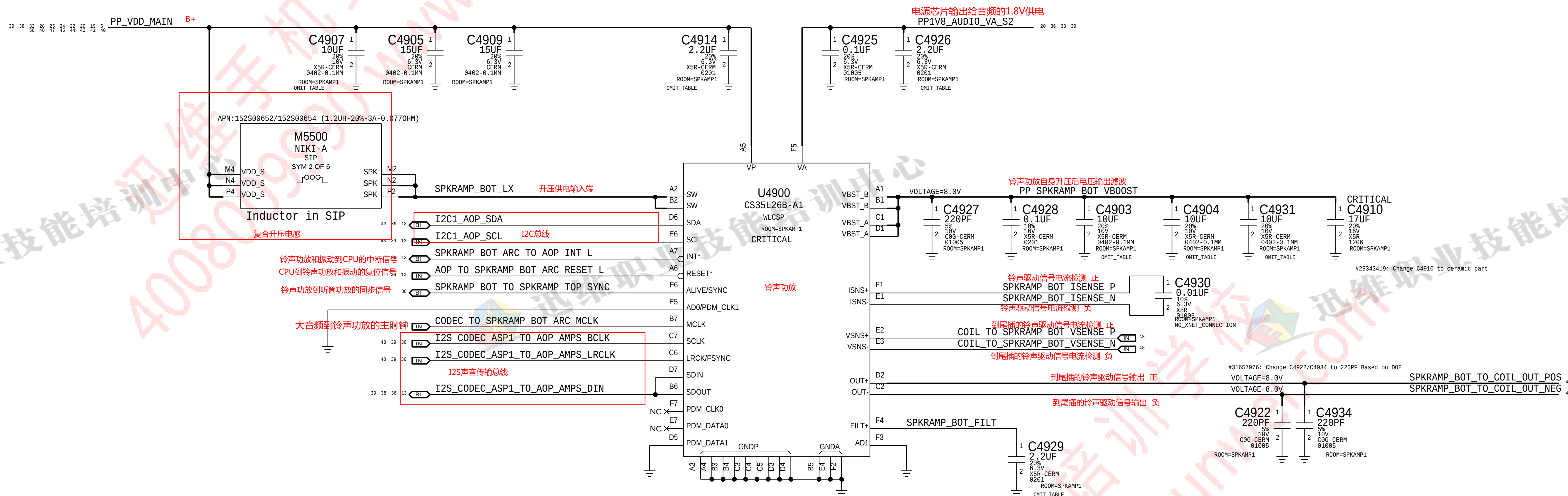
A



South Speaker Amplifier

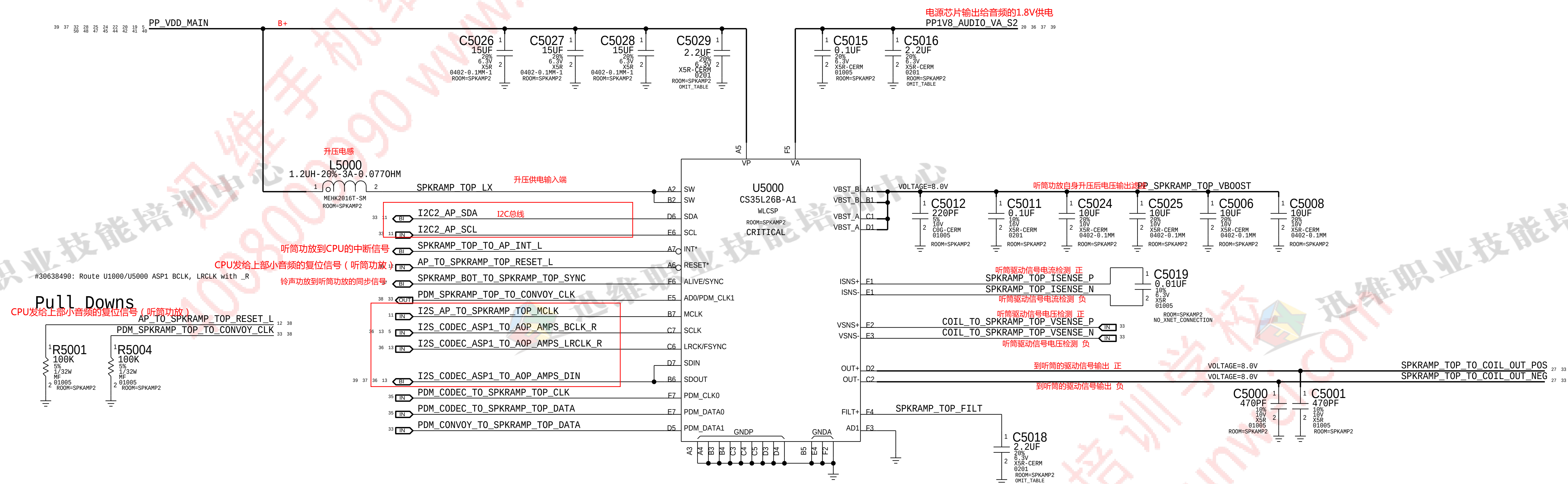
APN: 338S00143

I2C ADDRESS: 1000 000x
0x80



APN: 338S00295

I2C ADDRESS: 1000 000x (0x80)

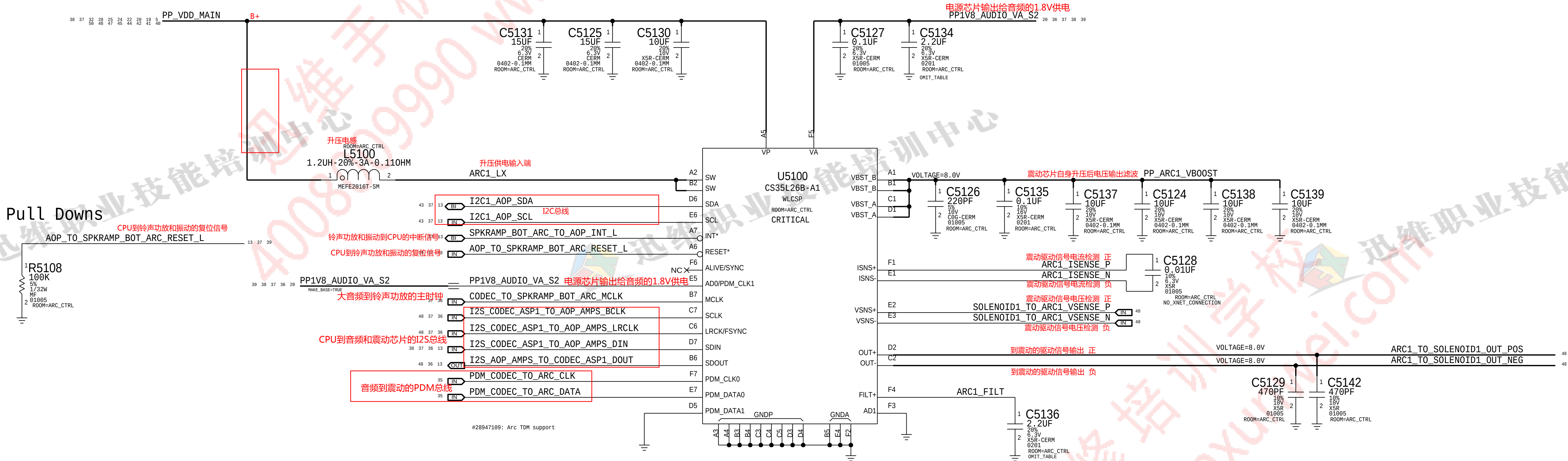


ARC DRIVER

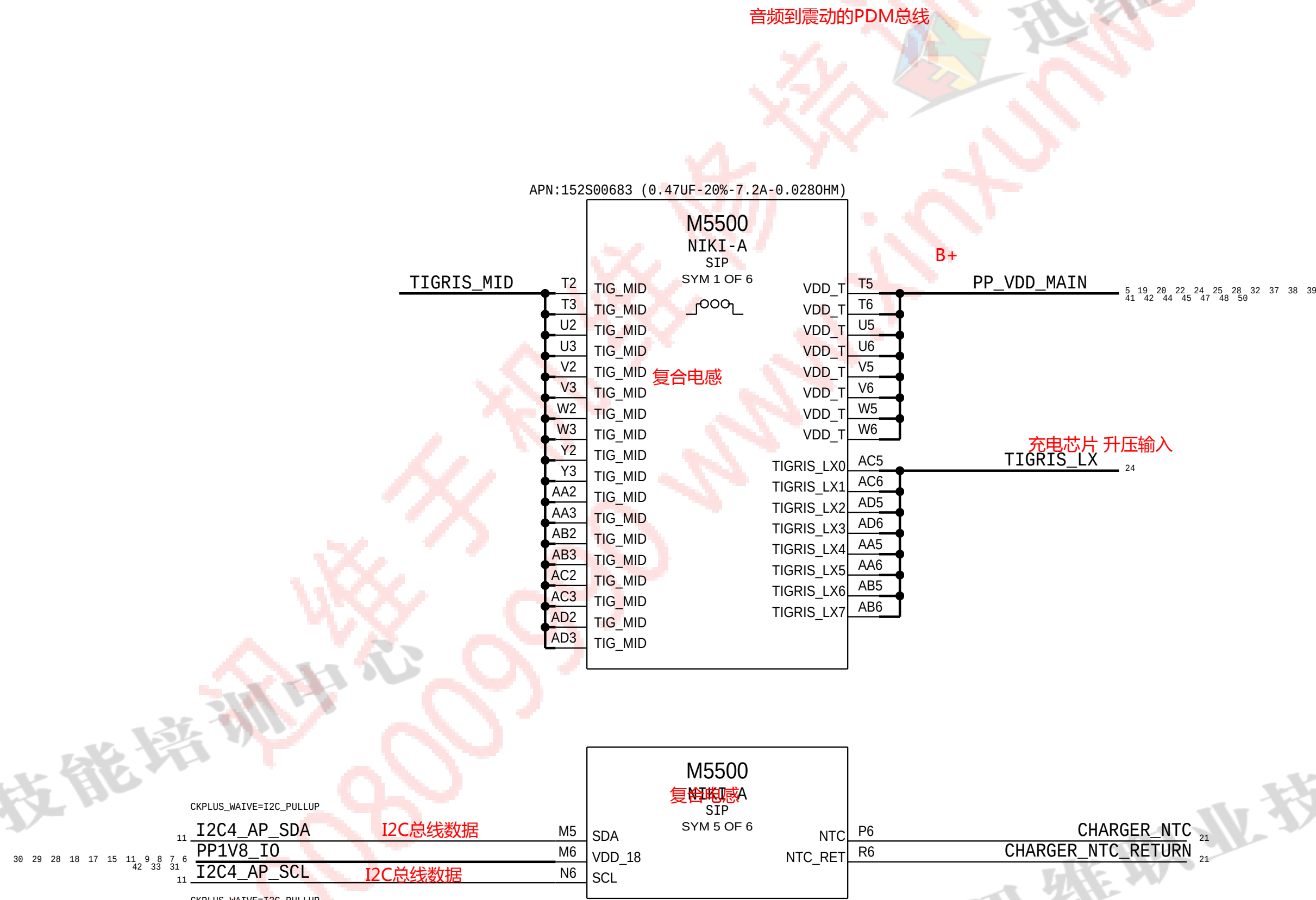
APN: 338S00295

I2C ADDRESS: 1000 001x (0x82)

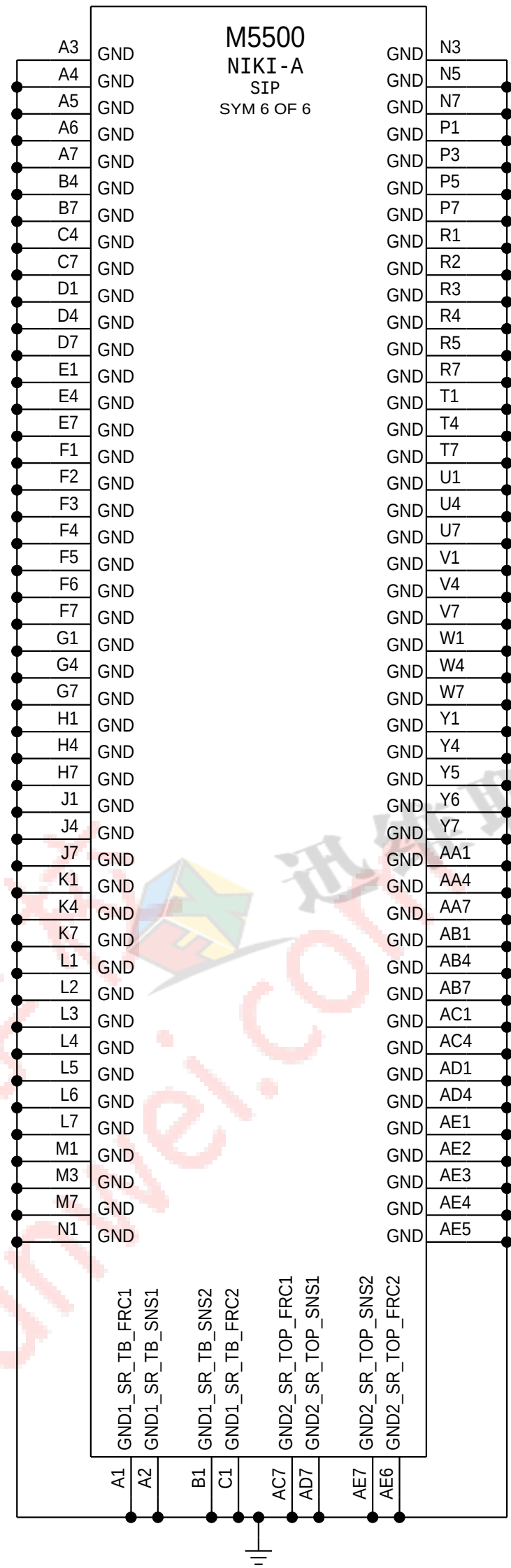
振动驱动芯片



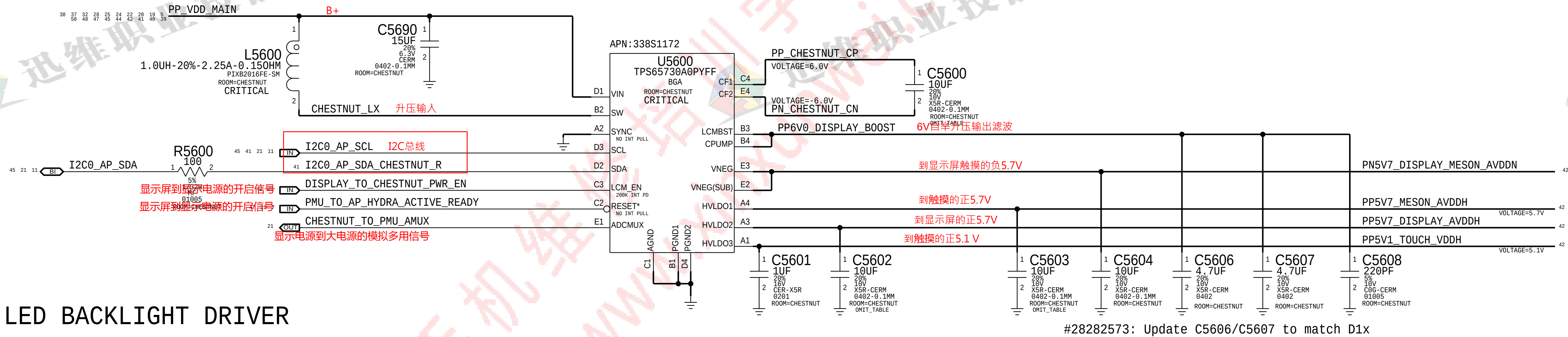
Niki SIP



Note: inductors in SIP used for bottom speaker amplifier and backlight
See pages 49, and 56

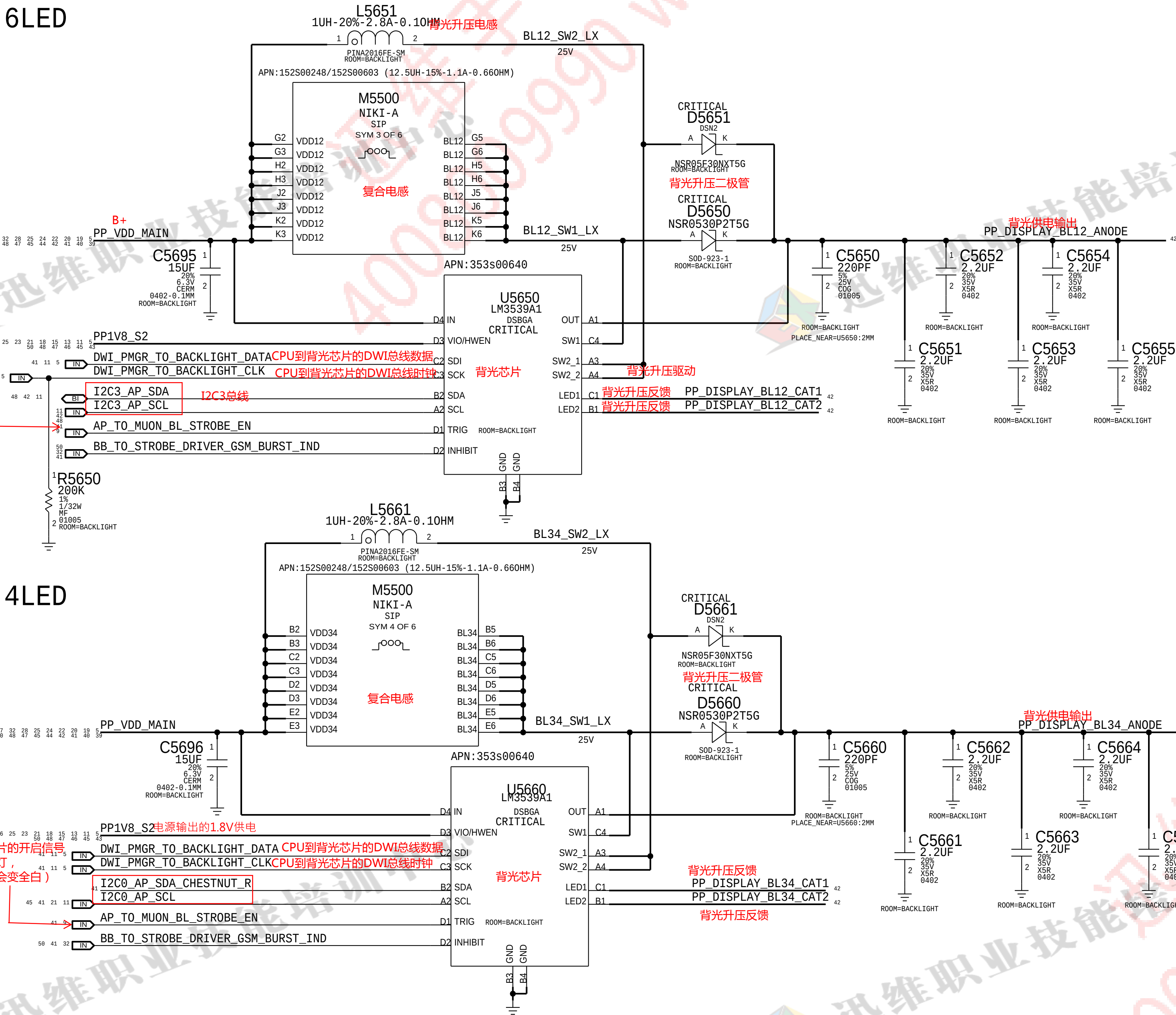


CHESTNUT DISPLAY PMU

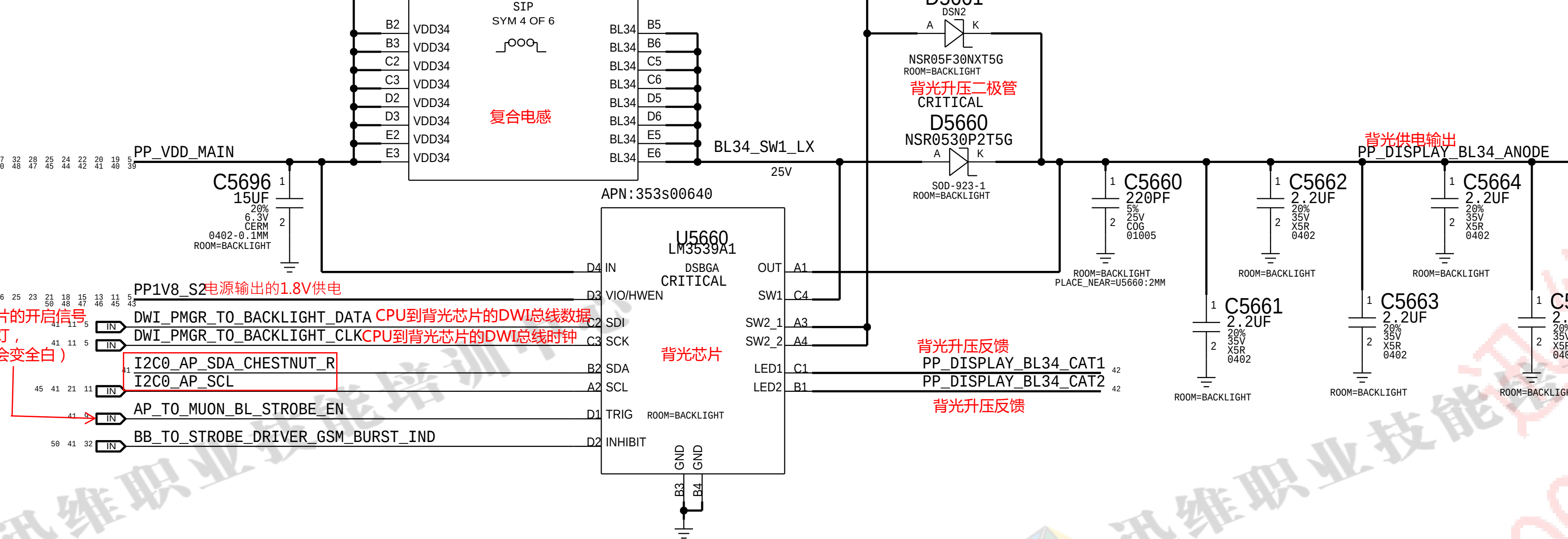


LED BACKLIGHT DRIVER

6LED

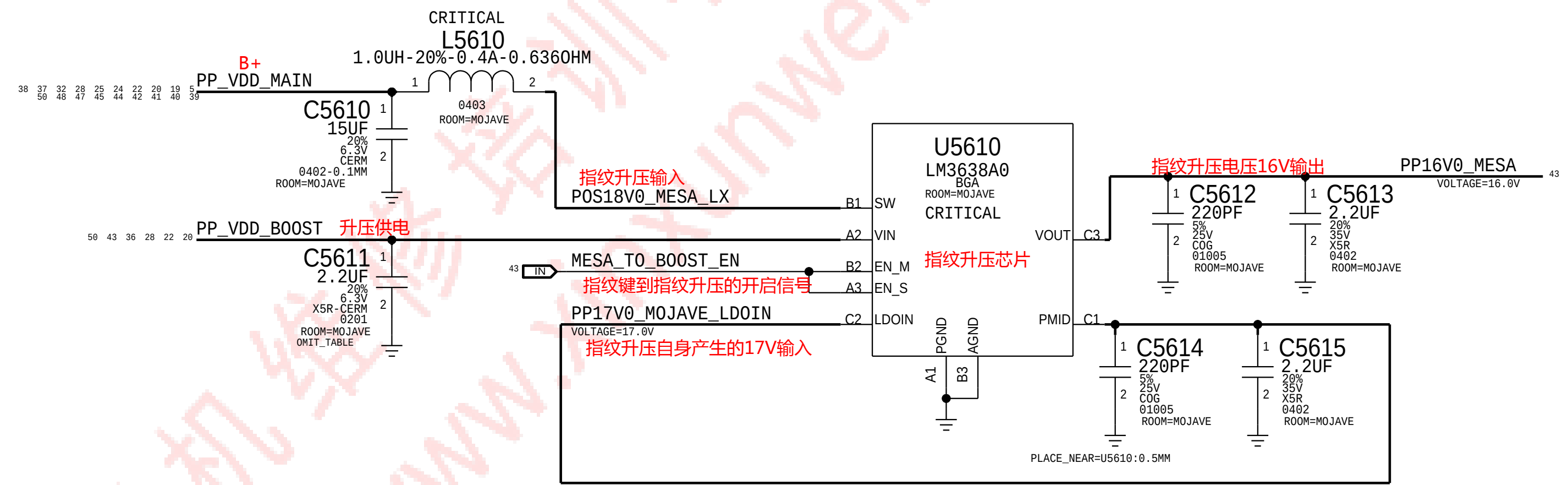


4LED

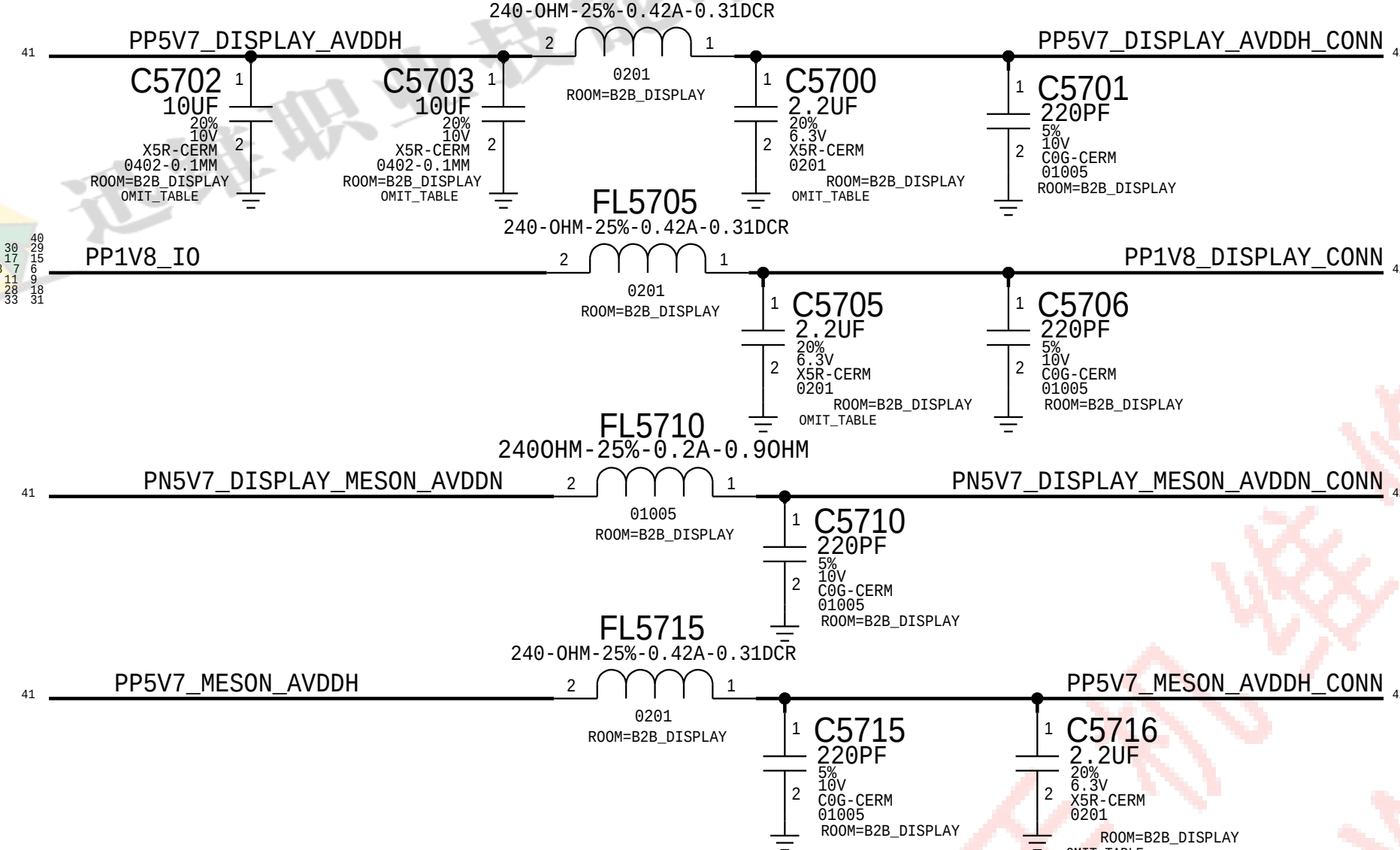


MOJAVE MESA BOOST

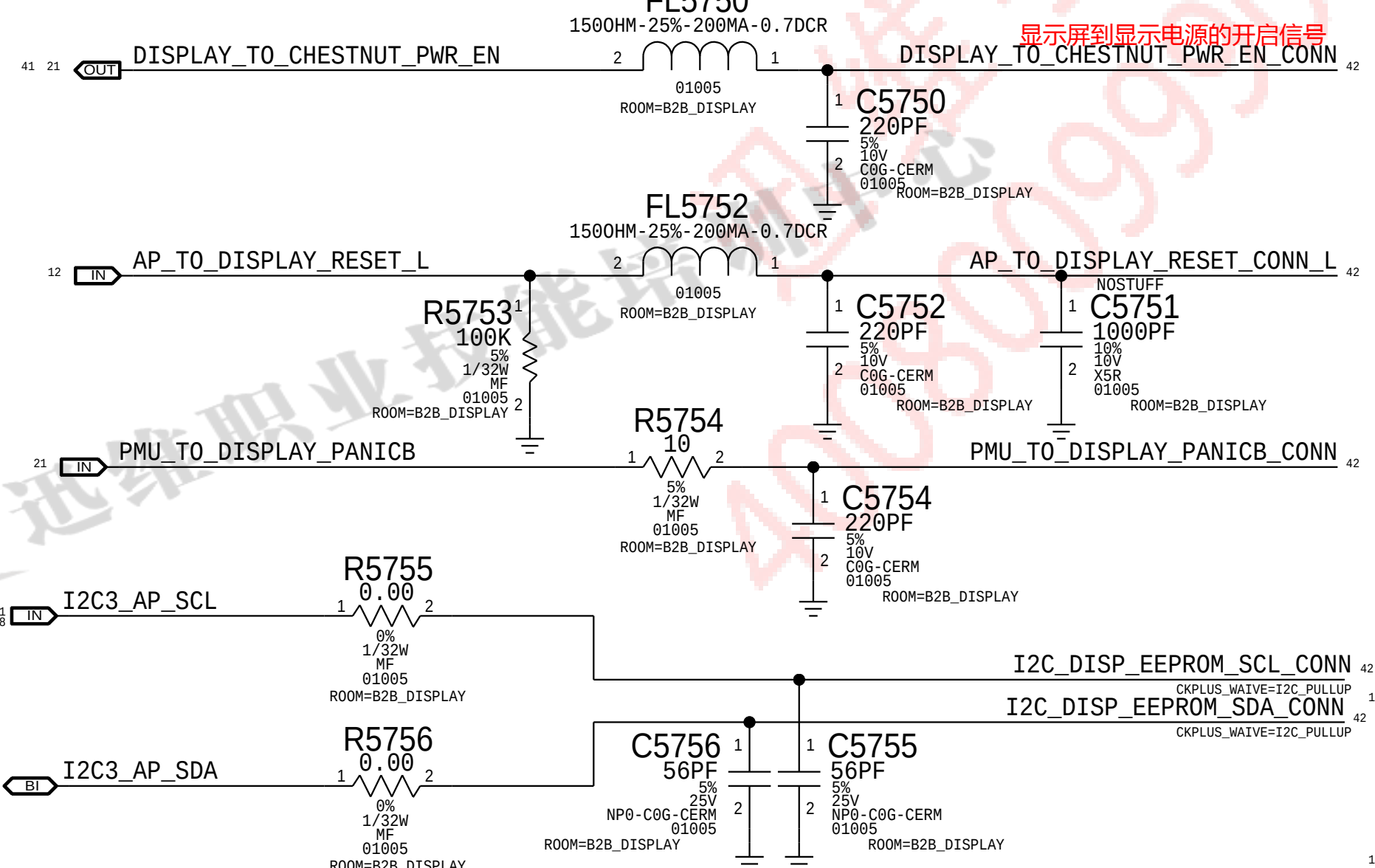
APN: 353S00671



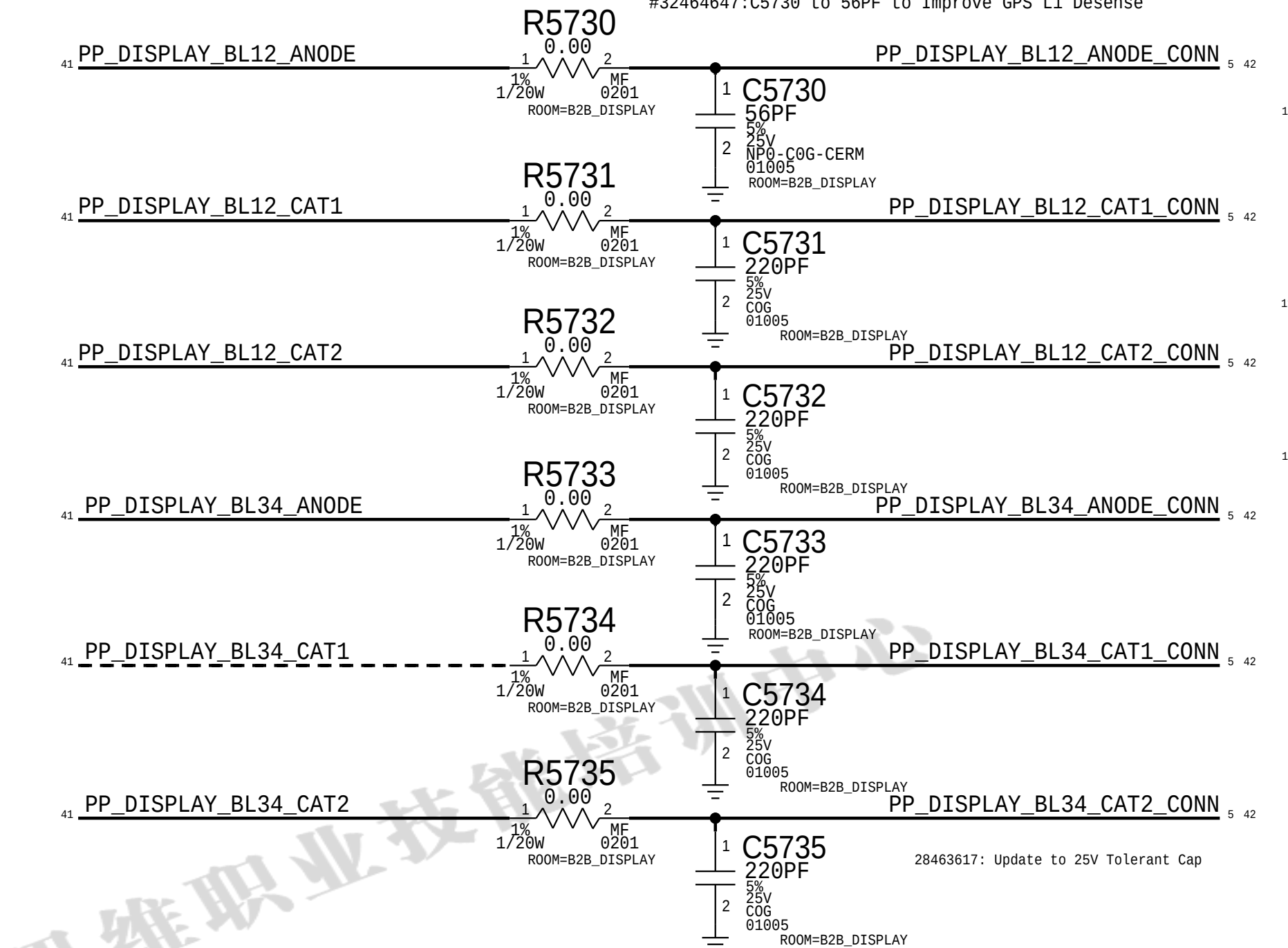
DISPLAY POWER



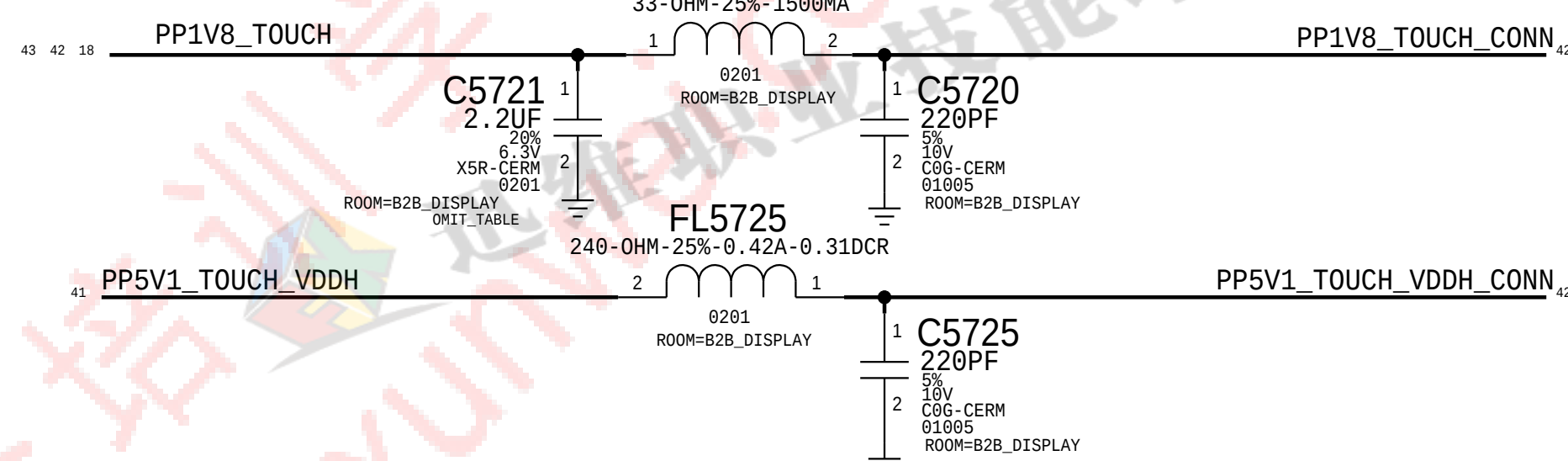
DISPLAY CONTROL SIGNALS



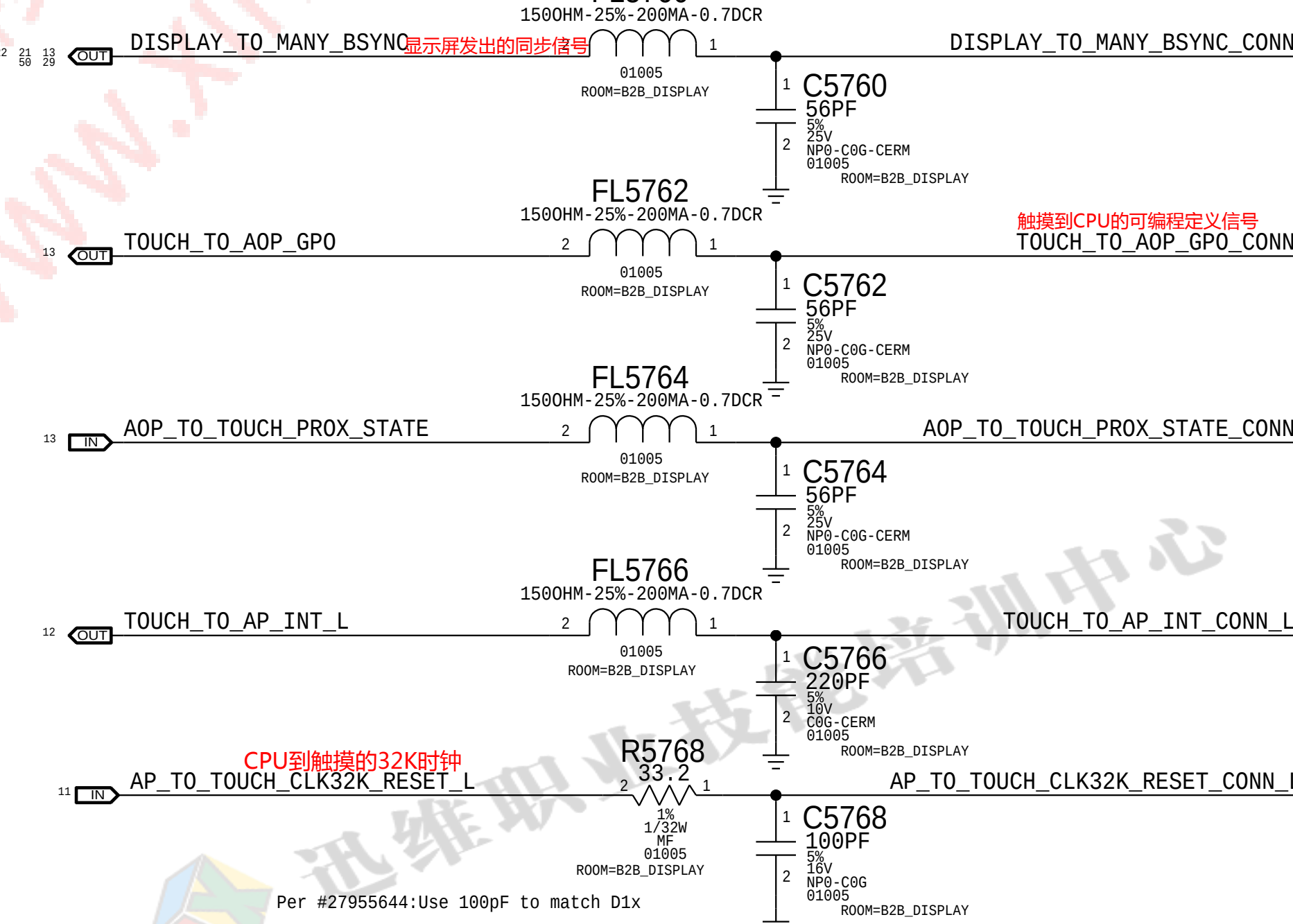
BACKLIGHT



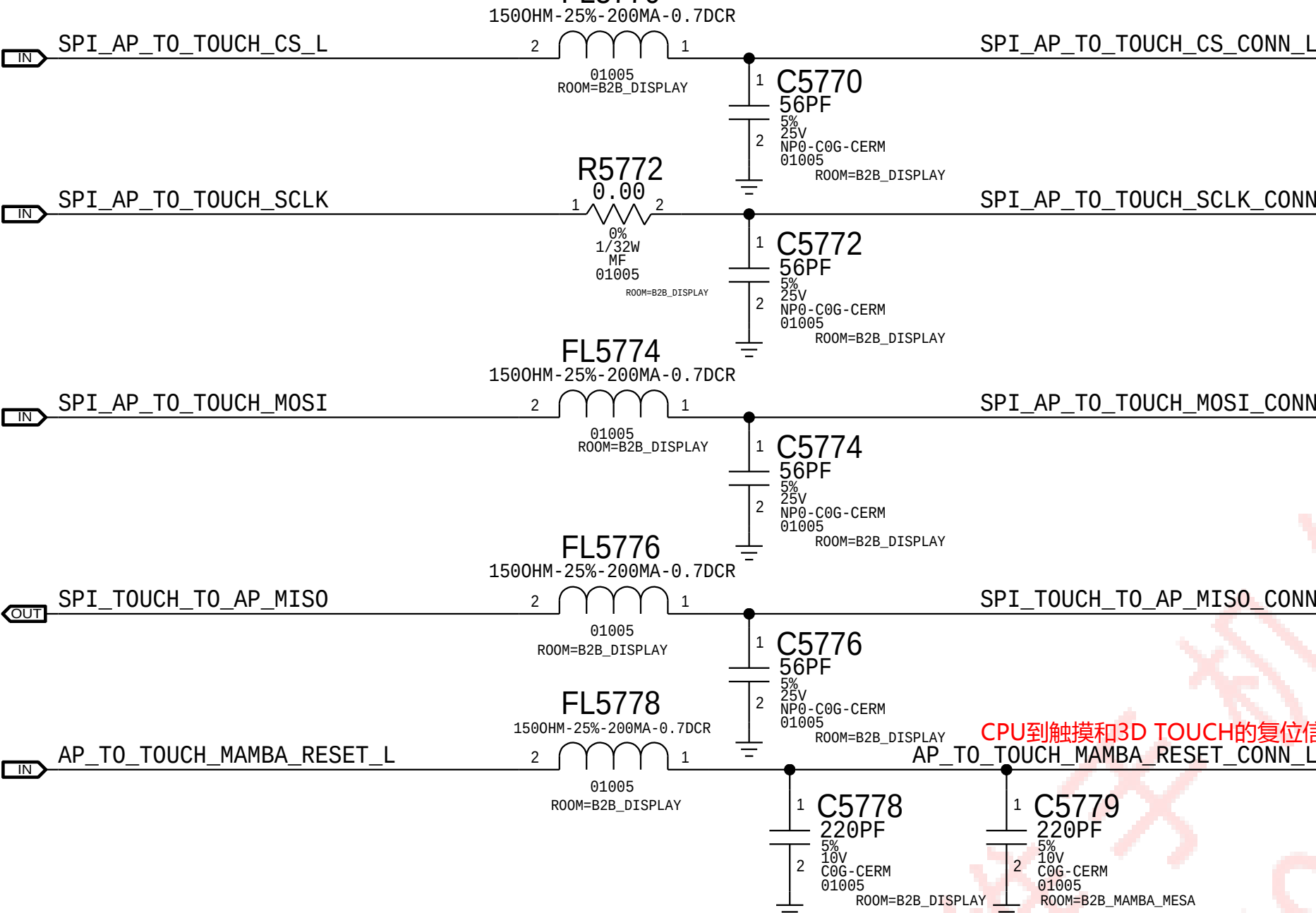
TOUCH POWER



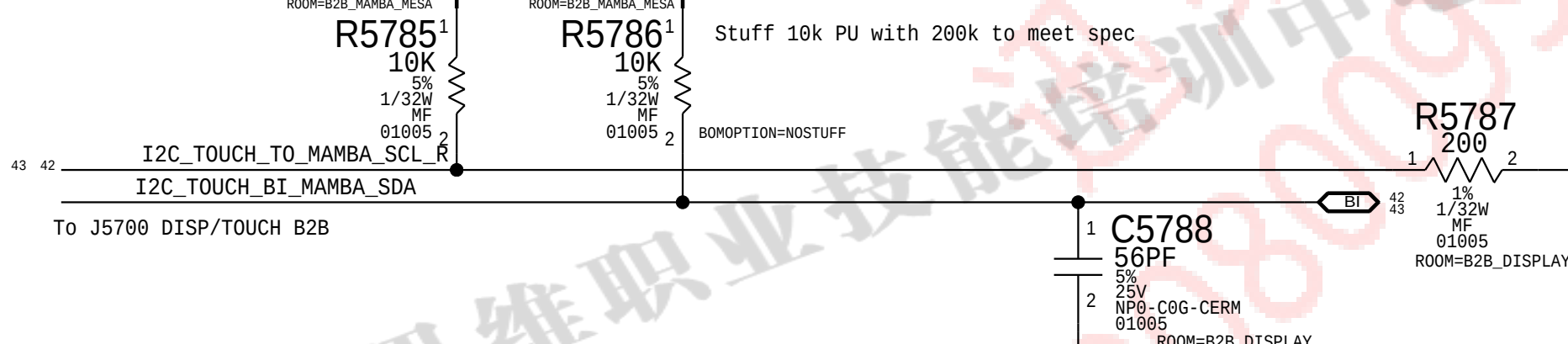
AOP/TOUCH INTERFACE



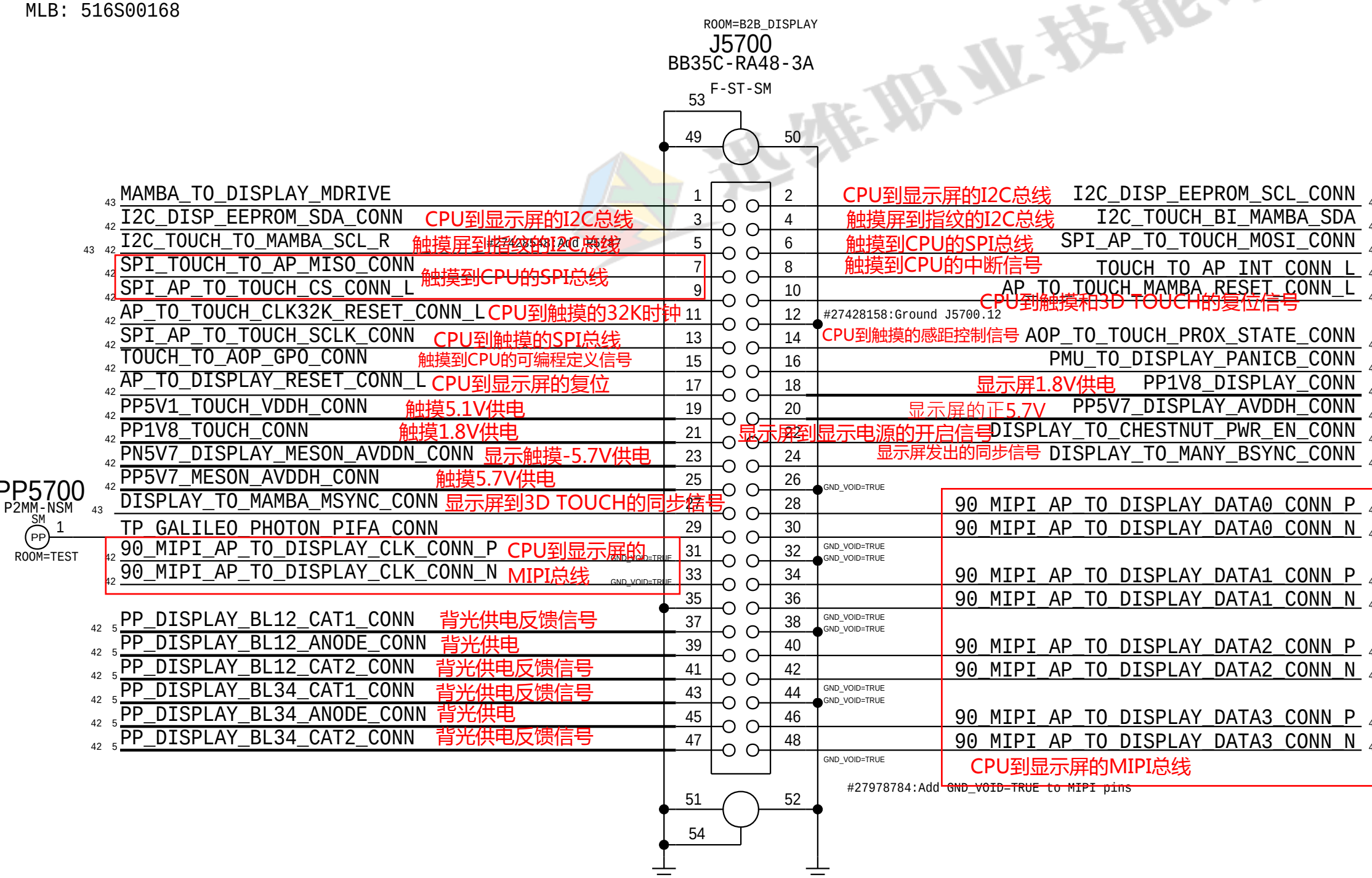
AP/TOUCH INTERFACE



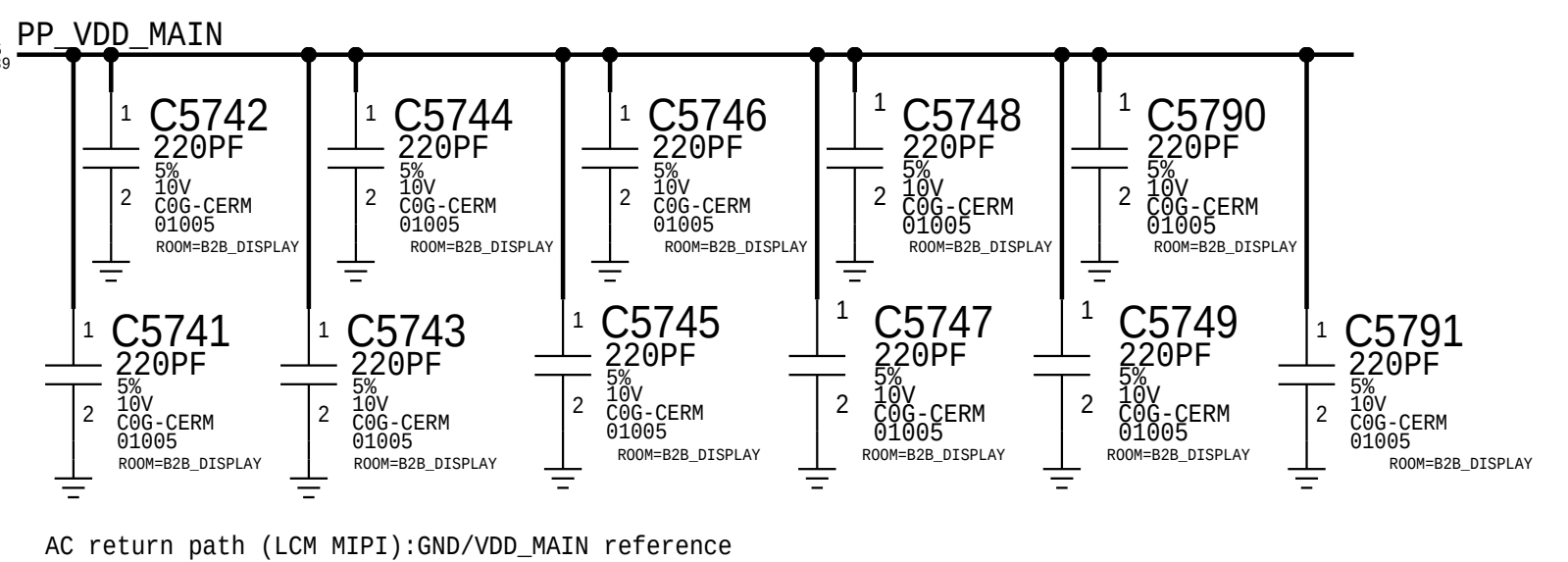
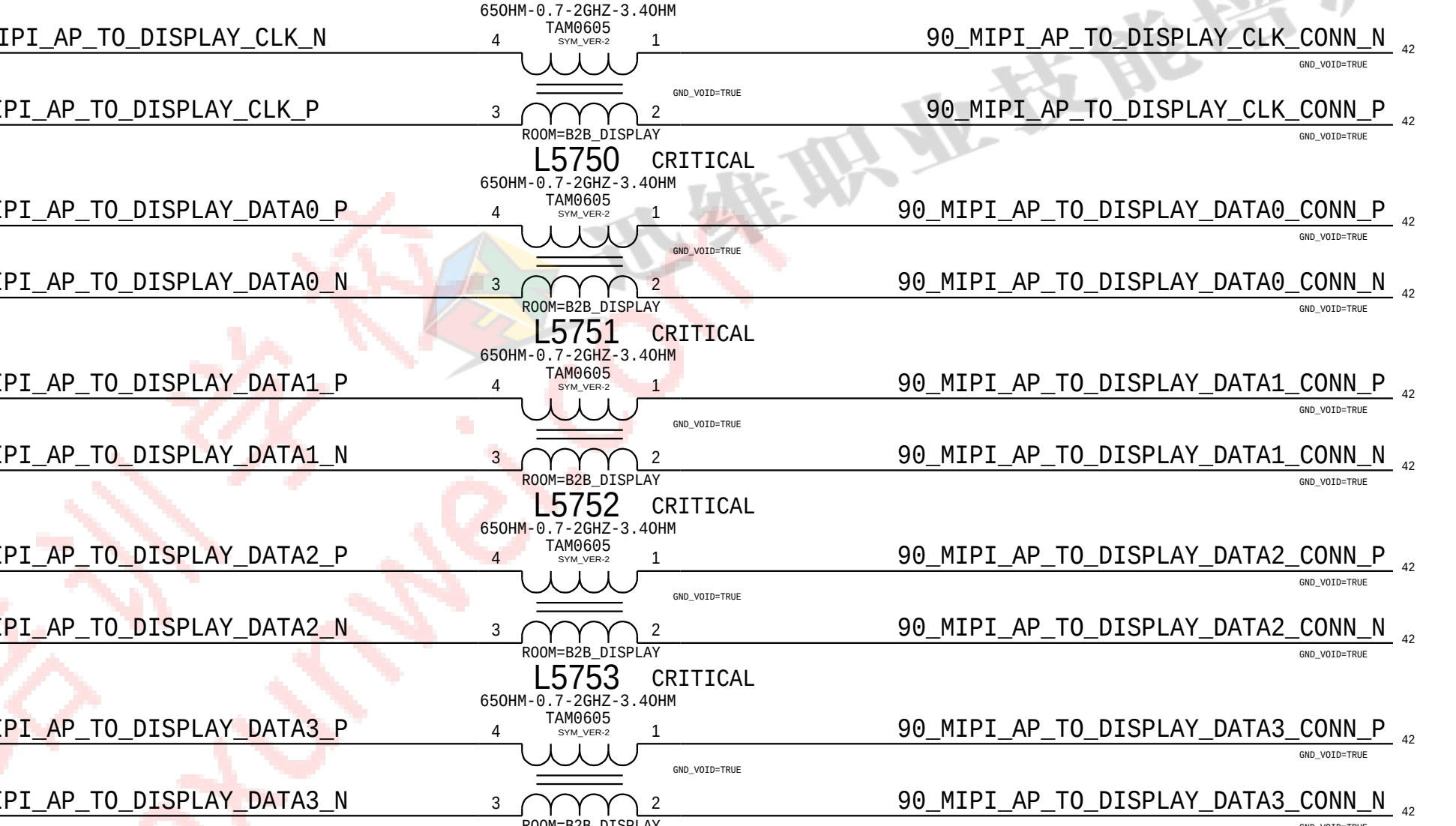
ORB/TOUCH INTERFACE



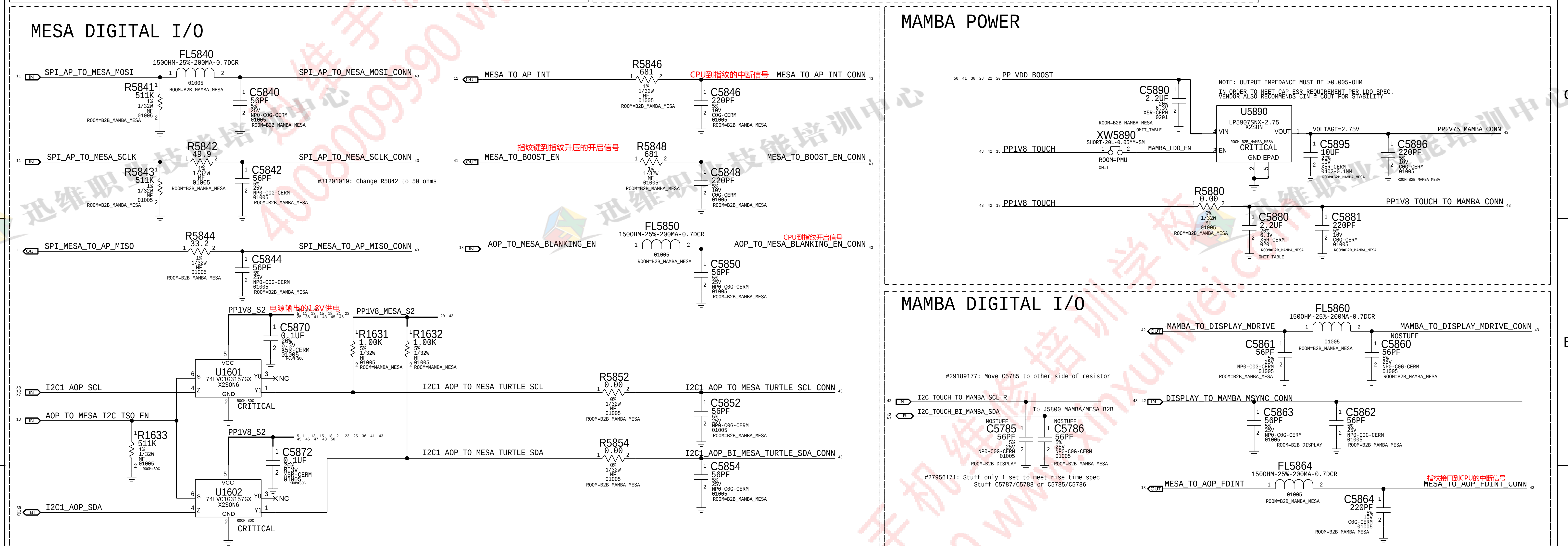
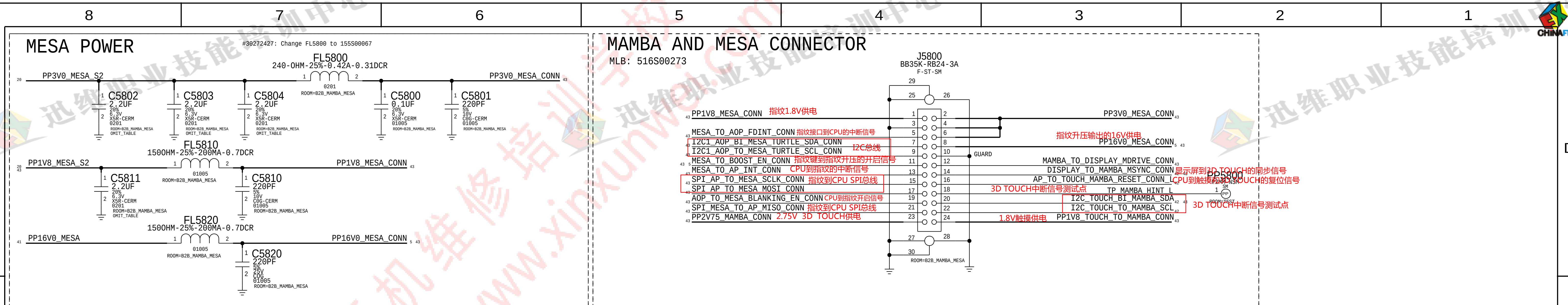
DISPLAY + TOUCH CONNECTOR



DISPLAY MIPI



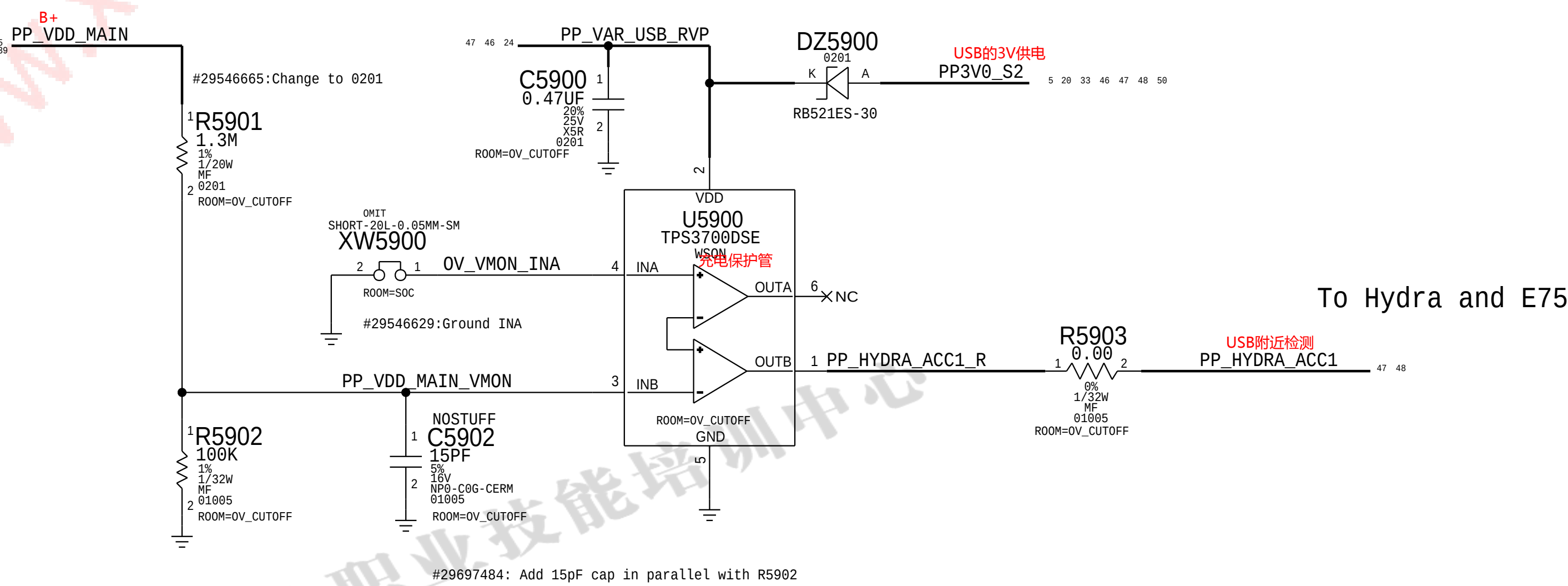
#28227044: Have cap on each side of R5787



CPU到指纹开启信号

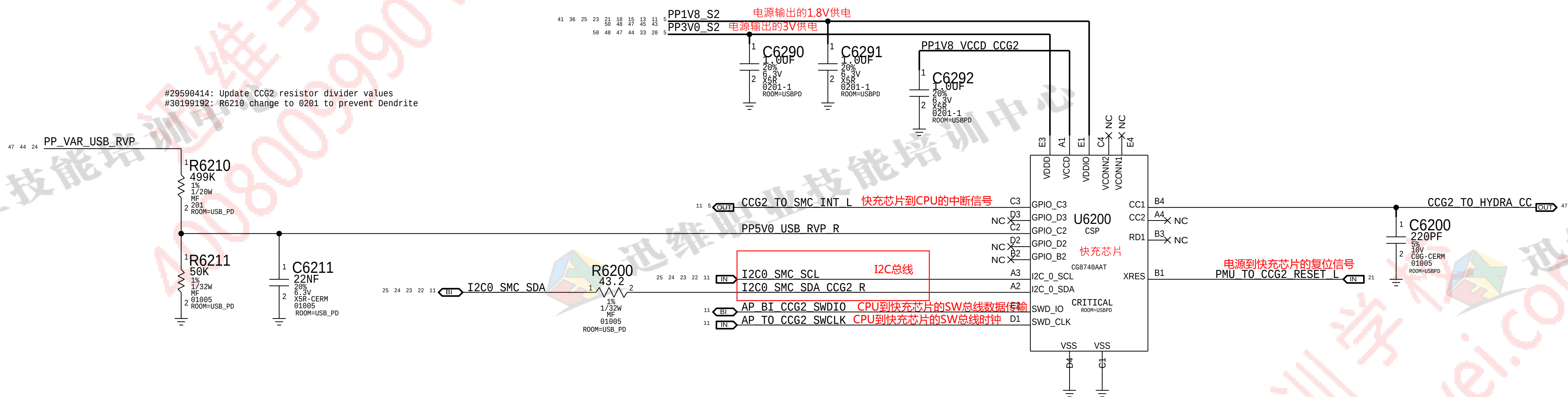


VDD_MAIN OV CUT-OFF CIRCUIT



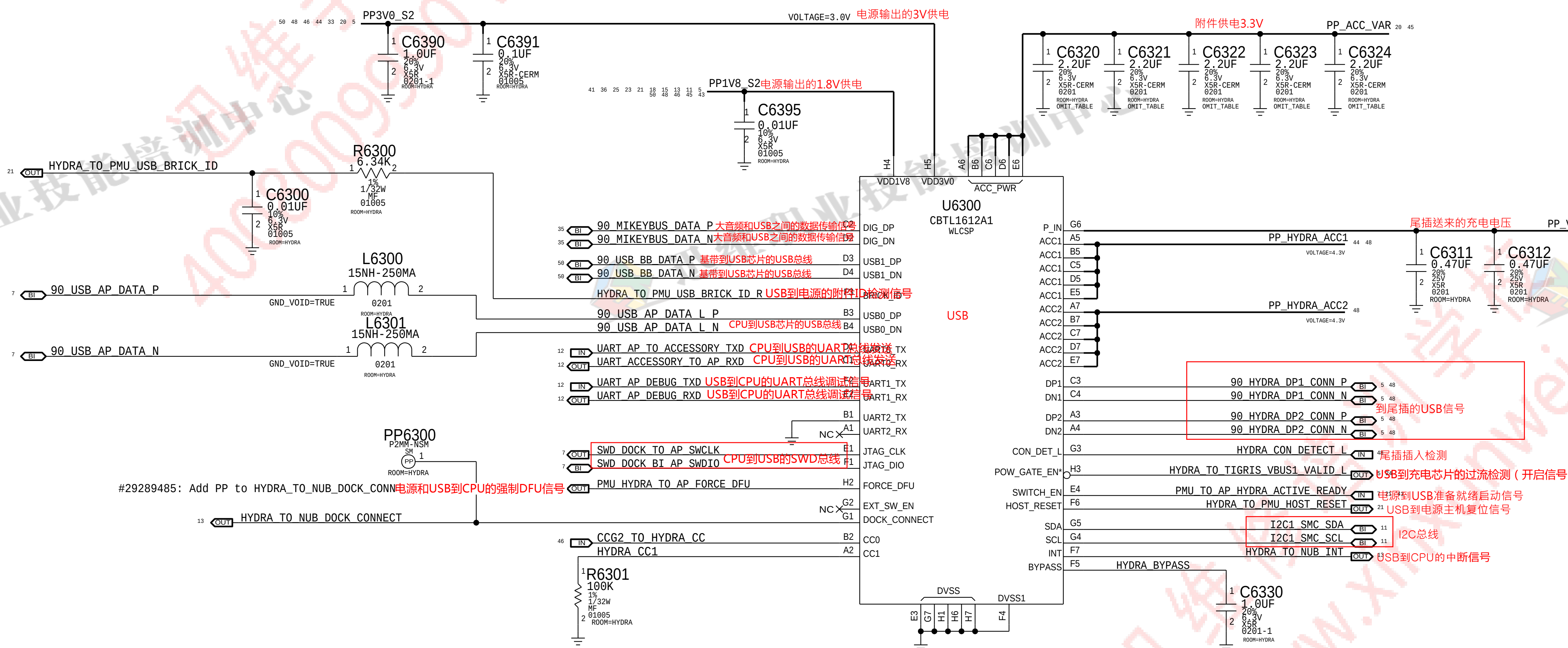


USB-PD

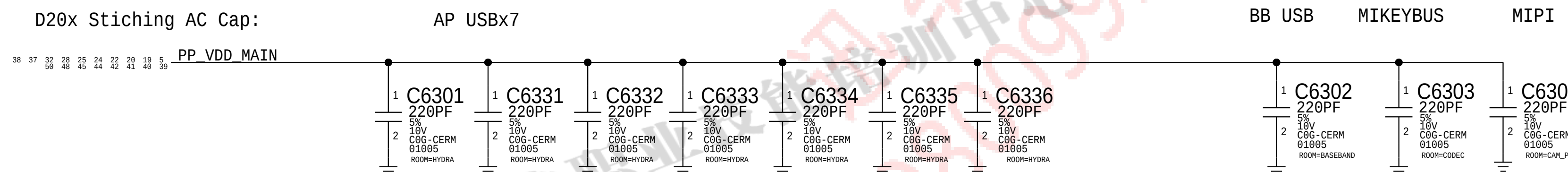


HYDRA

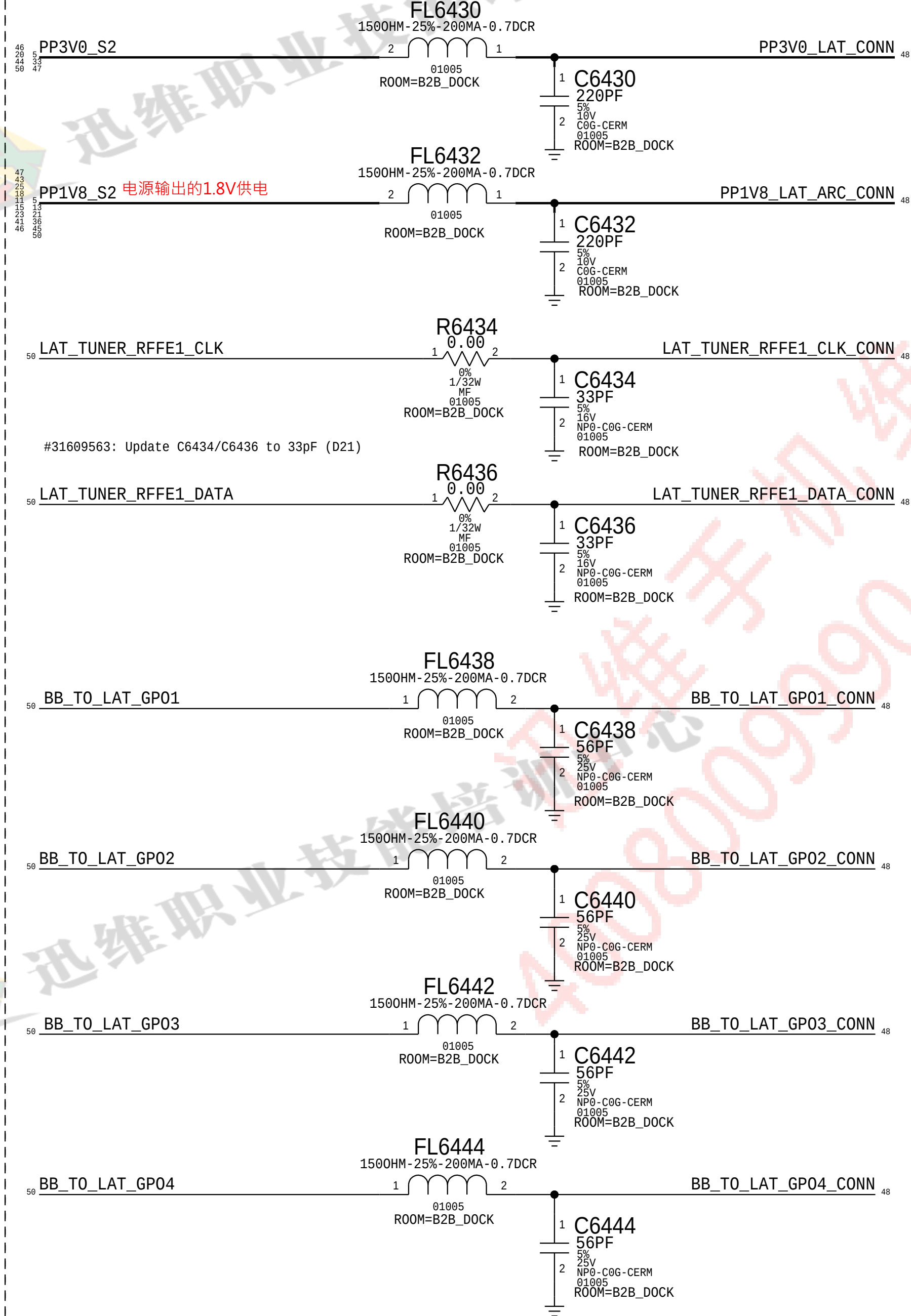
I2C ADDRESS: 0011010X



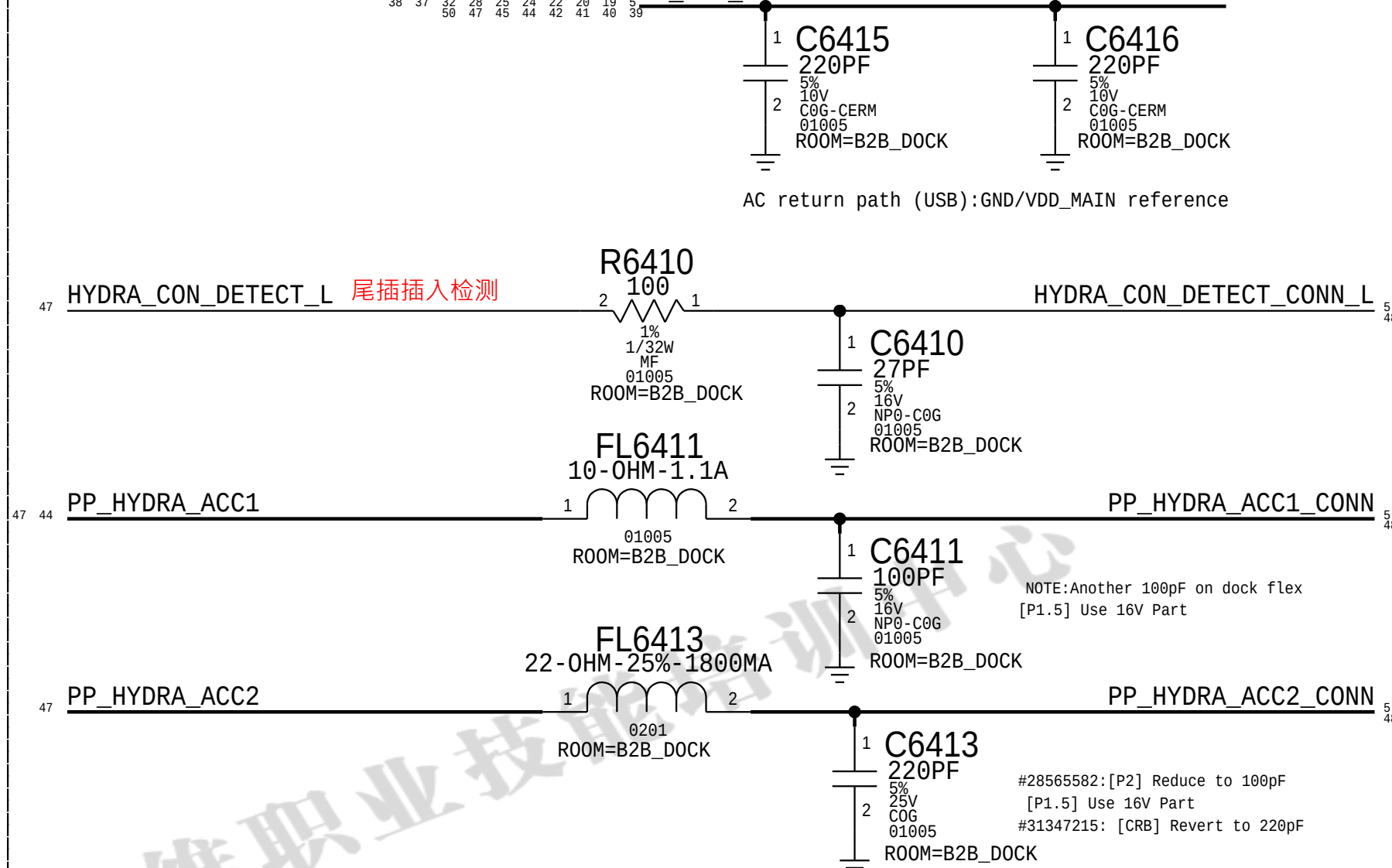
UART TX/RX pin names on Hydra match signal names (i.e. TX pin is input and connects to TX signal)



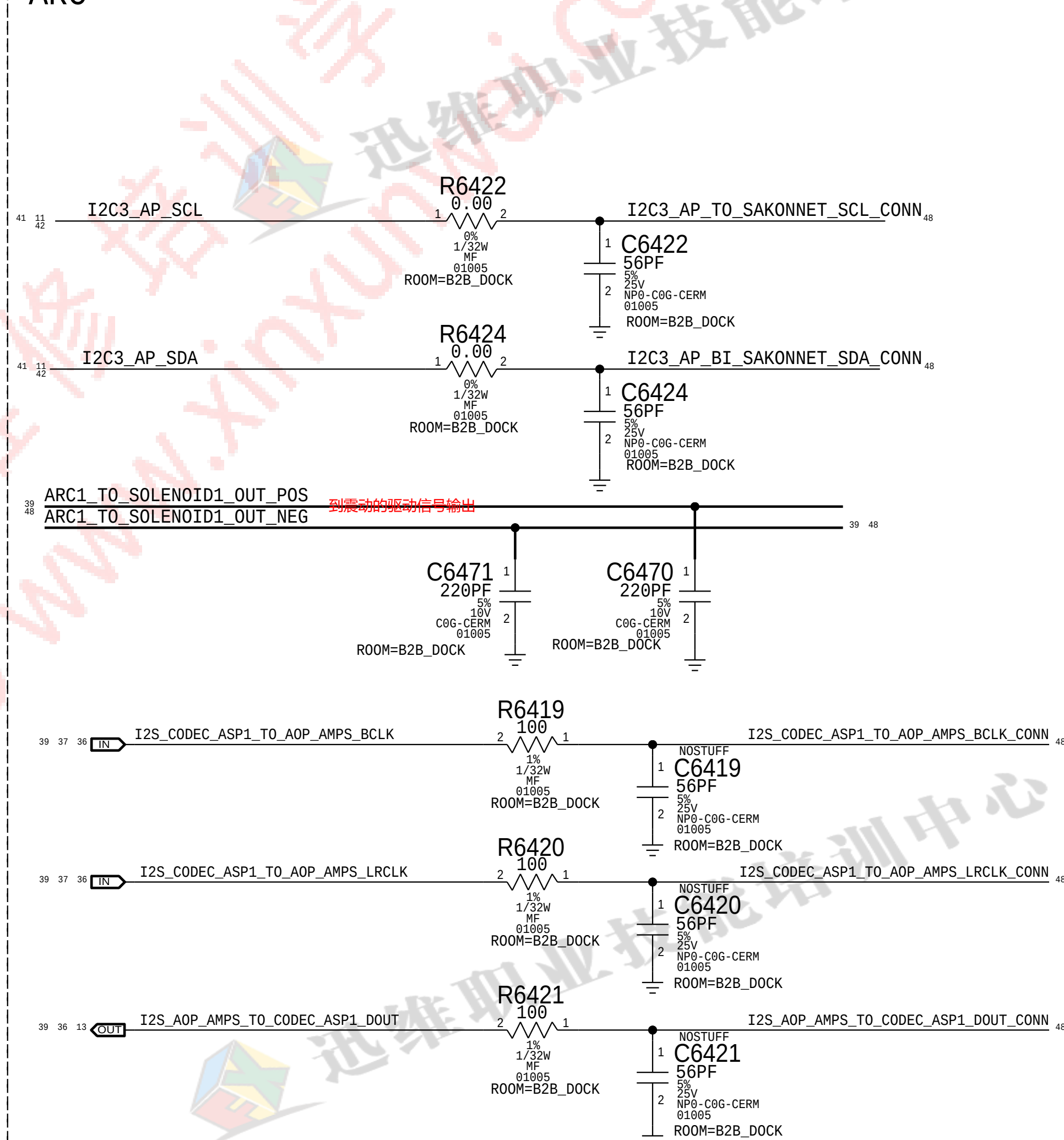
ANTENNA



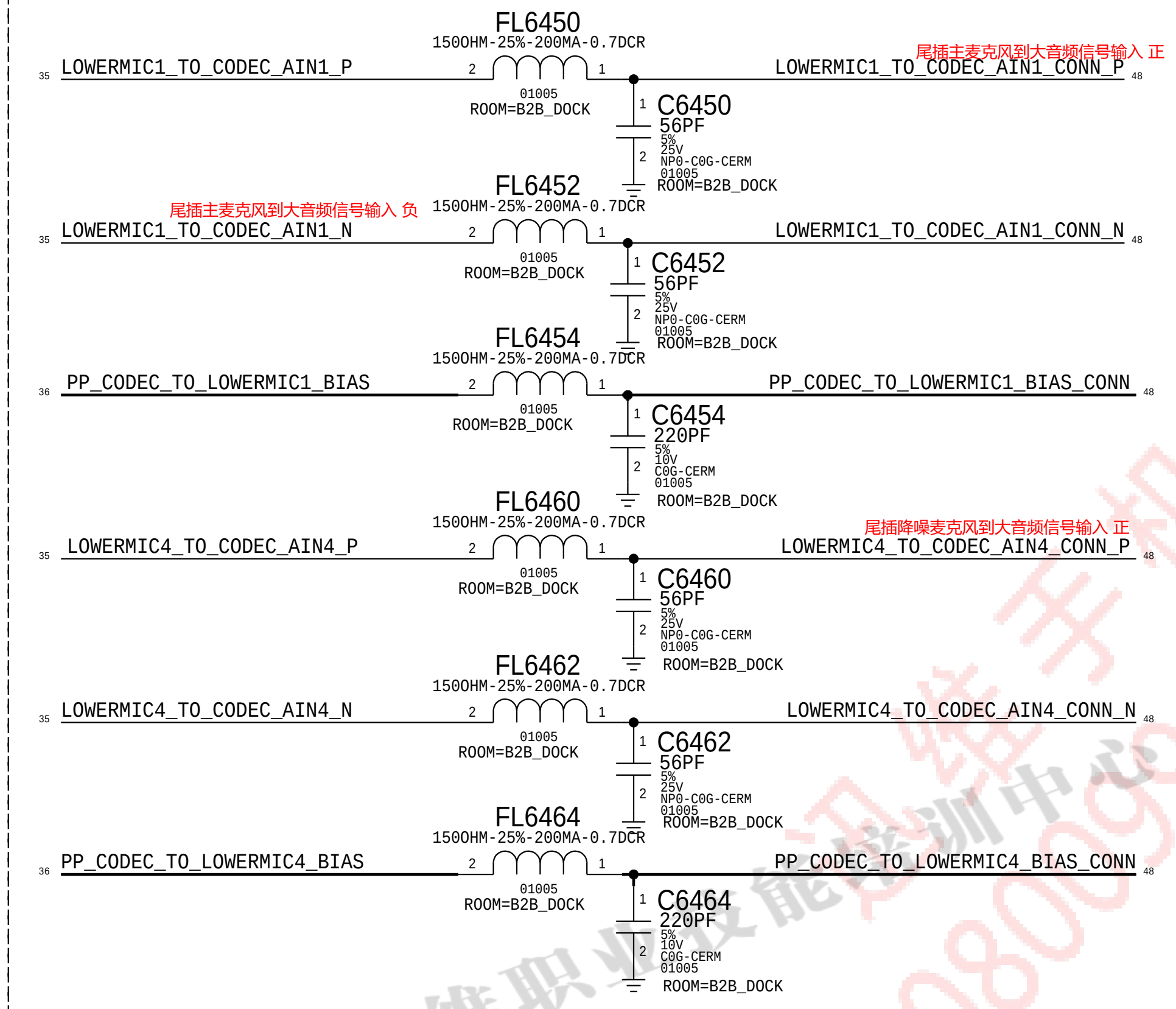
HYDRA



ARC

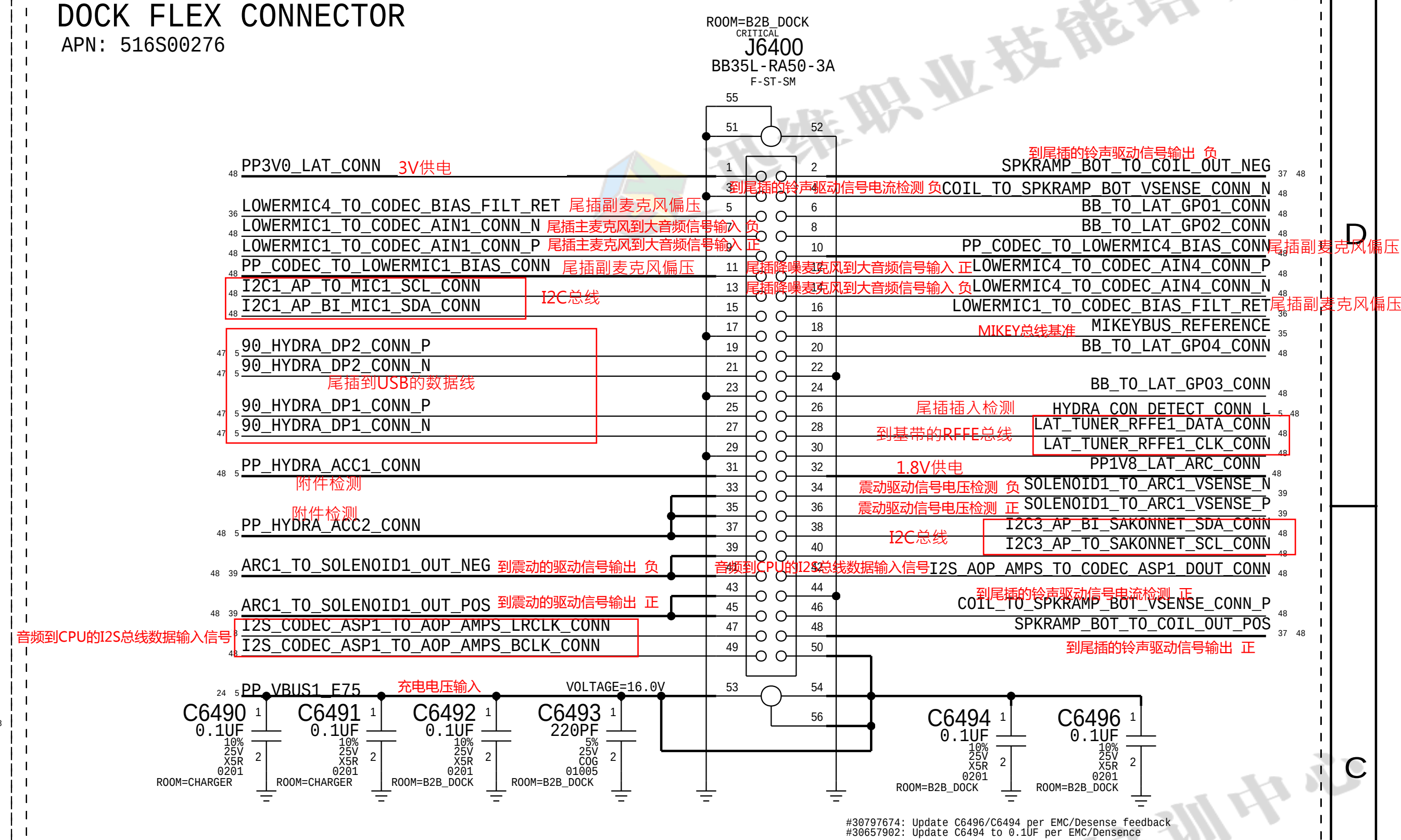


MICS

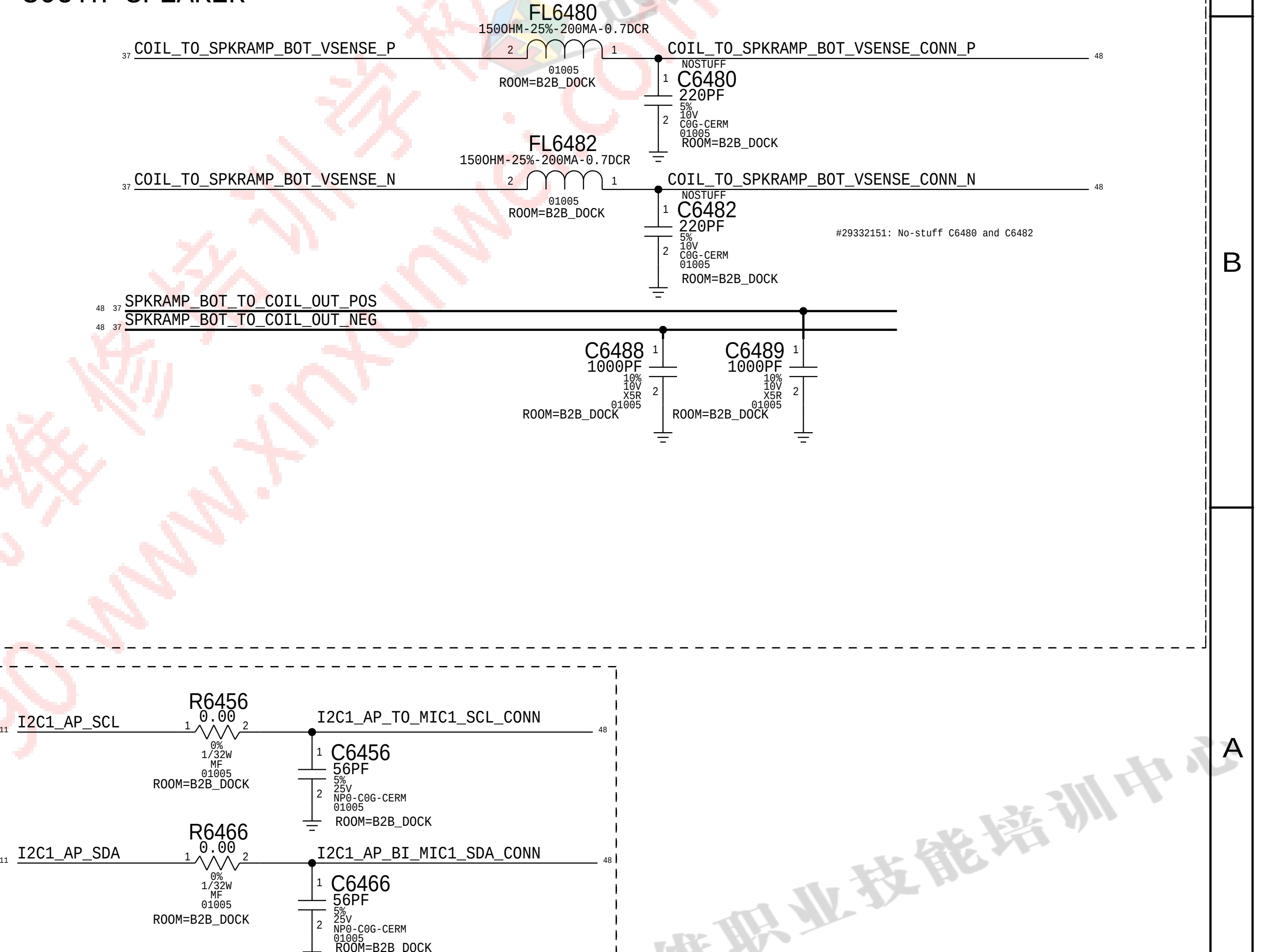


DOCK FLEX CONNECTOR

APN: 516S00276



SOUTH SPEAKER

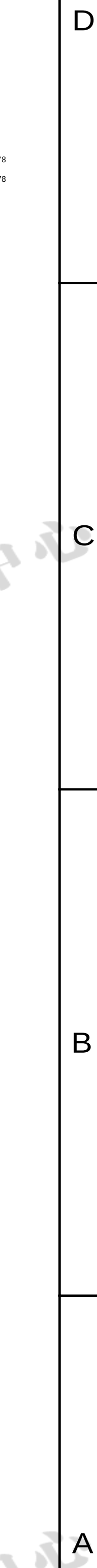




I2C MAP

(see #28857723 updates)

Diags ID	Bus	Device	7-Bit Address	8-Bit Address	Device Max Speed	Operating Speed	Net Name
0	AP I2C0	Agnes (System PMU)	0x74	0xE8	400 Kbps	400 Kbps	I2C0_AP
		Chestnut (Display PMU)	0x27	0x4E	400 Kbps		
		AccessoryBuck	0x52	0xA4	400 Kbps		
		Muon (Backlight Controller 2)	0x62	0xC4	1 Mbps		
1	AP I2C1	MIC1 Temp Sensor (South)	0x54	0xA8	1 Mbps	100 Kbps	I2C1_AP
		MIC2 Temp Sensor (North Rear)	0x56	0xAC	1 Mbps		
2	AP I2C2	Adare (Top Speaker Amp)	0x40	0x80	1 Mbps	1 Mbps	I2C2_AP
		Convoy (Top Speaker Cap Sense)	0x21	0x42	1 Mbps		
3	AP I2C3	EEPROM	0x51	0xA2	400 Kbps	400 Kbps	I2C3_AP
		Muon (Backlight Controller 1)	0x62	0xC4	400 Kbps		
		Sakonnet (Arc: Hall Effect Sensor)	0x08	0x10	1 Mbps		
		EEPROM	0x50	0xA0	400 Kbps		
	ISP I2C0	Casper (Rear Wide Camera)	0x10	0x20	1 Mbps	1 Mbps	I2C0_ISP
		Grunberg+ (AF/OIS Wide Camera)	0x3C	0x78	1 Mbps		
	ISP I2C1	Billings (Rear Tele Camera)	0x20	0x40	1 Mbps	1 Mbps	I2C1_ISP
		Edwin (AF Tele Camera)	0x0D	0x1A	1 Mbps		
	ISP I2C2	Concord (Front Camera)	0x10	0x20	1 Mbps	400 Kbps	I2C2_ISP
	ISP I2C3	Neon (Strobe - CW : LM3566)	0x63	0xC6	1 Mbps	400 Kbps	I2C3_ISP
		Ansel (Camera PMU)	0x40	0x80	1 Mbps		
		Neon (Strobe - AM : LM35662)	0x67	0xCE	1 Mbps		
	SEP I2C	EEPROM	0x51	0xA2	400 Kbps	400 Kbps	I2C4_AP
4	AOP I2C0	Doppler (Prox Sensor)	0x58	0xB0	1 Mbps	400 Kbps	I2C0_AOP
		CT709 (ALS : Firefish)	0x29	0x52	400 Kbps		
5	AOP I2C1	Adare (Bottom Speaker Amp)	0x40	0x80	1 Mbps	1 Mbps	I2C1_AOP
		Adare (Arc : Driver)	0x41	0x82	1 Mbps		
		Apache3 (Mesa Power Supply)	0x61	0xC2	1 Mbps		I2C1_AOP_ISO
		Turtle	0x2C	0x58	1 Mbps		
6	SMC I2C0	Tigris2 (Charger)	0x71	0xE2	400 Kbps	400 Kbps	I2C0_SMC
		TPS61280 (VDD Boost)	0x75	0xEA	400 Kbps		
		CCG2 (USB PD)	0x12 (0x08)	0x24 (0x10)	1 Mbps		
		Iktara	0x39	0x72	400 Kbps		
		SN45546 (Gas Gauge)	0x55	0xAA	400 Kbps		
7	SMC I2C1	Hydra	0x1A	0x34	400 Kbps	400 Kbps	I2C1_SMC



NFC_MLB D20/D21

MAY 15, 2017

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	3	CAP, SOFT TERM, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	TYPICAL_CAP



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00882	1	390PF, 0201, 2%, 25V	C7514_S	D20_JPN
131S00026	1	820PF, 0201, 2%, 25V	C7515_S	D20_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D20_JPN
131S00883	1	220PF, 0201, 2%, 50V	C7518_S	D20_JPN
131S00025	1	1000PF, 0201, 2%, 25V	C7510_S	D20_JPN
131S00025	1	1000PF, 0201, 2%, 25V	C7513_S	D20_JPN
157S00022	1	TDK, BALUN, 11T	T7500_S	D20_JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D20_ROW
131S00016	1	470PF, 0201, 2%, 25V	C7514_S	D20_ROW
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D20_ROW
131S00825	1	560PF, 0201, 2%, 25V	C7516_S	D20_ROW
131S00883	1	220PF, 0201, 2%, 50V	C7518_S	D20_ROW
131S00026	1	820PF, 0201, 2%, 25V	C7510_S	D20_ROW
131S00026	1	820PF, 0201, 2%, 25V	C7513_S	D20_ROW
157S00022	1	TDK, BALUN, 11T	T7500_S	D20_ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D201
131S00016	1	470PF, 0201, 2%, 25V	C7514_S	D201
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D201
131S00825	1	560PF, 0201, 2%, 25V	C7516_S	D201
131S00883	1	220PF, 0201, 2%, 50V	C7518_S	D201
131S00026	1	820PF, 0201, 2%, 25V	C7510_S	D201
131S00026	1	820PF, 0201, 2%, 25V	C7513_S	D201
157S00022	1	TDK, BALUN, 11T	T7500_S	D201

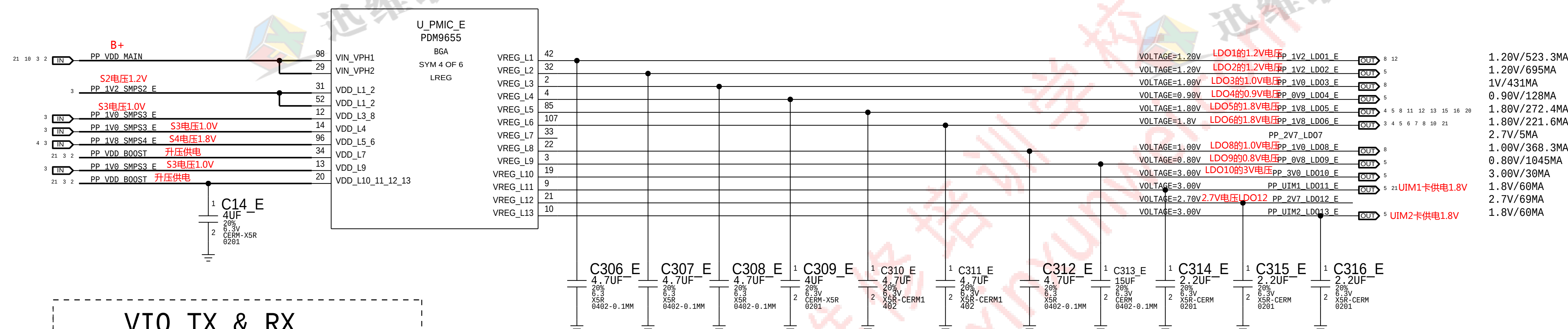
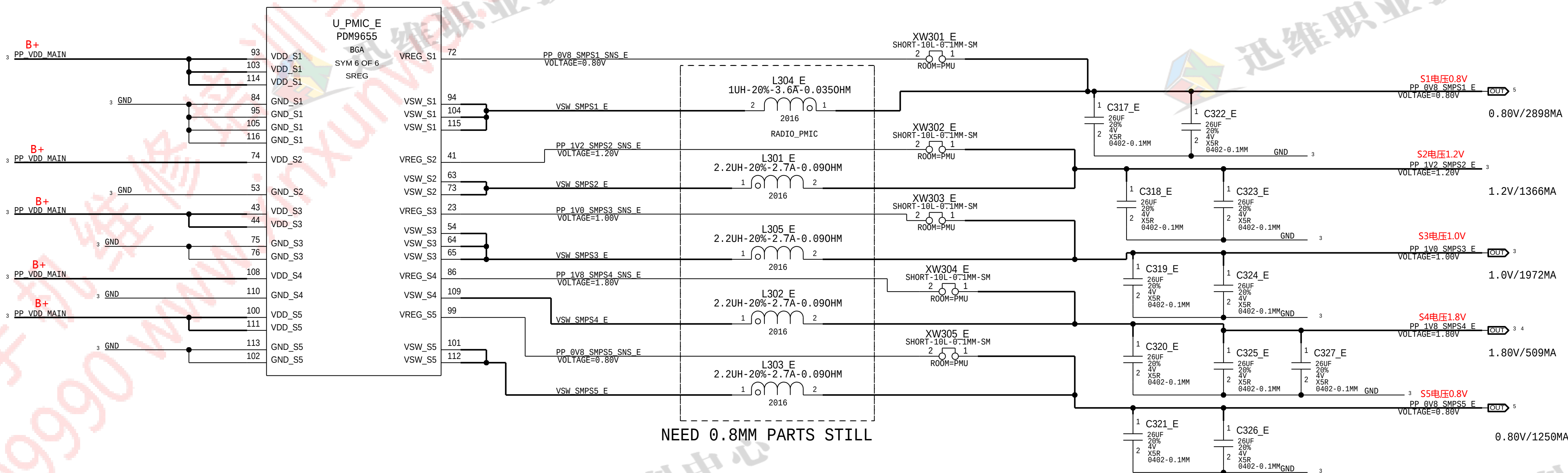
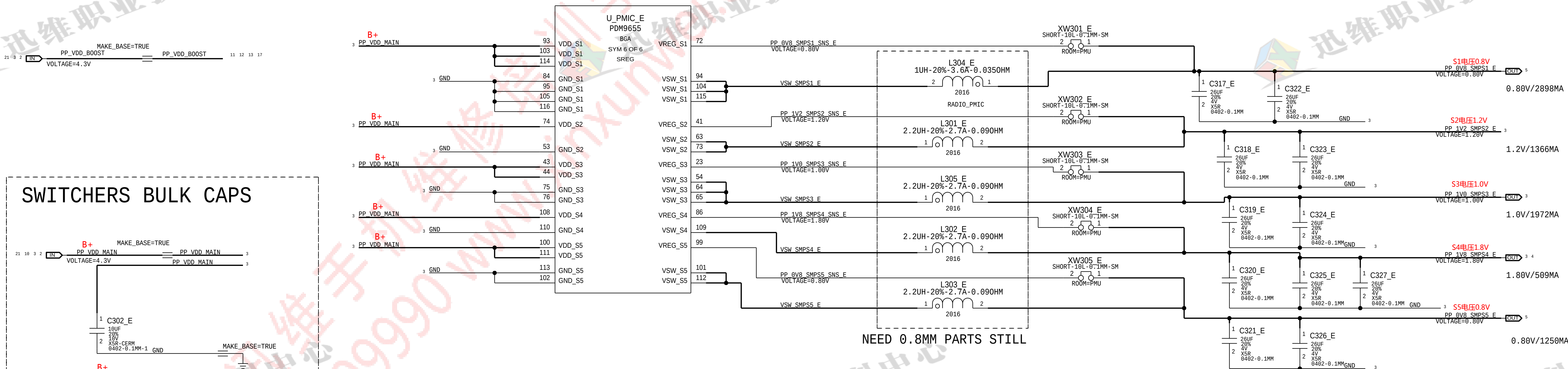
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131S00081	1	270PF, 0201, 2%, 25V	C7514_S	D21_JPN
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D21_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D21_JPN
131S00019	1	150PF, 0201, 2%, 50V	C7518_S	D21_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7510_S	D21_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7513_S	D21_JPN
157S00023	1	MURATA, BALUN, 11T	T7500_S	D21_JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D21_ROW
131S00081	1	270PF, 0201, 2%, 25V	C7514_S	D21_ROW
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D21_ROW
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D21_ROW
131S00019	1	150PF, 0201, 2%, 50V	C7518_S	D21_ROW
131S00033	1	680PF, 0201, 2%, 25V	C7510_S	D21_ROW
131S00033	1	680PF, 0201, 2%, 25V	C7513_S	D21_ROW
157S00023	1	MURATA, BALUN, 11T	T7500_S	D21_ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D211
131S00081	1	270PF, 0201, 2%, 25V	C7514_S	D211
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D211
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D211
131S00019	1	150PF, 0201, 2%, 50V	C7518_S	D211
131S00033	1	680PF, 0201, 2%, 25V	C7510_S	D211
131S00033	1	680PF, 0201, 2%, 25V	C7513_S	D211
157S00023	1	MURATA, BALUN, 11T	T7500_S	D211

RF PMIC: SWITCHERS & LDOS

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RF PMIC: XTAL, CLOCKS AND MISCELLANEOUS

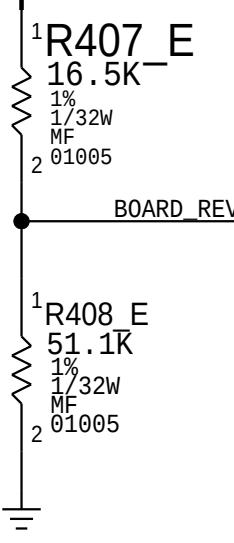
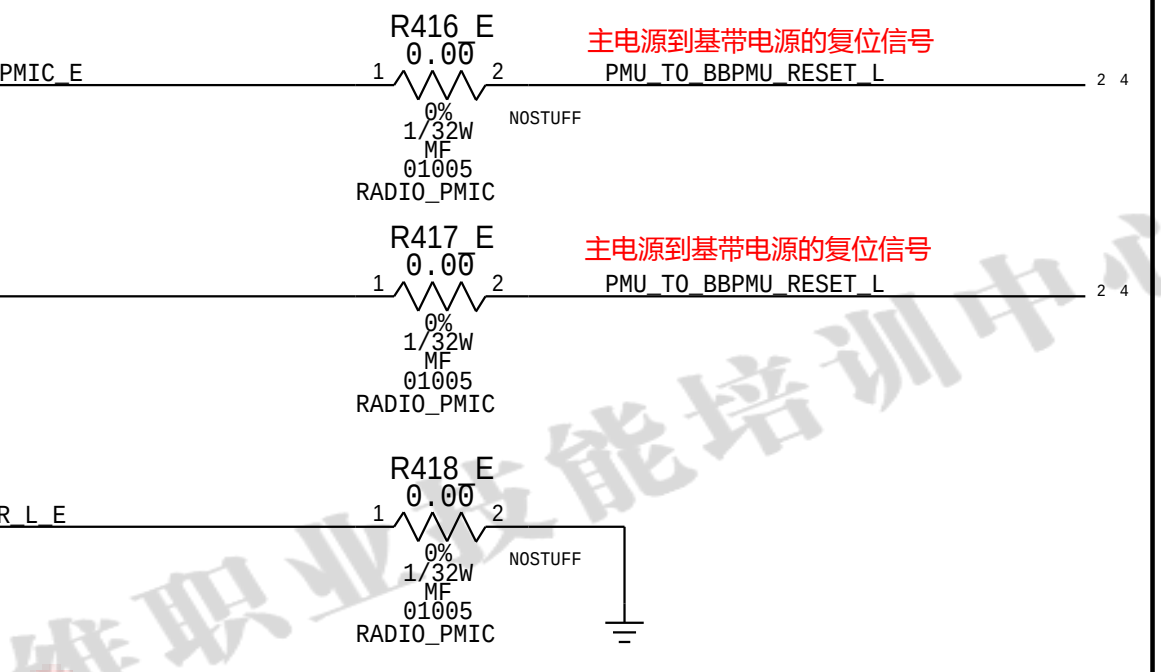
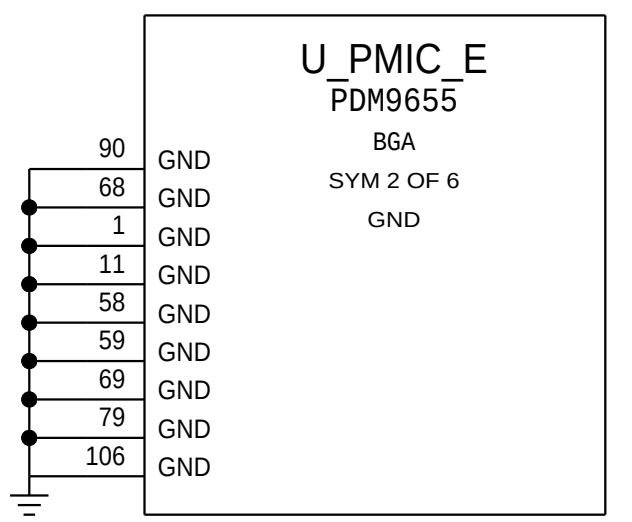
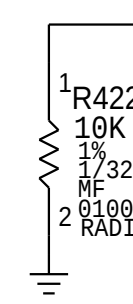
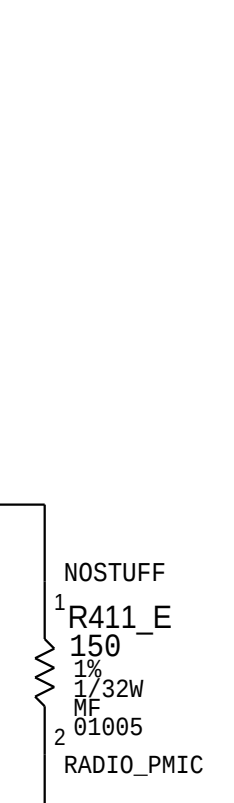
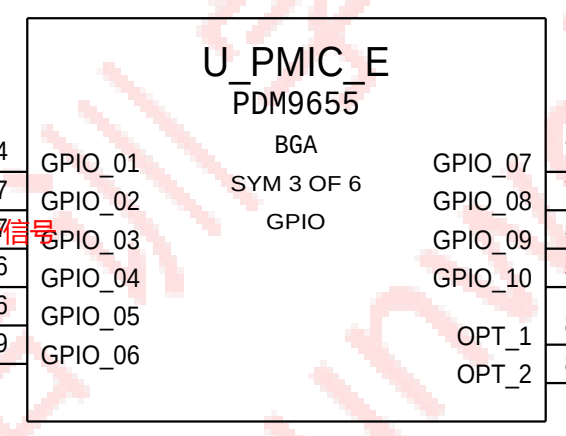
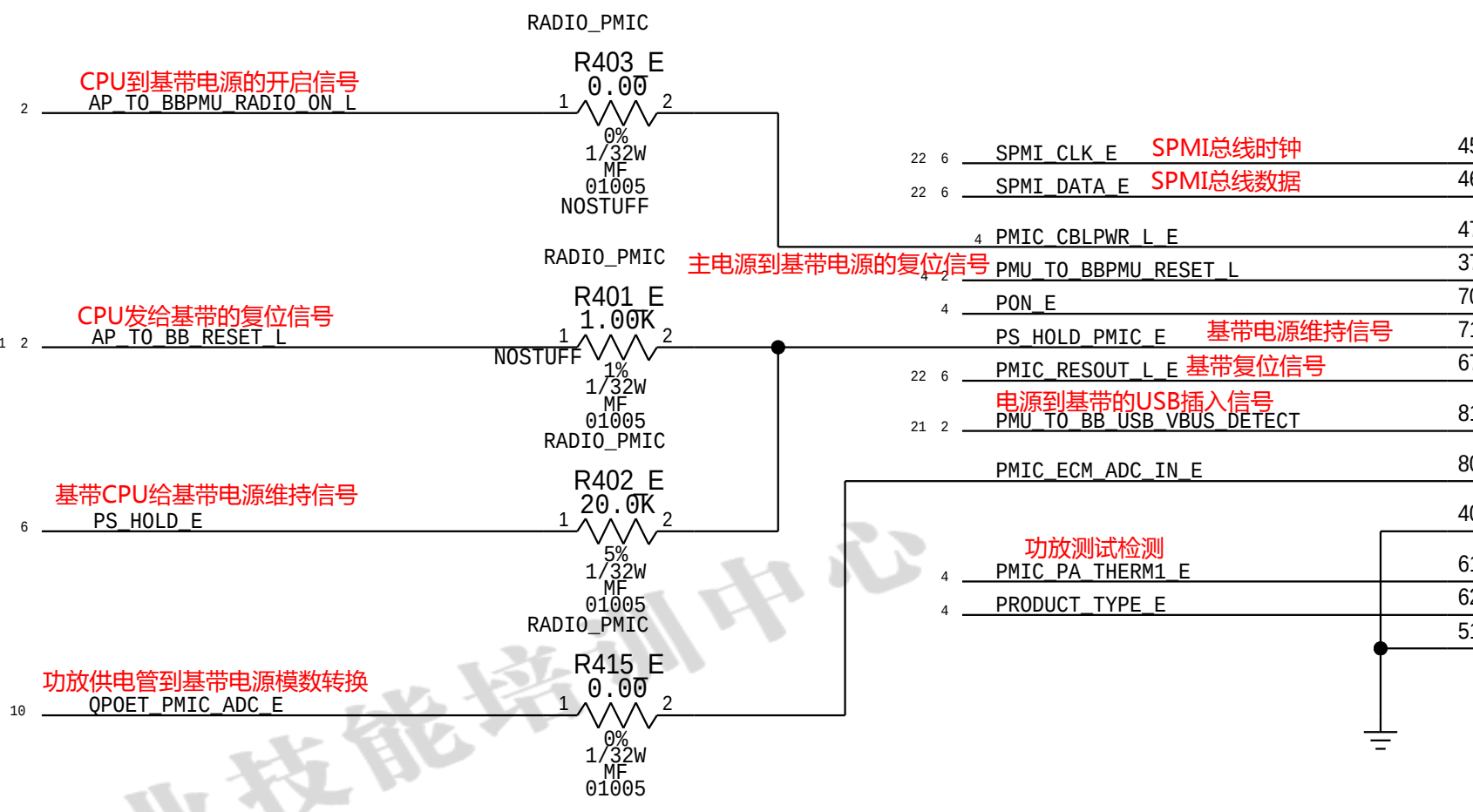
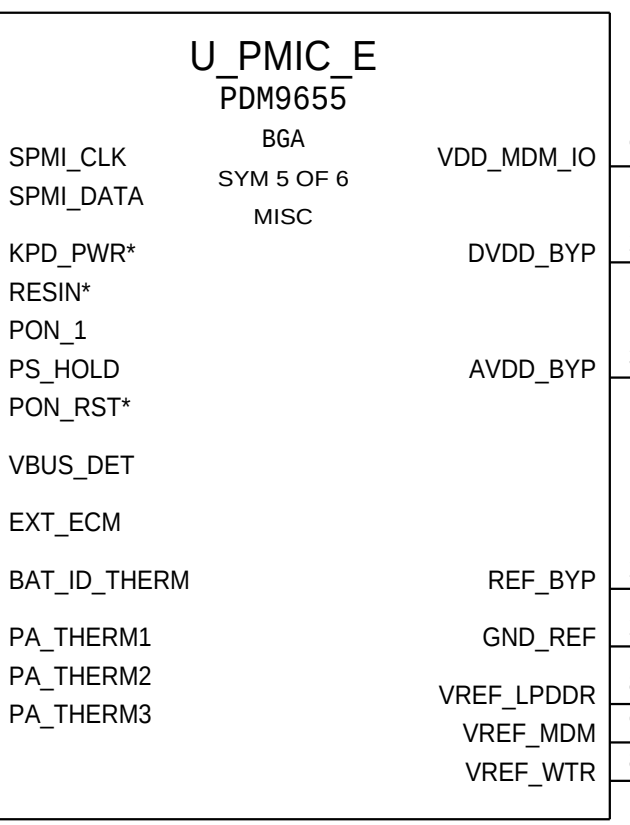
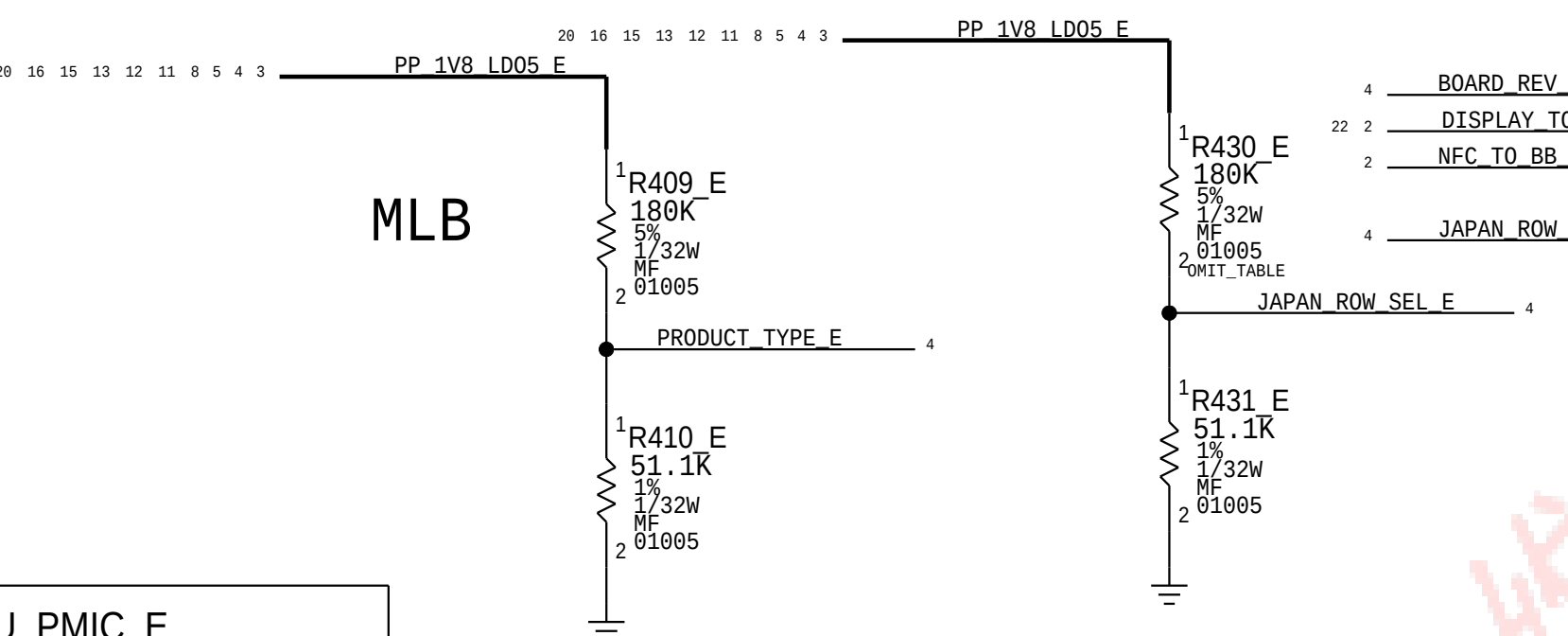
HW_REV_ID	R407	R408	MLB/RADIO_DEV
<0.10V	NOSTUFF	51.1K	T500/DEV1
0.15V-0.25V	422K	51.1K	PROT01.5
0.35V-0.45V	180K	51.1K	PROT02.5/DEV2
0.55V-0.65V	105K	51.1K	PROT01/DEV3
0.75V-0.85V	66.5K	51.1K	PROT02/DEV4
0.95V-1.05V	44.2K	51.1K	EVT
1.15V-1.25V	28K	51.1K	CARRIER/DEV5
1.35V-1.45V	16.5K	51.1K	DVT/DEV6
1.55V-1.65V	7.87K	51.1K	PVT/DEV7
1.75V-1.85V	10K	NOSTUFF	SPARE

NOTE XTAL SYMBOL MIRRORED COMPARED TO PREVIOUS SCH

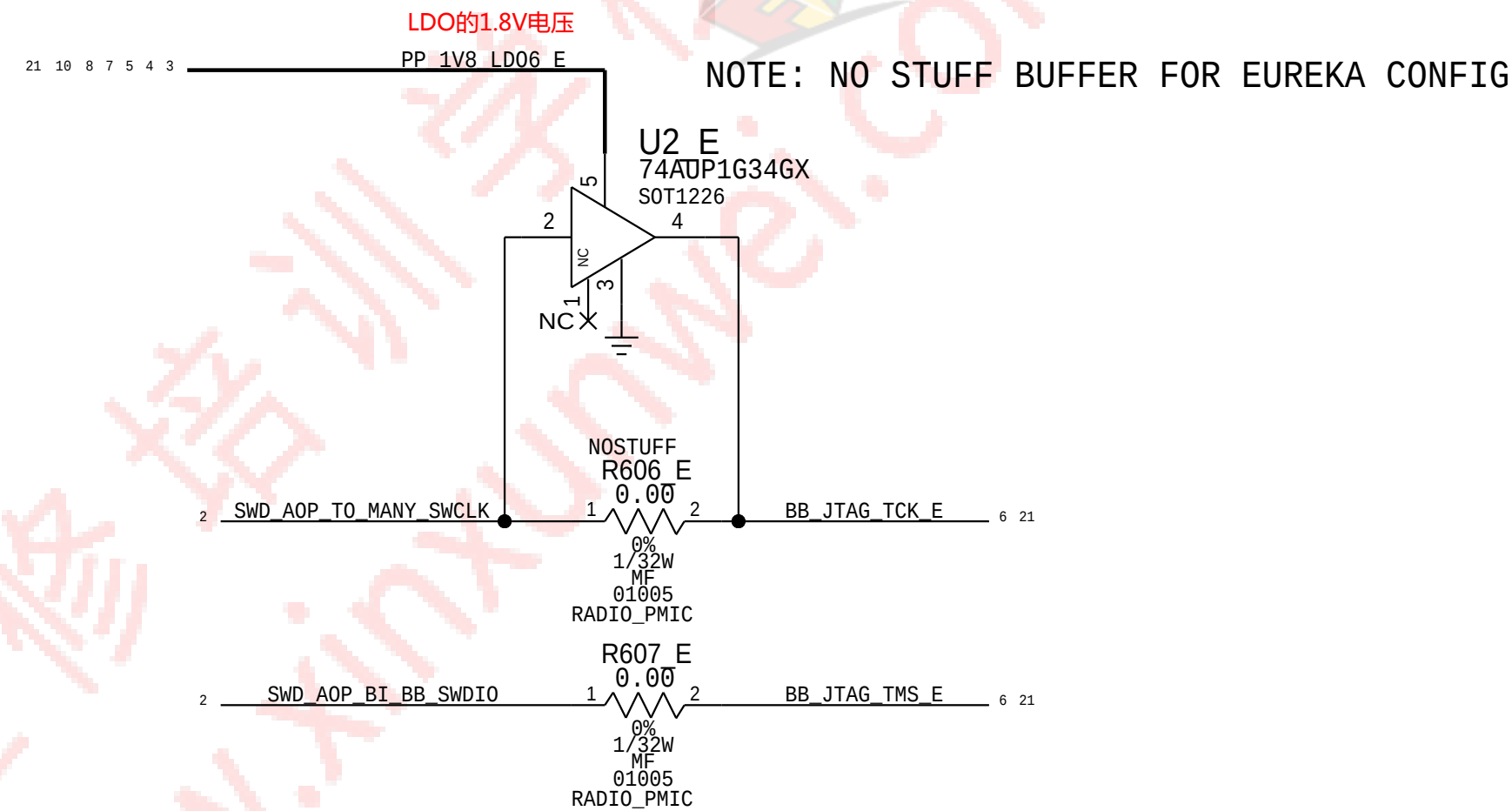
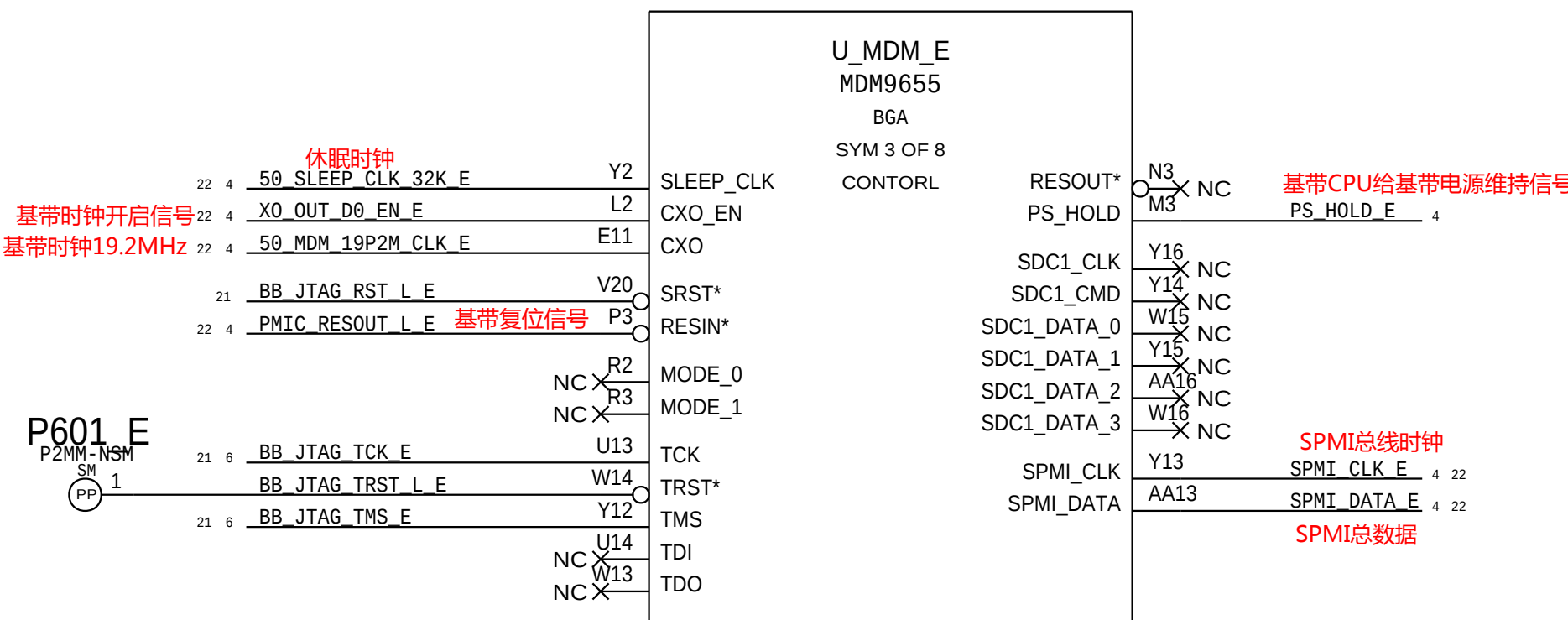
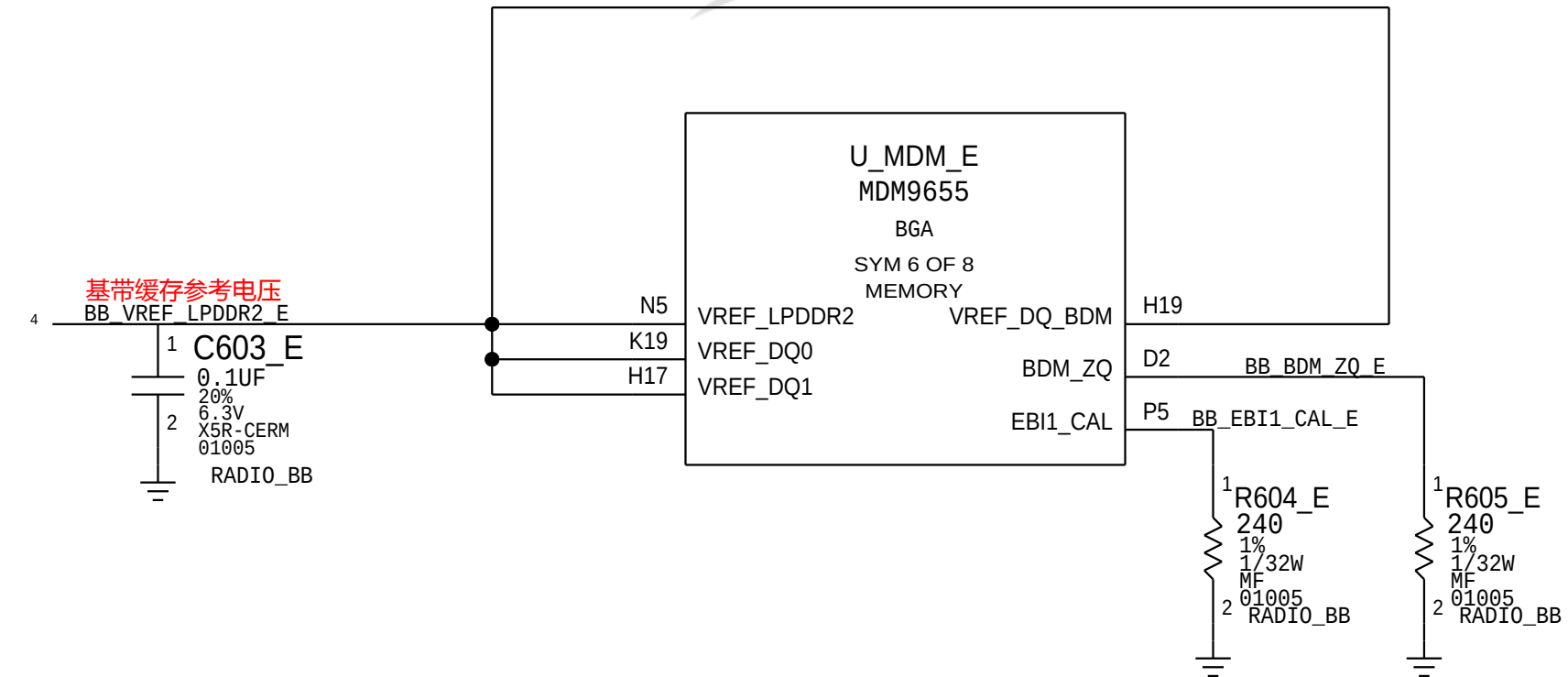
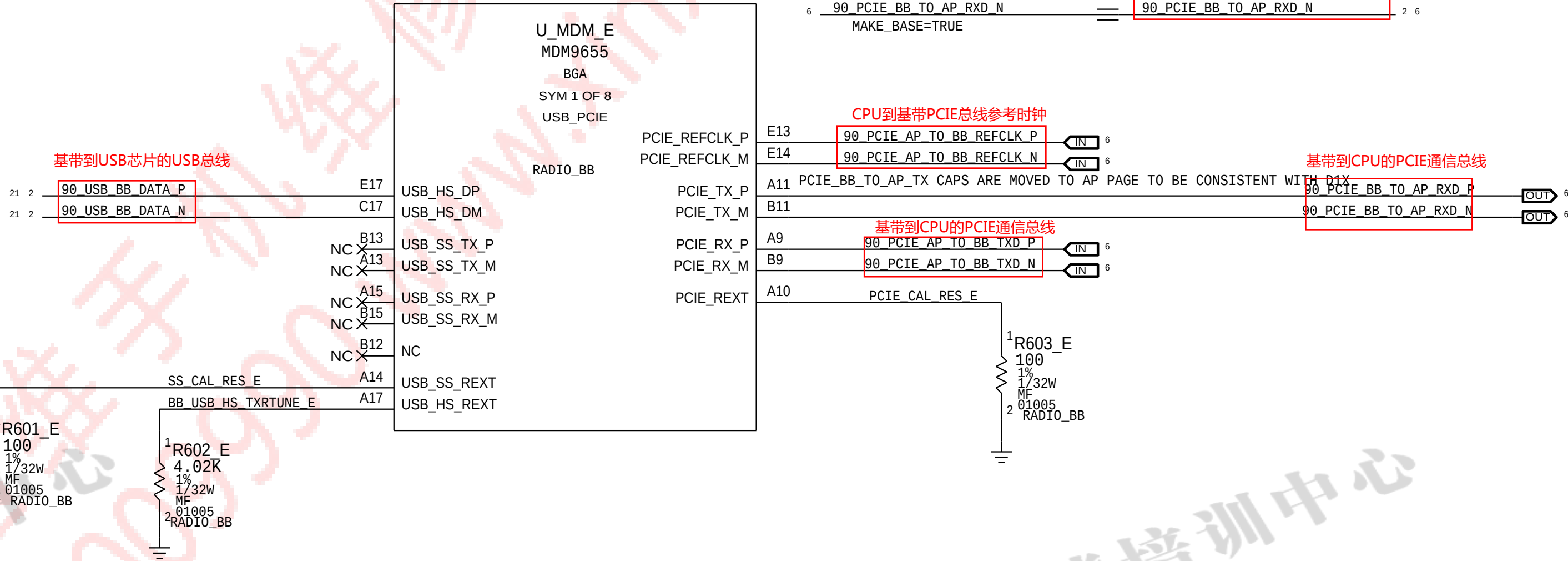
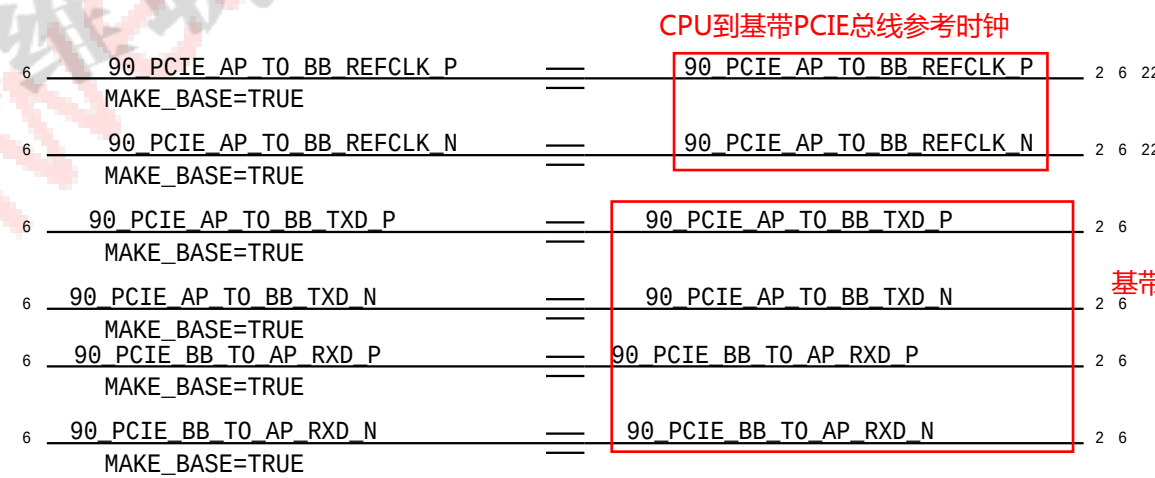
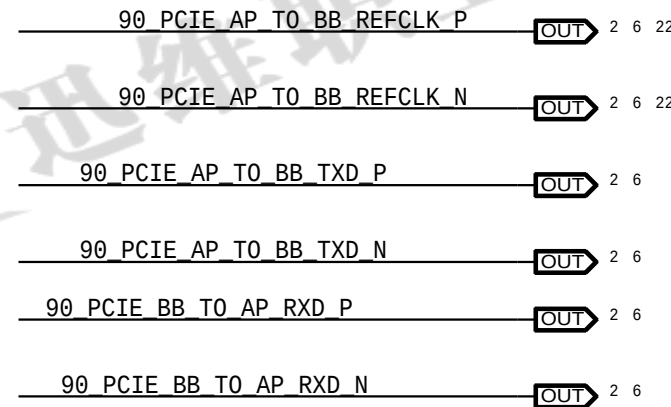
PA_THERM2	R409	R410	PRODUCT TYPE	Y
0.00V-0.10V	NOSTUFF	51.1K	RADIO DEV	1
0.15V-0.25V	422K	51.1K	3RD TYPE	2
0.35V-0.45V	180K	51.1K	POR MLB (FF)	3
0.55V-0.65V	105K	51.1K	SPARE	4
0.75V-0.85V	66.5K	51.1K	SPARE	5
0.95V-1.05V	44.2K	51.1K	SPARE	6
1.15V-1.25V	28K	51.1K	SPARE	7
1.35V-1.45V	16.5K	51.1K	SPARE	8
1.55V-1.65V	7.87K	51.1K	SPARE	9
1.75V-1.85V	10K	NOSTUFF	SPARE	A

JPN_ROW_SEL	R430	R431	MAV PLATFORM	Y	SKU
<0.10V	NOSTUFF	51.1K	MAV17.0	0	ROW
0.15V-0.25V	422K	51.1K	MAV17.1	1	JPN
0.35V-0.45V	180K	51.1K	MAV17.2	2	ROW
0.55V-0.65V	105K	51.1K	MAV17.3	3	JPN
0.75V-0.85V	66.5K	51.1K	MAV17.4	4	ROW
0.95V-1.05V	44.2K	51.1K	MAV17.5	5	JPN
1.15V-1.25V	28K	51.1K	MAV17.6	6	ROW
1.35V-1.45V	16.5K	51.1K	MAV17.7	7	JPN
1.55V-1.65V	7.87K	51.1K	MAV17.8	8	ROW
1.75V-1.85V	10K	NOSTUFF	MAV17.9	9	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0606	1	RESISTOR, 01005, 105K OHMS	R430_E	CRITICAL	JPN
117S0197	1	RESISTOR, 01005, 100K OHMS	R430_E	CRITICAL	ROW

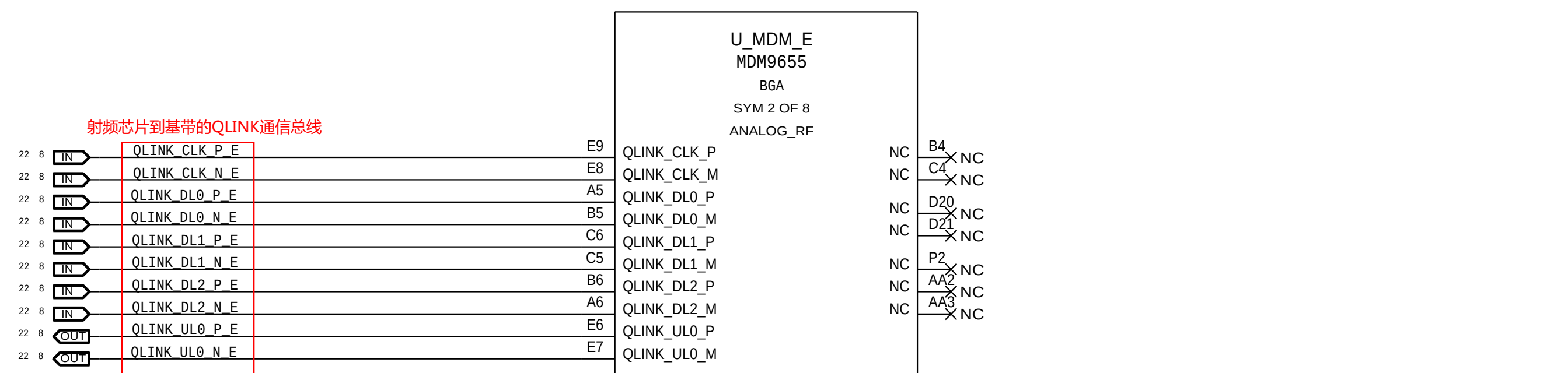
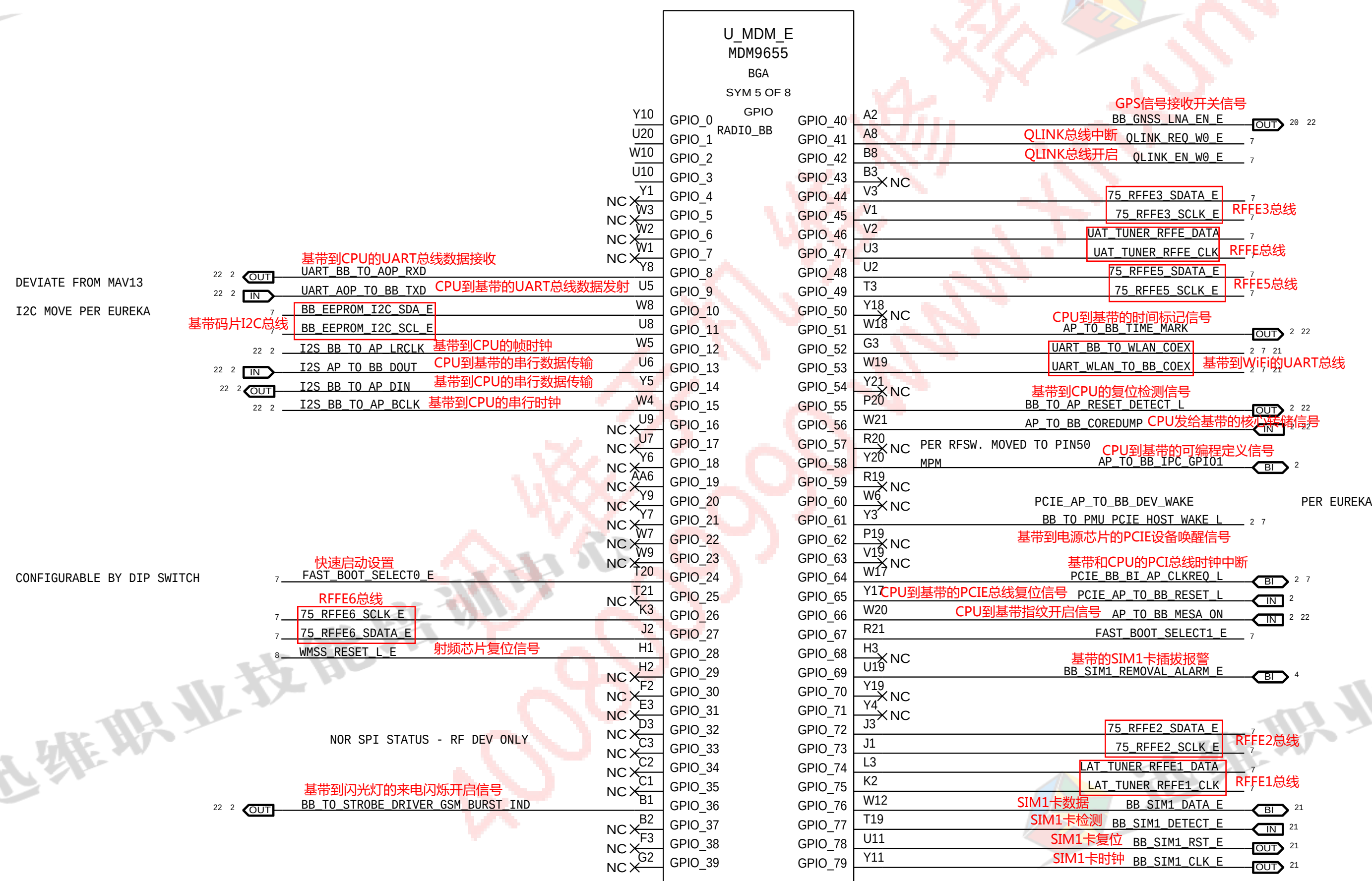


BB: CONTROL & HS PERIPHERALS

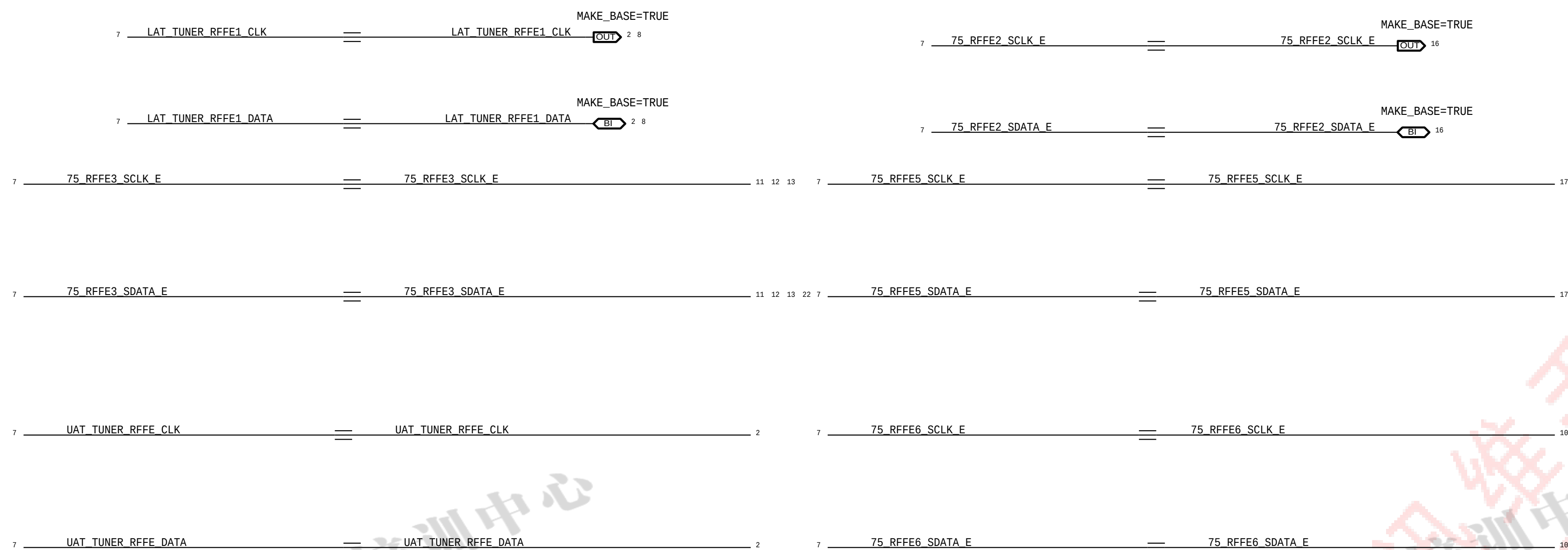
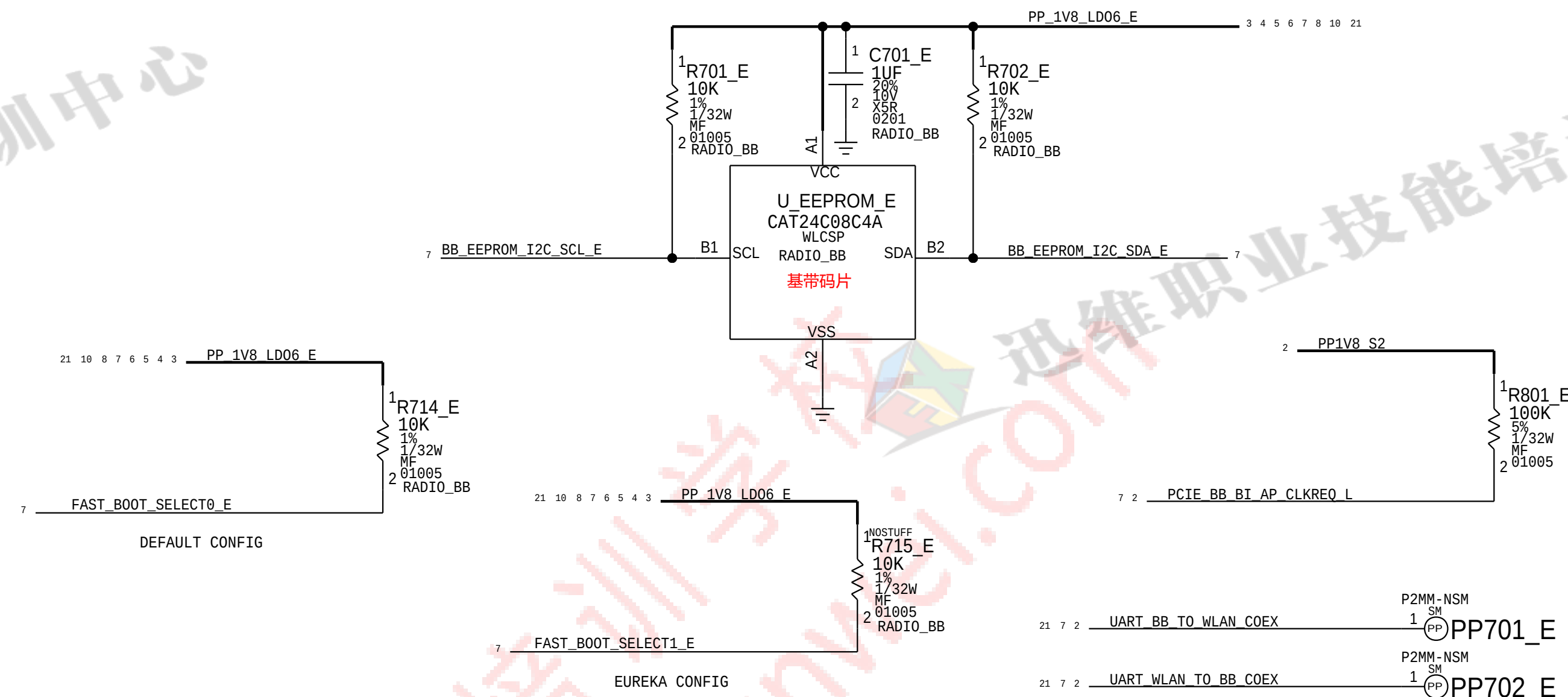


BB: GPIOS & QLINK

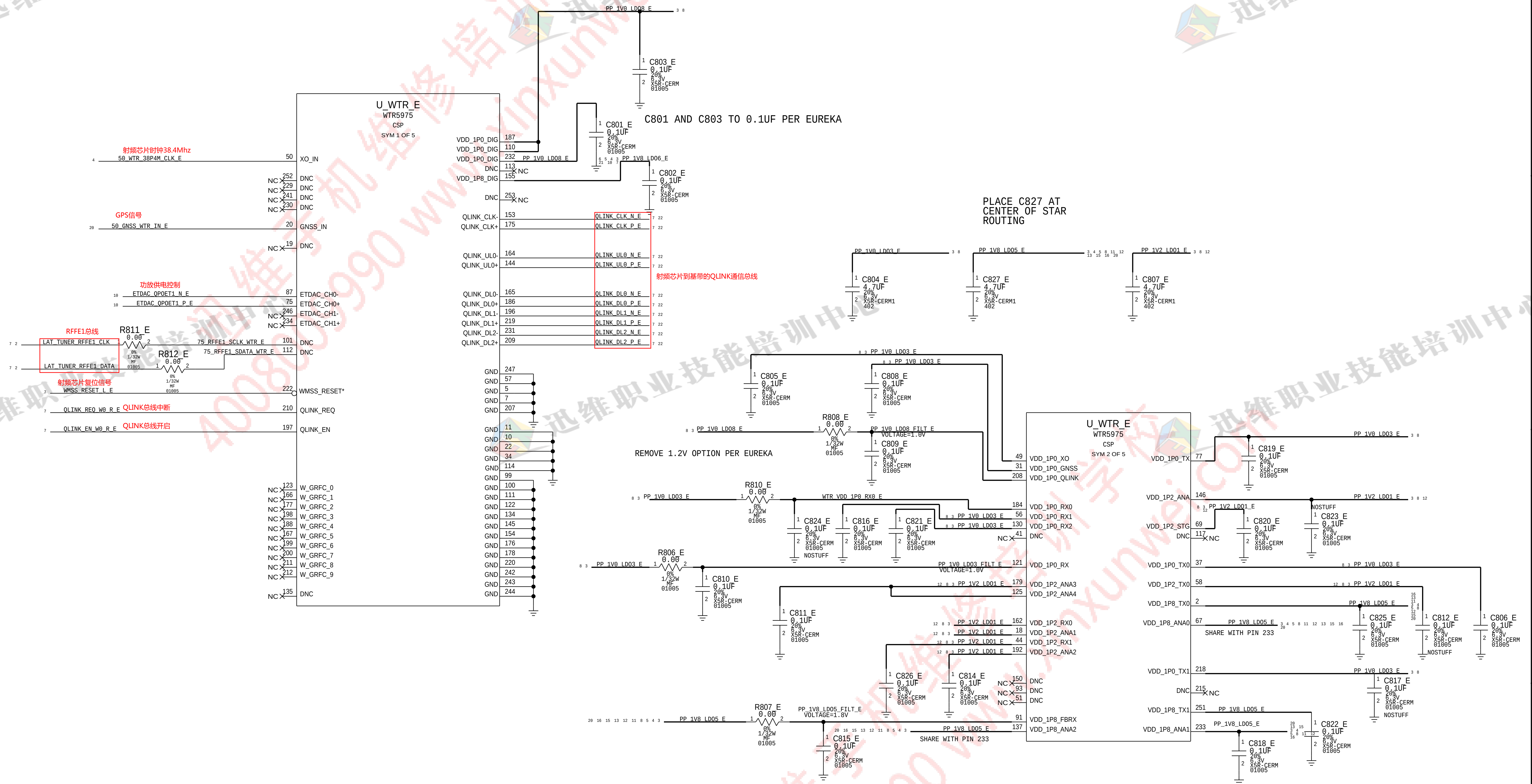
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BB EEPROM



TRANSCEIVER: QLINK & POWER



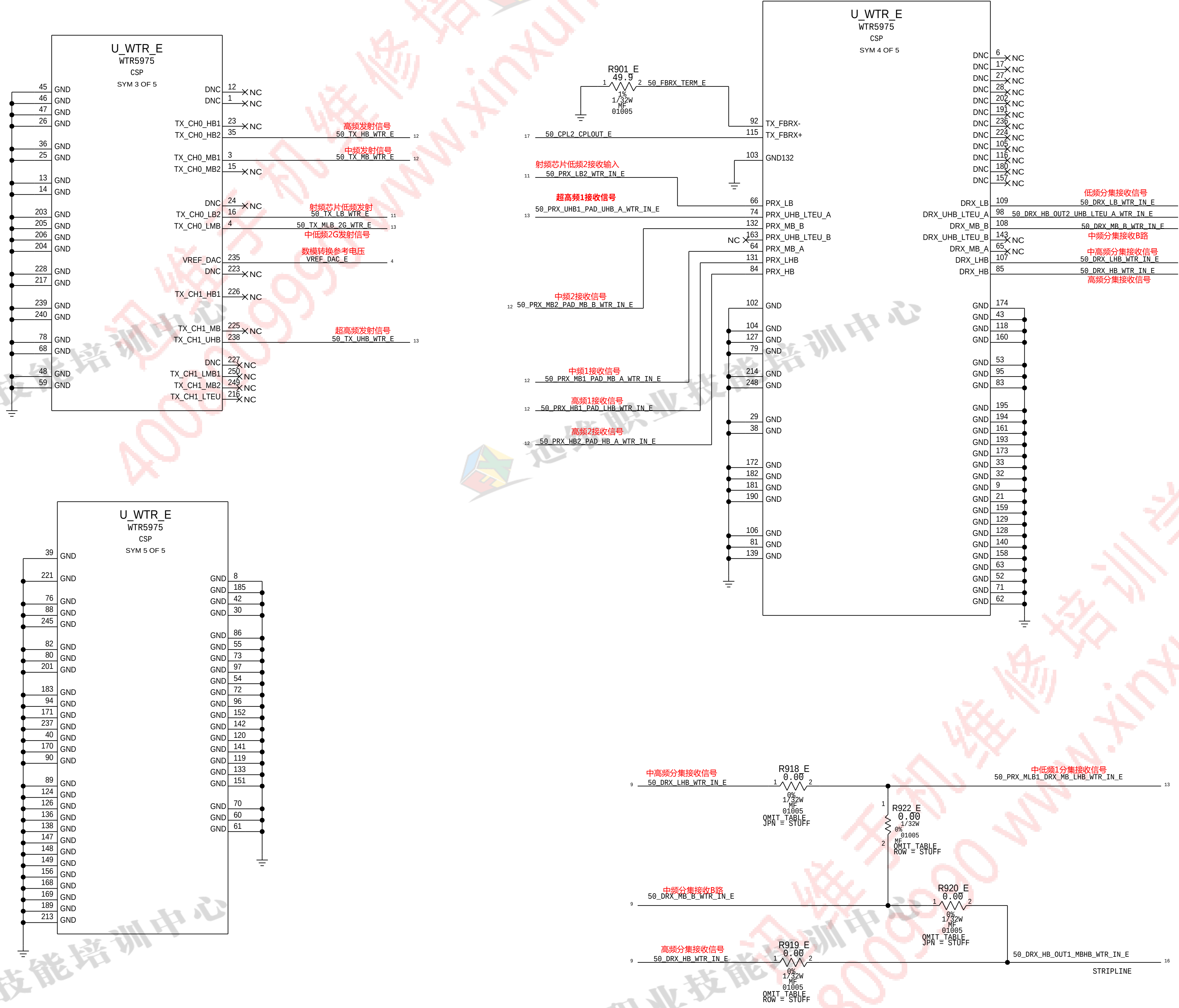


TRANSCEIVER: TX & RX

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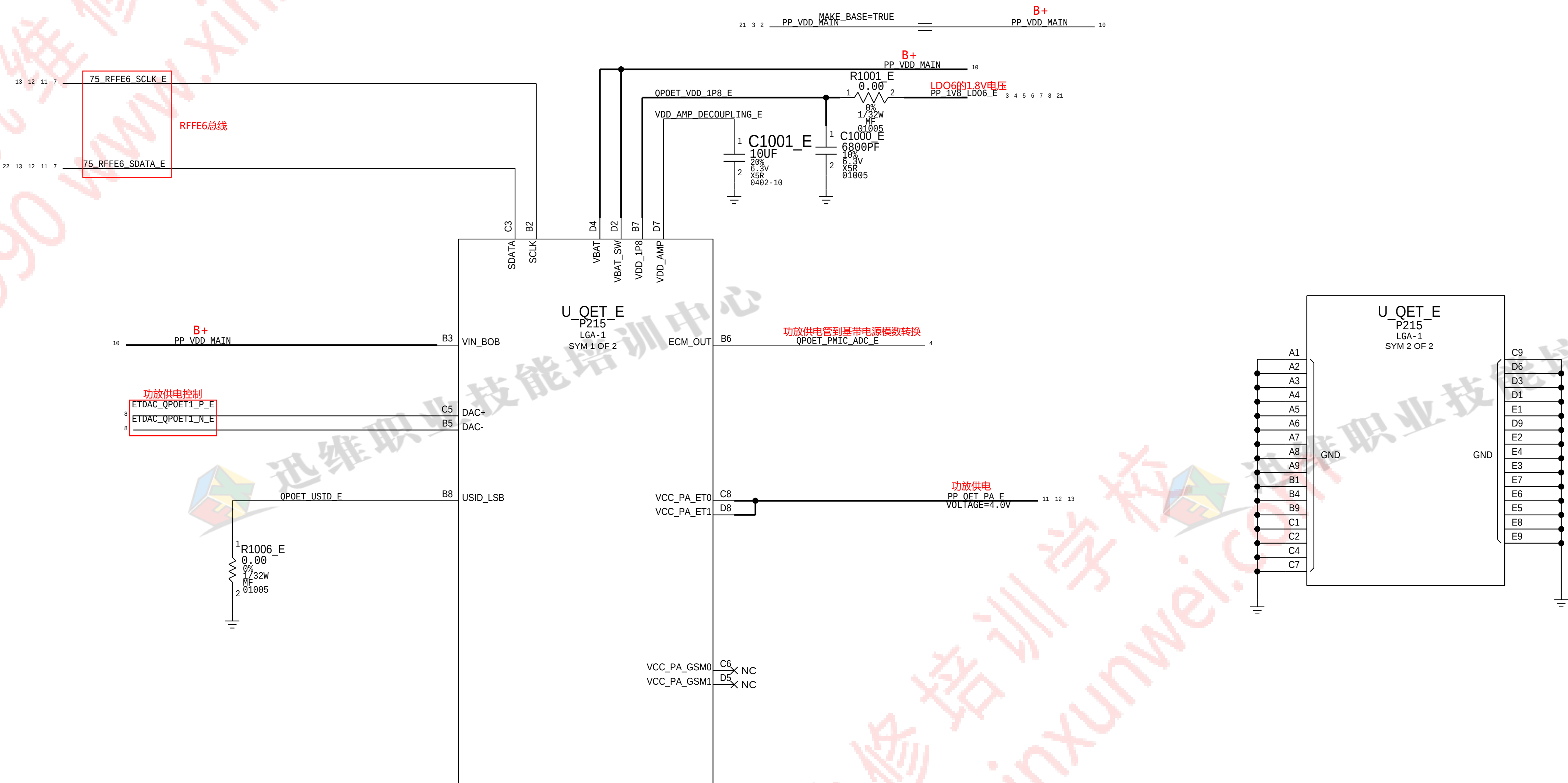
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 OR	R919_E	CRITICAL	ROW
117S0161	1	01005 OR	R922_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 OR	R918_E	CRITICAL	JPN
117S0161	1	01005 OR	R920_E	CRITICAL	JPN



QET:MODULE

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LB PAD

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0724	1	2G LB MATCH, 0 OHMS	R1108_E	CRITICAL	JPN
152S08021	1	6.8NH UHQ, LB ANT MATCH	R1104_E	CRITICAL	JPN
131S0555	1	1PF, LB ANT MATCH	L1101_E	CRITICAL	JPN
131S0176	1	2.4PF, 2G LB MATCH	L1102_E	CRITICAL	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S2086	1	6.2NH UHQ, LB ANT MATCH	R1104_E	CRITICAL	ROW

D

C

B

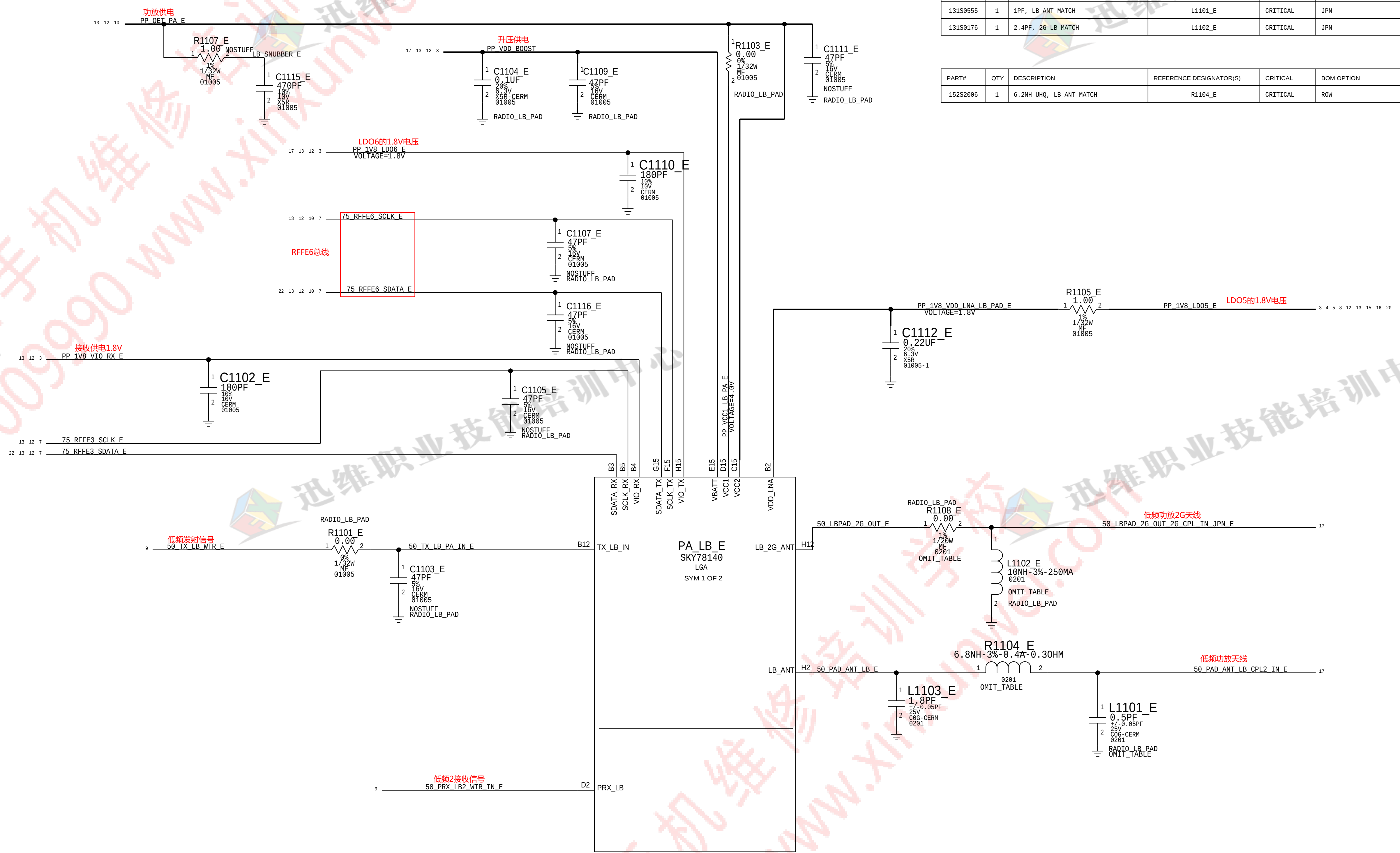
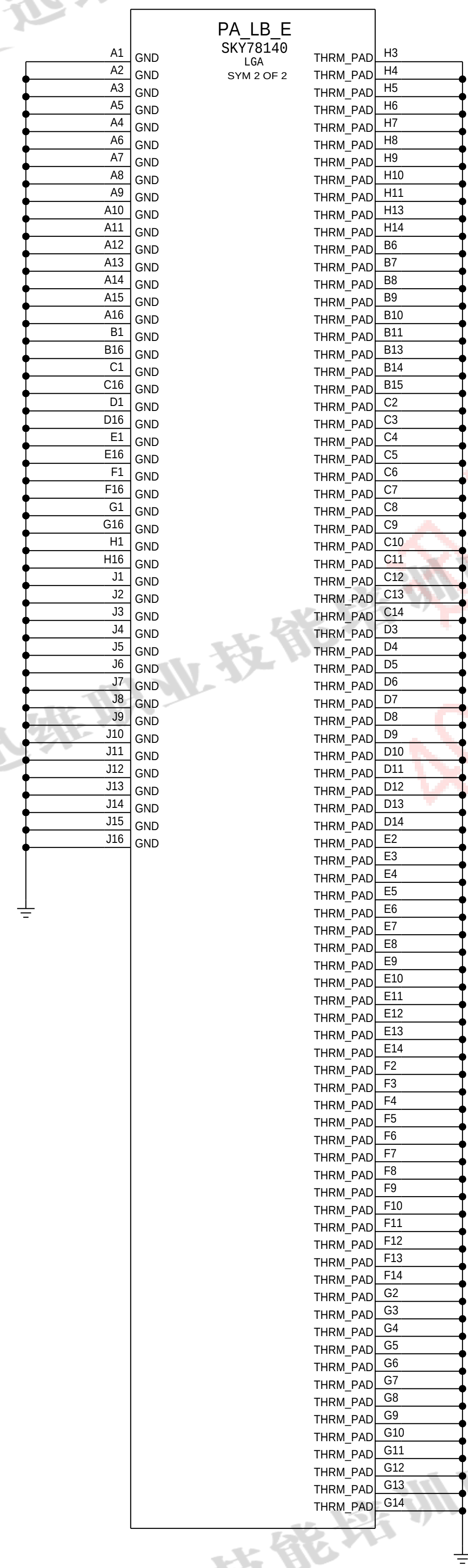
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D

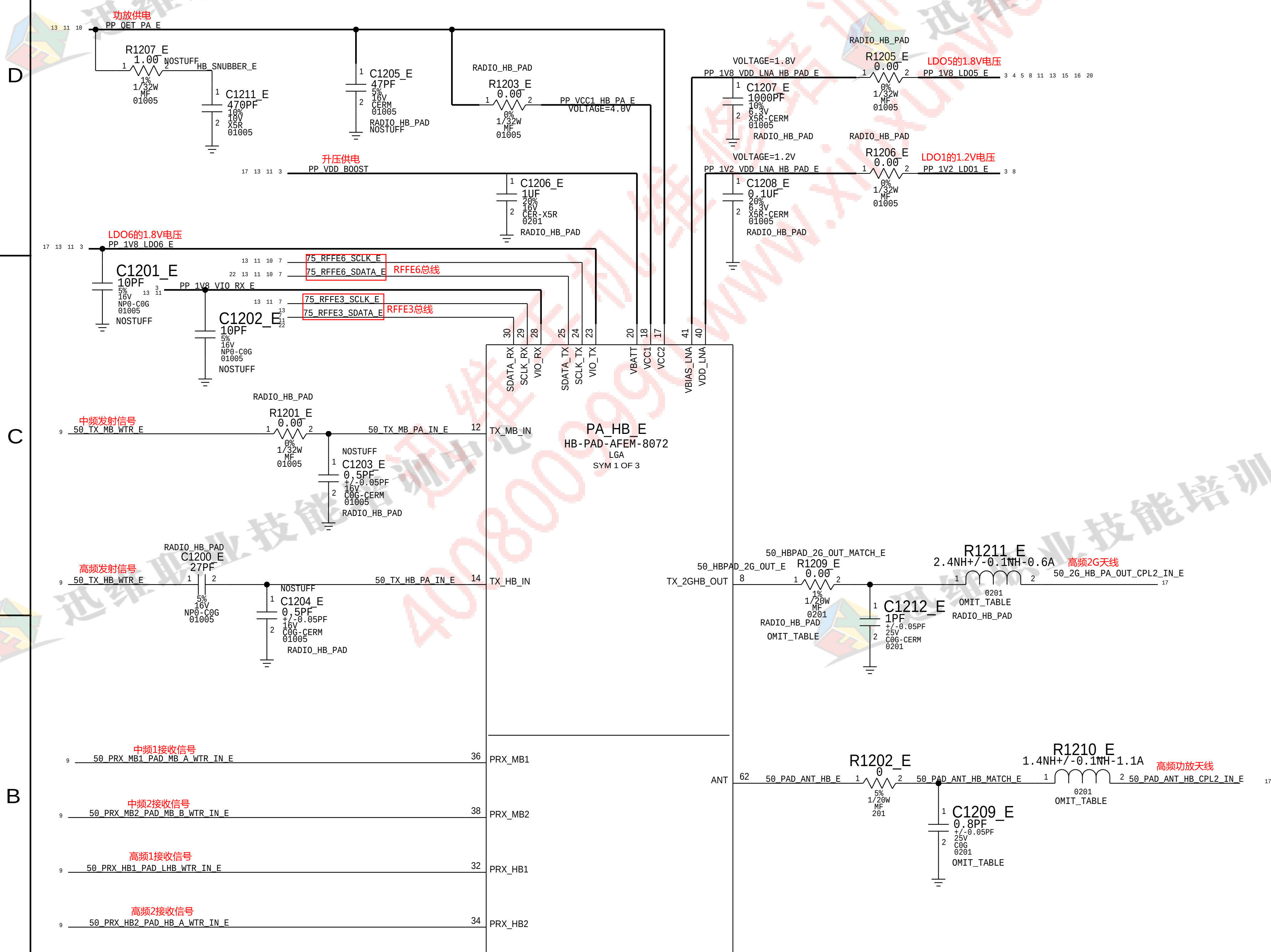
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A



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Part#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S2023	1	3.2NH UHQ, 2G HB PAD MATCH	R1209_E	CRITICAL	JPN
152S00153	1	1NH UHQ, 2G HB PAD MATCH	R1211_E	CRITICAL	JPN
131S0425	1	0.5PF, HB ANT MATCH	C1209_E	CRITICAL	JPN
152S2042	1	1.8NH UHQ, HB ANT MATCH	R1210_E	CRITICAL	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S9329	1	0.4PF, HB ANT MATCH	C1209_E	CRITICAL	ROW
152S2800	1	2.0NH UHQ, HB ANT MATCH	R1210_E	CRITICAL	ROW

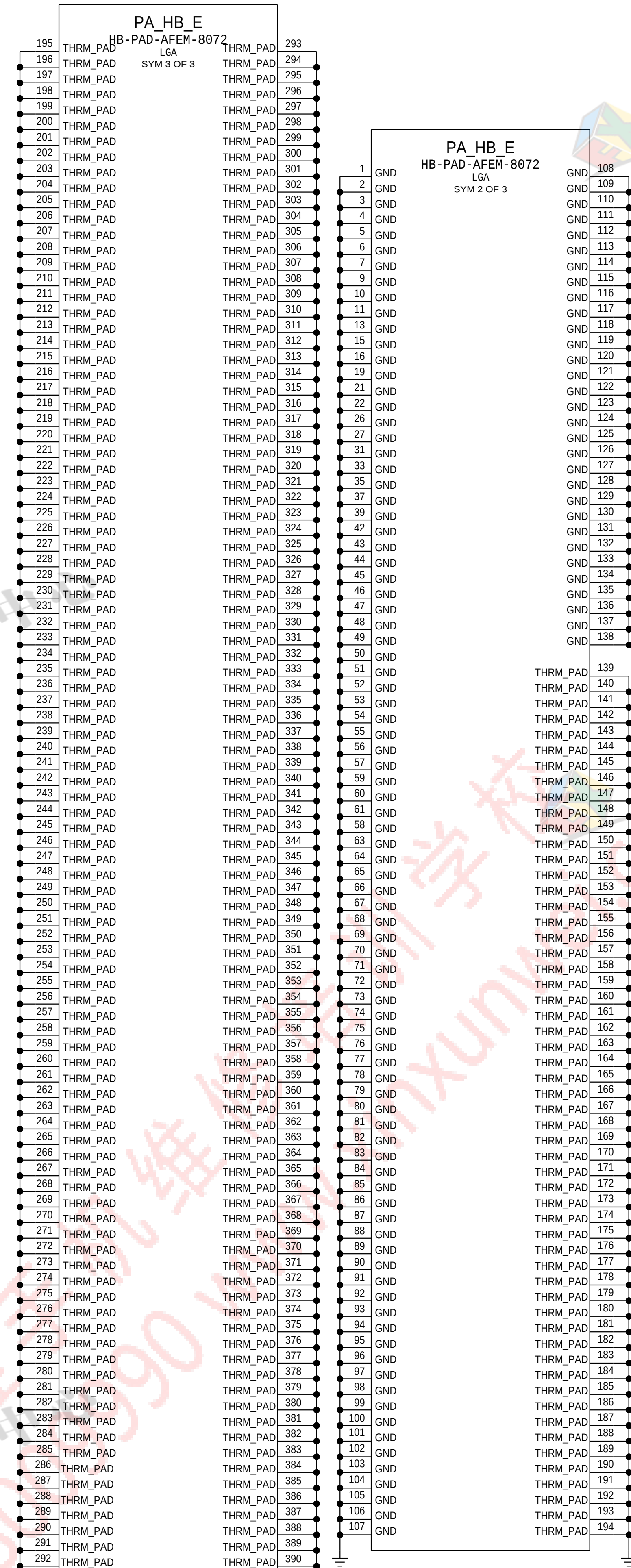


TABLE 2. (cont.)	D
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1



THESE ARE ONLY STUFFED FOR ROW CONFIG



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2G PA - ROW

D

D

C

C

B

B

A

A

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01247	1	2G PA	PA_UHB_E	CRITICAL	ROW
117S0161	1	MLB/26TX IN MCH2, 0 OHMS	R1302_E	CRITICAL	ROW
117S0161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	ROW
118S0724	1	MLB/2G MB OUT MCH1, 0 OHMS	R1305_E	CRITICAL	ROW
118S0724	1	UHB2 OUT/DRX BP MCH1, 0 OHMS	R1307_E	CRITICAL	ROW
152S2053	1	4.7NH UHQ, 2G LB OUTPUT MCH2	R1309_E	CRITICAL	ROW
131S0214	1	VBAT DECOUPLING,18PF	C1311_E	CRITICAL	ROW
138S0692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	ROW
117S0161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	ROW
103S00089	1	50 OHM TERMINATION FOR ROW ONLY	C1319_E	CRITICAL	ROW

8

7

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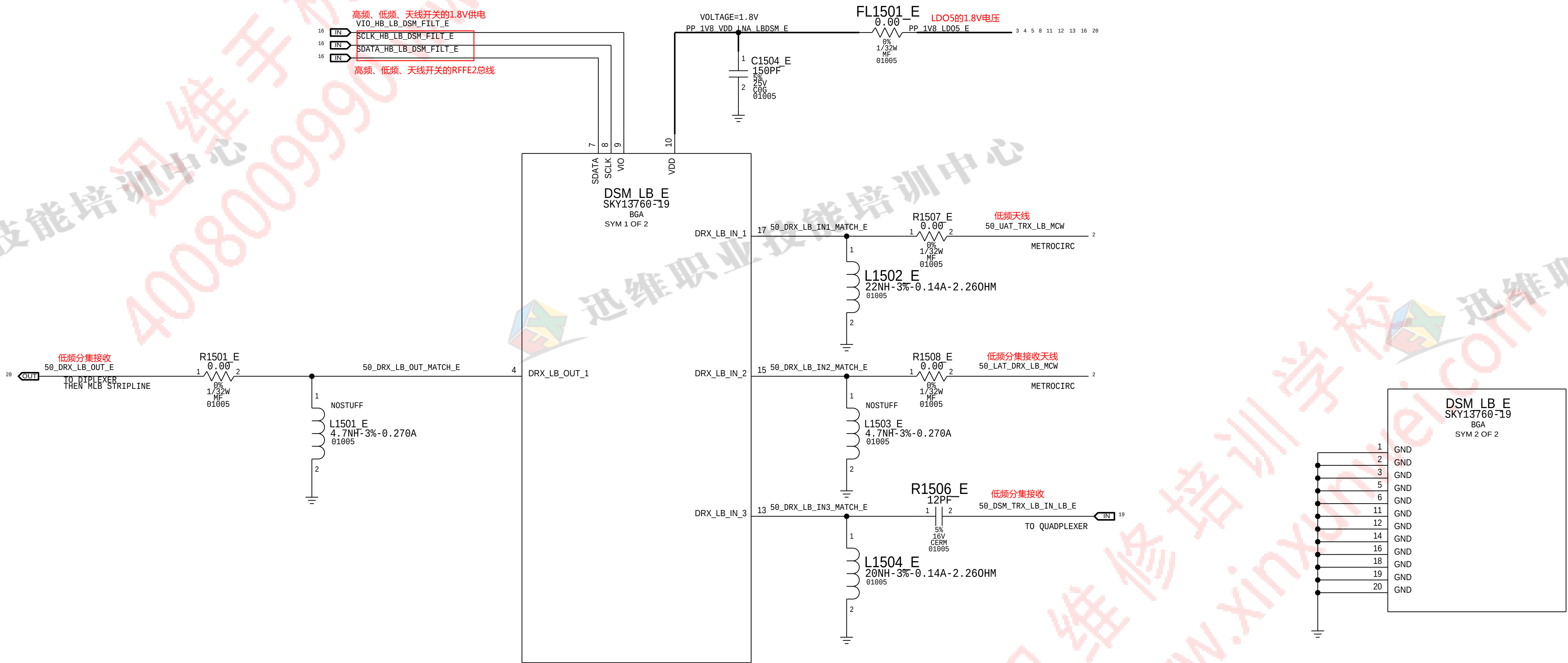
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3

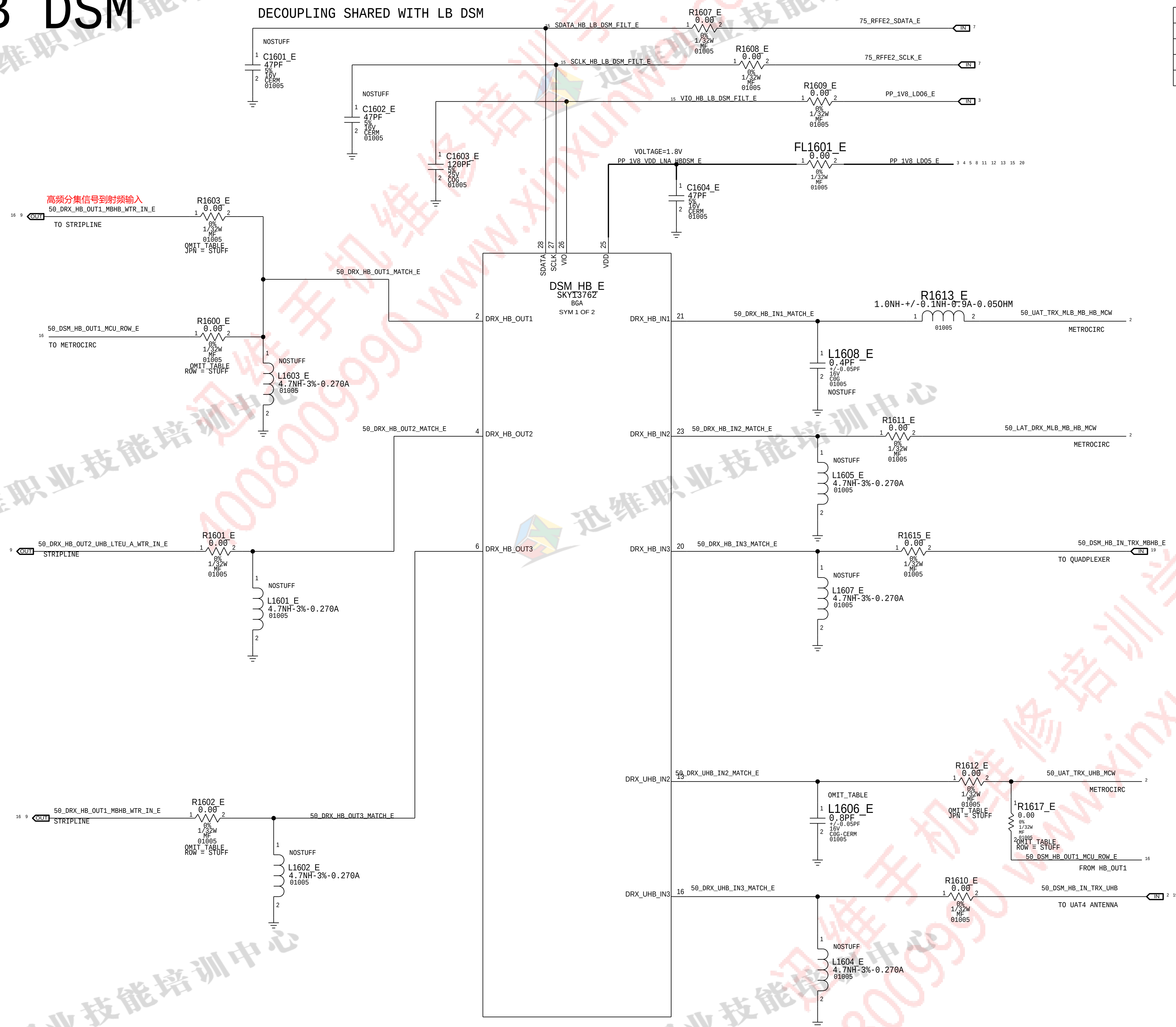
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1

LB DSM

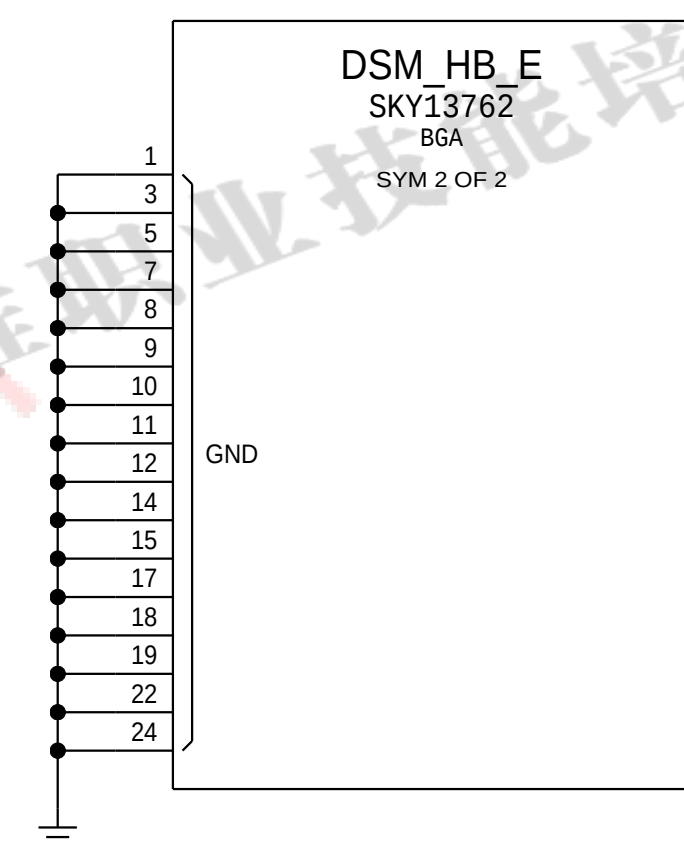


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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
11759161	1	01005 0R	R1600_E	CRITICAL	ROW
11759161	1	01005 0R	R1602_E	CRITICAL	ROW
11759161	1	01005 0R	R1617_E	CRITICAL	ROW
13159639	1	01005 0.8PF	L1606_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R1603_E	CRITICAL	JPN
152S1985	1	01005 1.7NH SHQ	R1612_E	CRITICAL	JPN
131S0616	1	01005 0.6PF	L1606_E	CRITICAL	JPN



COUPLER2

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S9724	1	0201 0R	R1810_E	CRITICAL	ROW
131S90142	1	150PF, DC BLOCK FOR GSM	R1813_E	CRITICAL	ROW
118S9724	1	0201 0R	R1814_E	CRITICAL	ROW
131S90901	1	0.1PF, CPLR HB LAT MATCH	L1801_E	CRITICAL	ROW
152S2044	1	2.2NH, CPLR HB LAT MATCH	R1807_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S9724	1	0201 0R	R1809_E	CRITICAL	JPN
152S2007	1	8.2NH 2G LB MATCH	R1812_E	CRITICAL	JPN
131S9431	1	0.2PF, CPLR HB LAT MATCH	L1801_E	CRITICAL	JPN
152S90955	1	2.6NH, CPLR HB LAT MATCH	R1807_E	CRITICAL	JPN
131S9249	1	2.2PF, 2G LB MATCH	C1810_E	CRITICAL	JPN

D

D

C

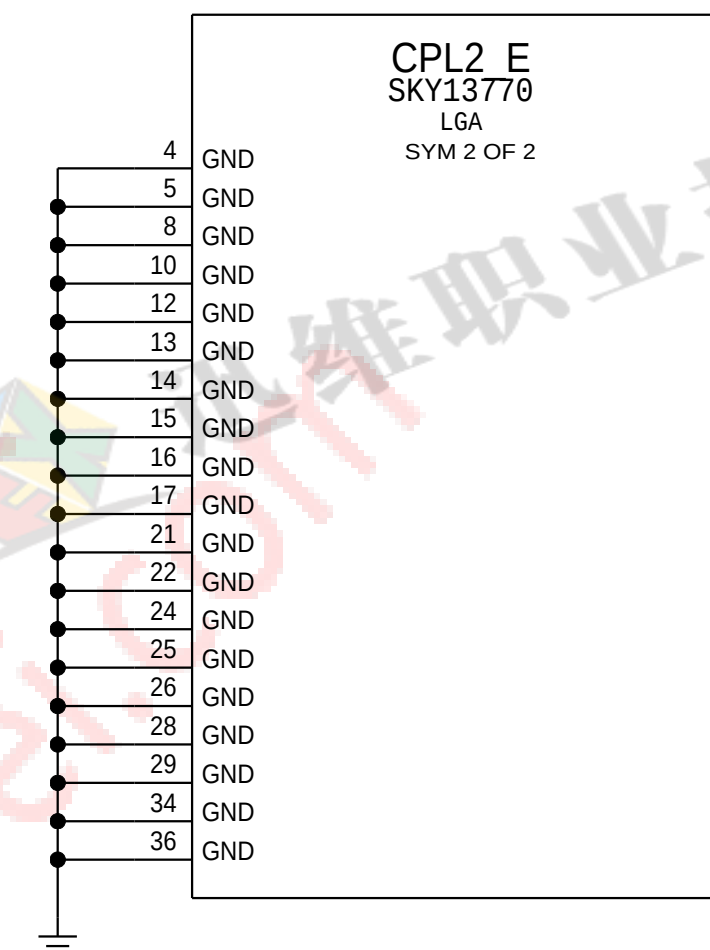
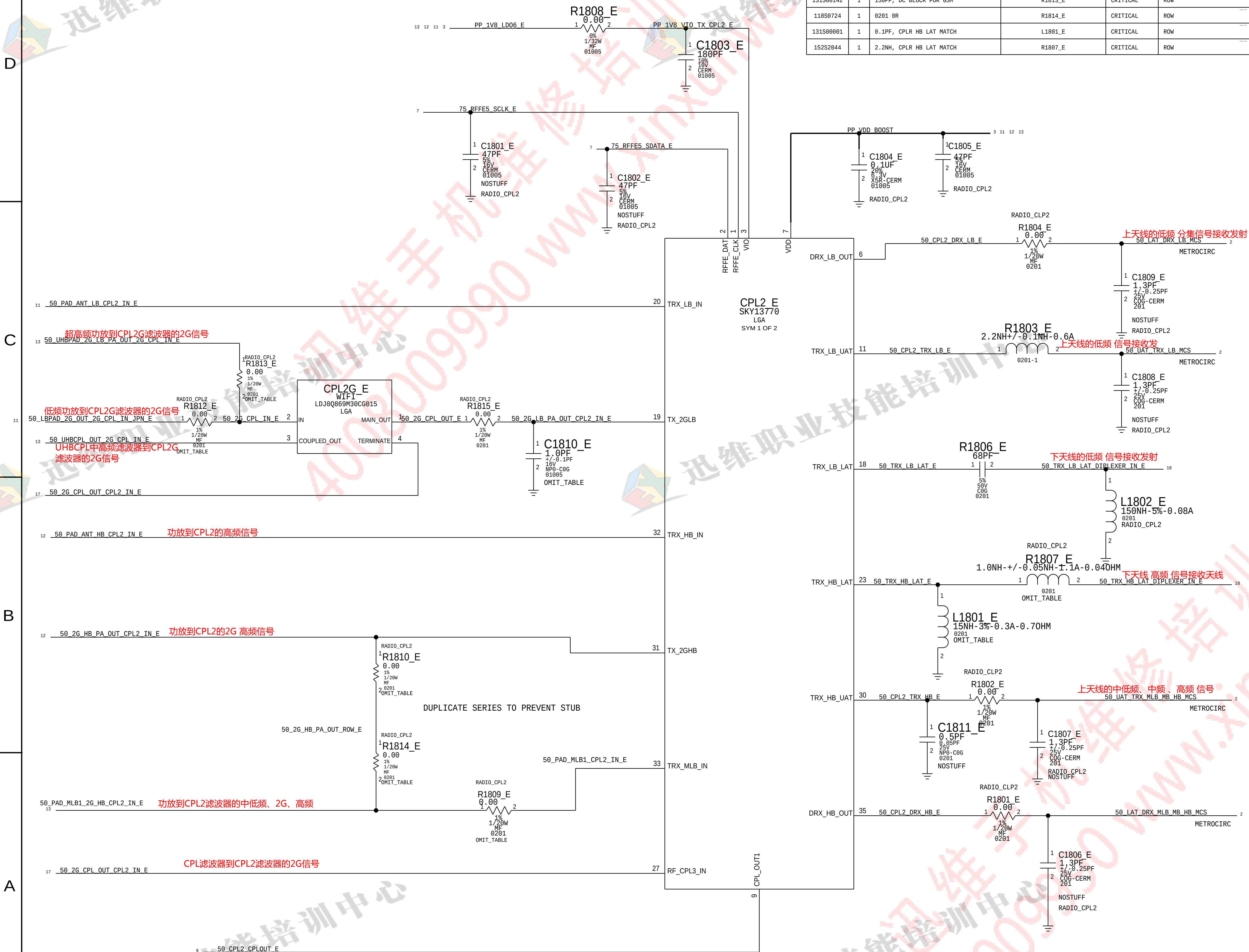
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B

B

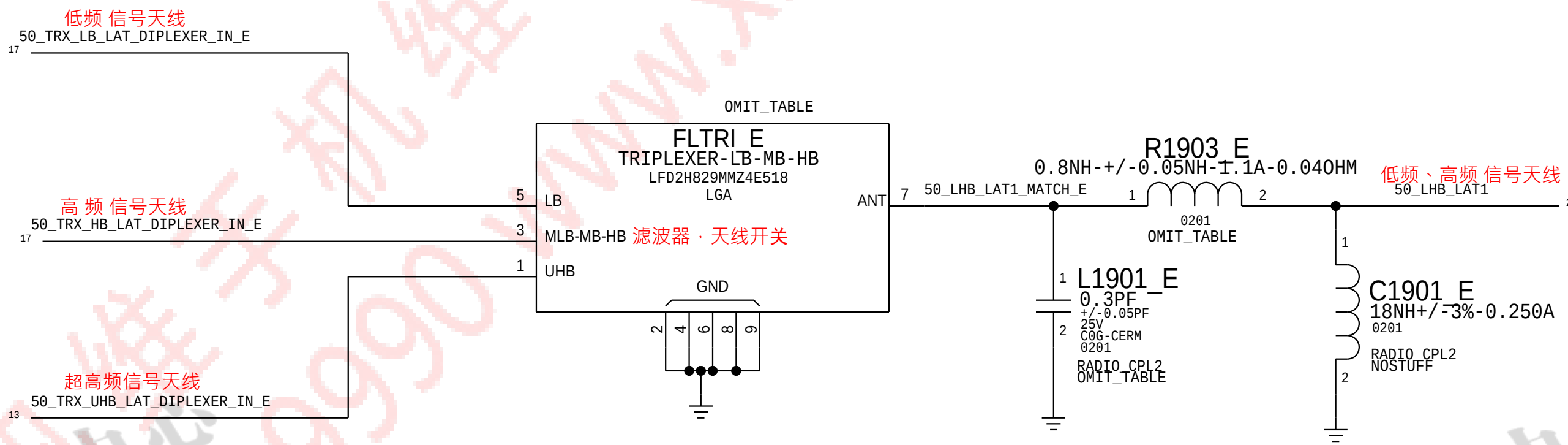
A

A



LOWER ANTENNA

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

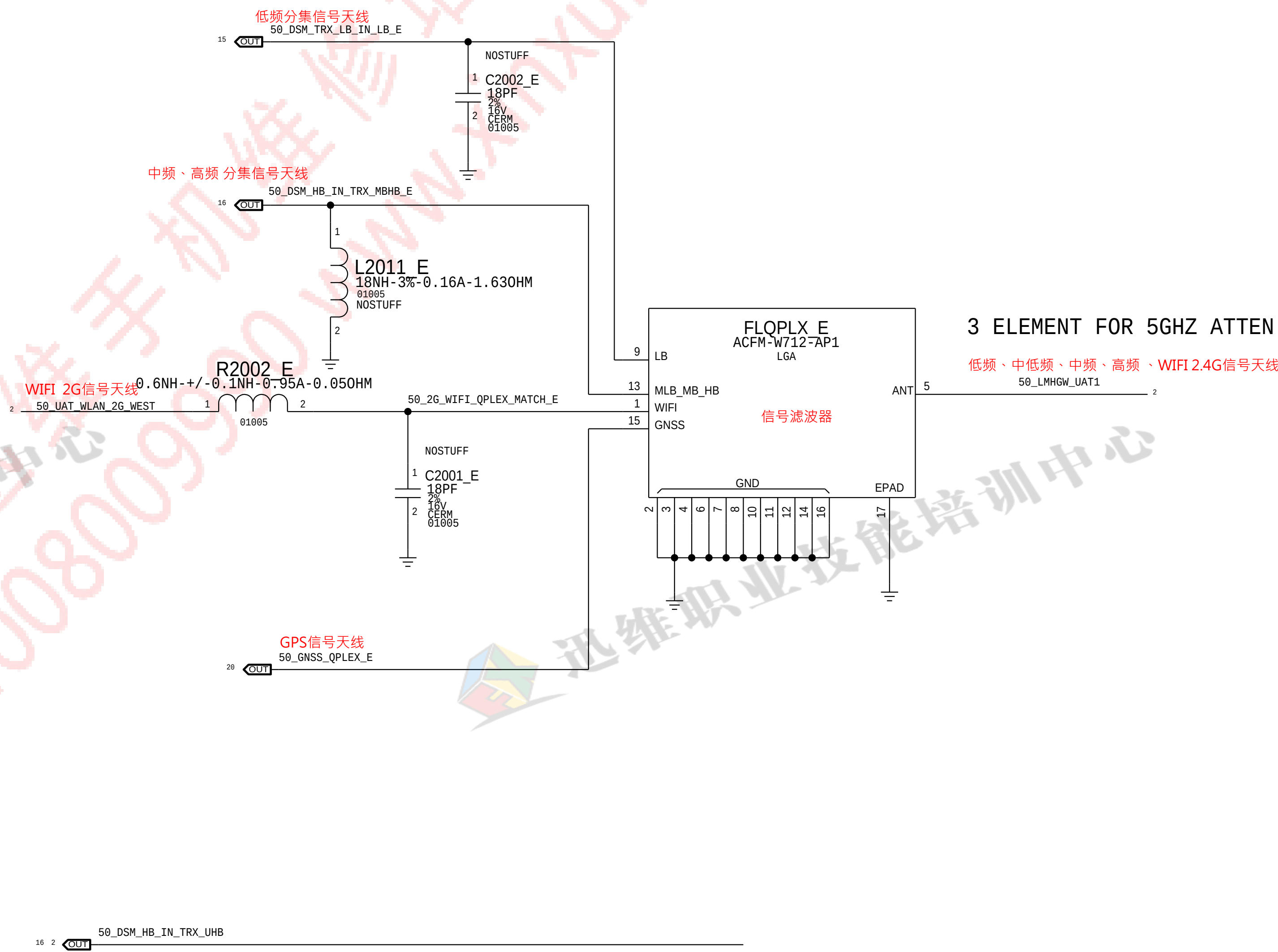


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
155S00298	1	TRIPLEXER, L+MLBMHB+UHB	FLTRI_E	CRITICAL	JPN
155S00297	1	DIPLEXER, L+MLBMHB	FLTRI_E	CRITICAL	ROW
152S00147	1	0.6NH UHQ, TRI ANT MATCH	R1903_E	CRITICAL	JPN
152S00151	1	0.8NH UHQ, TRI ANT MATCH	R1903_E	CRITICAL	ROW
131S00001	1	0.1PF, TRI ANT MATCH	L1901_E	CRITICAL	ROW

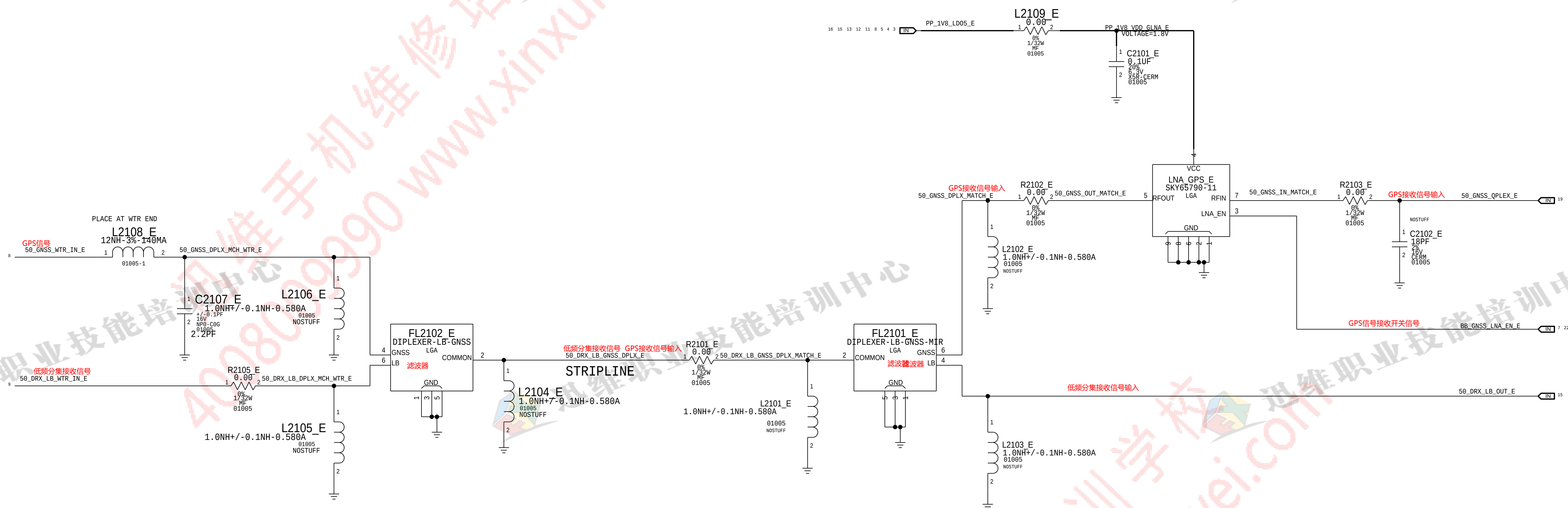


UAT & LB/UHB DIPLEXER

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

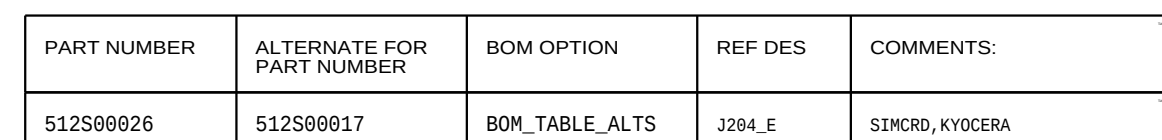


GPS & GPS/LB DIPLEXER



CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST



MLB 516S1185
FLEX 618S1184

J_DEBUG_E
20-5857-036-001-829
F-ST-SM

PP_VDD_MAIN

PMU_TO_BB_VBUS_DETECT

主板上诊断接口，在SIM卡槽下面，工厂测试用的

AP_TO_BB_RESET_L

BB_JTAG_TCK_E

BB_JTAG_TMS_E

PP_VDD_BOOST

90_USB_BB_DATA_P

90_USB_BB_DATA_N

PP_UTIM1_LD011_E

BB_SIM1_RST_E

BB_SIM1_CLK_E

BB_SIM1_DATA_E

NFC_SWP

BB_SIM1_DETECT_E

BB_JTAG_RST_L_E

UART_BB_TO_WLAN_COEX

UART_WLAN_TO_BB_COEX

NOSTUFF

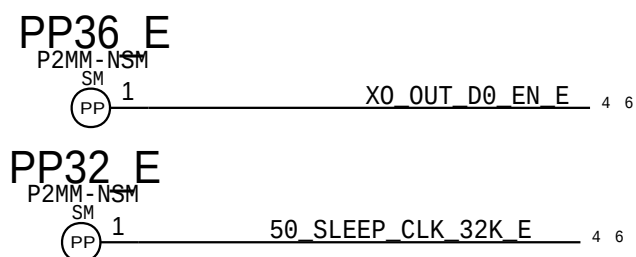
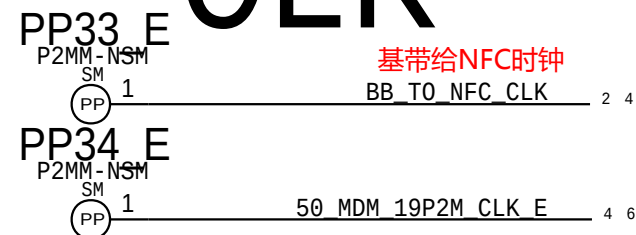


BB PROBE POINTS

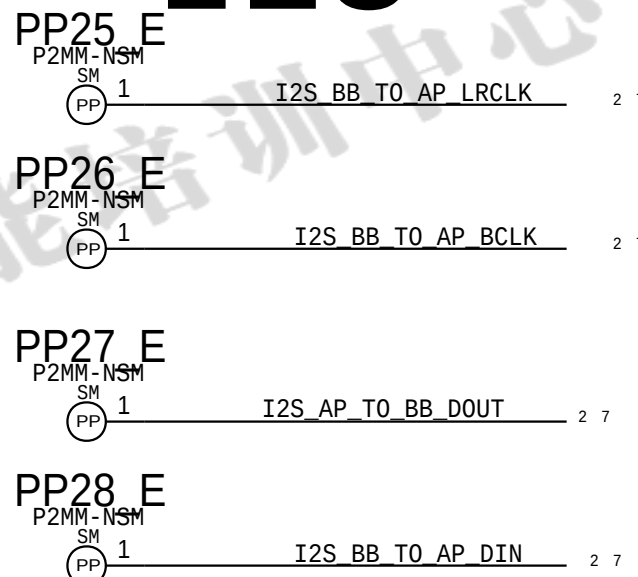
QLINK

BASEBAND

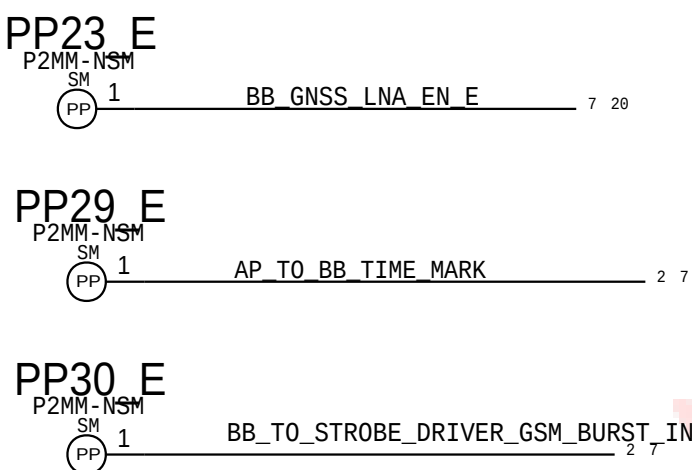
CLK



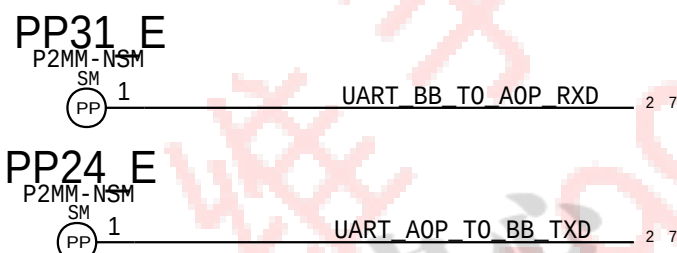
I2S



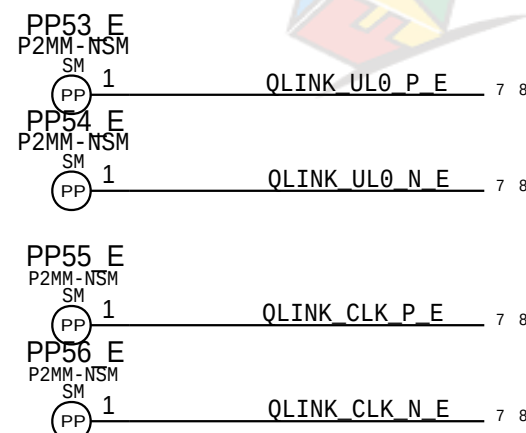
CTRL



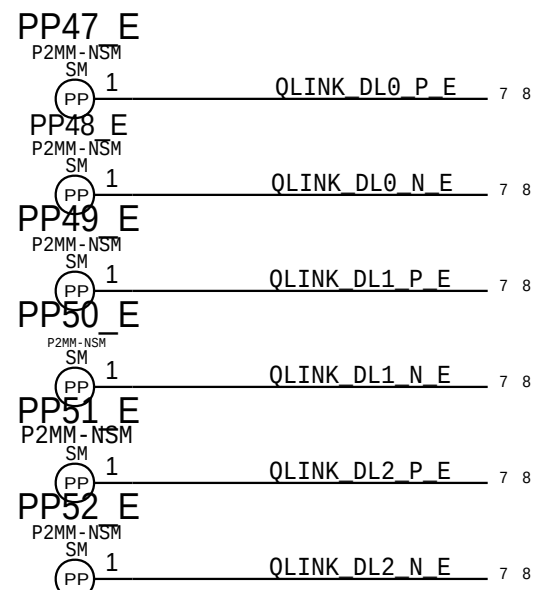
UART



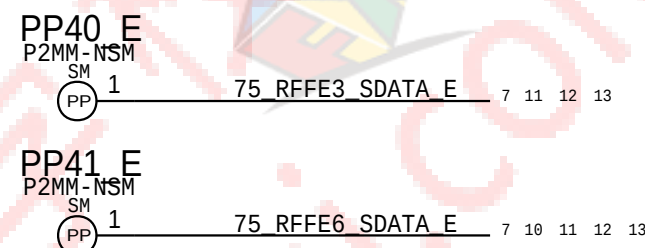
LAYOUT: VIA STUB AT WTR PINS



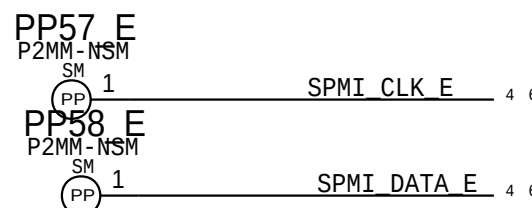
LAYOUT: VIA STUB AT BB PINS



RFFE

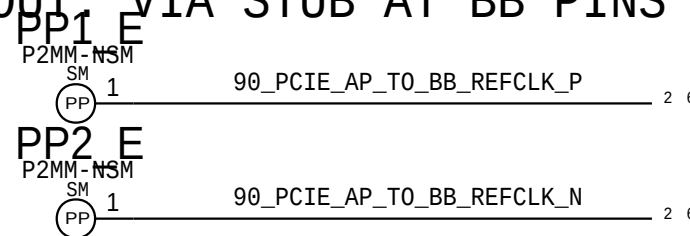


SPMI

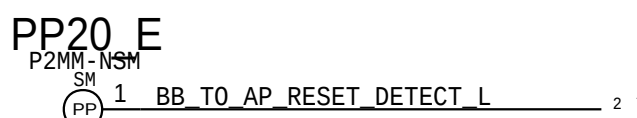


PCIE

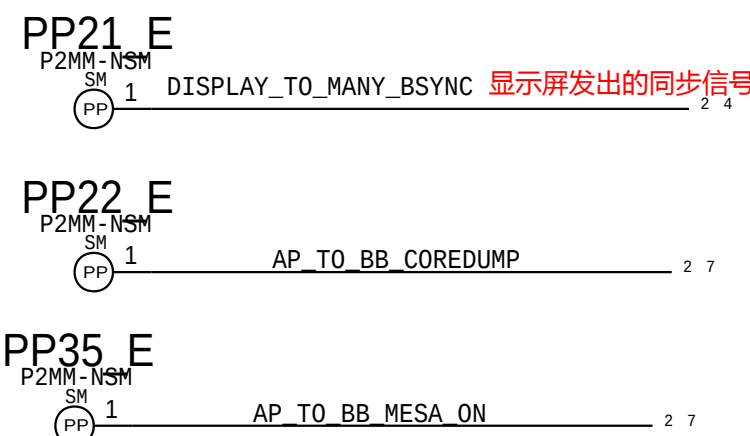
LAYOUT: VIA STUB AT BB PINS



PWR SEQ



OTHERS



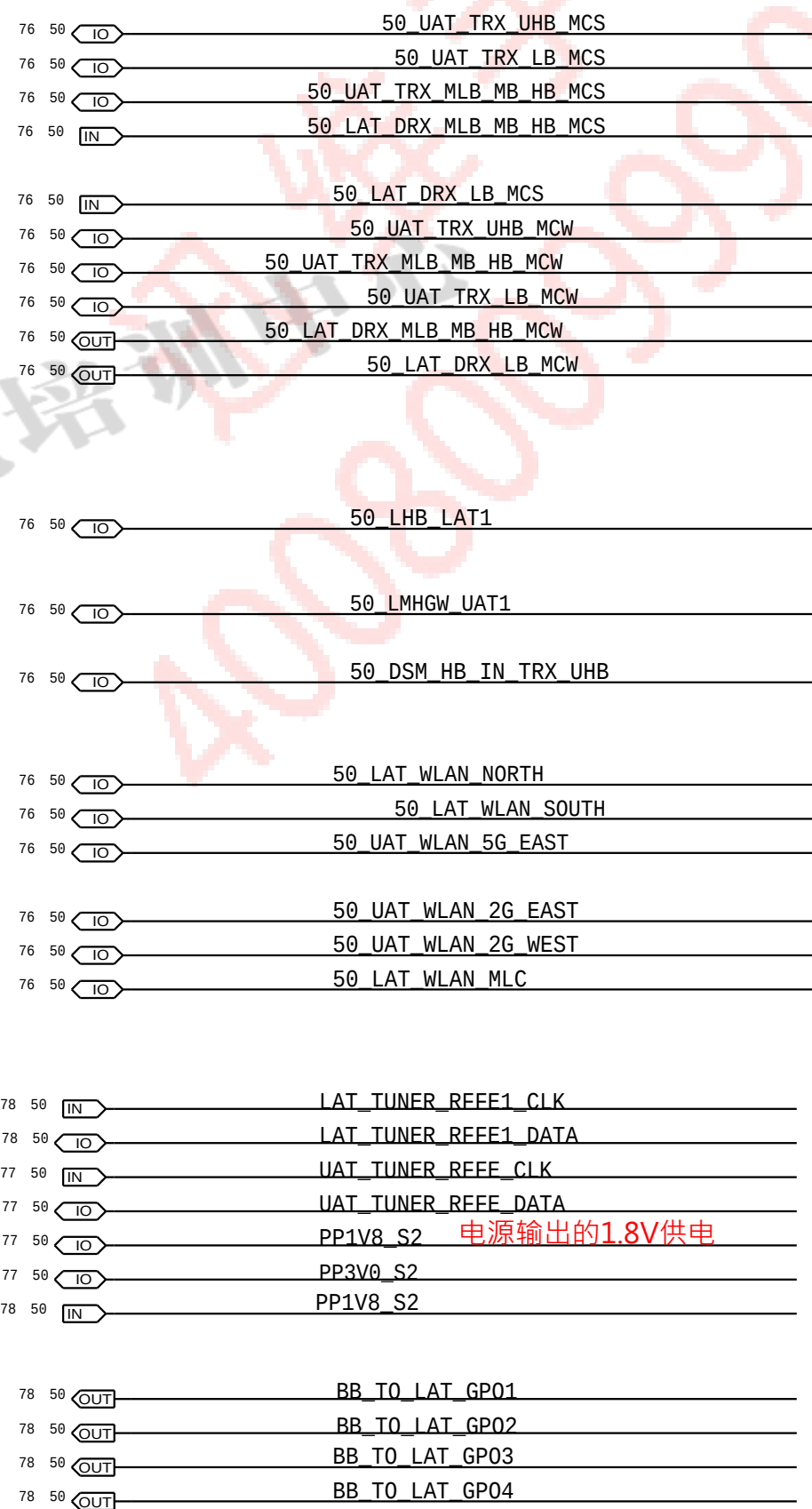


REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
10	0008927012	ENGINEERING RELEASED		2017-06-06

D21 RADIO_MLB_FF

5/22/2017

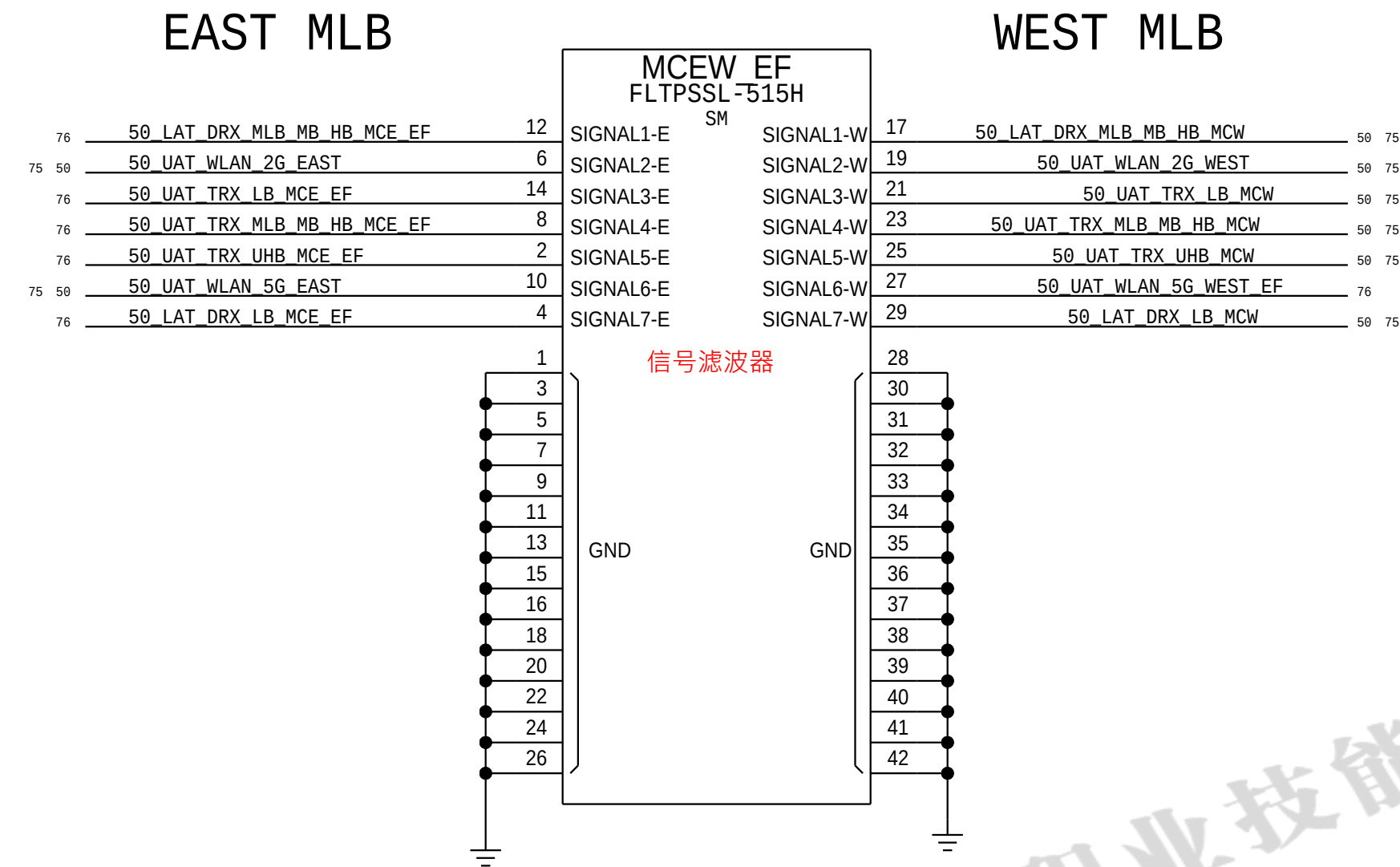
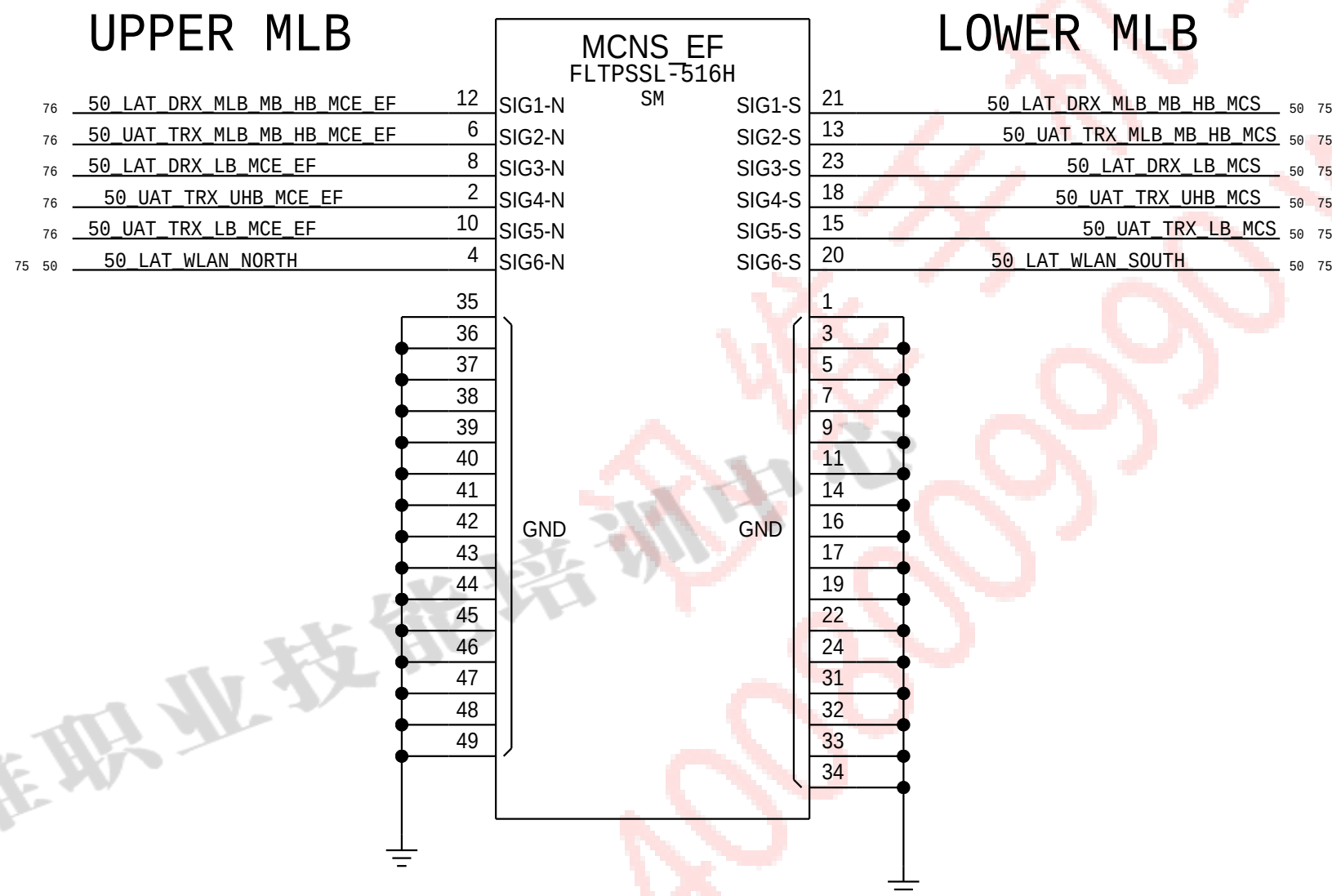
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.



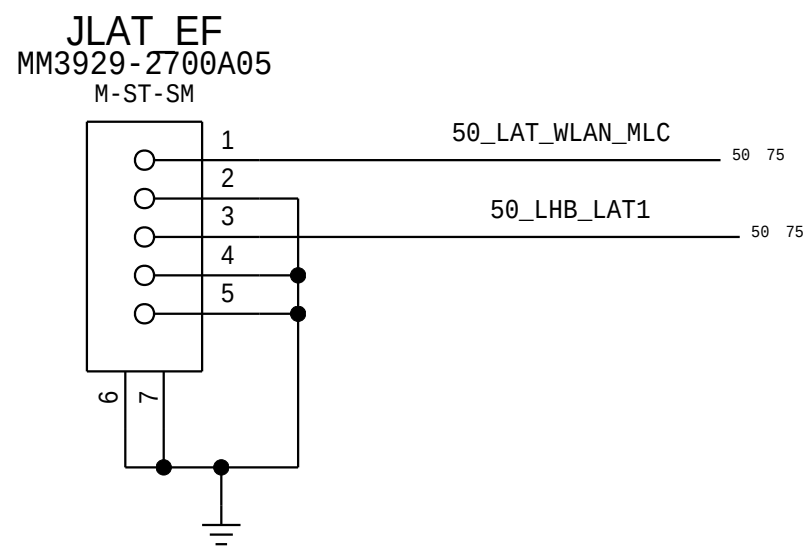
METROCIRC

NORTH-SOUTH METROCIRC 339S00297

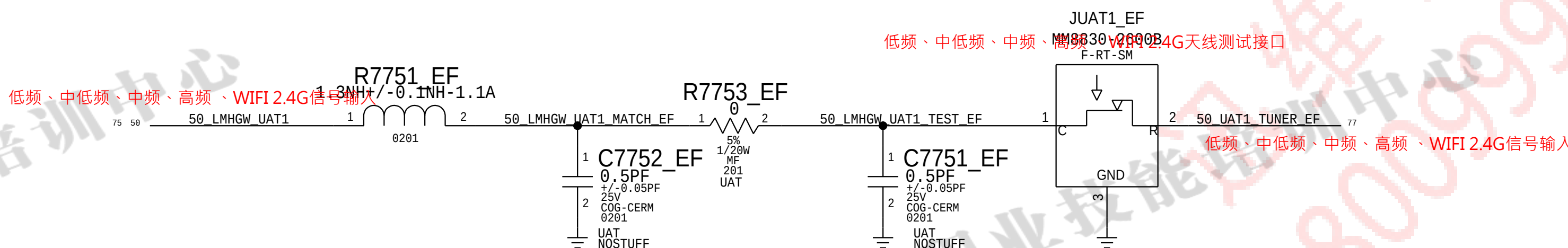
EAST-WEST METROCIRC 339S00296



LAT MLC

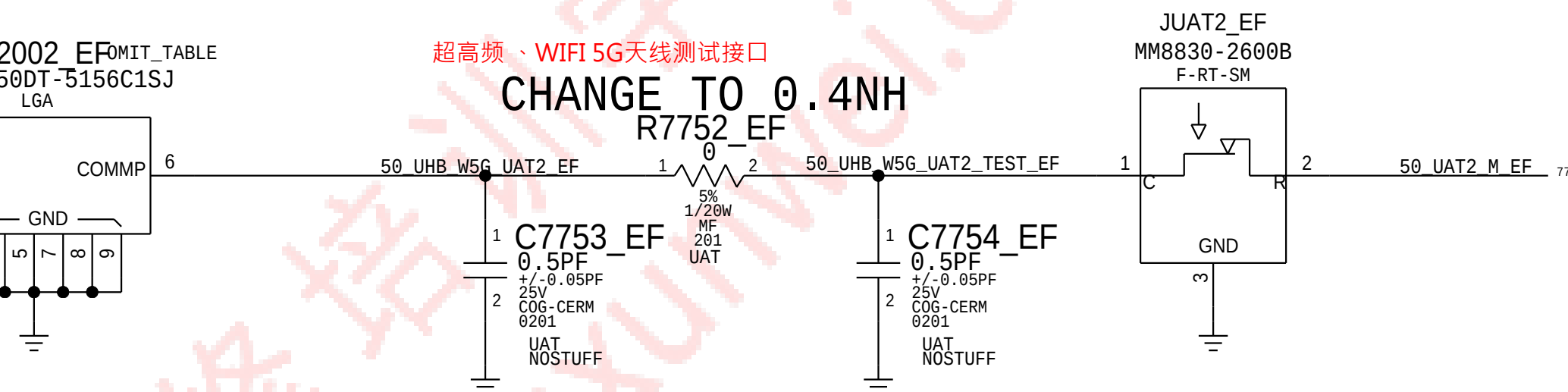


LB/MLB/MB/HB WIFI 2.4GHZ TEST



UHB/WIFI 5GHZ TEST

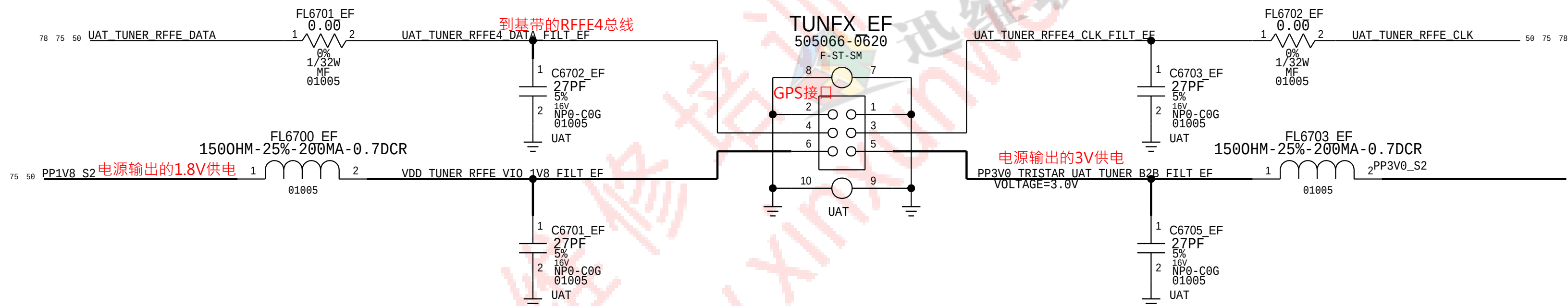
超高频、WIFI 5G天线测试接口
CHANGE TO 0.4NH



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
155S00274	1	B42/5G DIPLEXER	FL2002_EF	CRITICAL	JPN
155S00275	1	5G PASSTHROUGH	FL2002_EF	CRITICAL	ROW
131S00598	1	B42/5G DIPLEXER 5G_IN_MATCH	L7751_EF	CRITICAL	JPN
131S00385	1	5G PASSTHROUGH INPUT MATCH	L7751_EF	CRITICAL	ROW

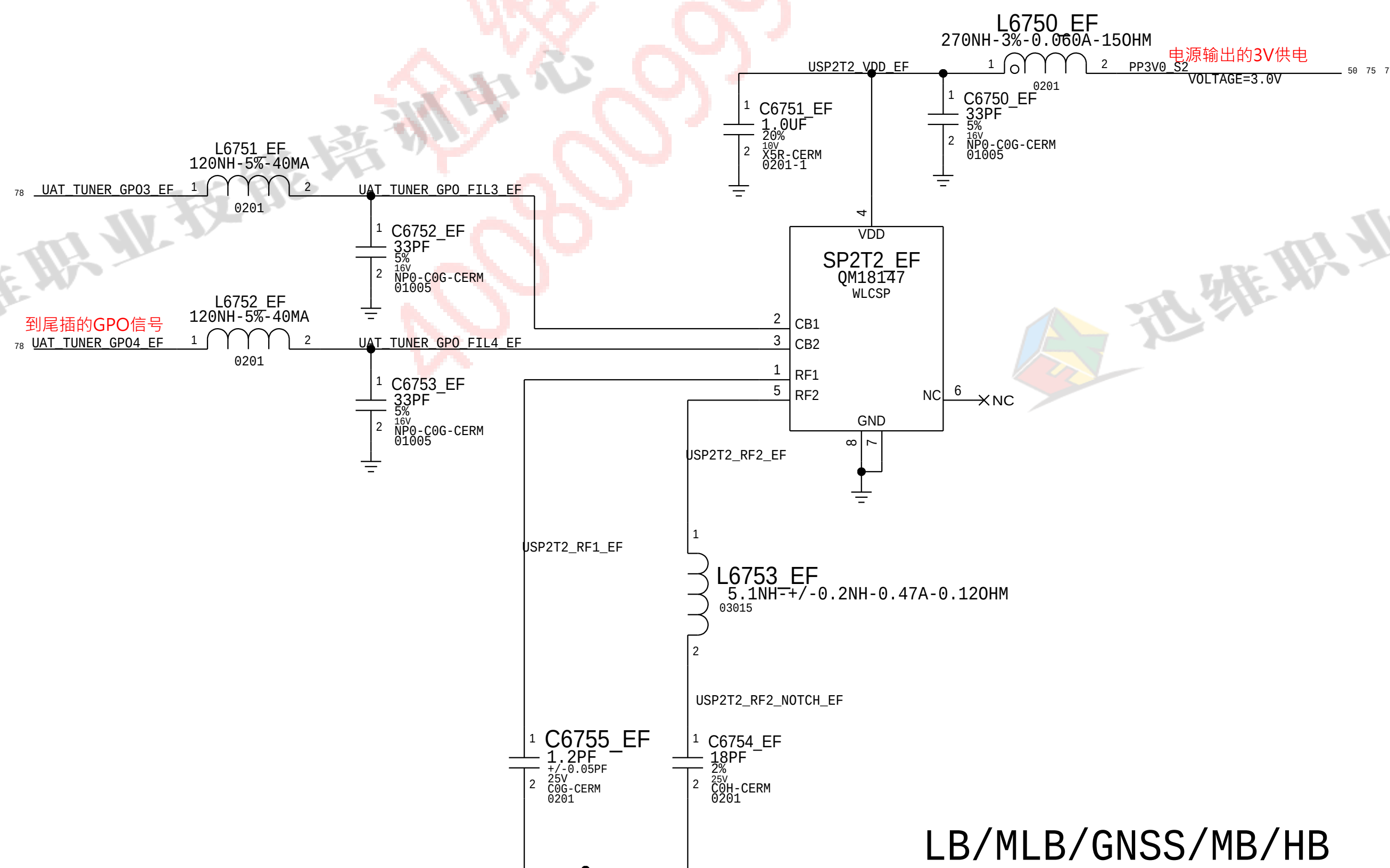
UAT TUNER FLEX

PLEASE CONTACT ANTENNA (MATT MOW)
FOR ANY COMPONENT CHANGE.

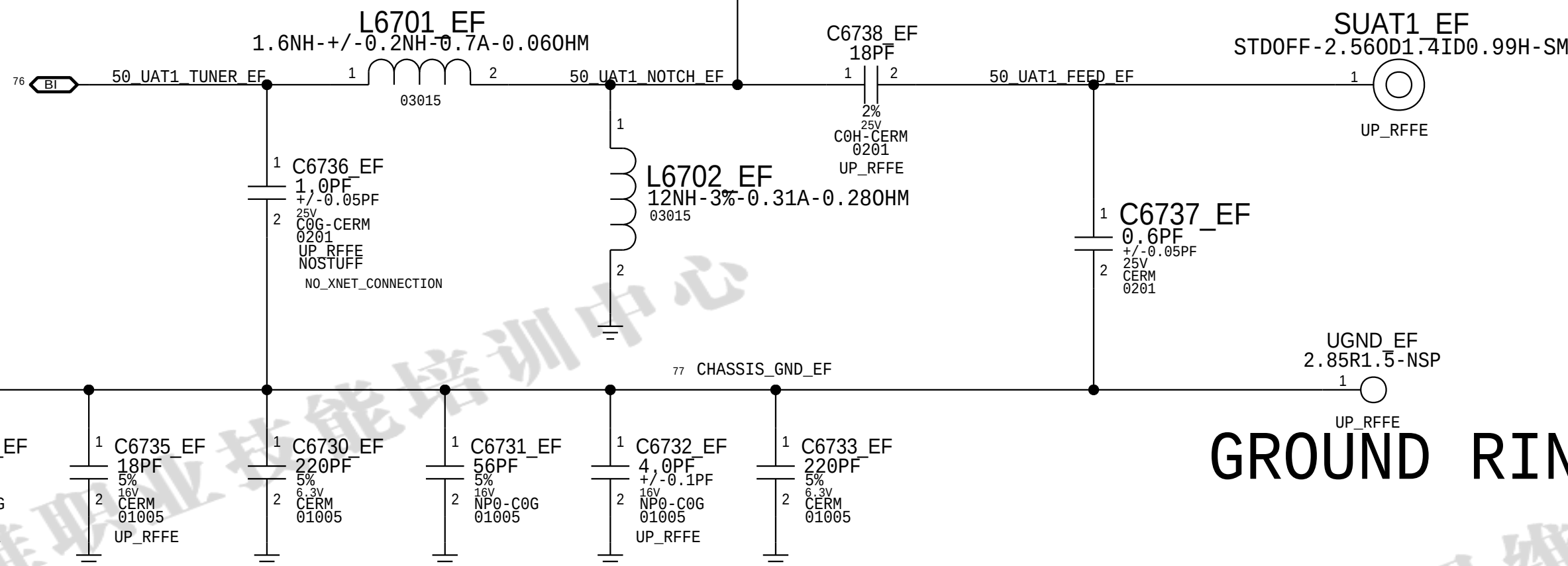


PINOUT IS ROTATED 180
COMPARED TO D20 & D201

UAT TUNER

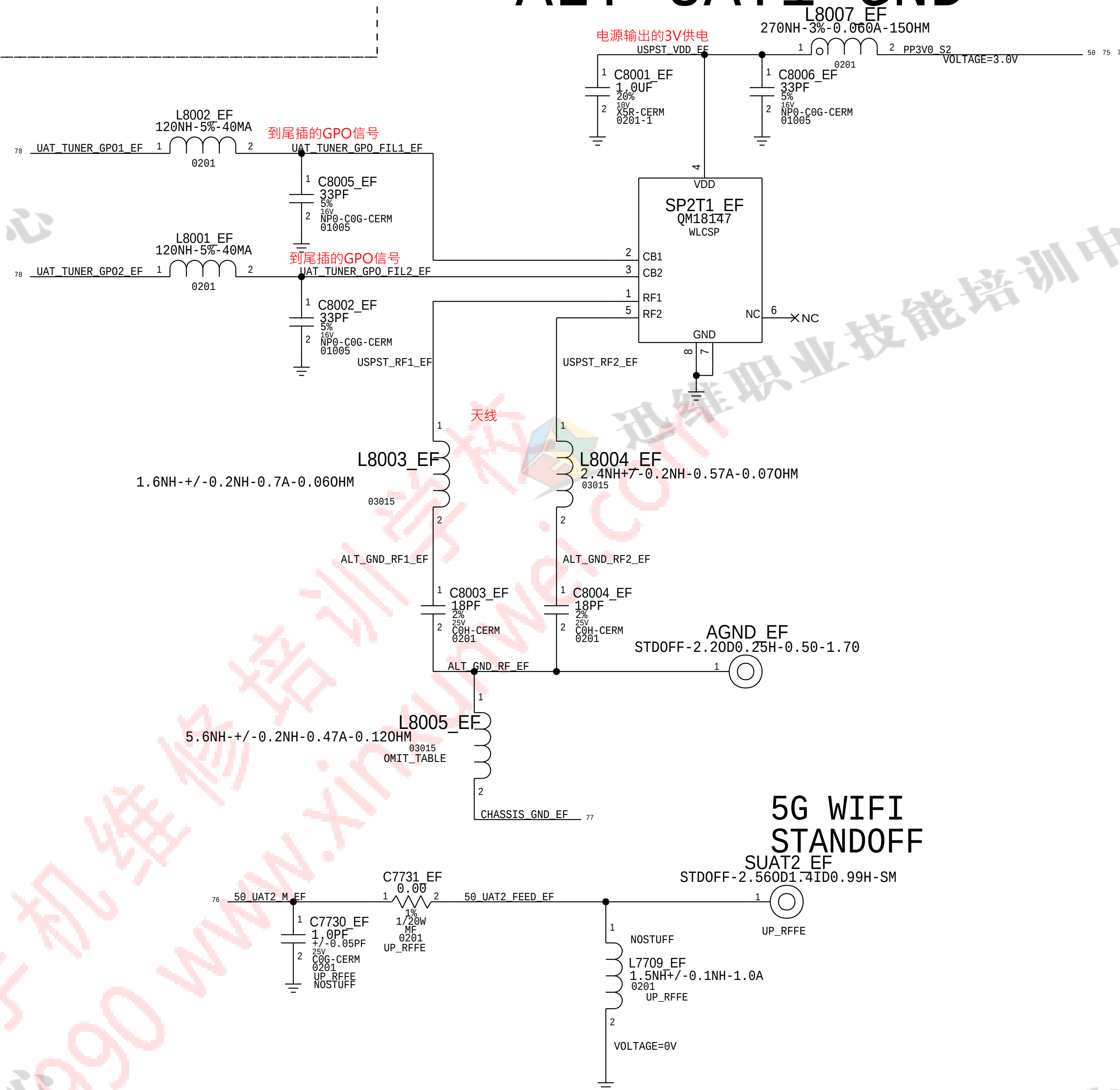


LB/MLB/GNSS/MB/HB
STANDOFF

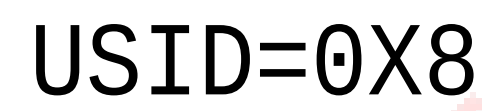


GROUND RING

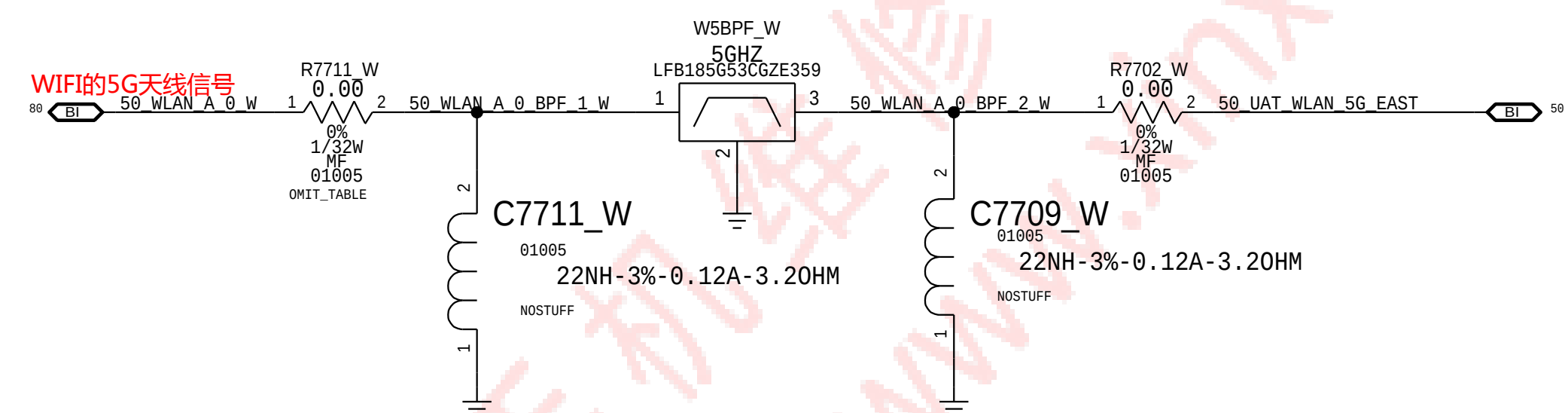
ALT UAT1 GND



5G WIFI
STANDOFF



WIFI UPPER ANTENNA FEED



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00498	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	D211
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	JPN
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	ROW